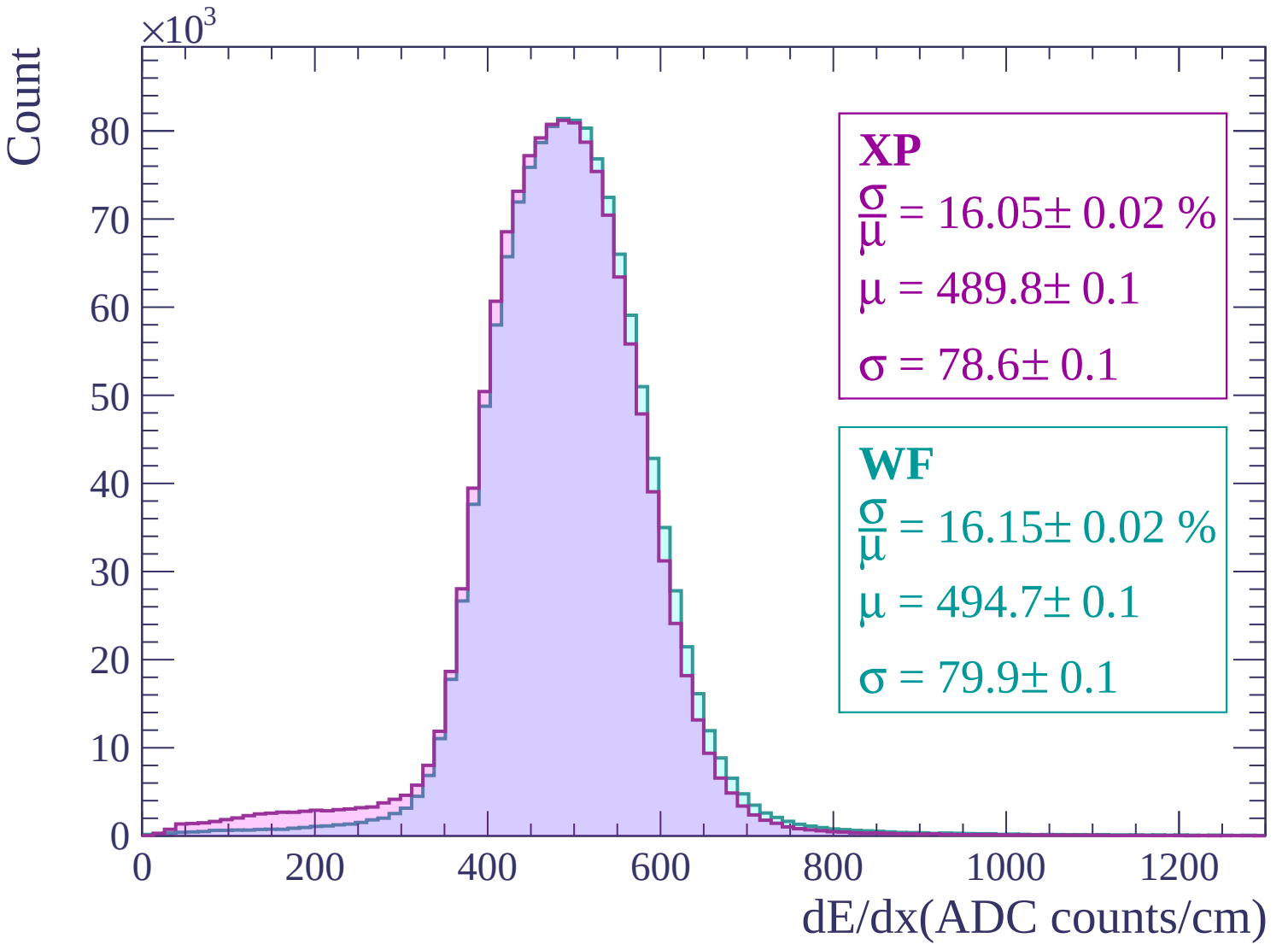
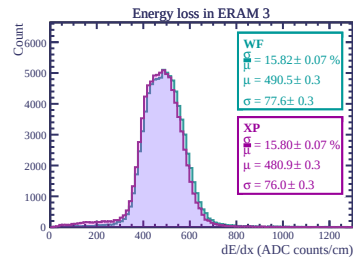
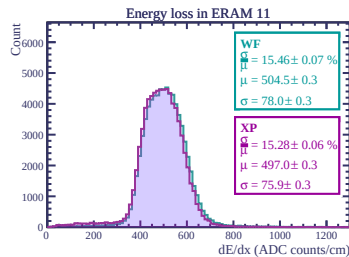
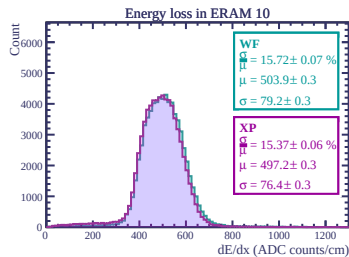
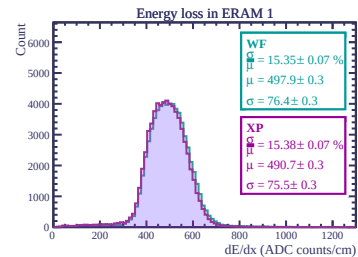
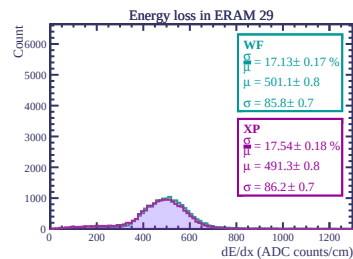
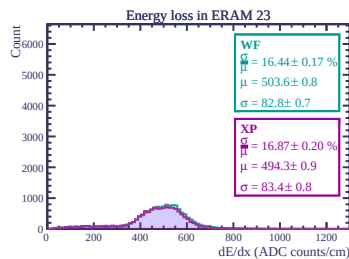
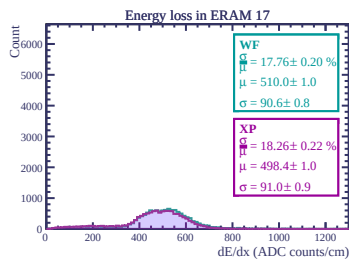
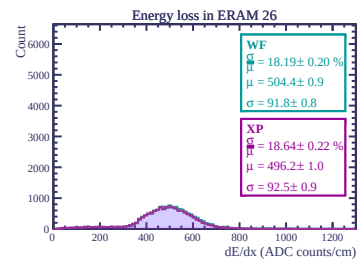
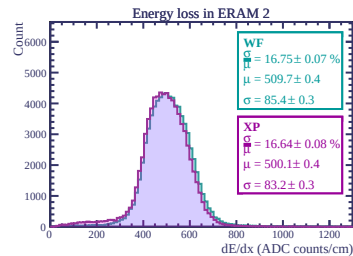
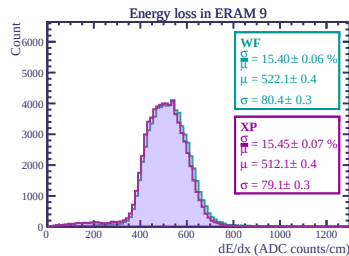
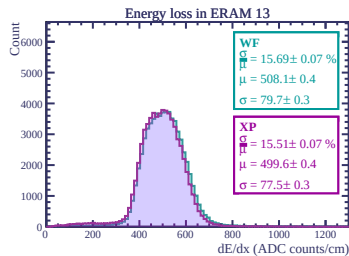
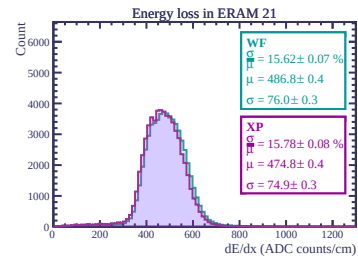
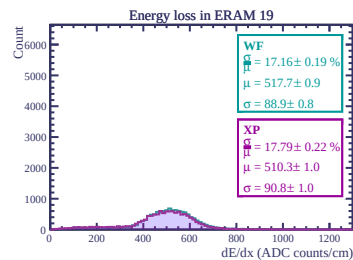
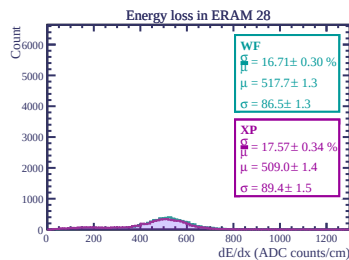
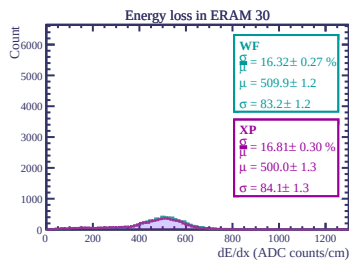
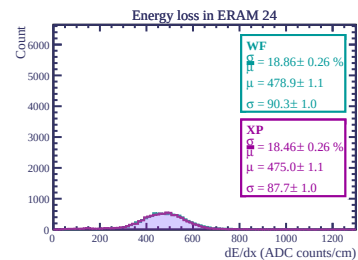
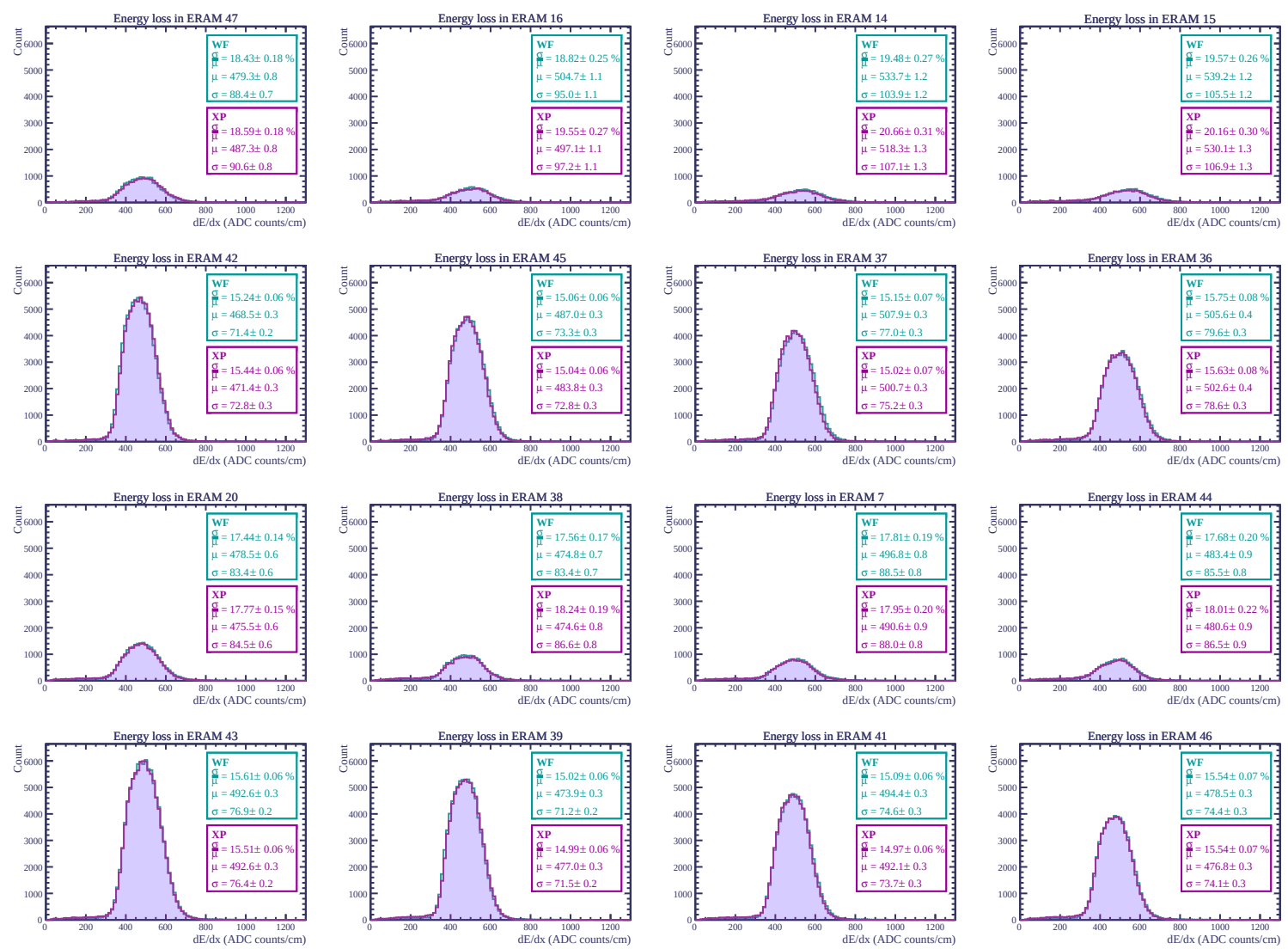
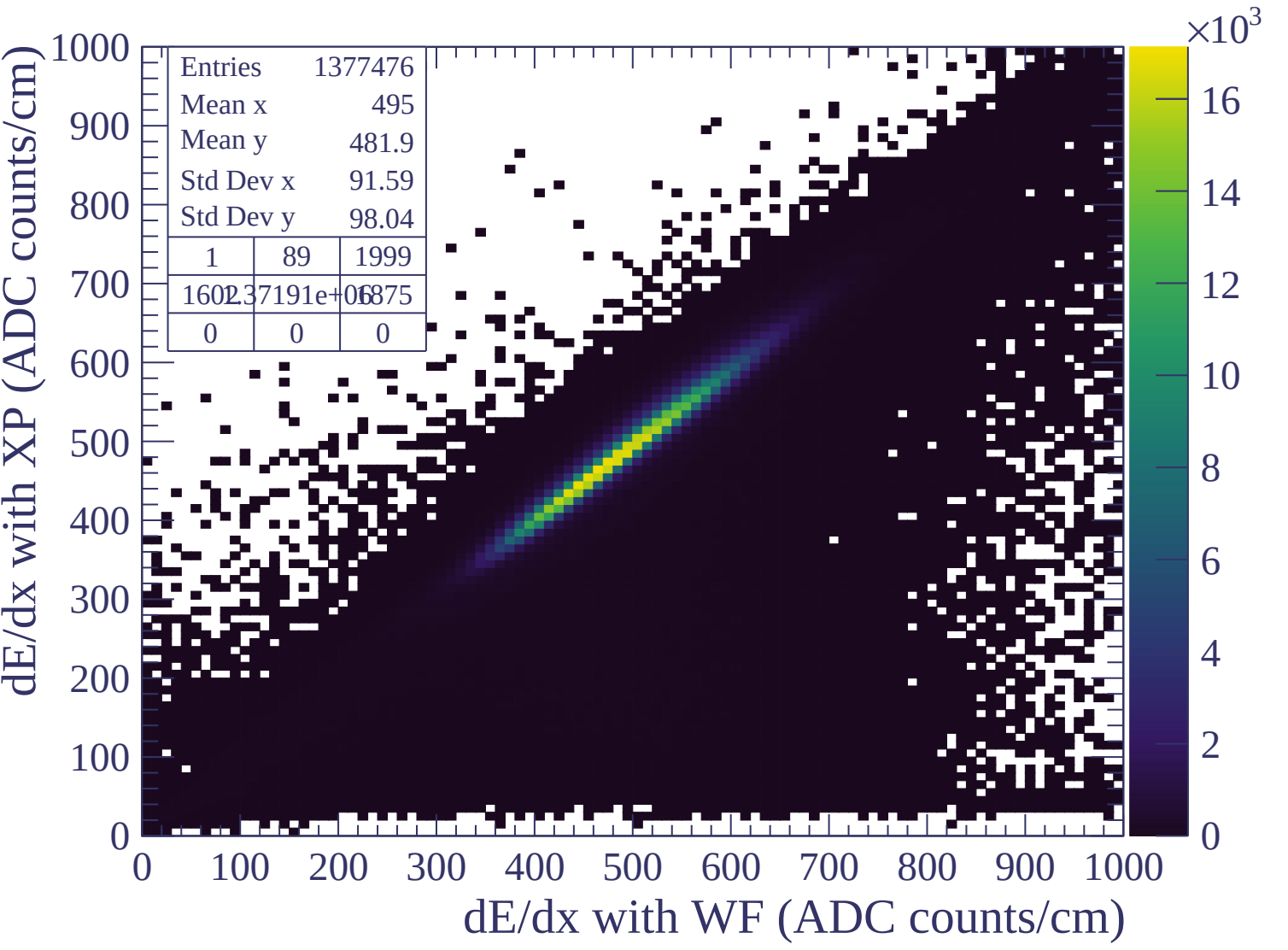


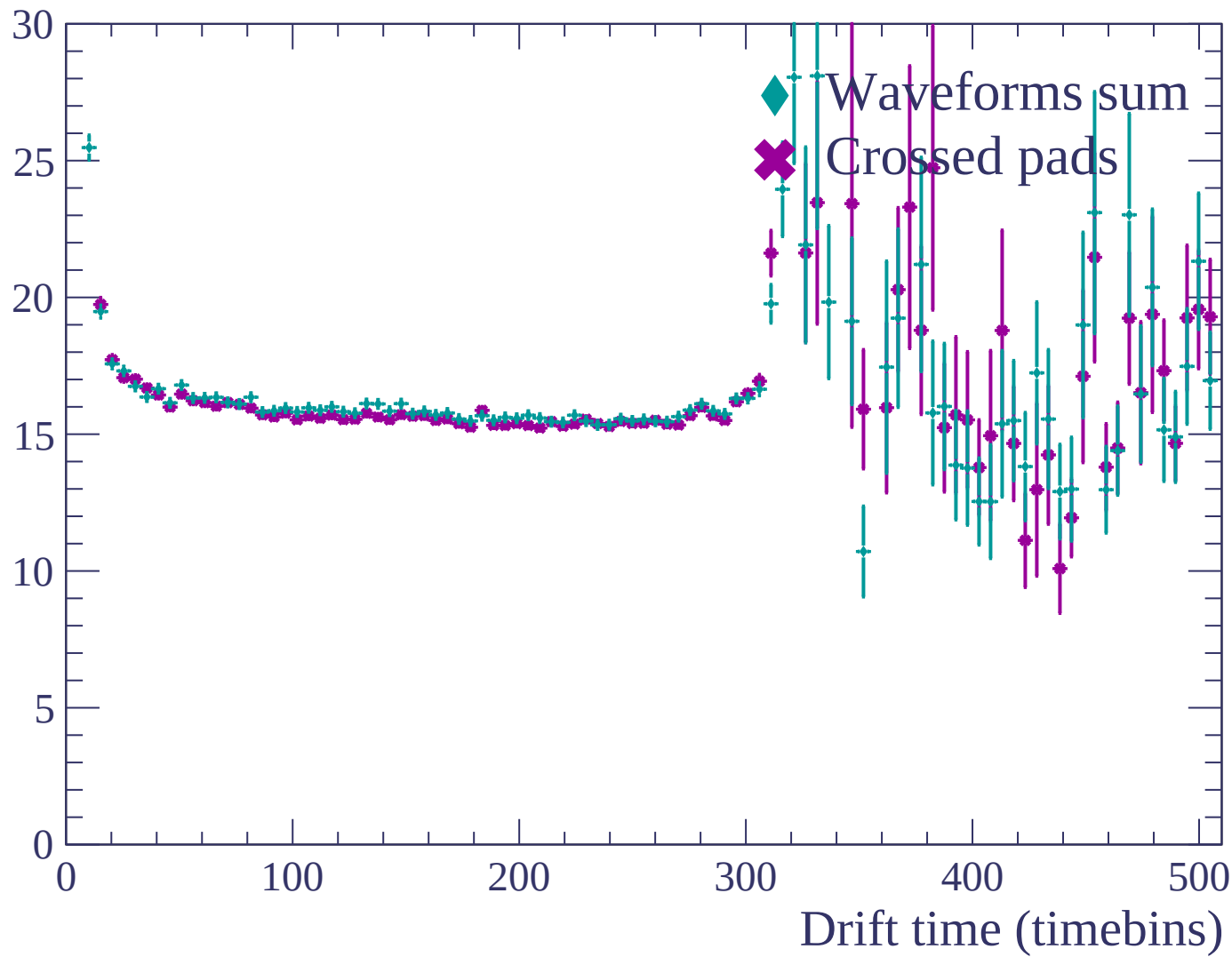


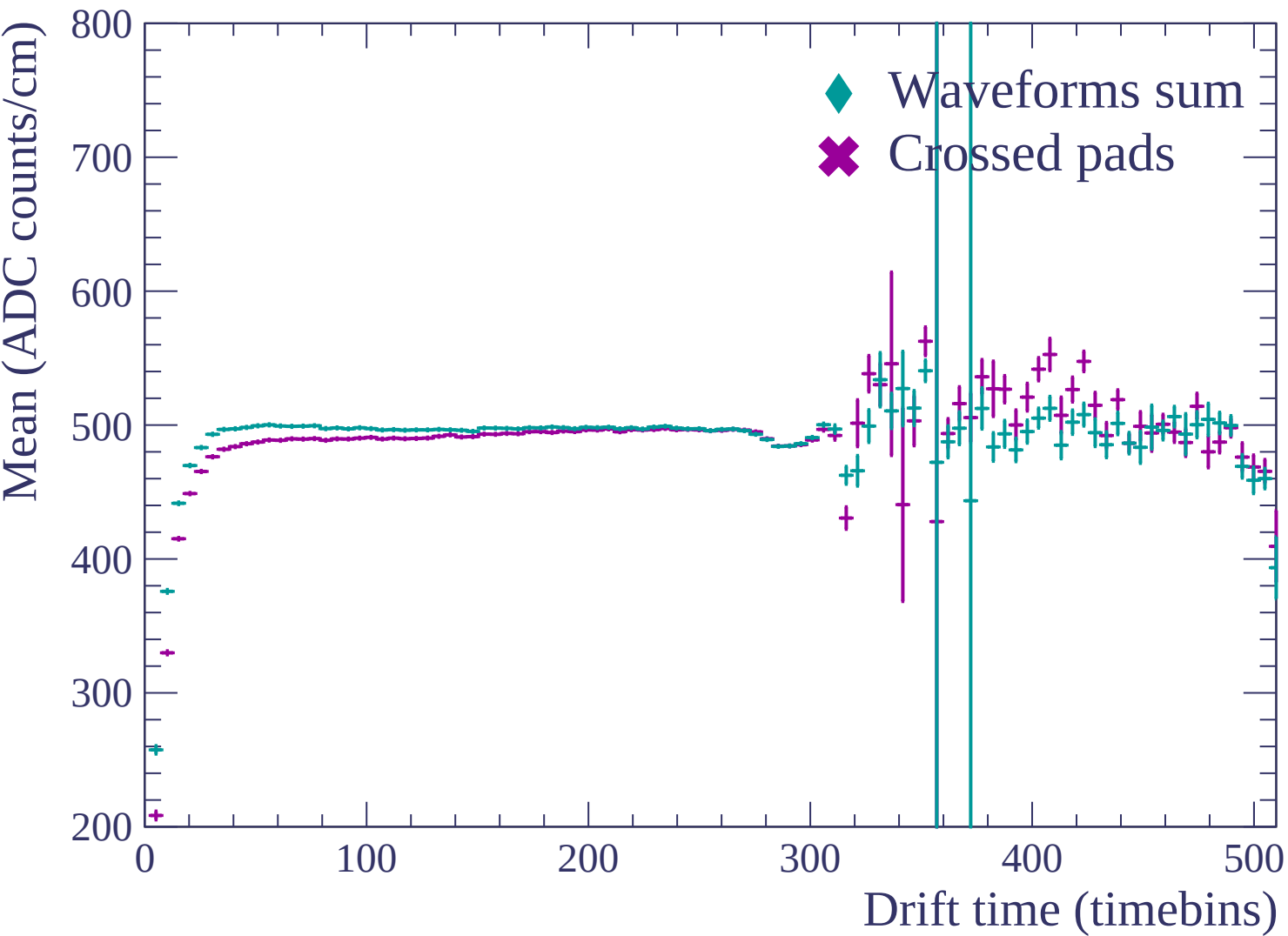


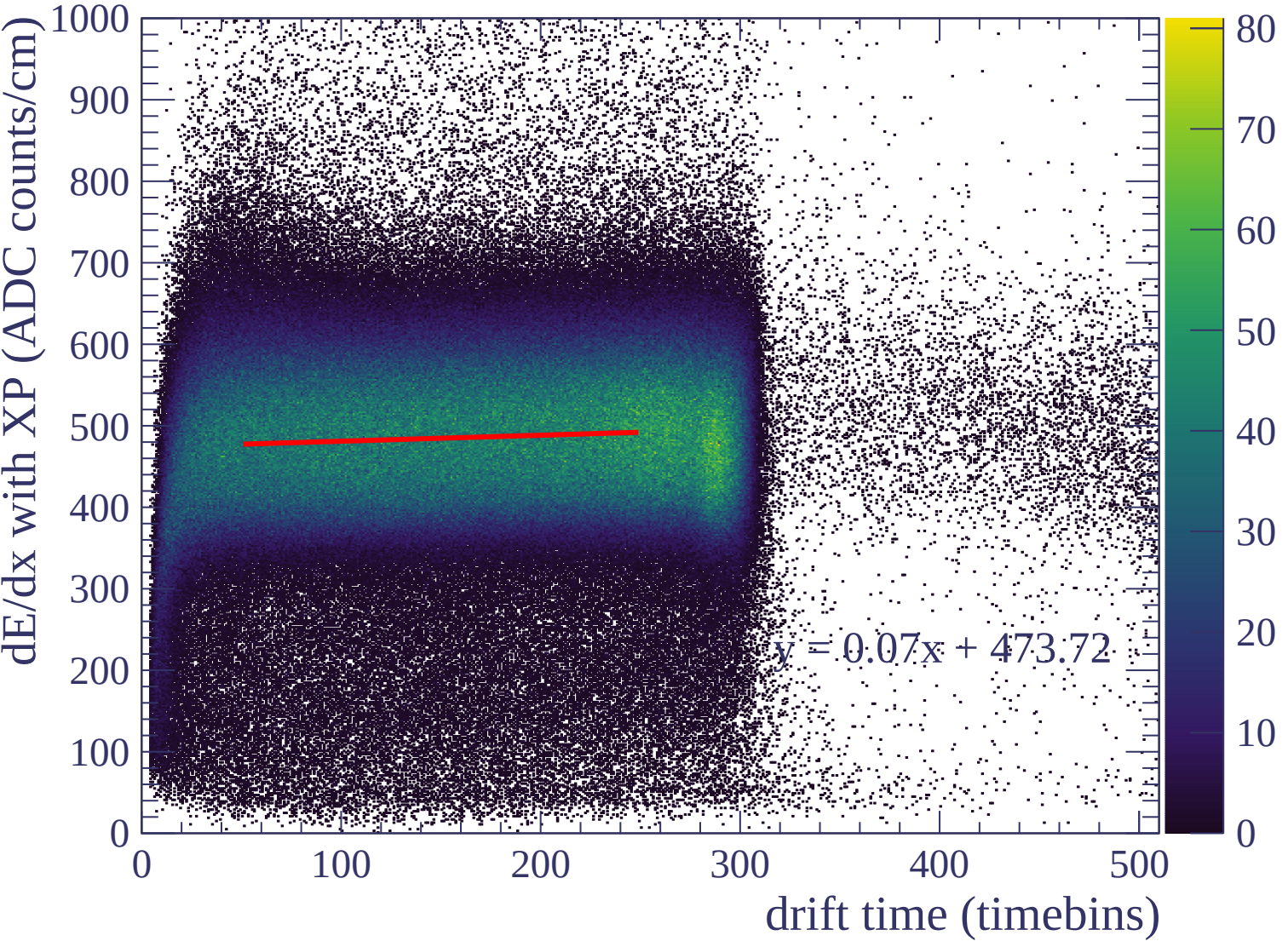


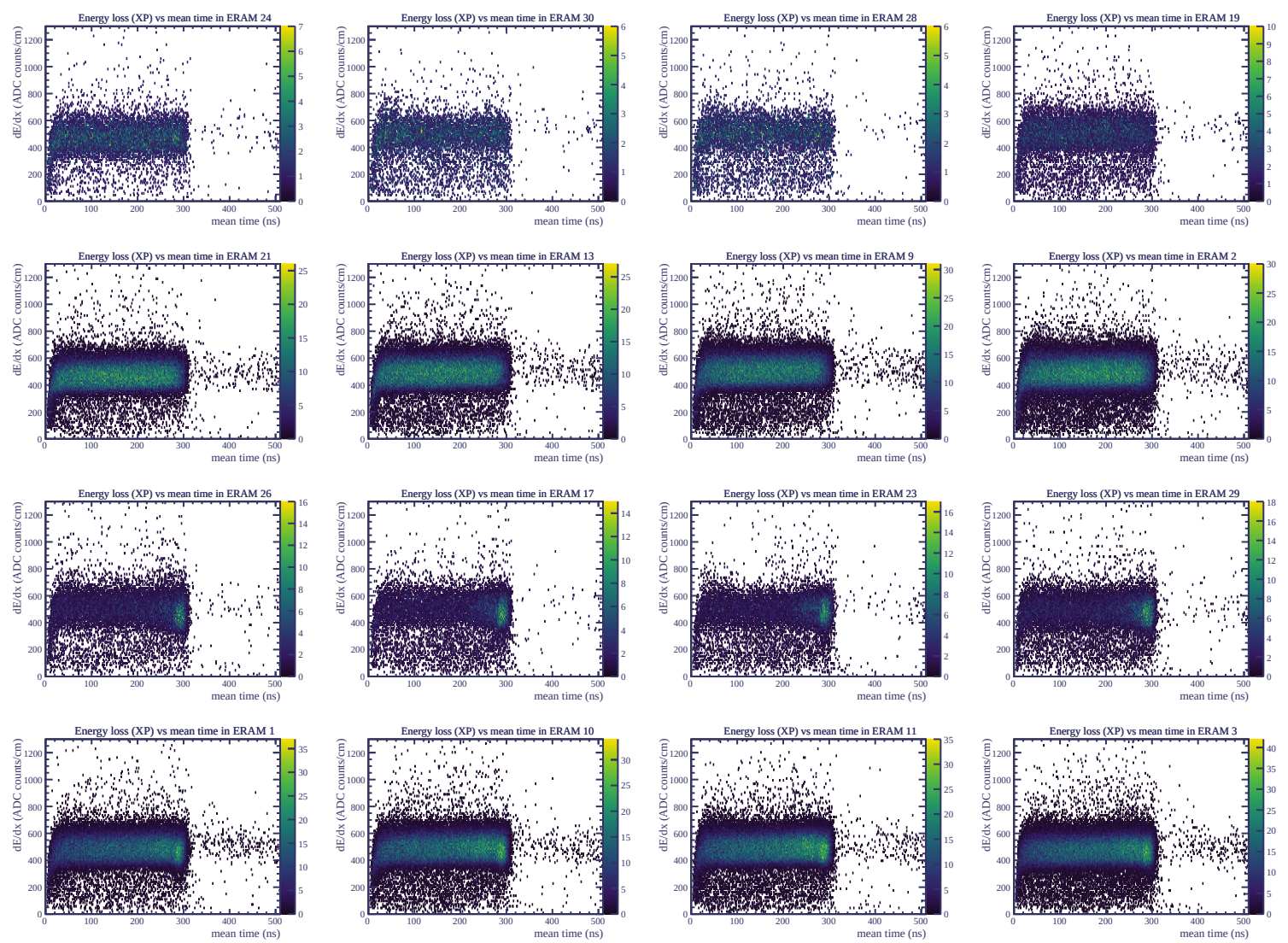


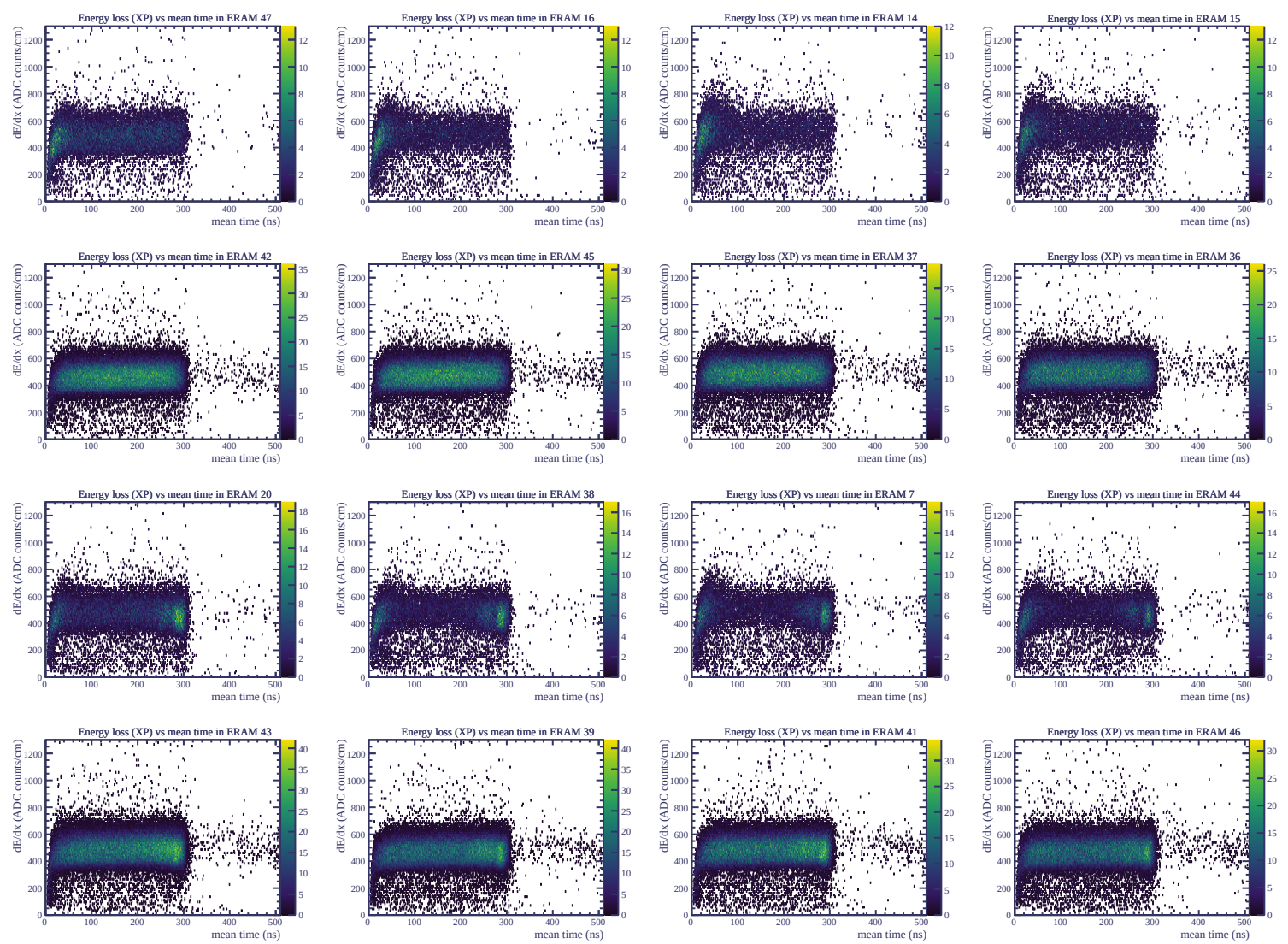
Resolution (%)

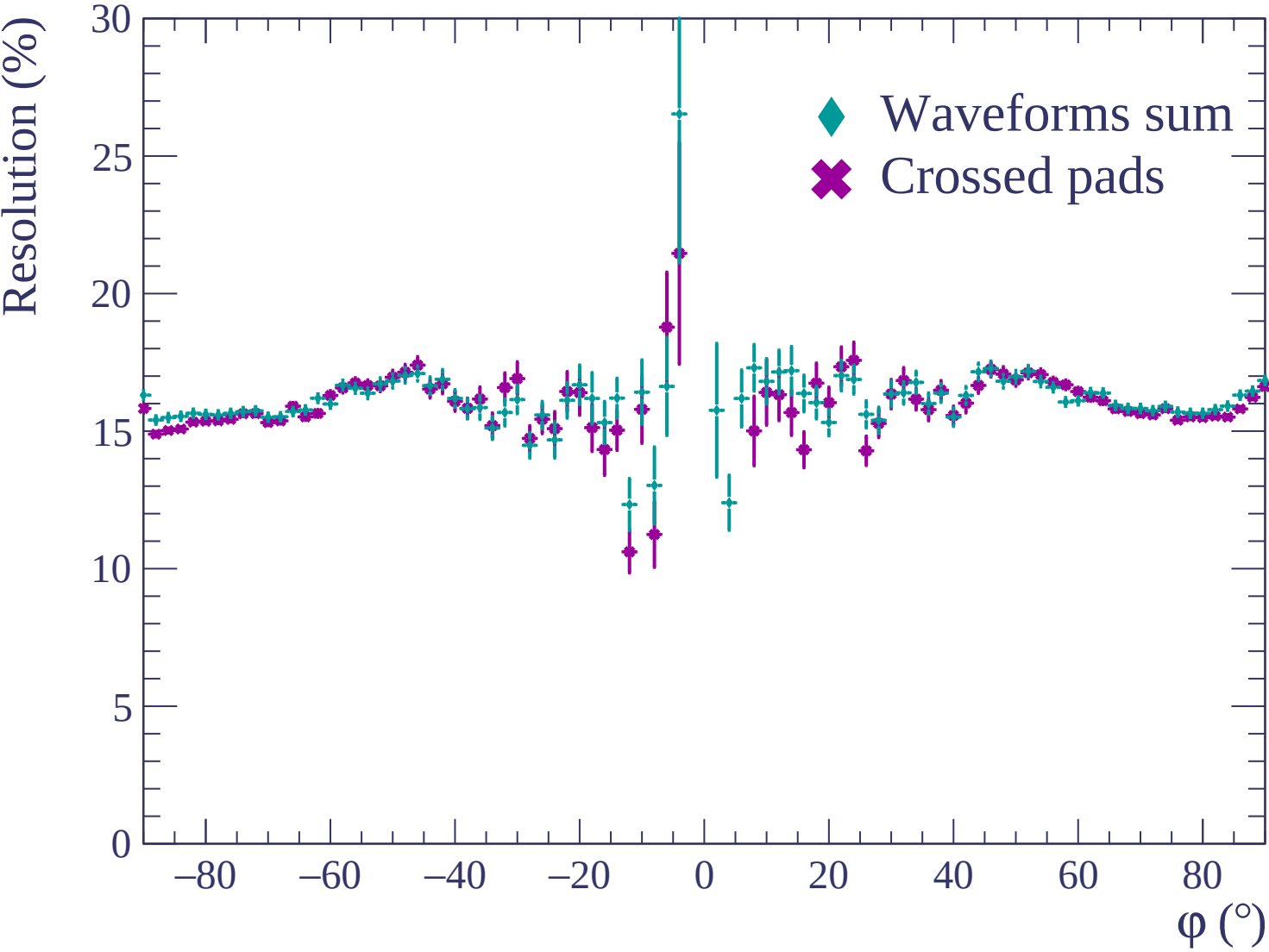


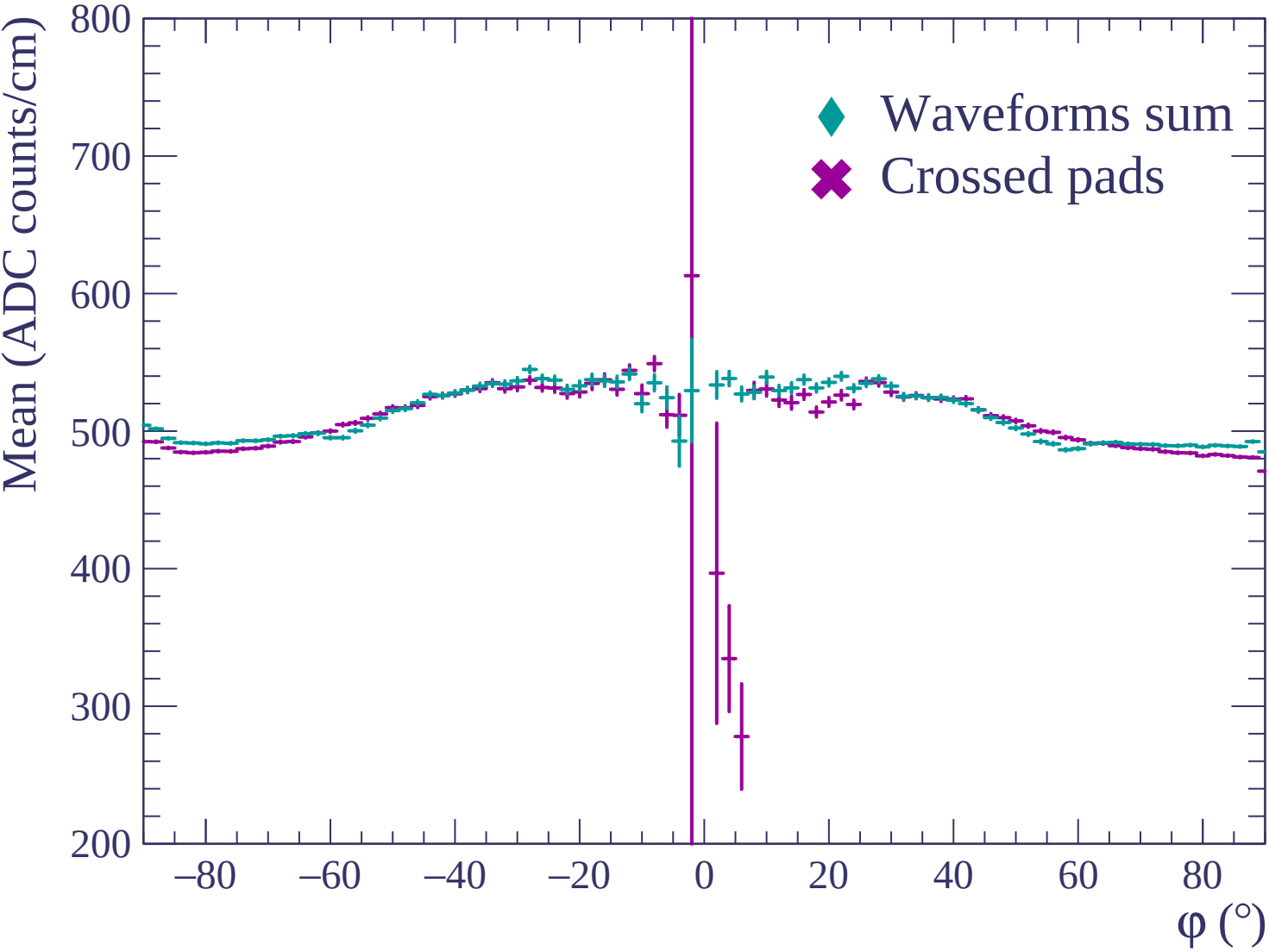


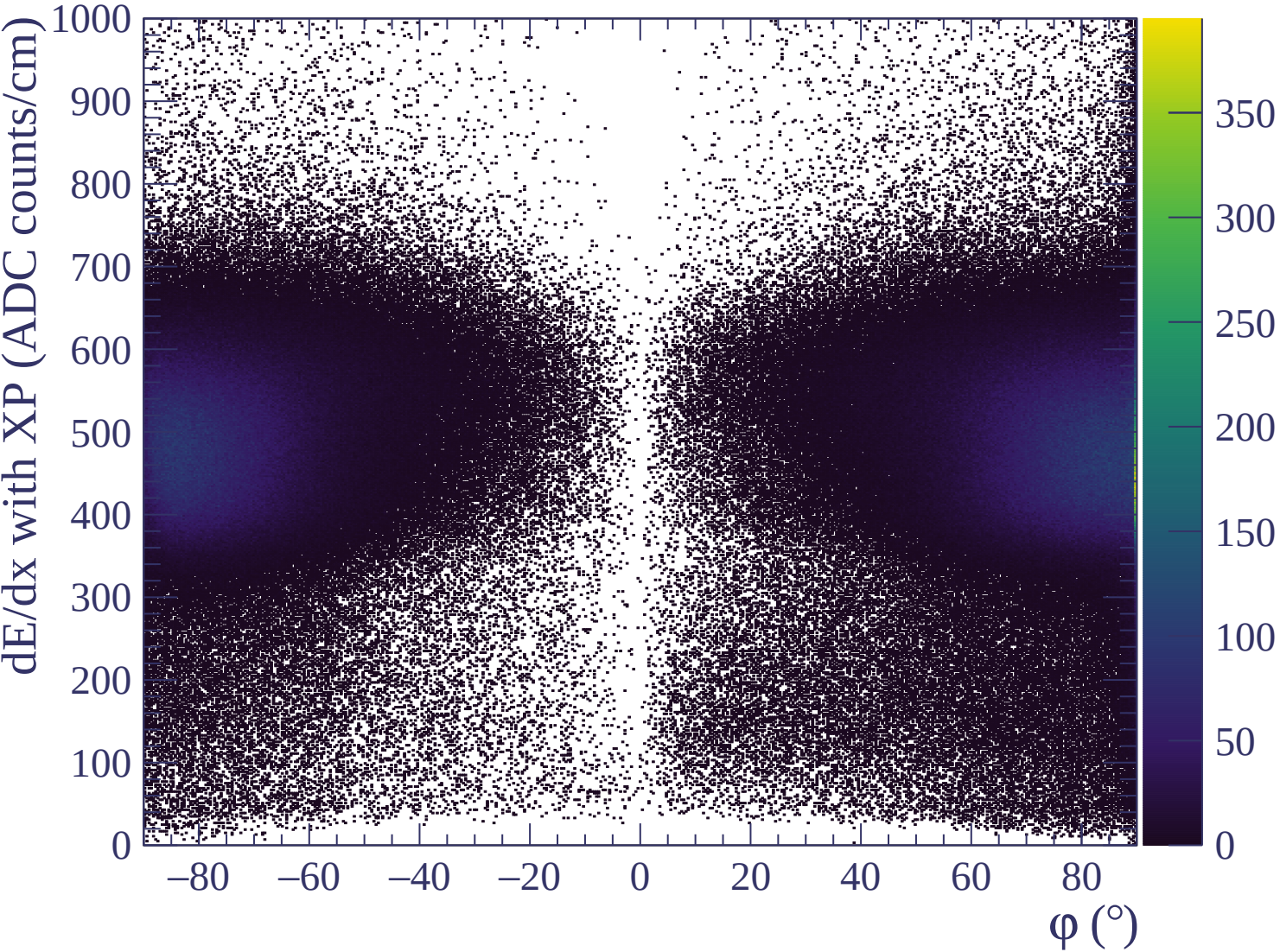


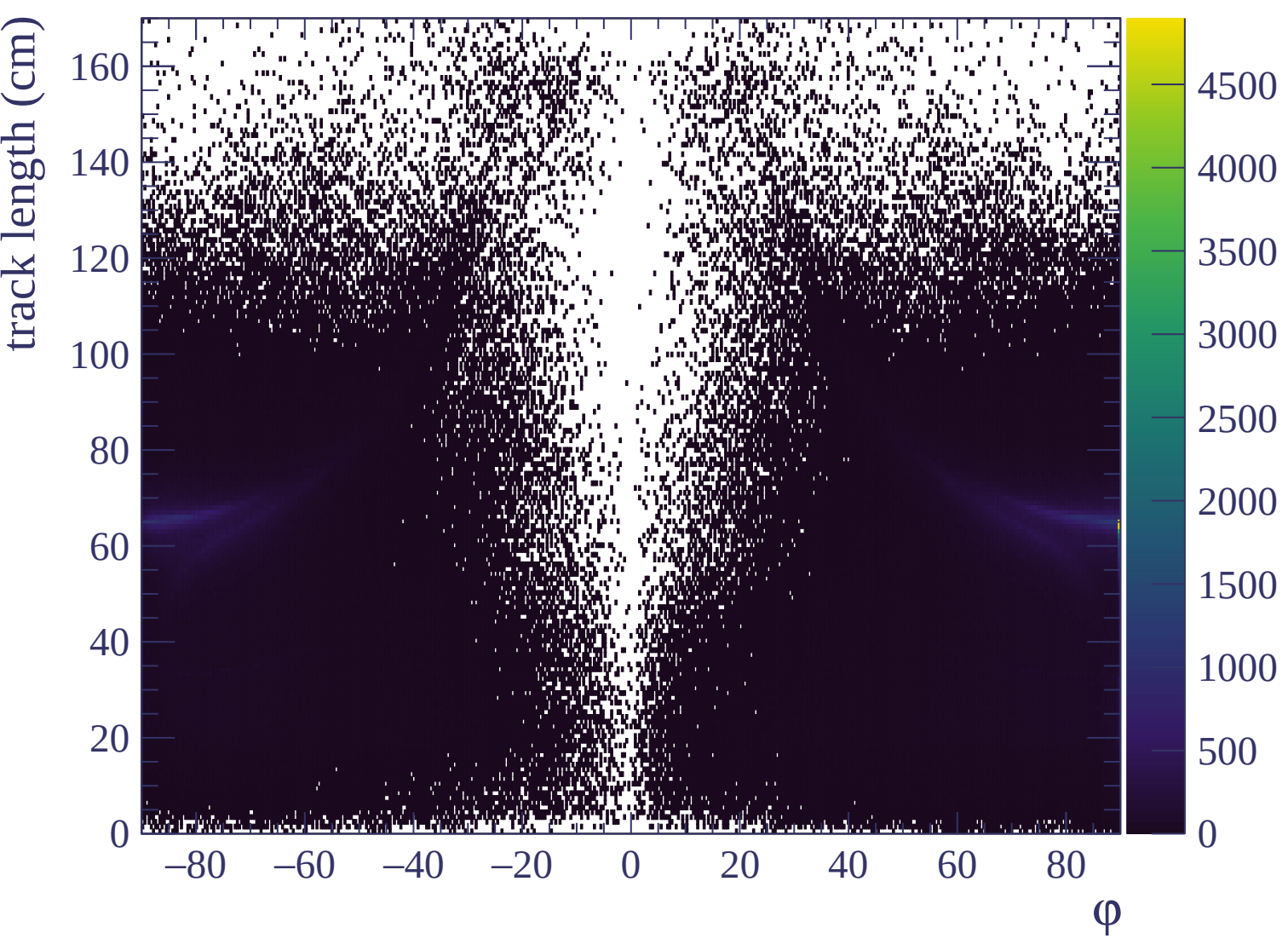


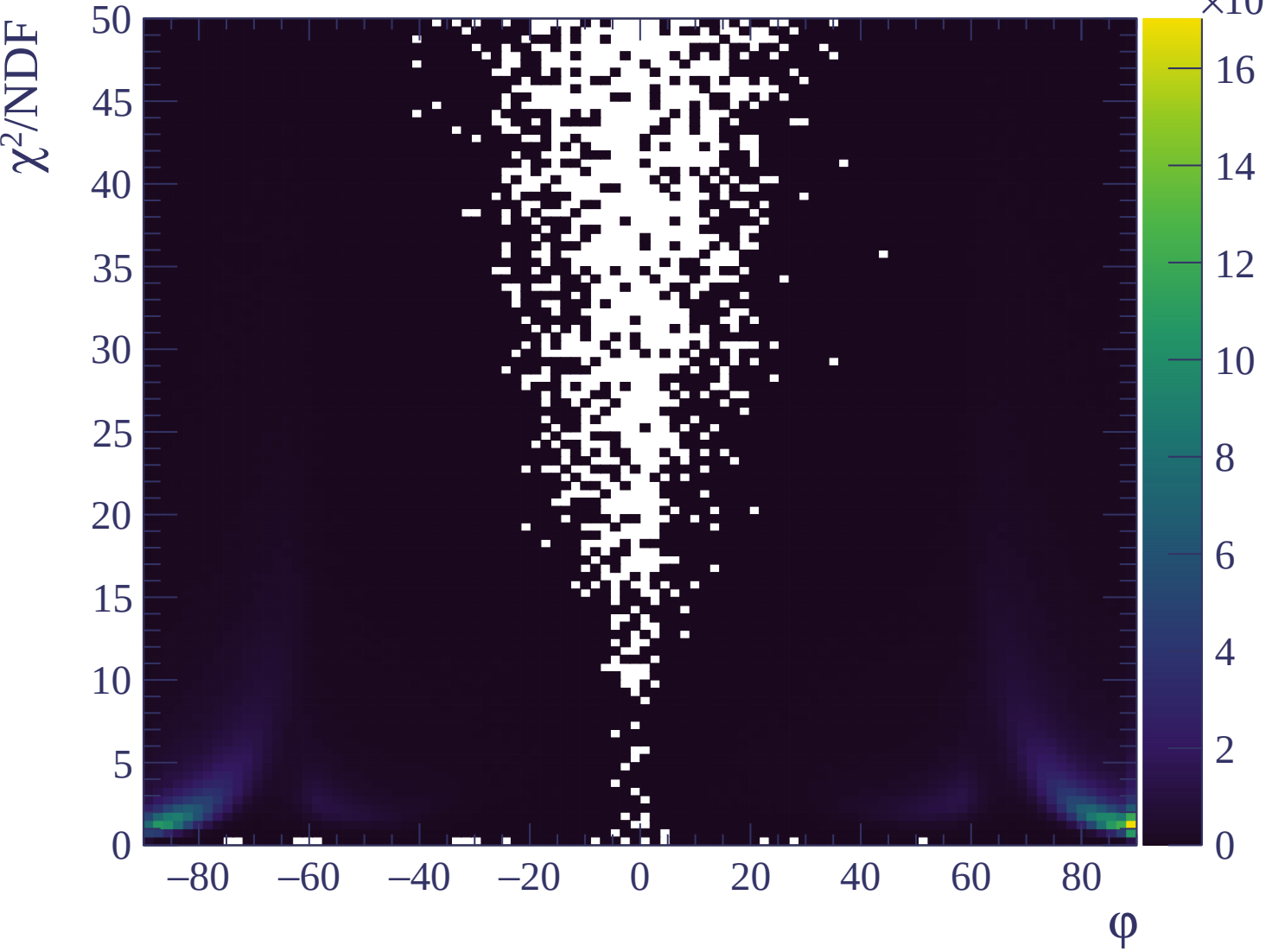


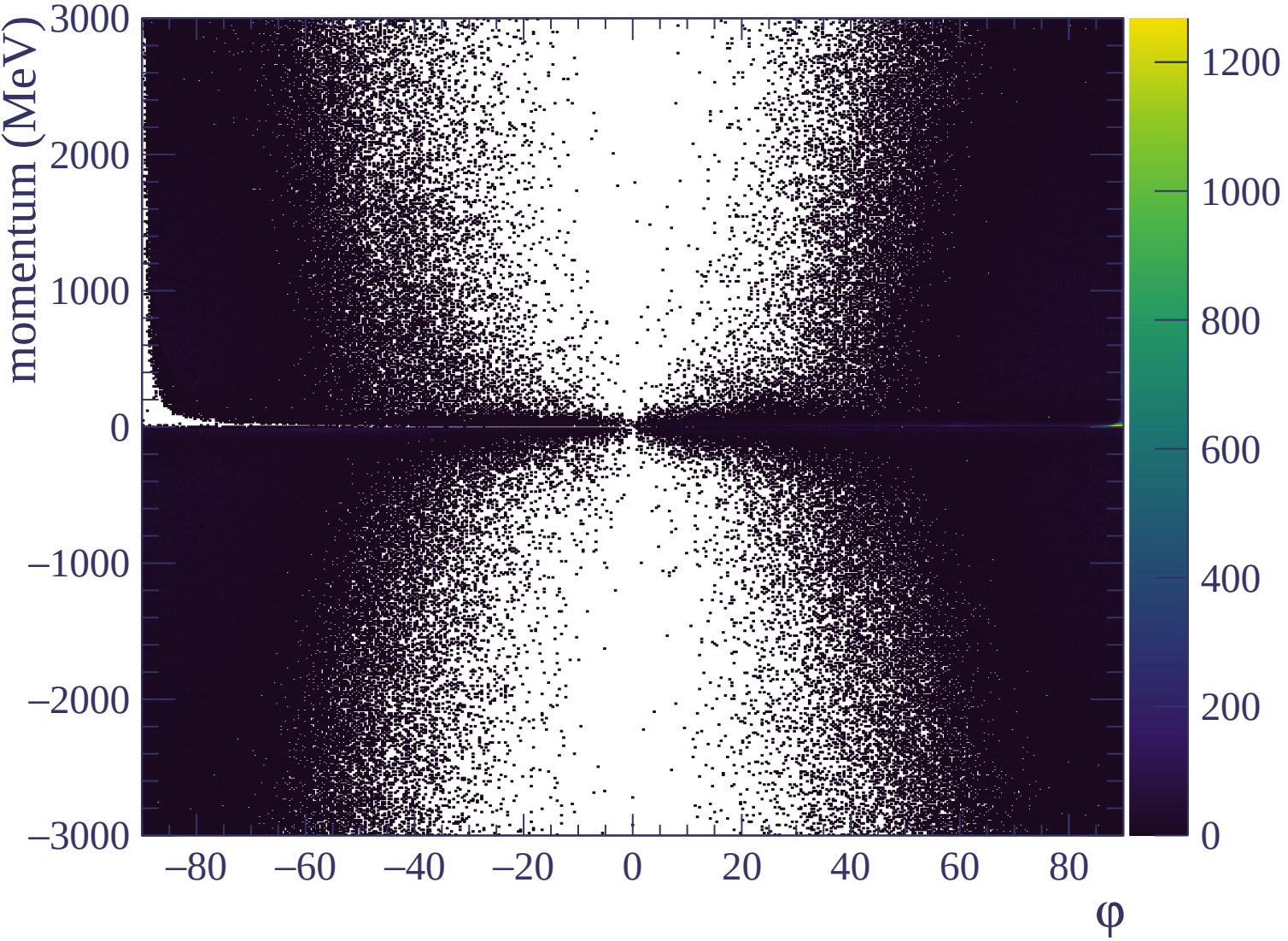


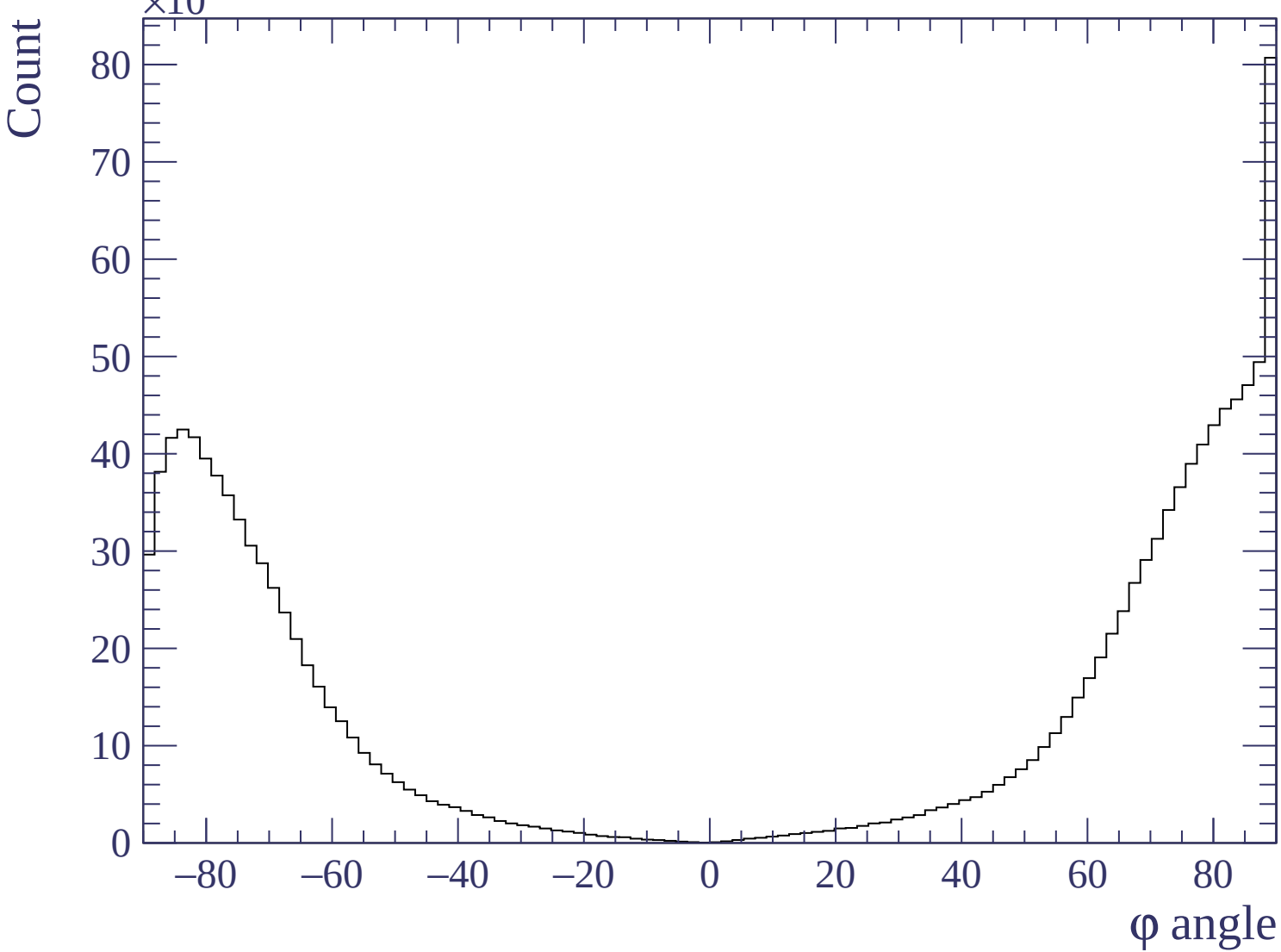


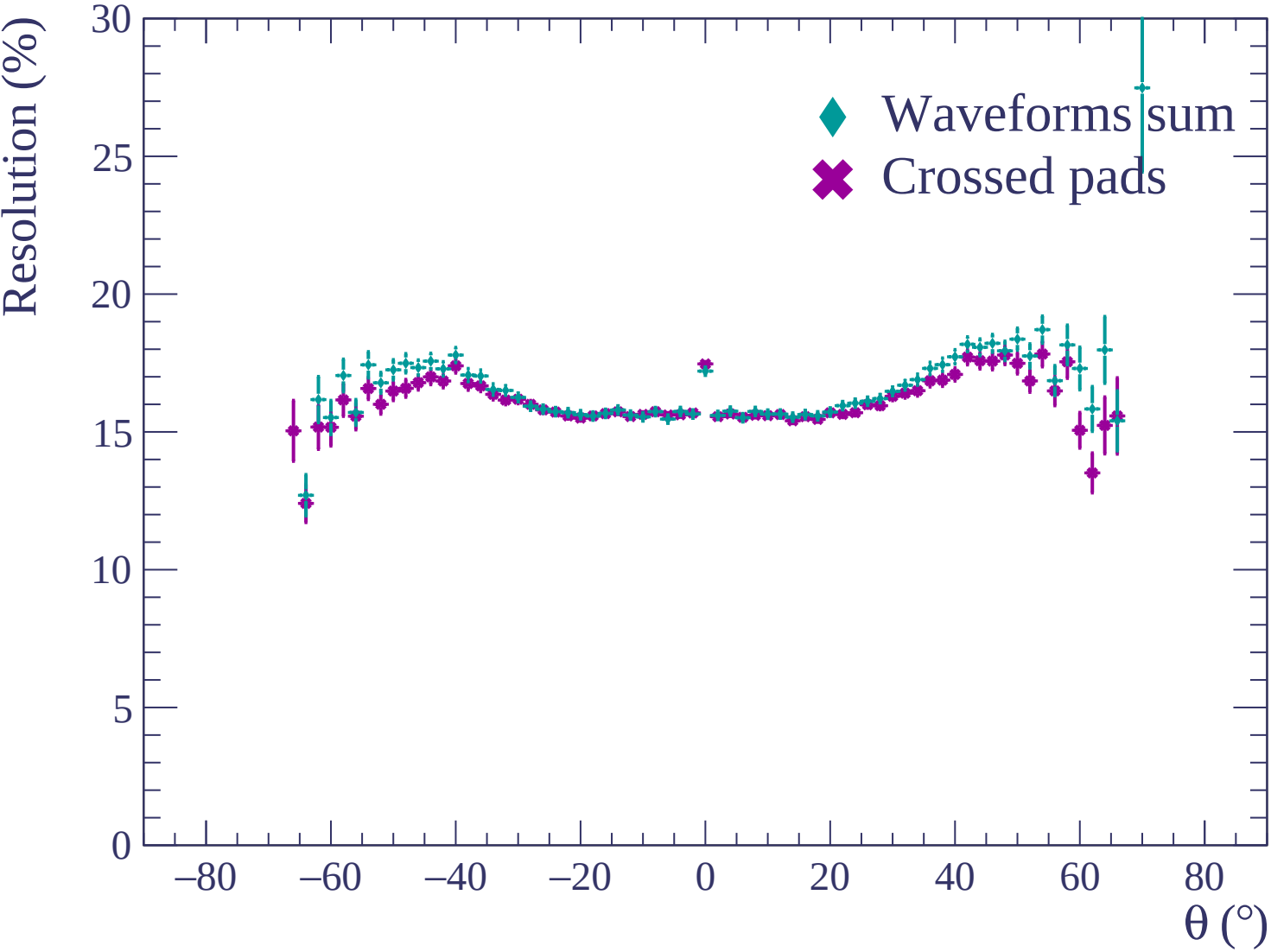


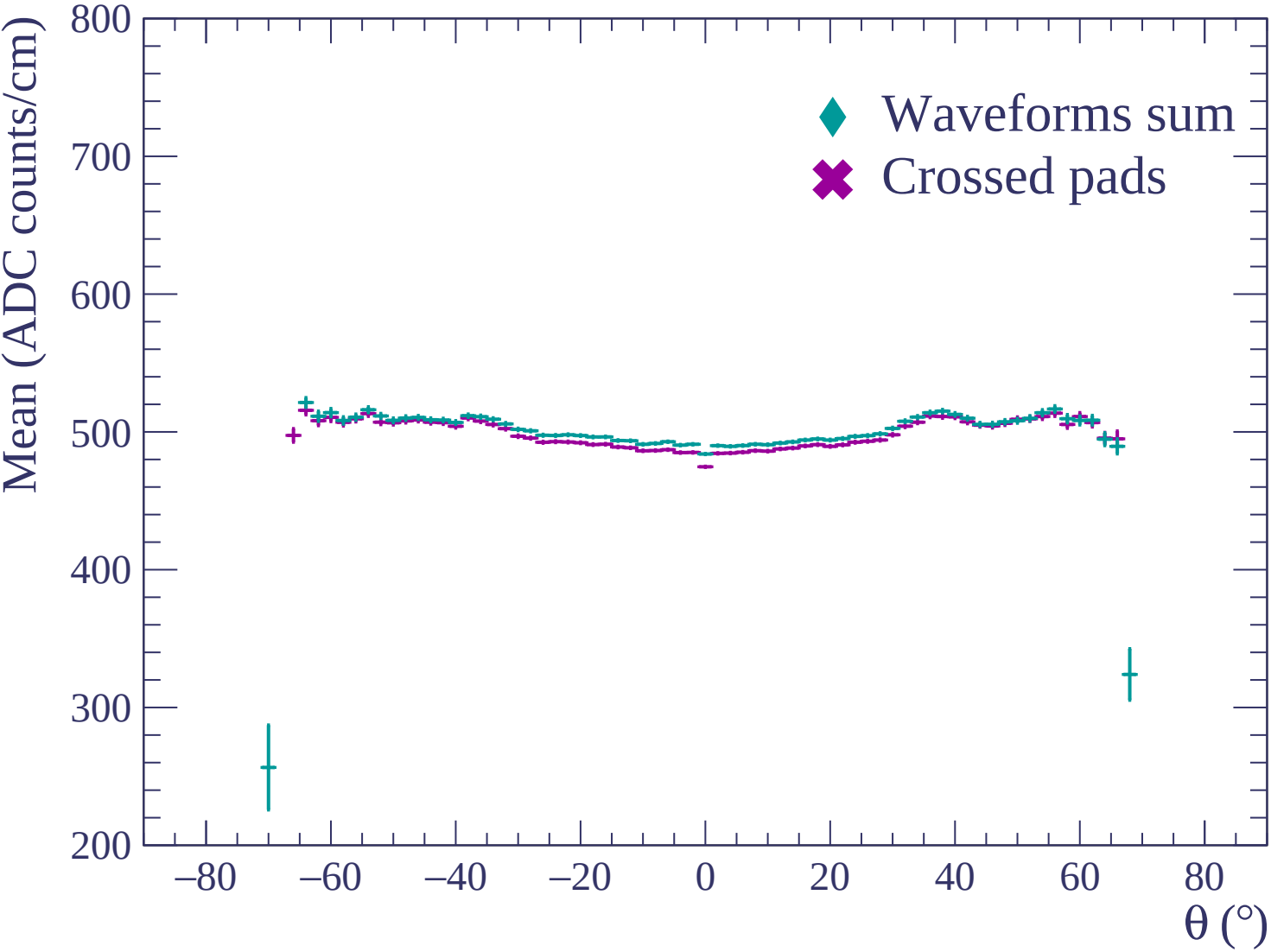


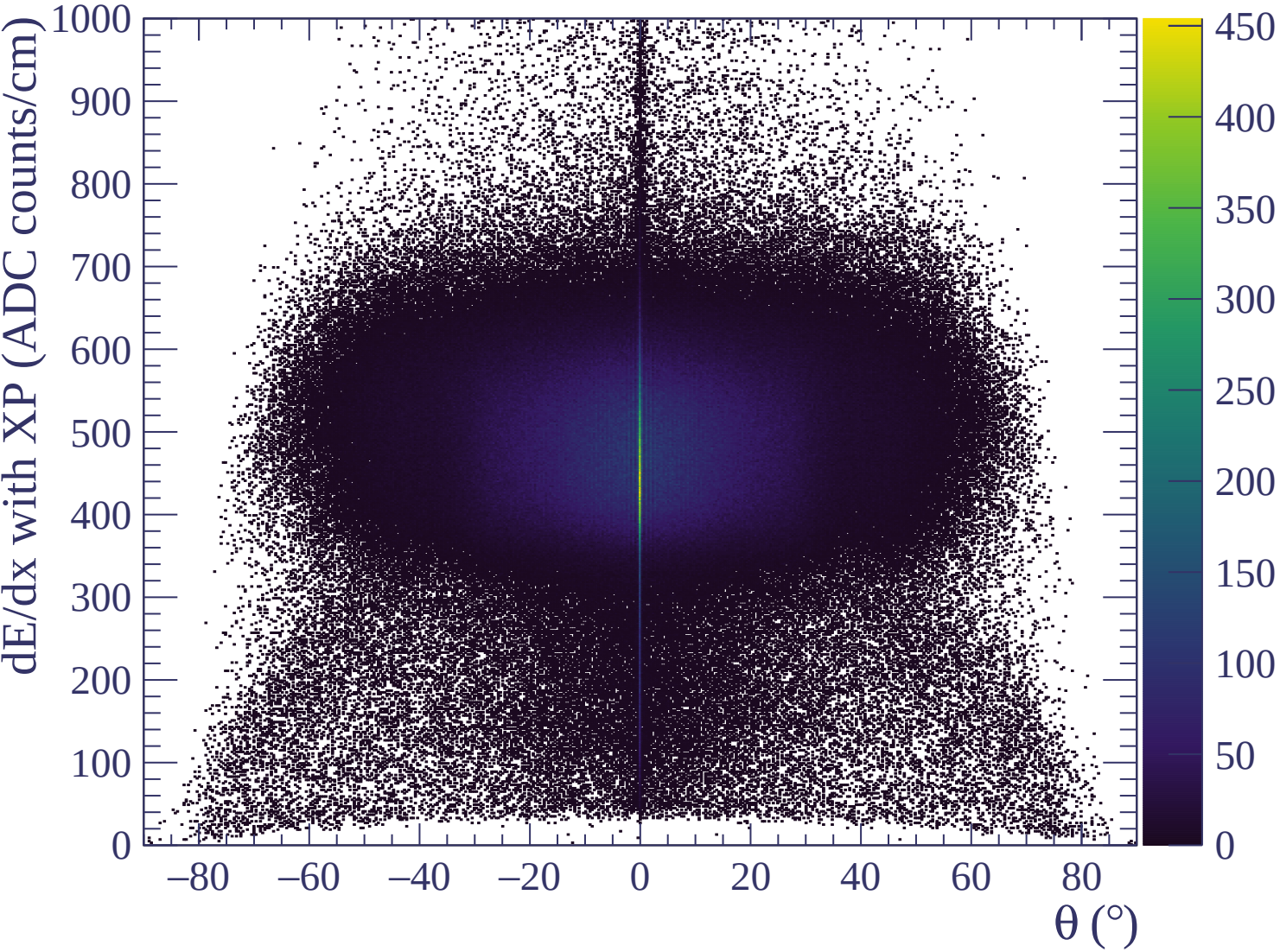


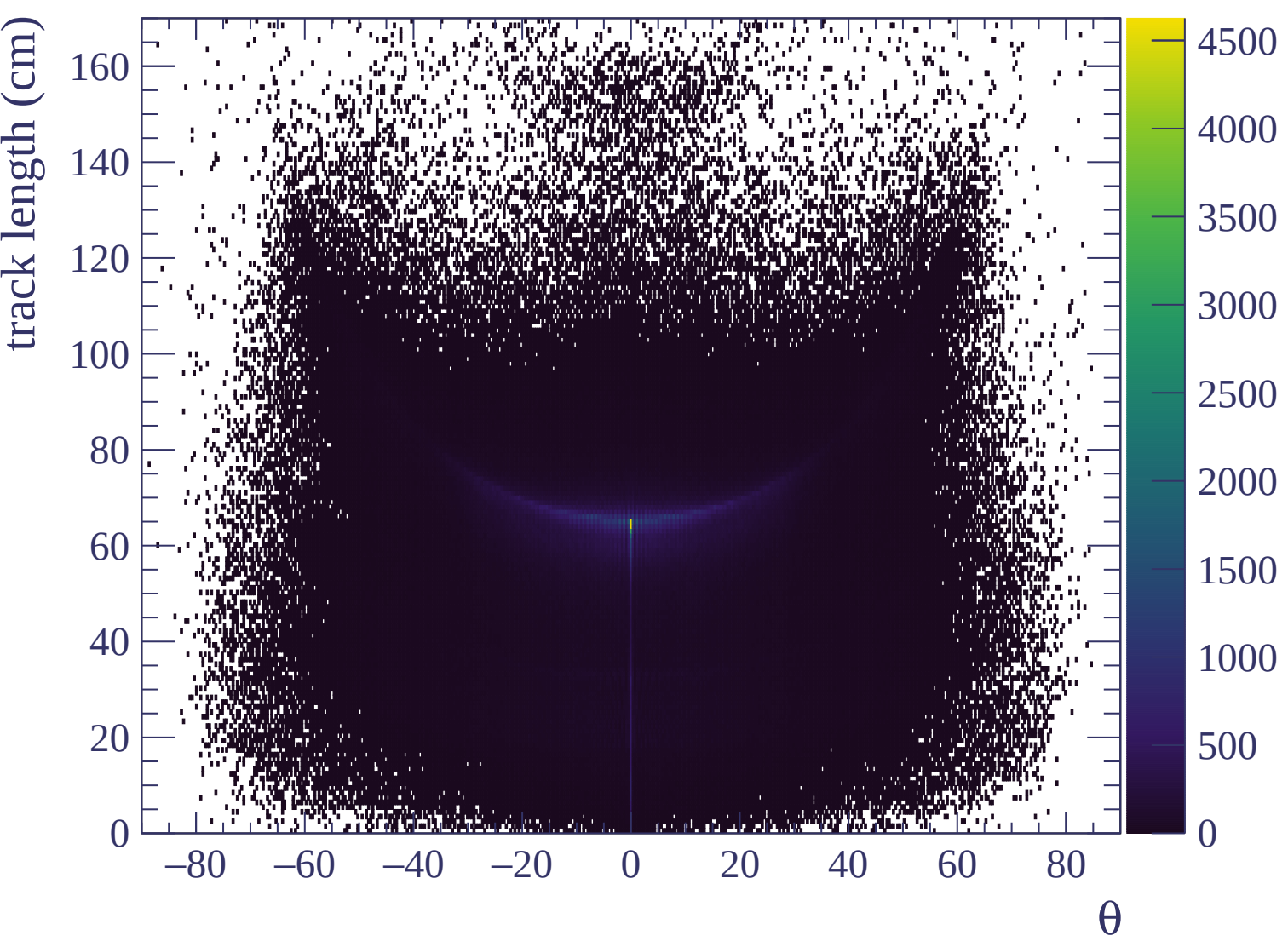


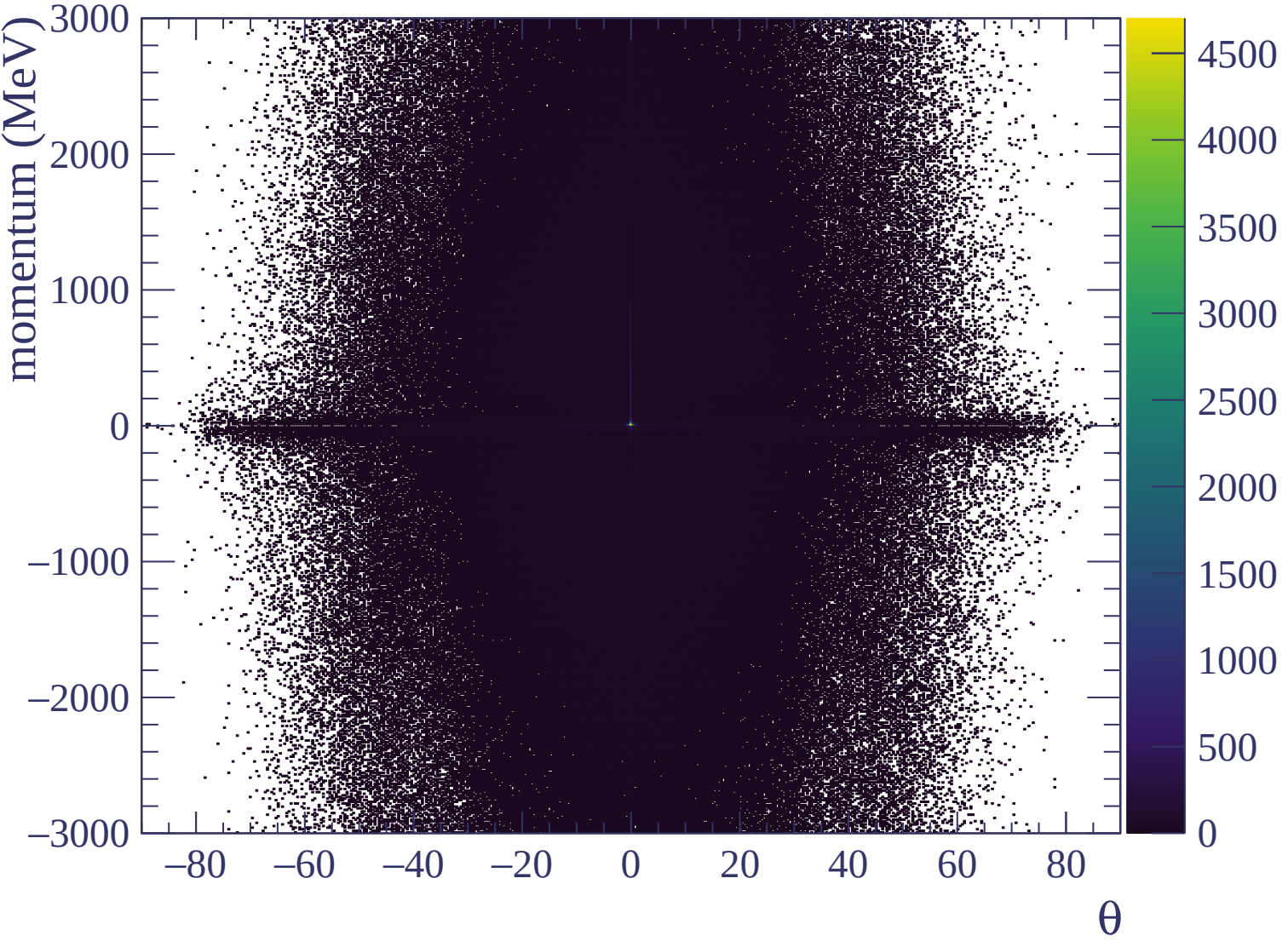


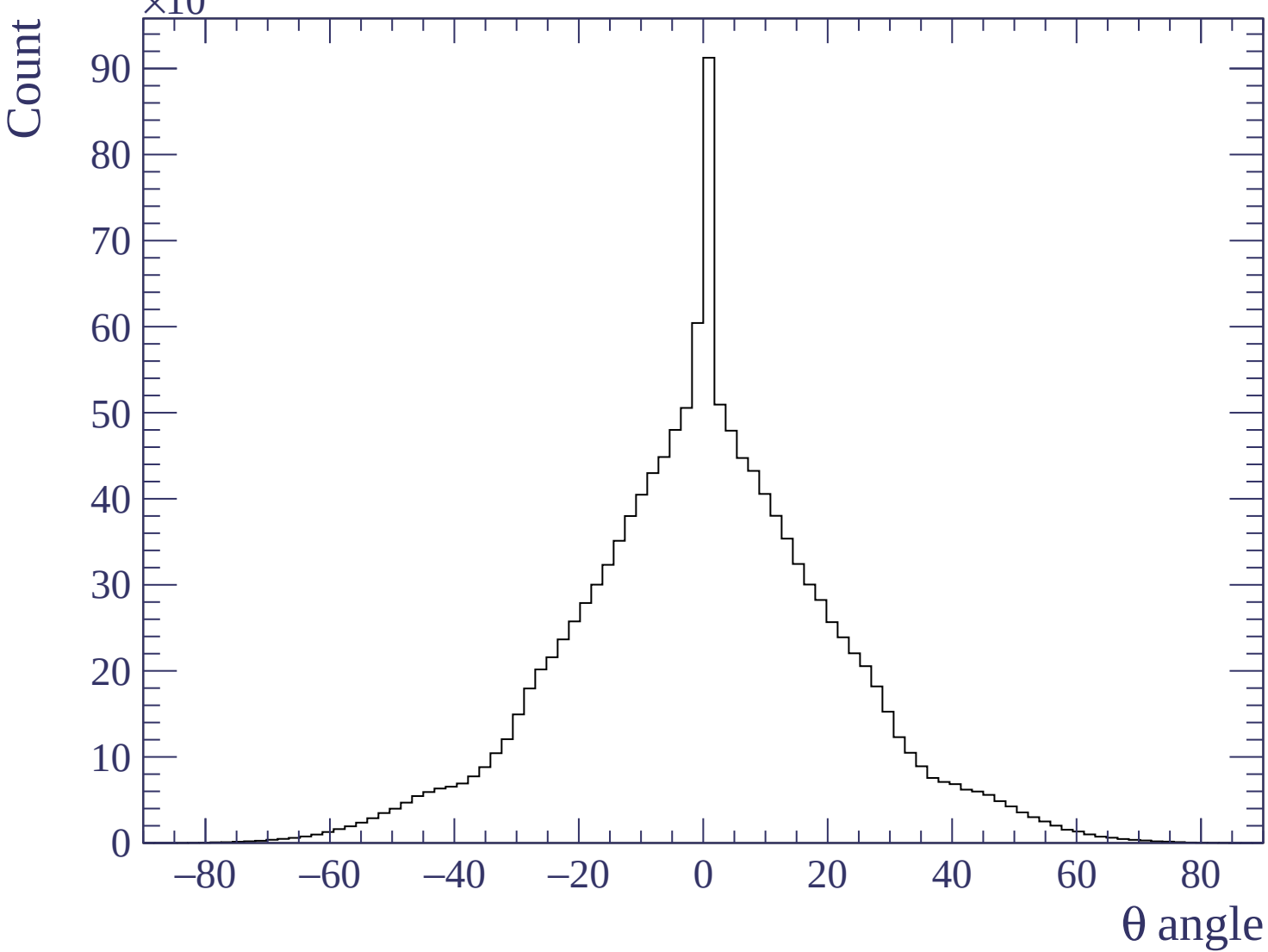


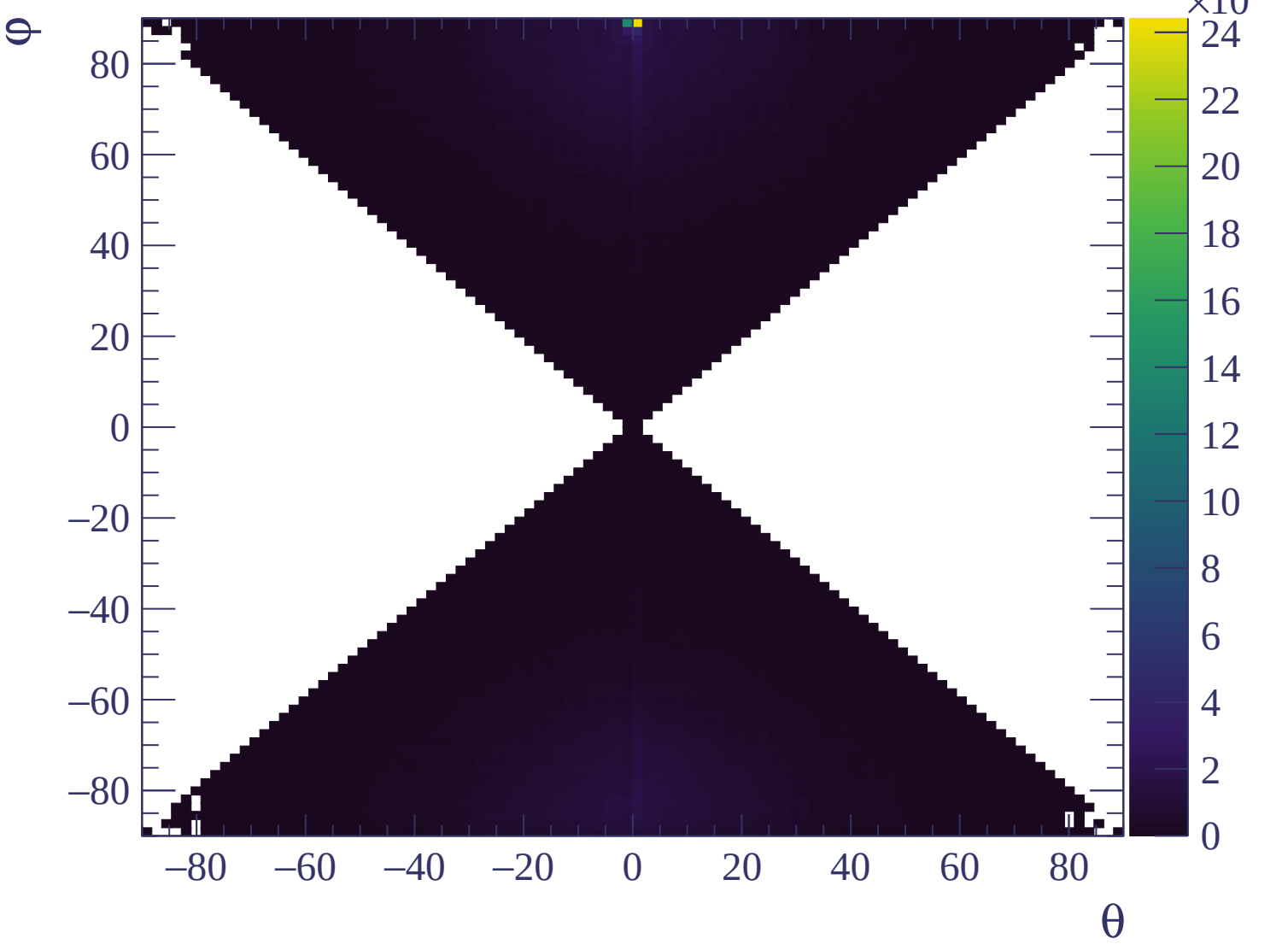


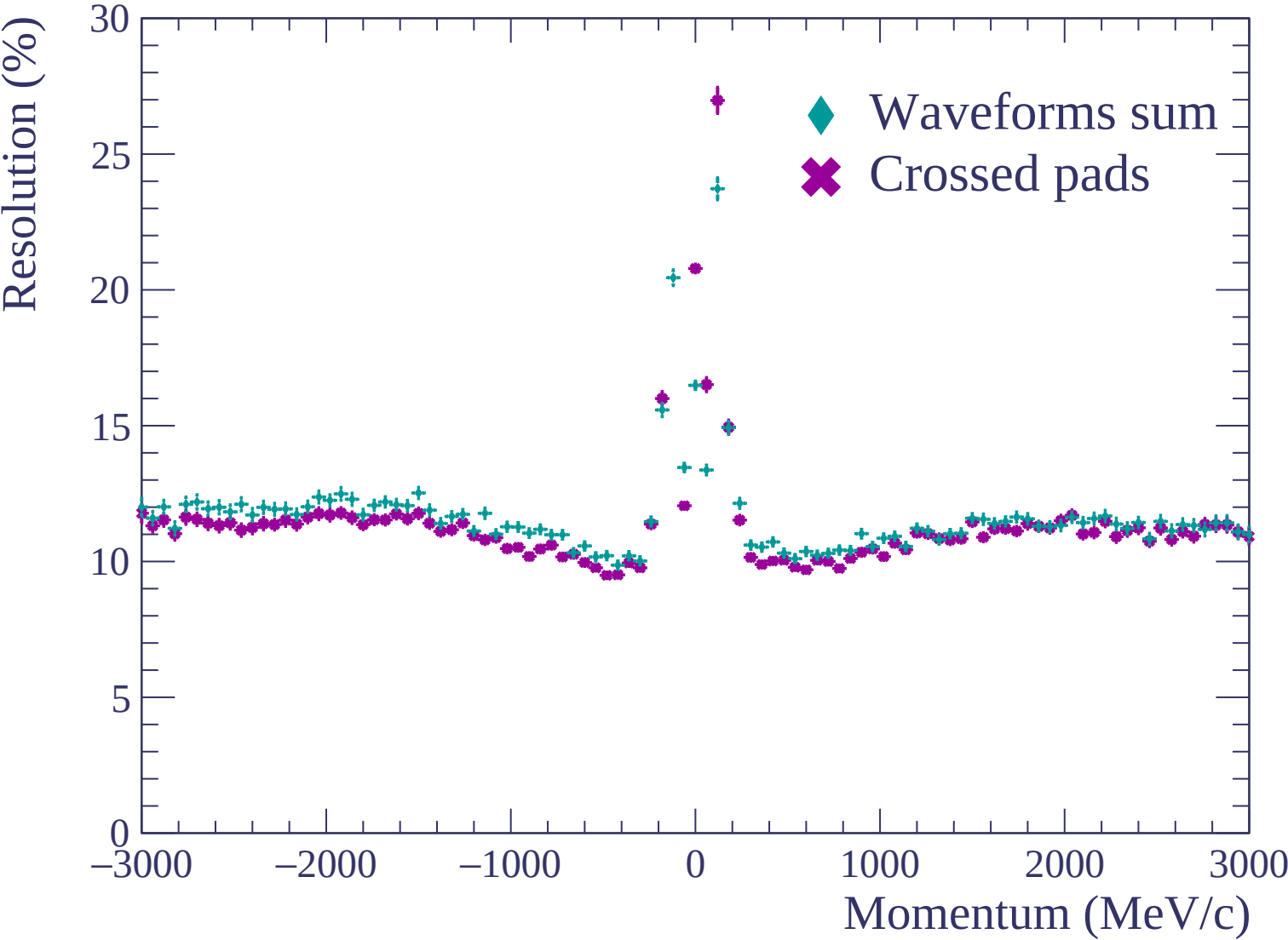


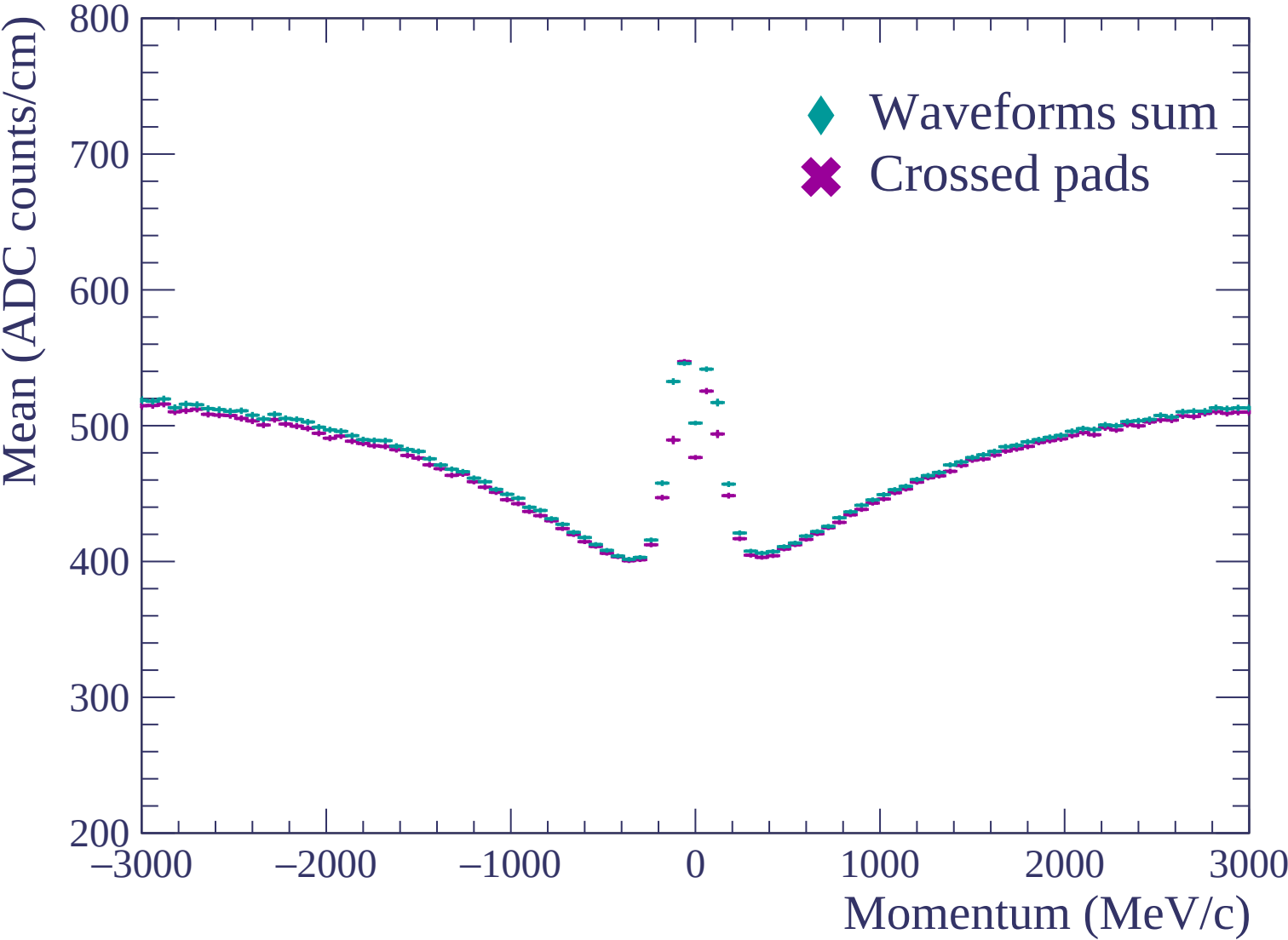


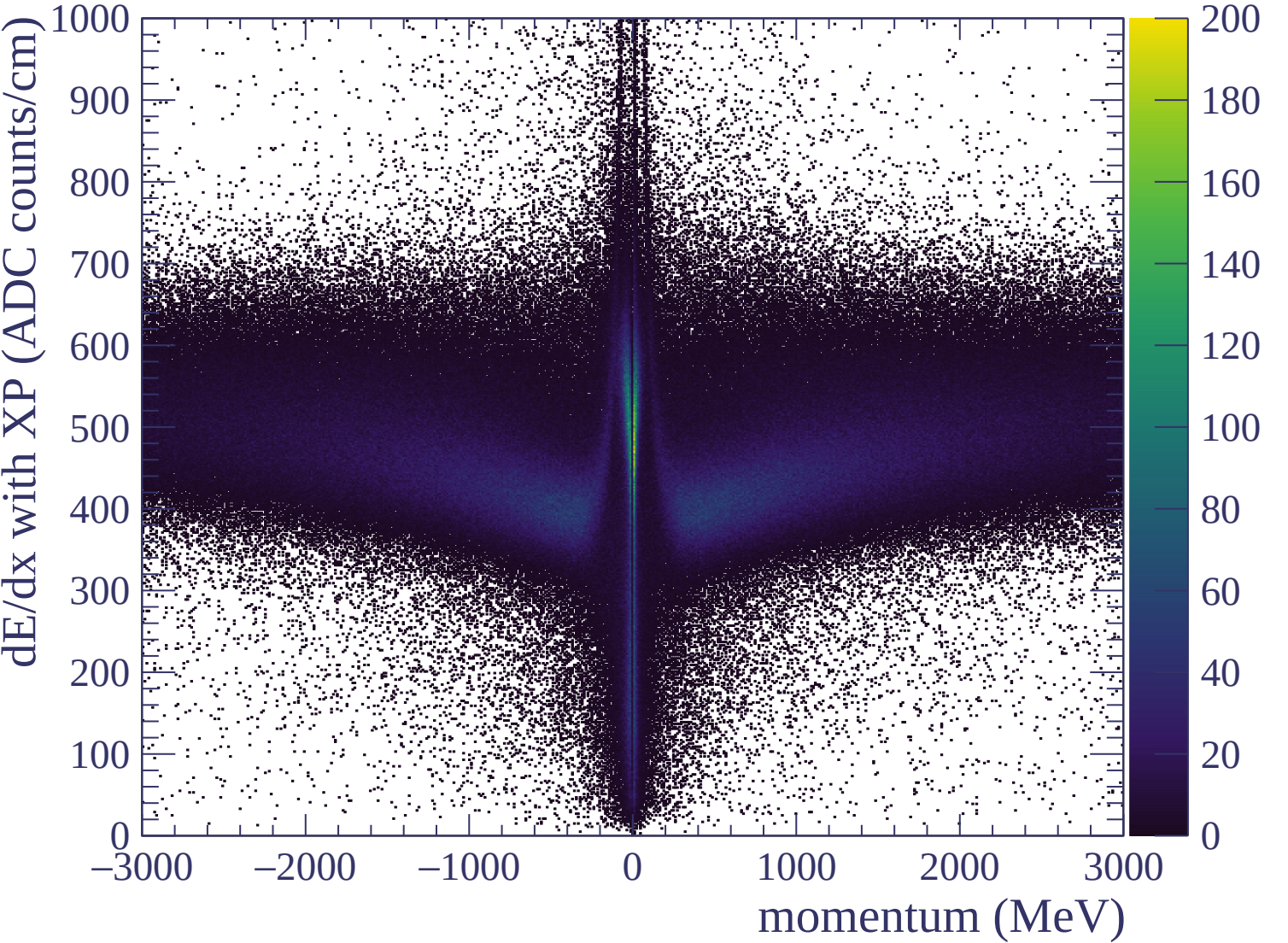


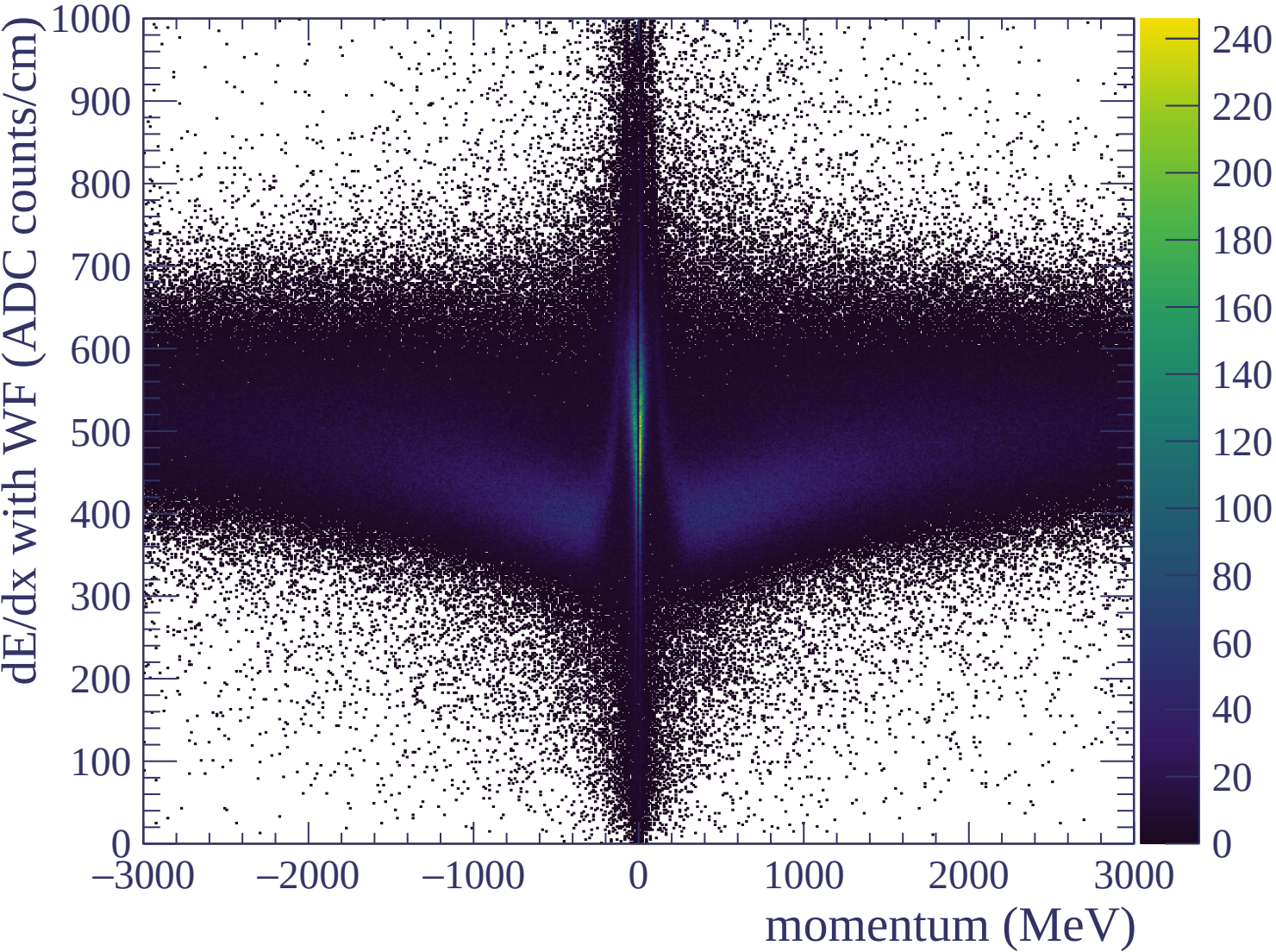


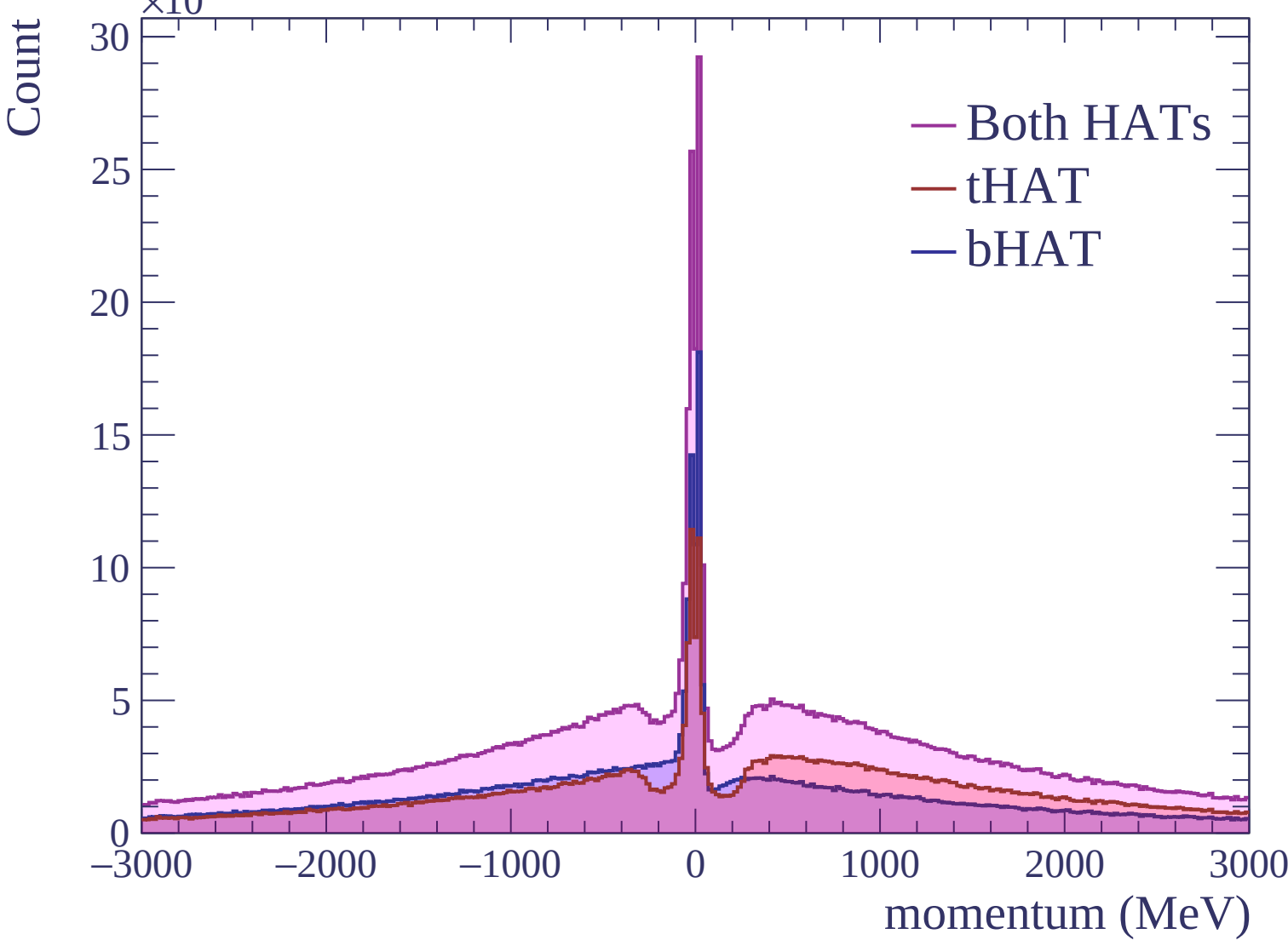


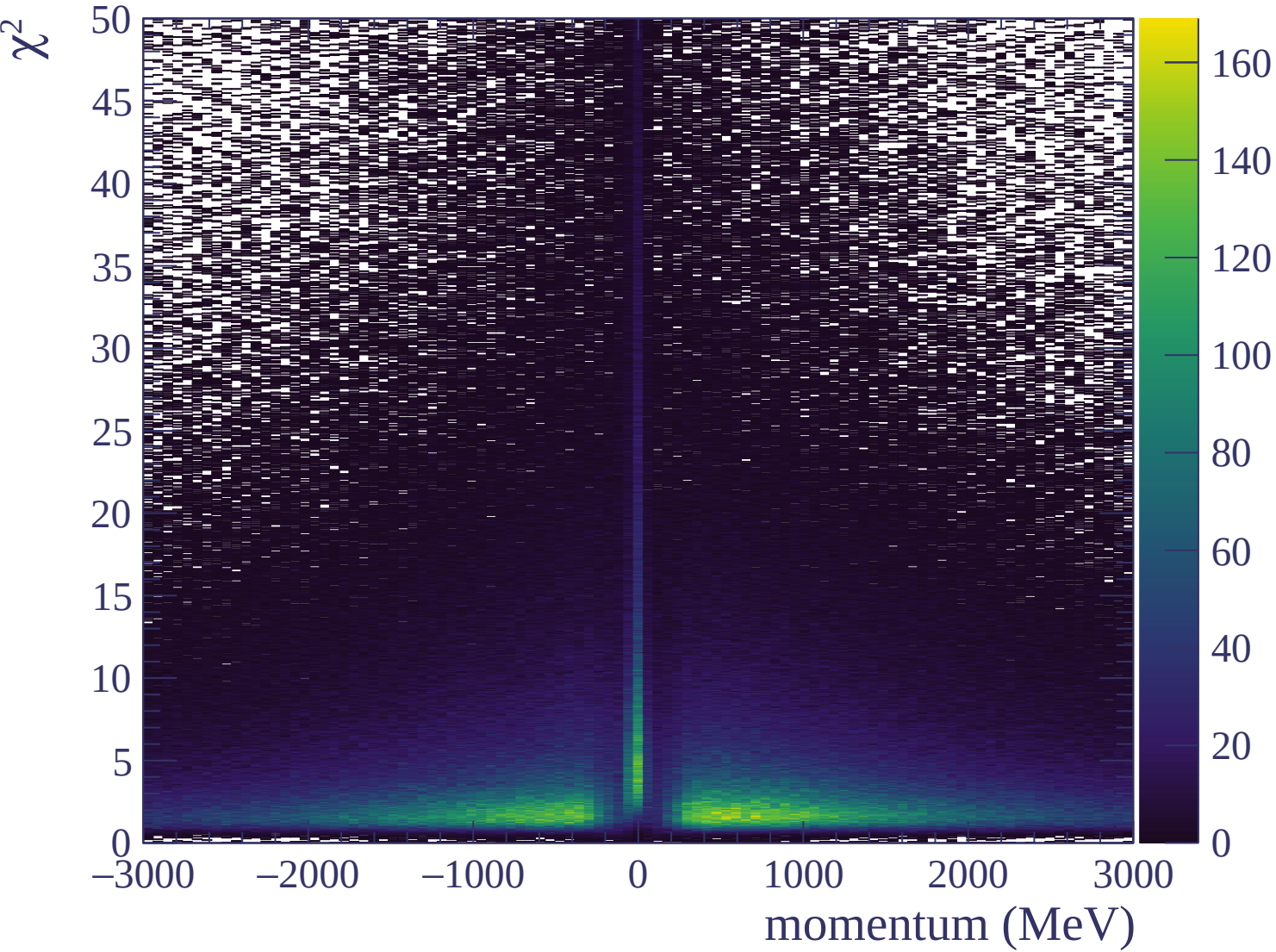


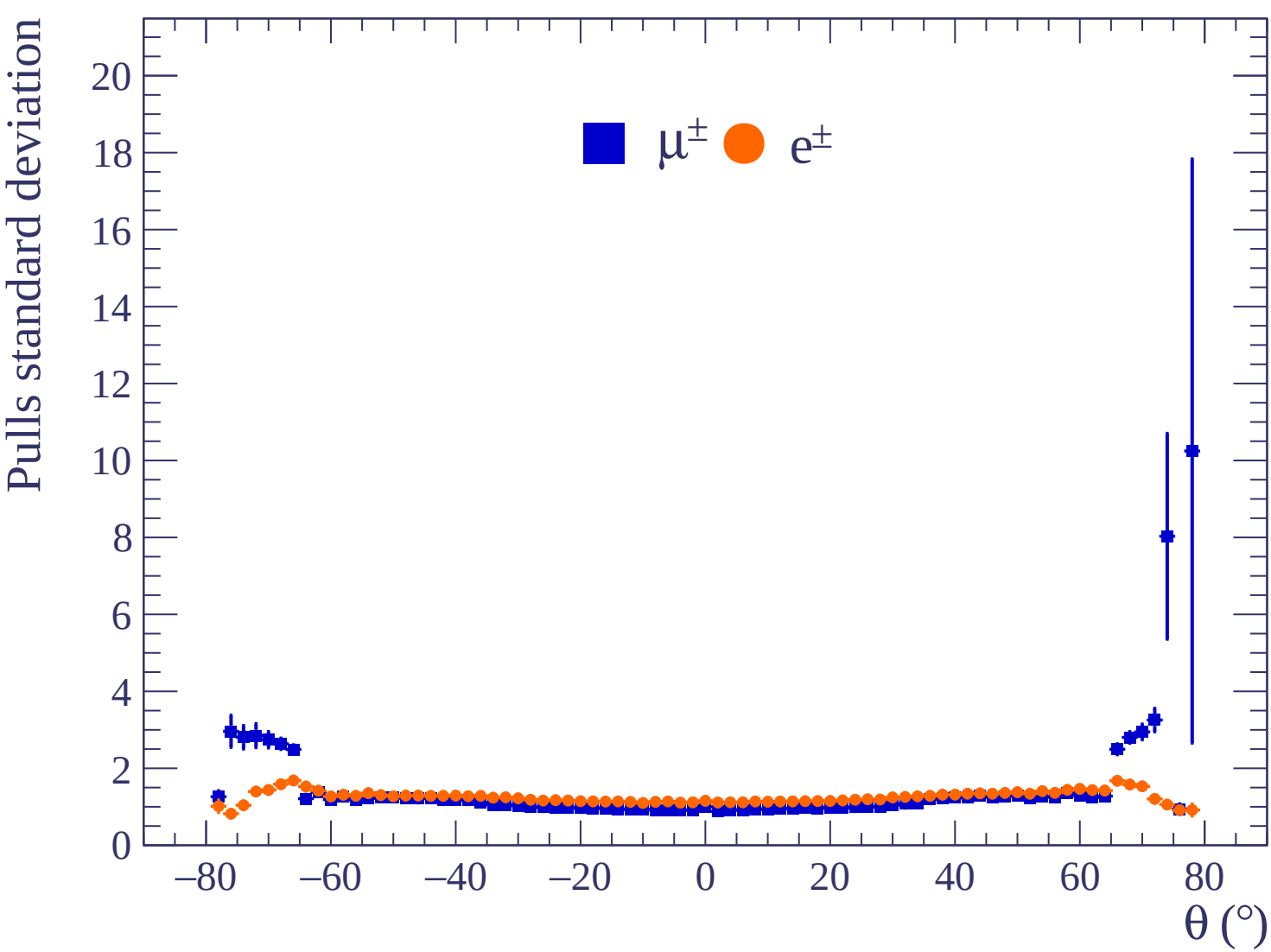


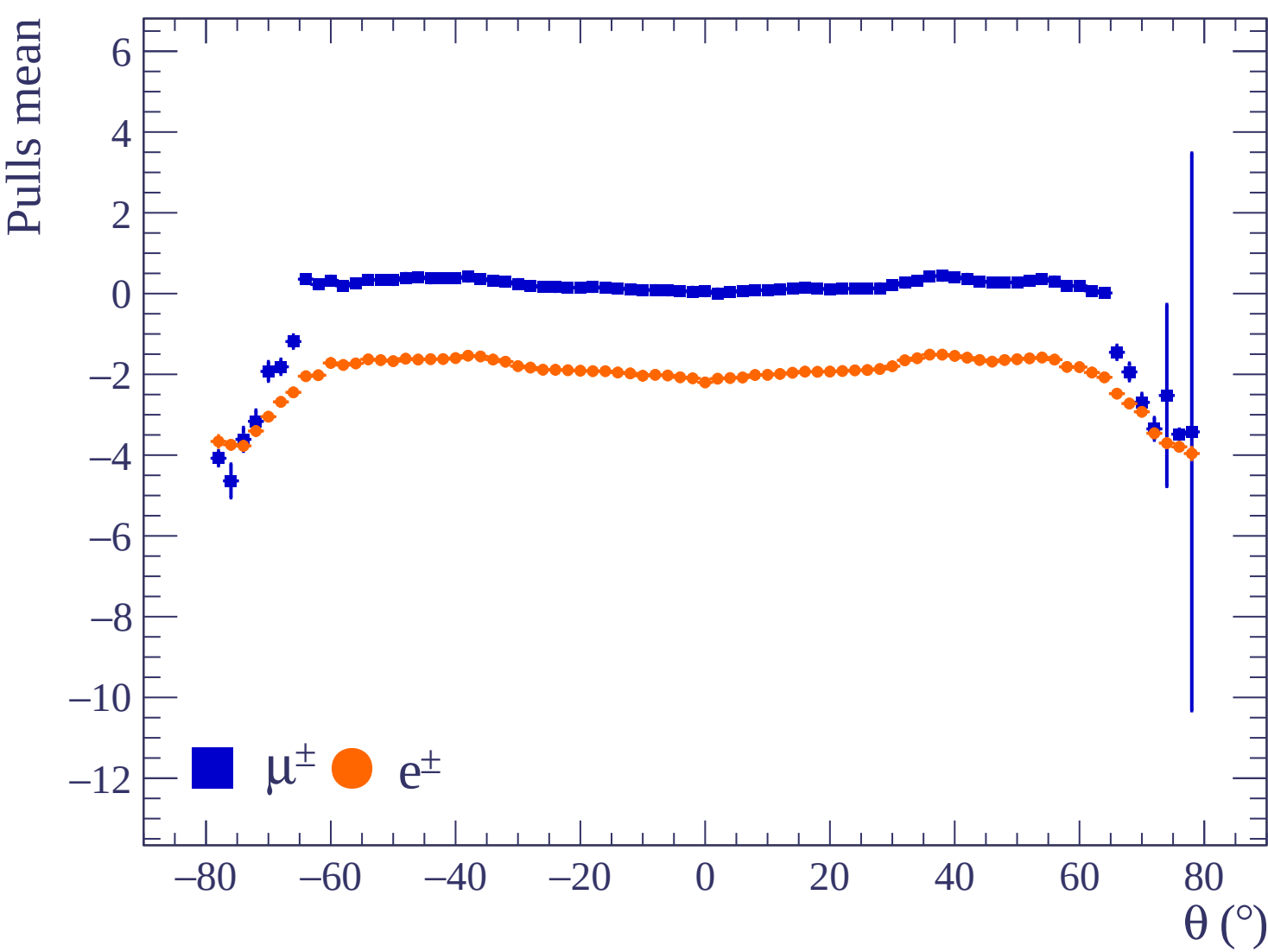


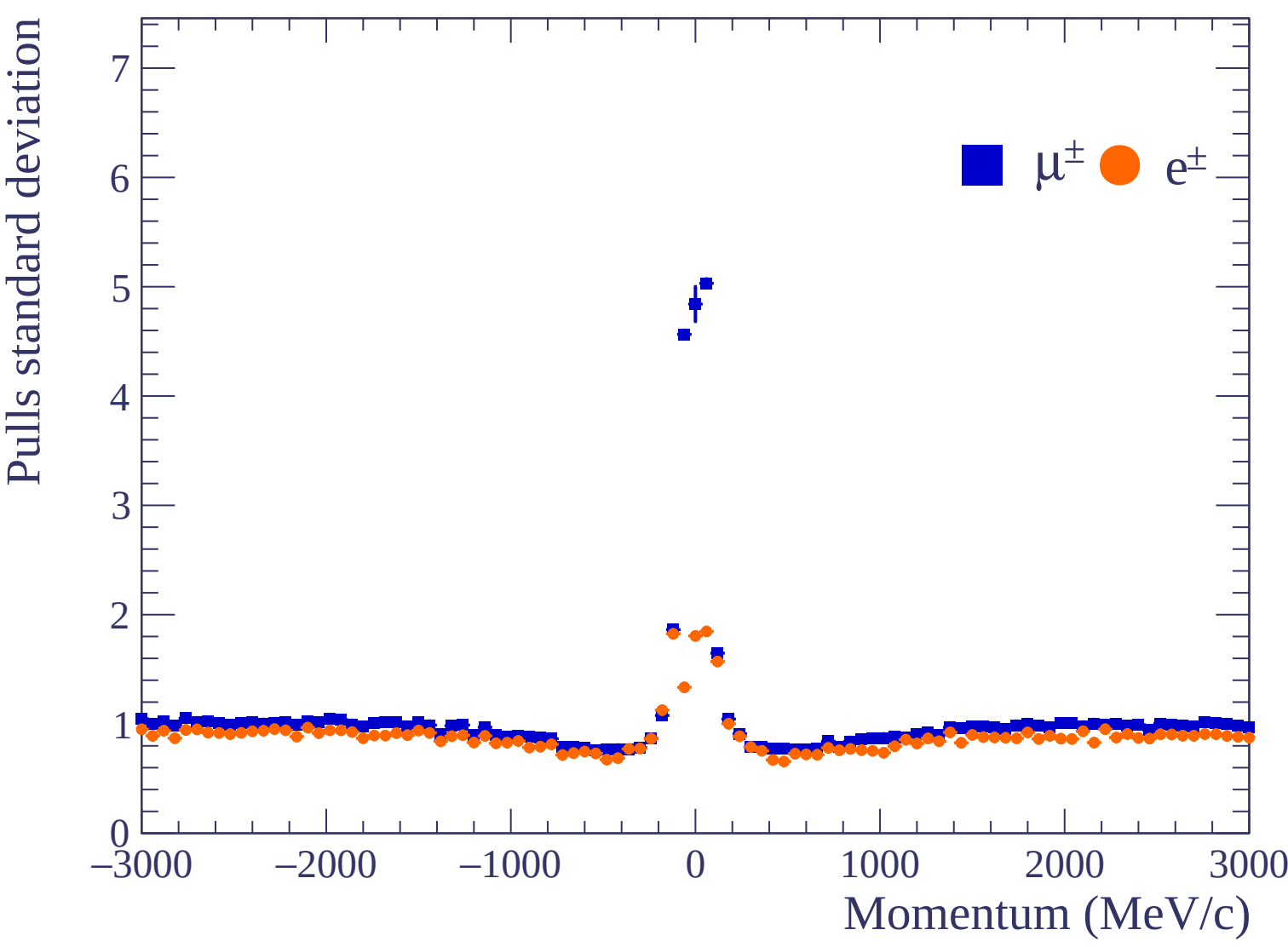


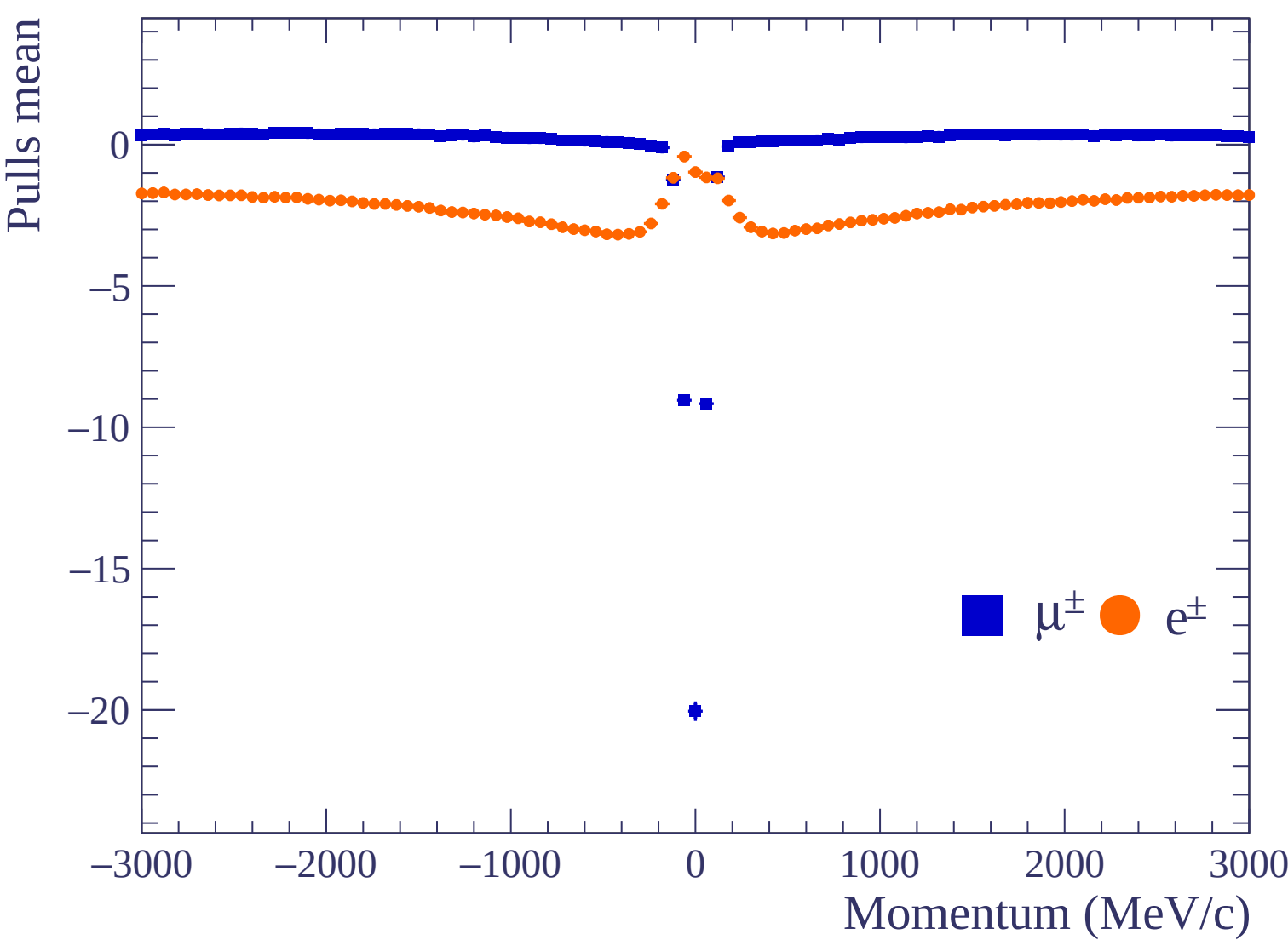


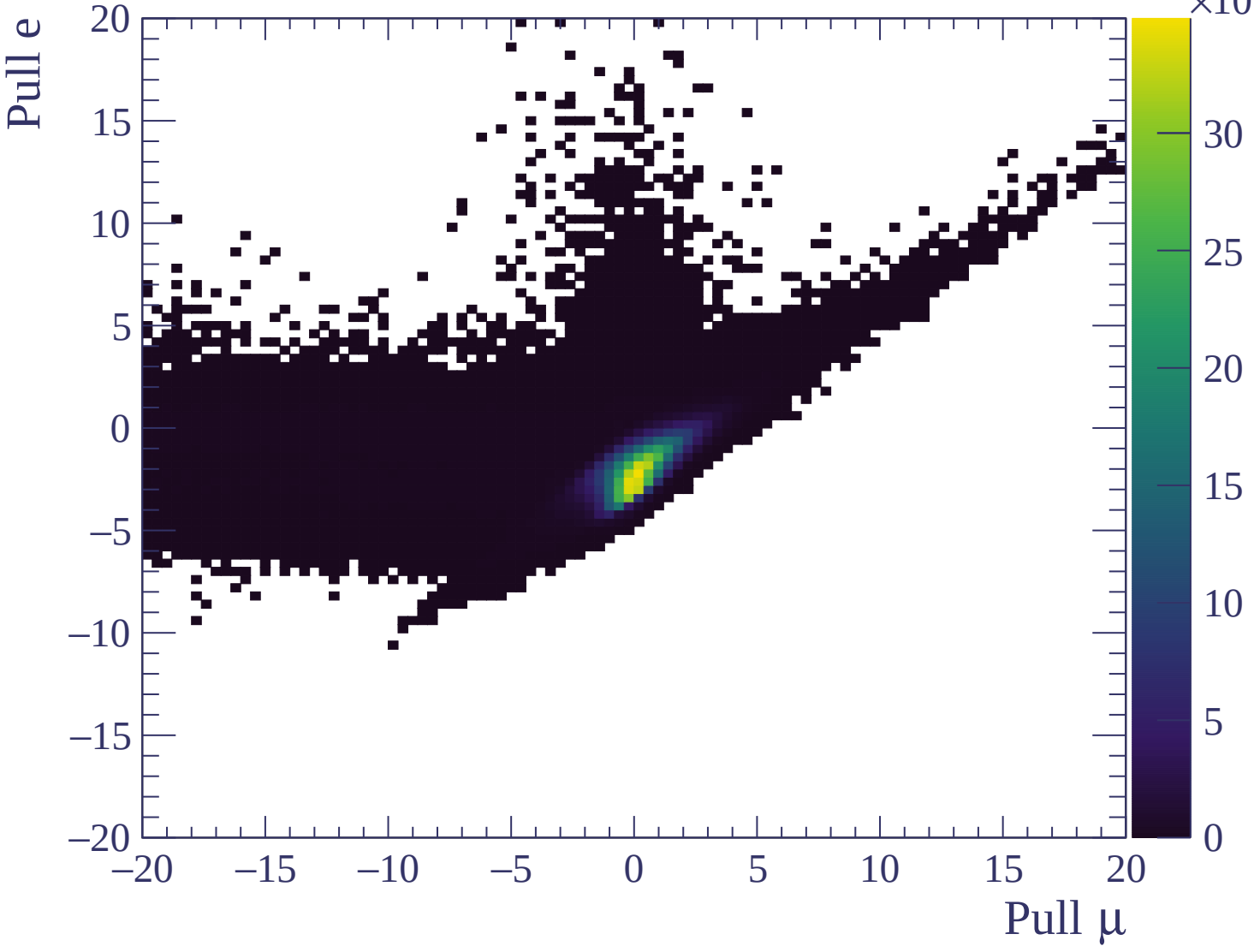


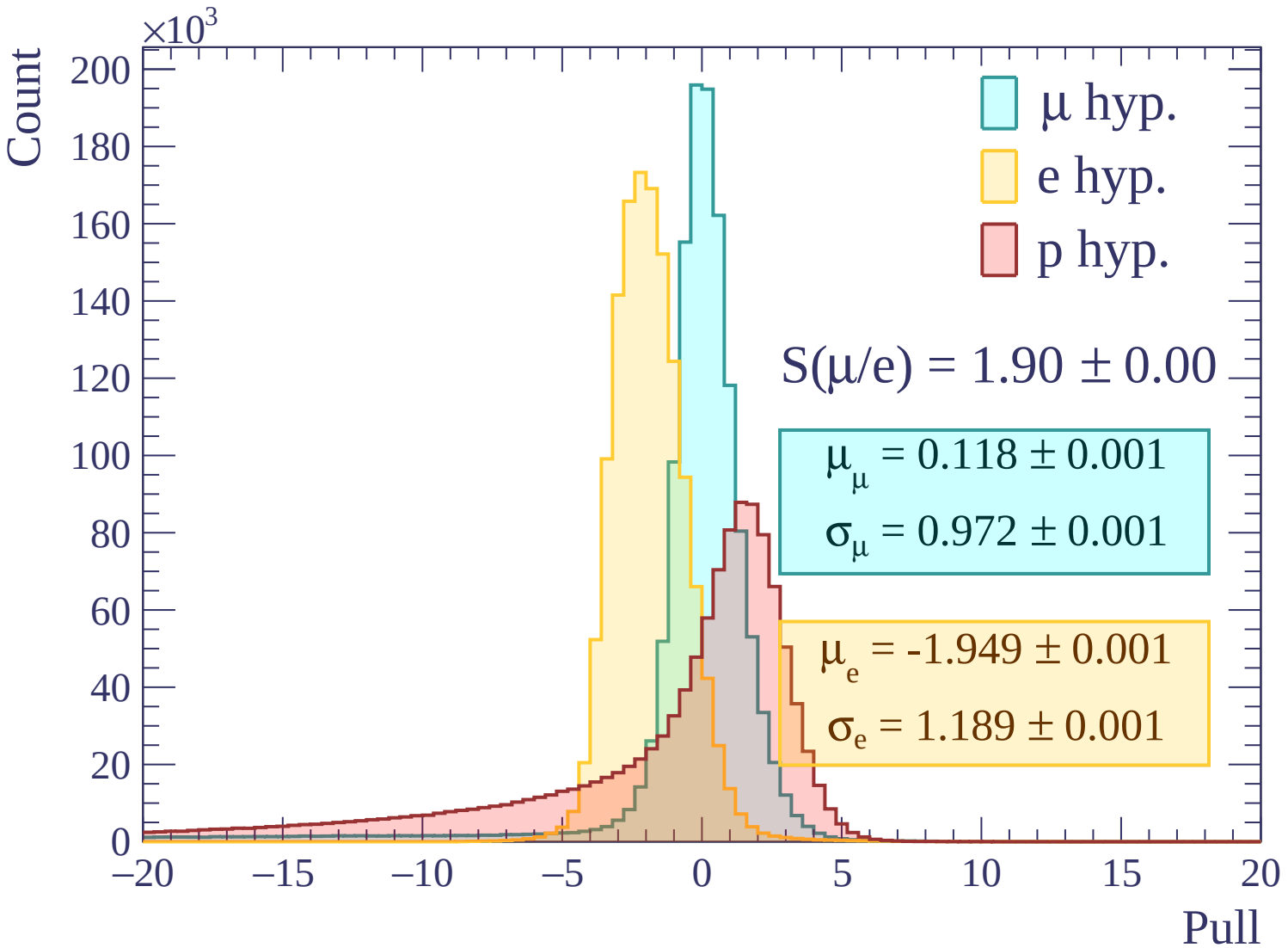


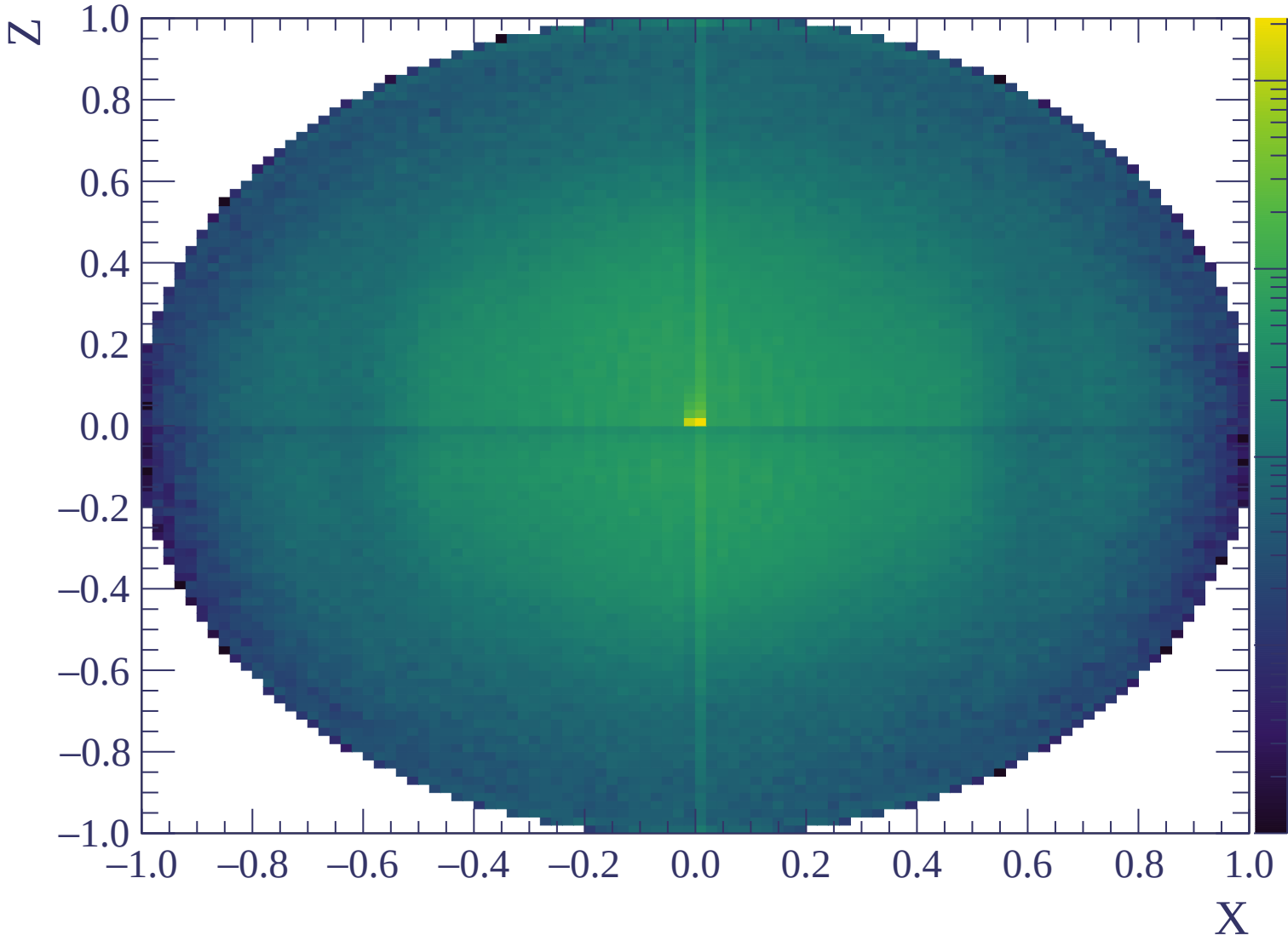


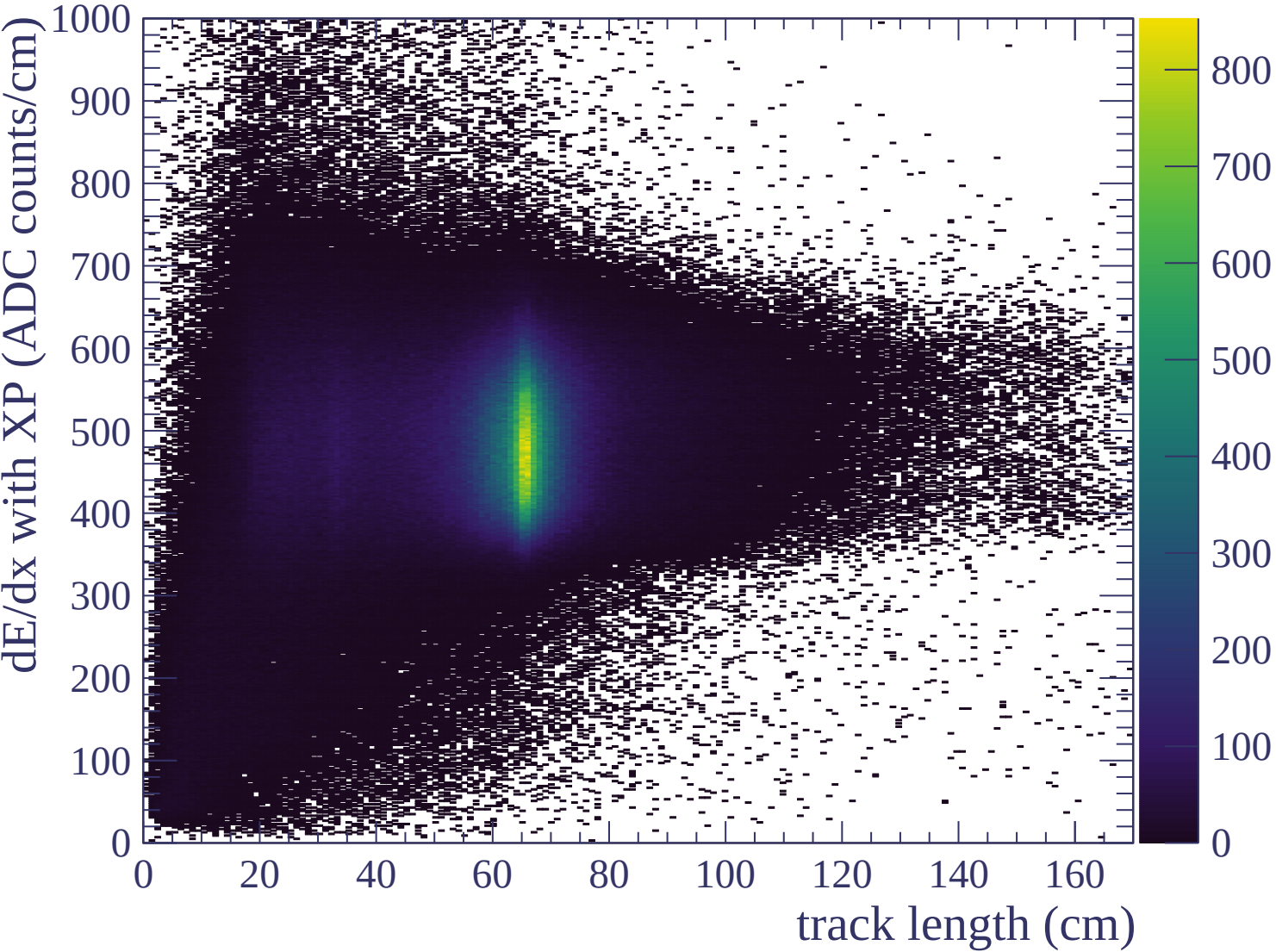


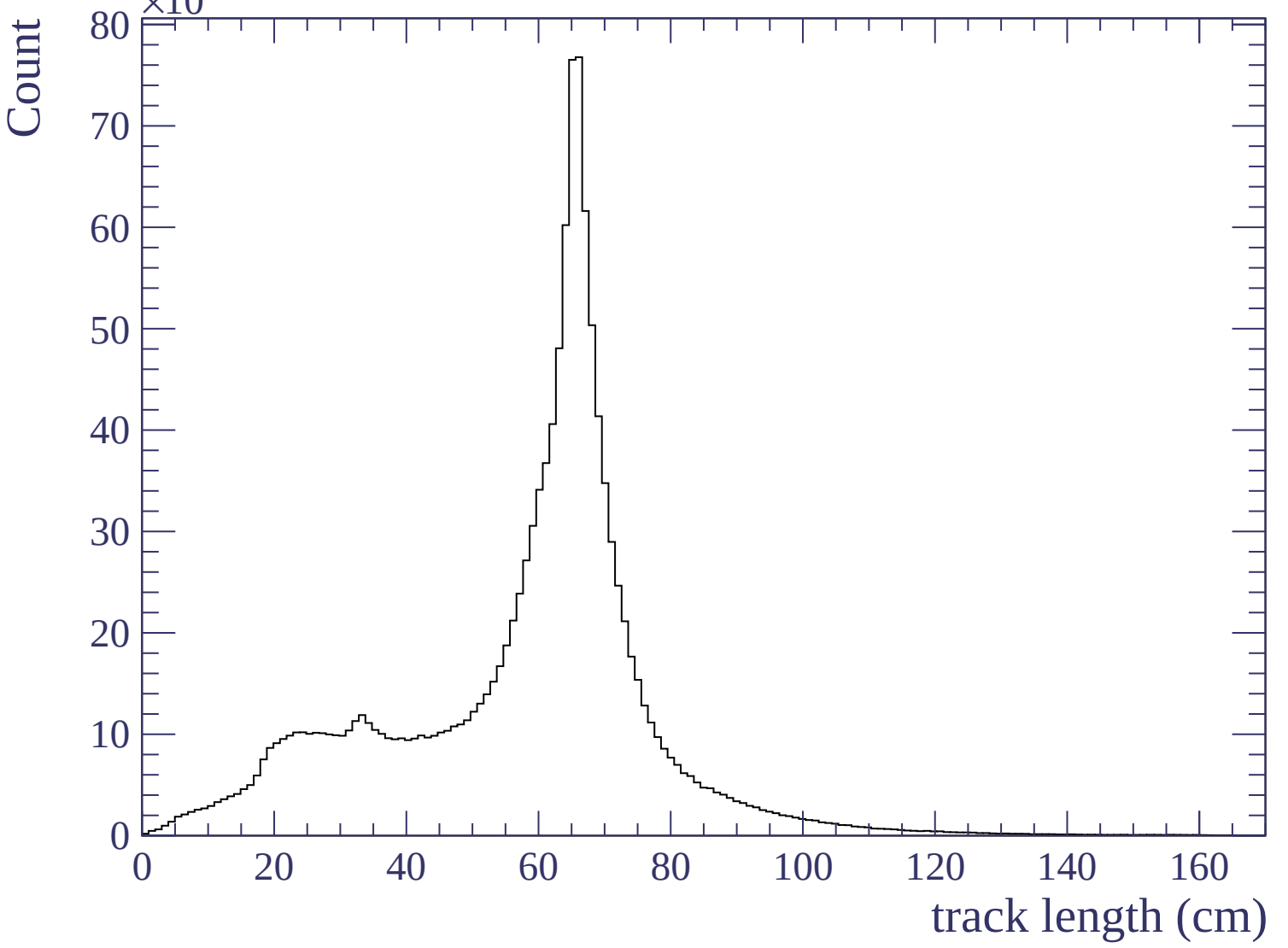


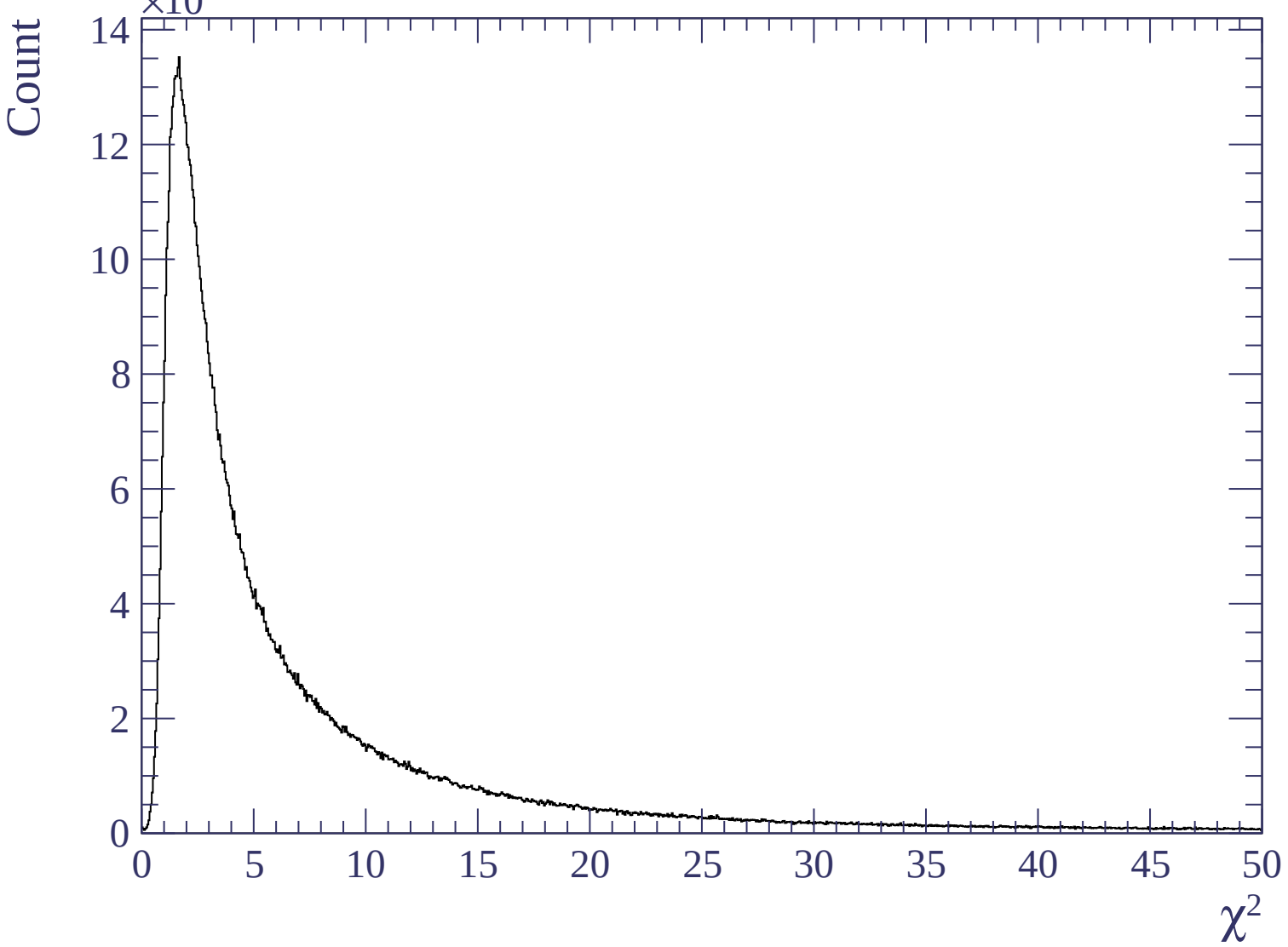


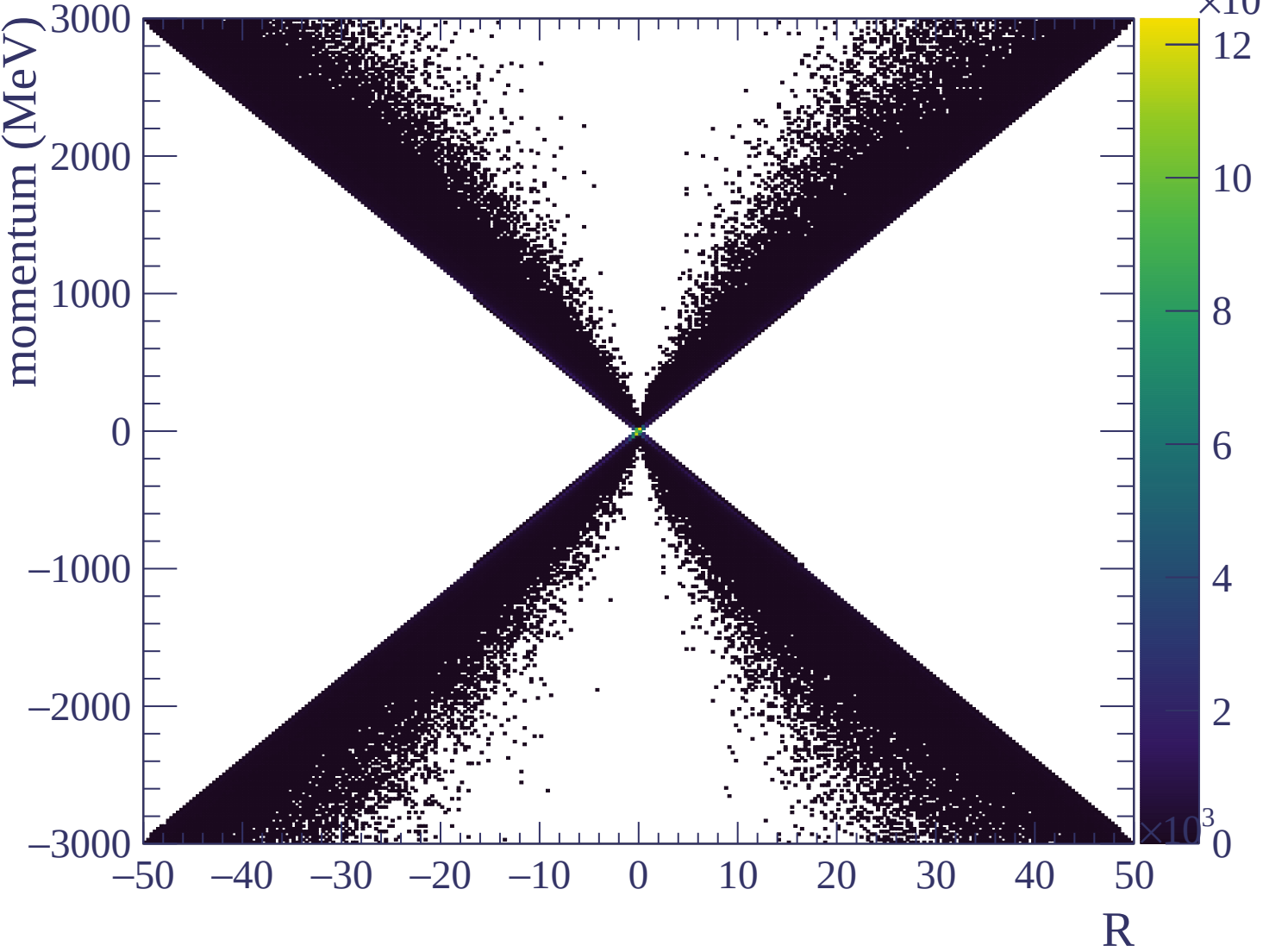


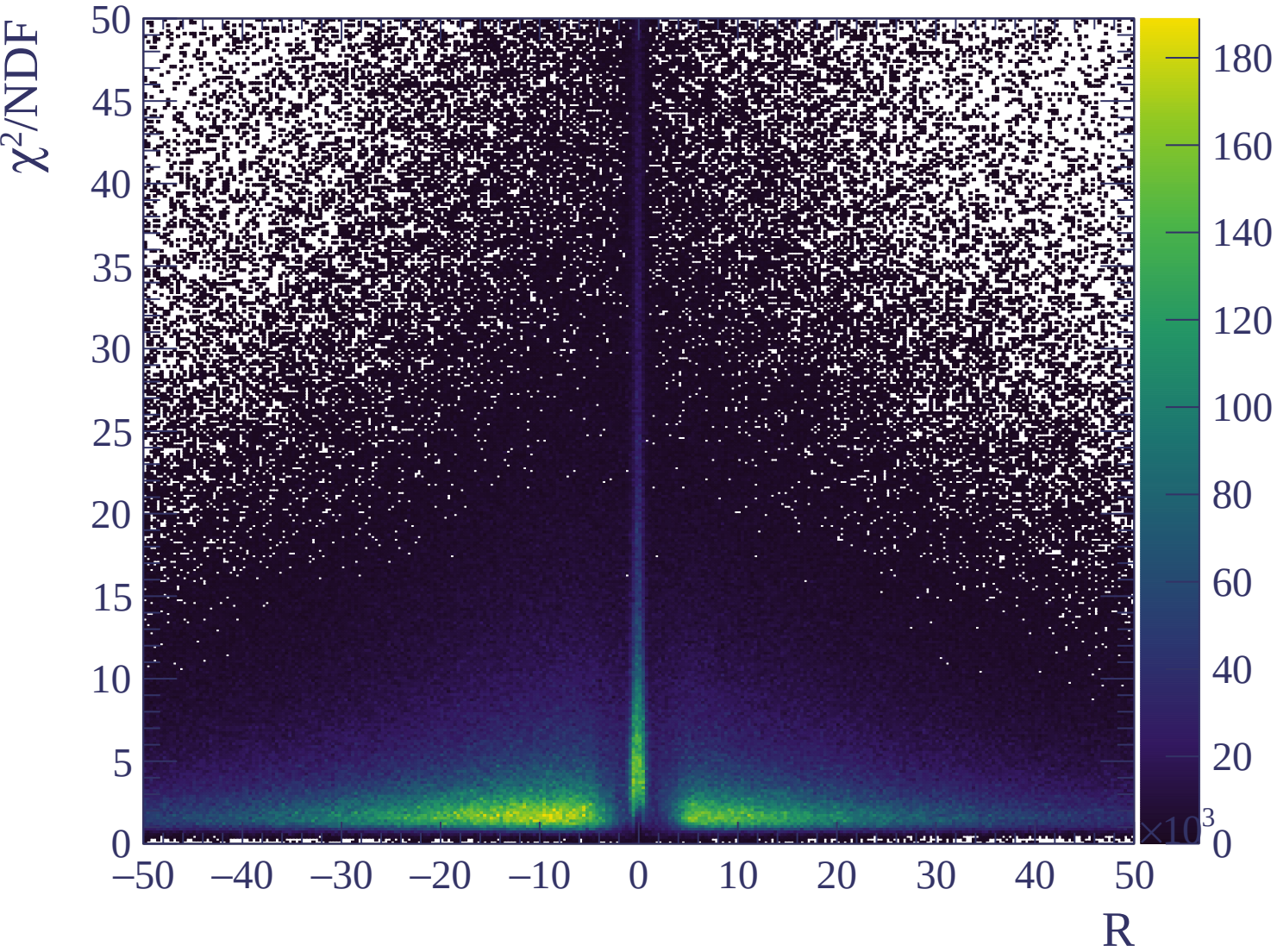






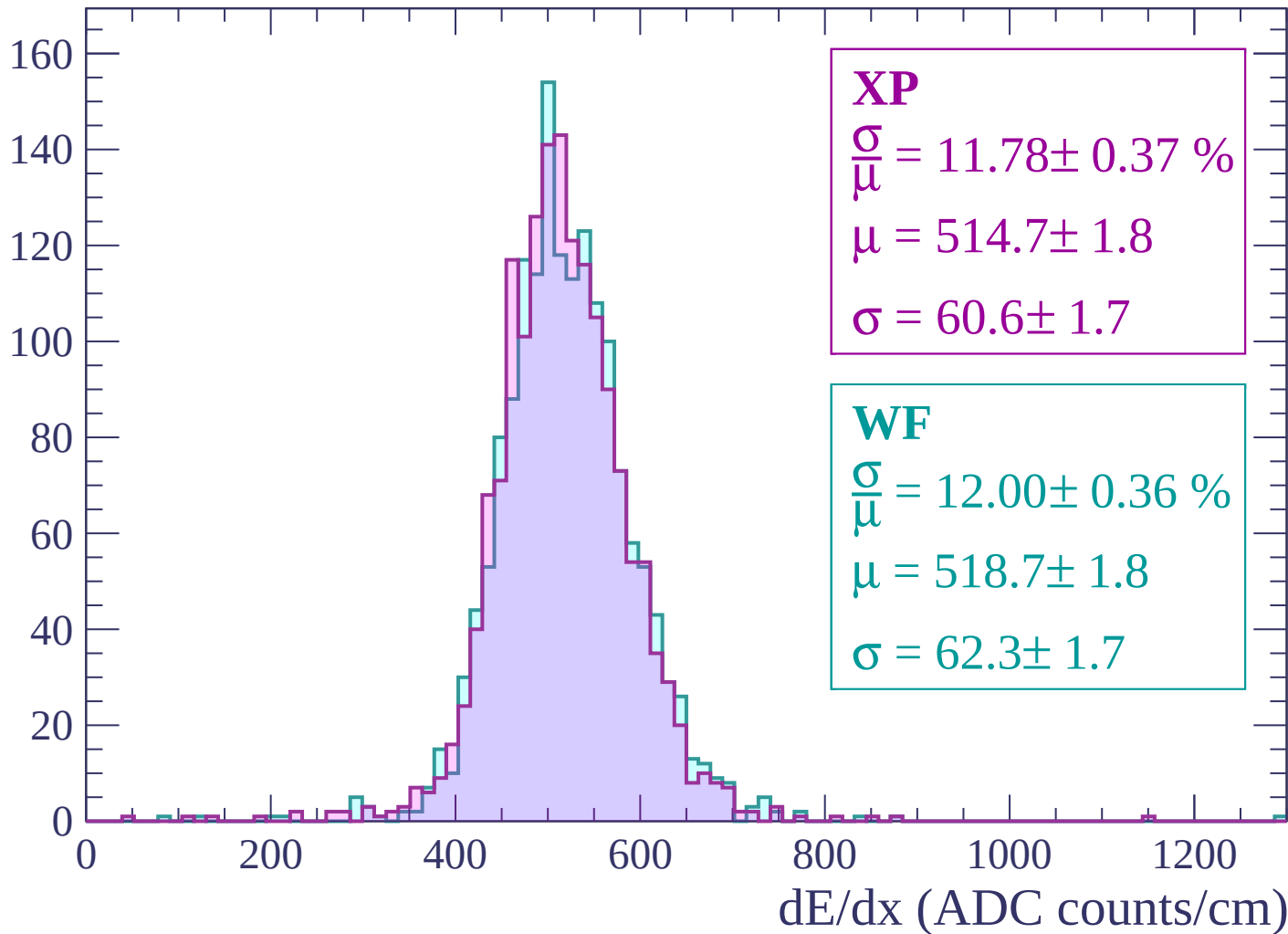






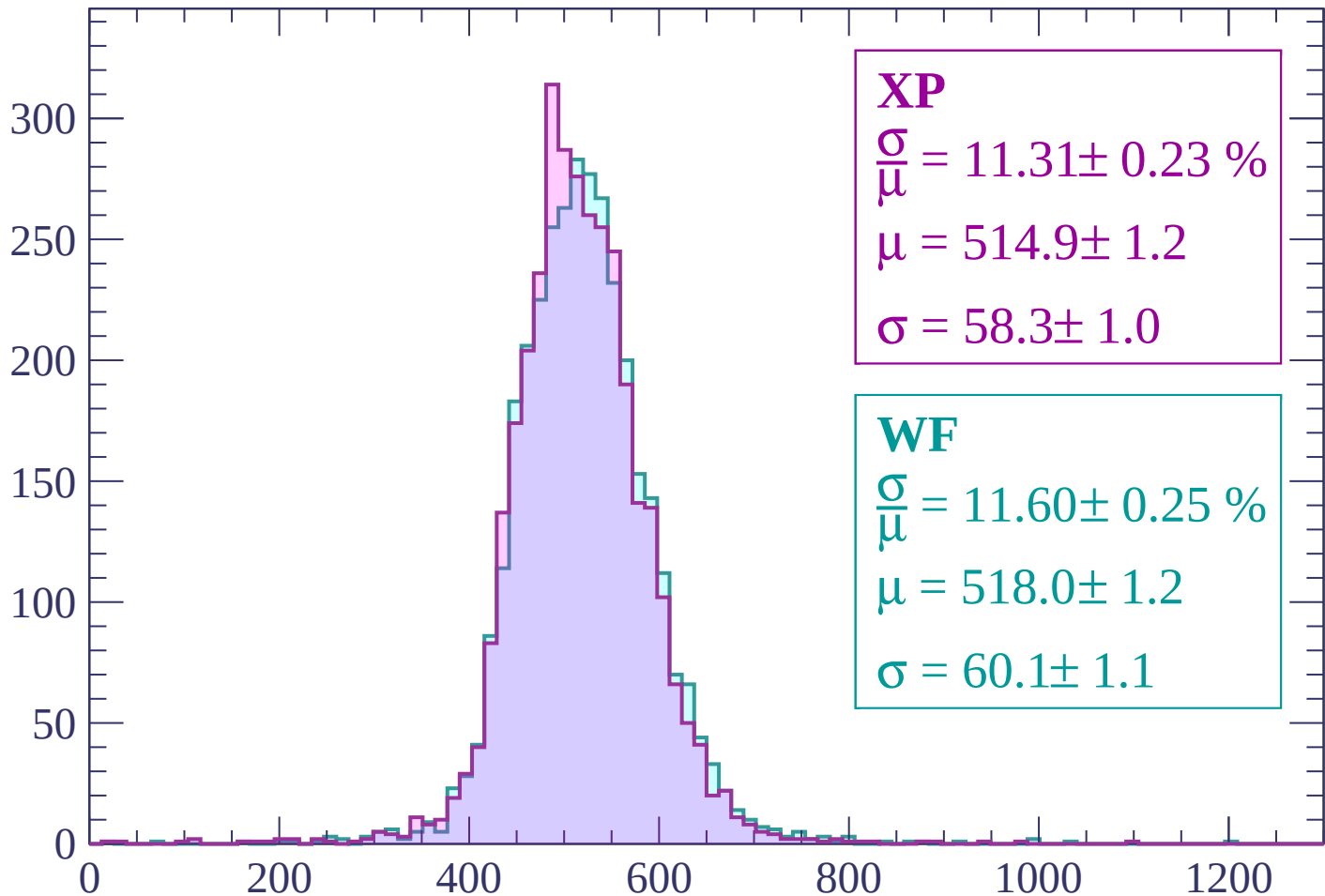
Energy loss | -3000 < p < -2940

Count



Energy loss | $-2940 < p < -2880$

Count



XP

$$\frac{\sigma}{\mu} = 11.31 \pm 0.23 \%$$

$$\mu = 514.9 \pm 1.2$$

$$\sigma = 58.3 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 11.60 \pm 0.25 \%$$

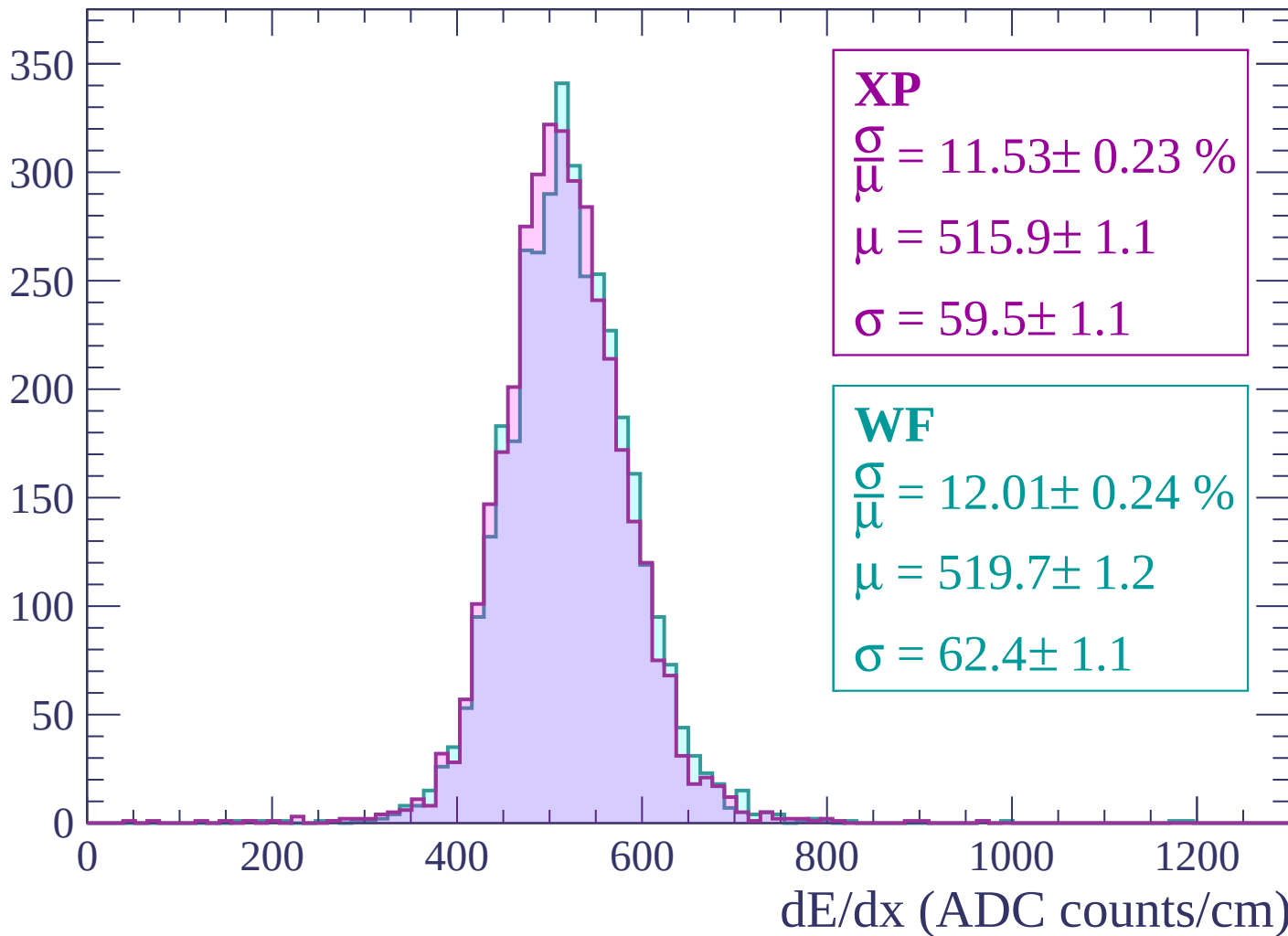
$$\mu = 518.0 \pm 1.2$$

$$\sigma = 60.1 \pm 1.1$$

dE/dx (ADC counts/cm)

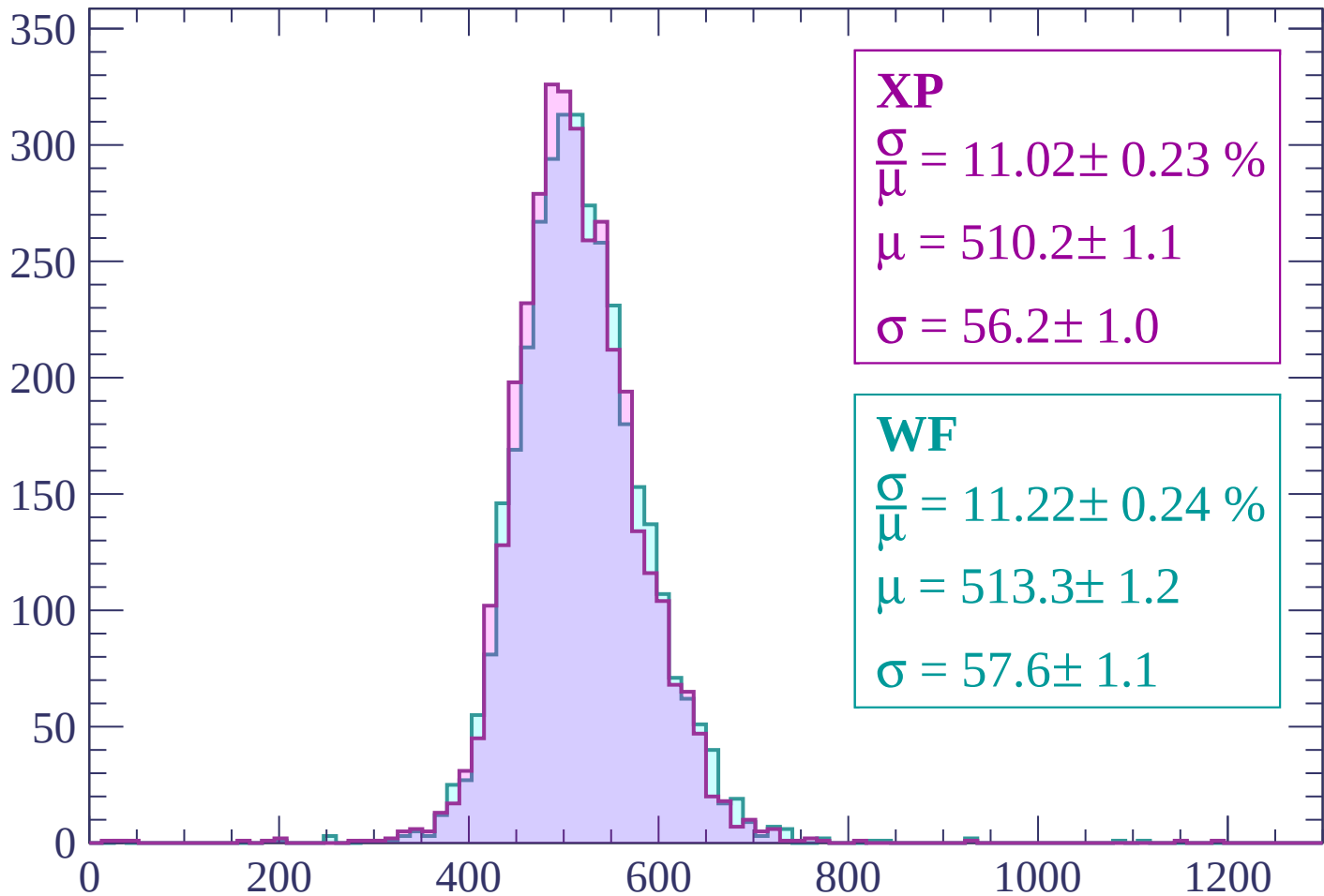
Energy loss | $-2880 < p < -2820$

Count



Energy loss | $-2820 < p < -2760$

Count



XP

$$\frac{\sigma}{\mu} = 11.02 \pm 0.23 \%$$

$$\mu = 510.2 \pm 1.1$$

$$\sigma = 56.2 \pm 1.0$$

WF

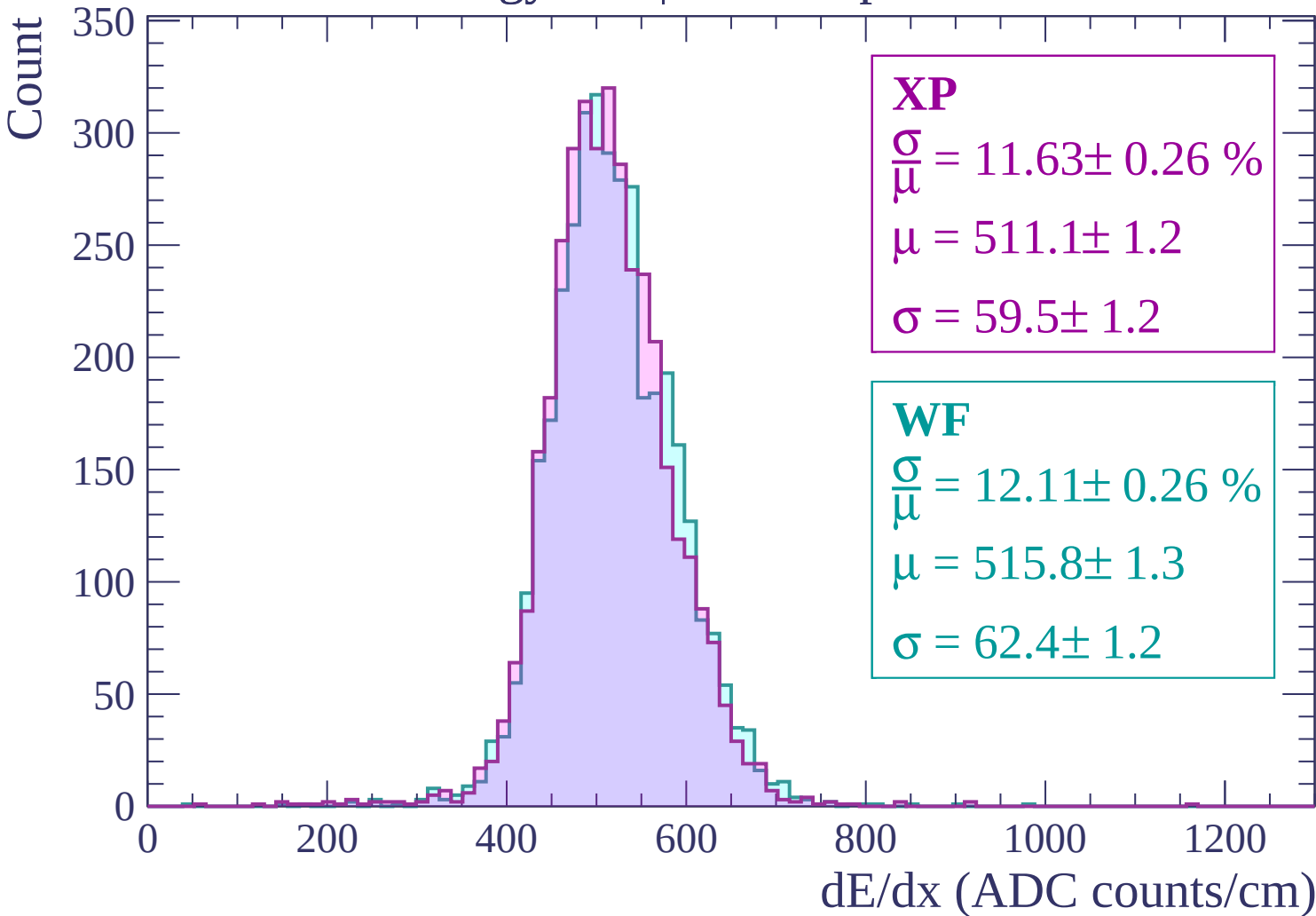
$$\frac{\sigma}{\mu} = 11.22 \pm 0.24 \%$$

$$\mu = 513.3 \pm 1.2$$

$$\sigma = 57.6 \pm 1.1$$

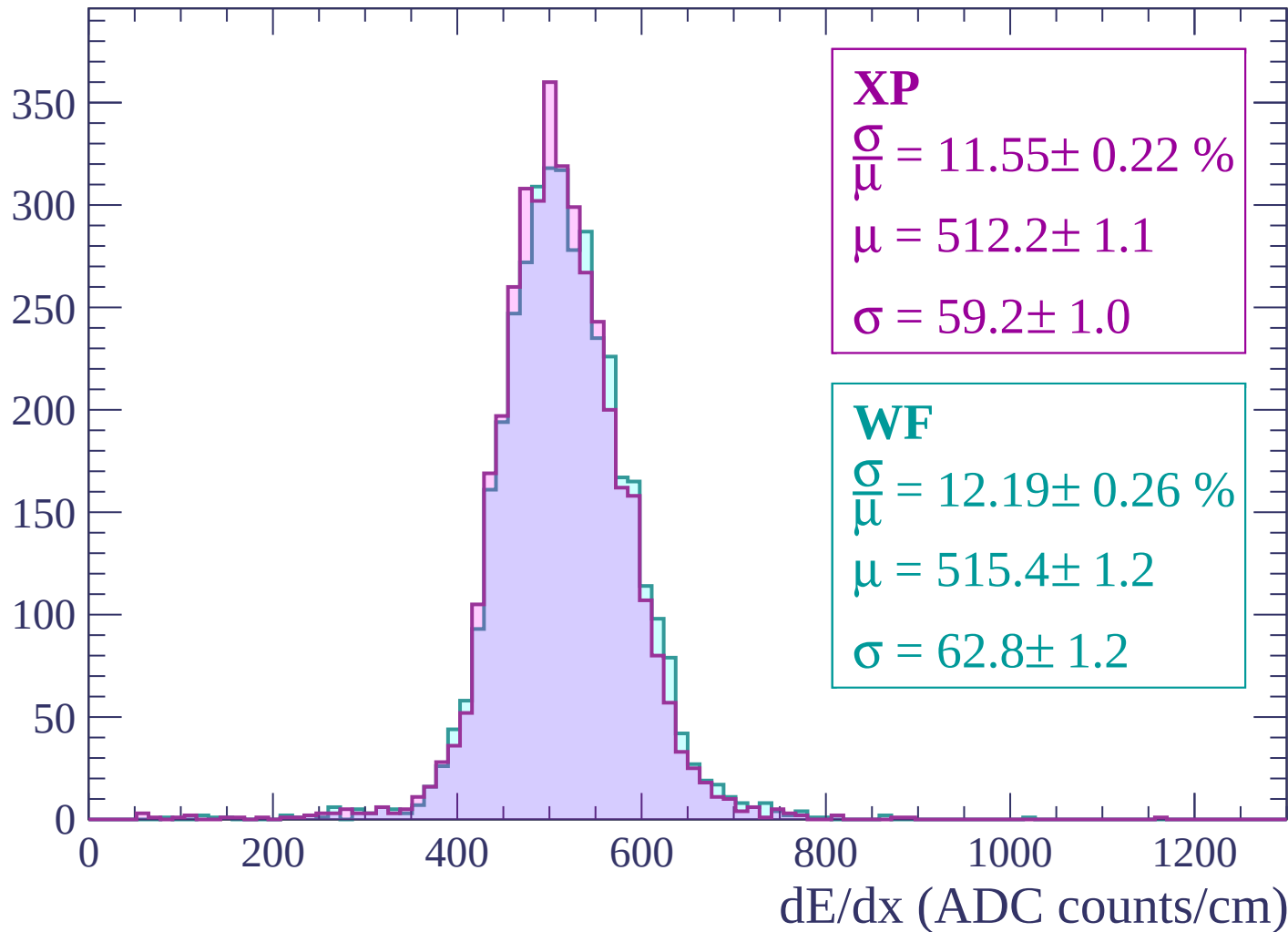
dE/dx (ADC counts/cm)

Energy loss | -2760 < p < -2700



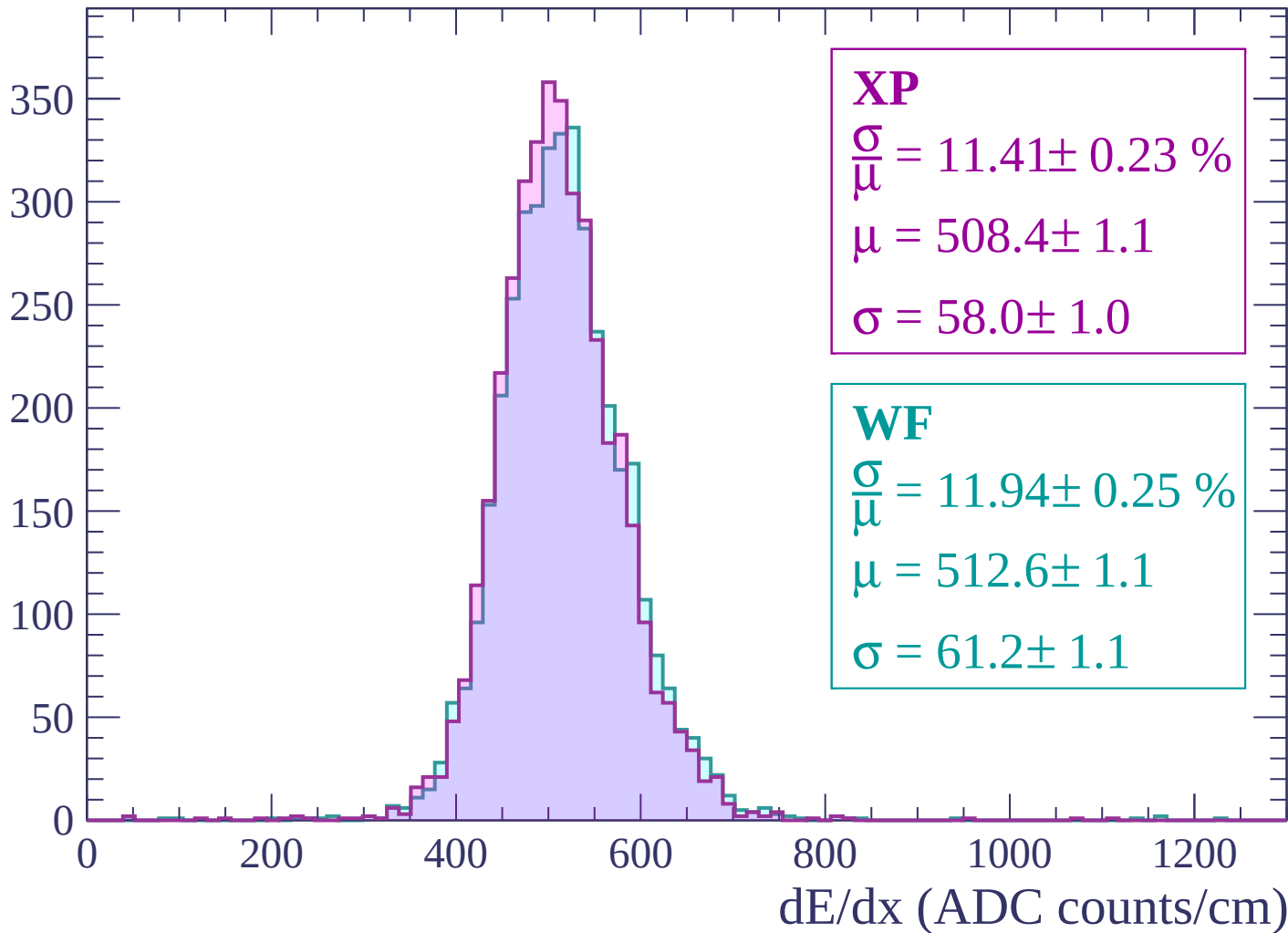
Energy loss | $-2700 < p < -2640$

Count



Energy loss | -2640 < p < -2580

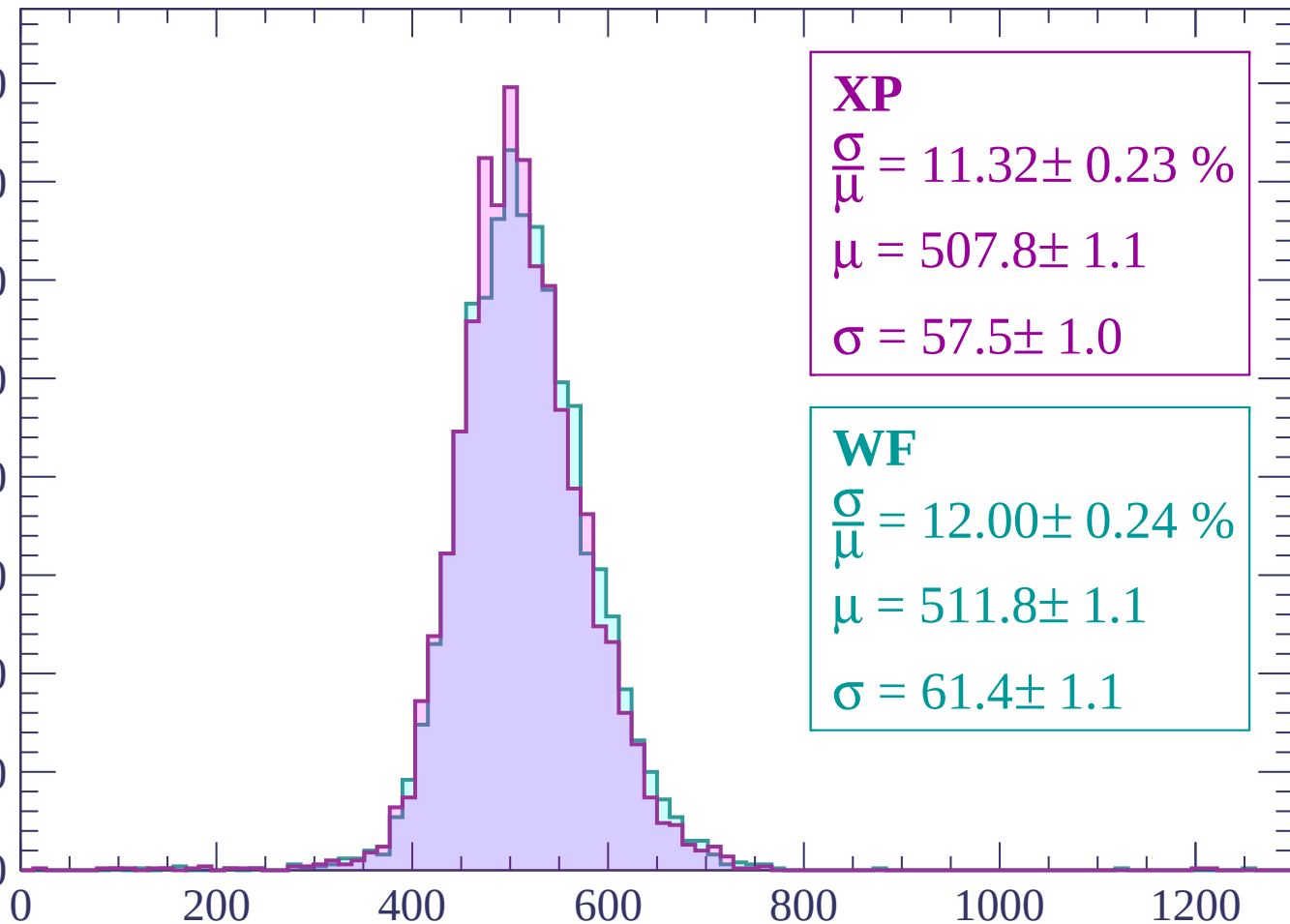
Count



Energy loss | $-2580 < p < -2520$

Count

400
350
300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 11.32 \pm 0.23 \%$$

$$\mu = 507.8 \pm 1.1$$

$$\sigma = 57.5 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 12.00 \pm 0.24 \%$$

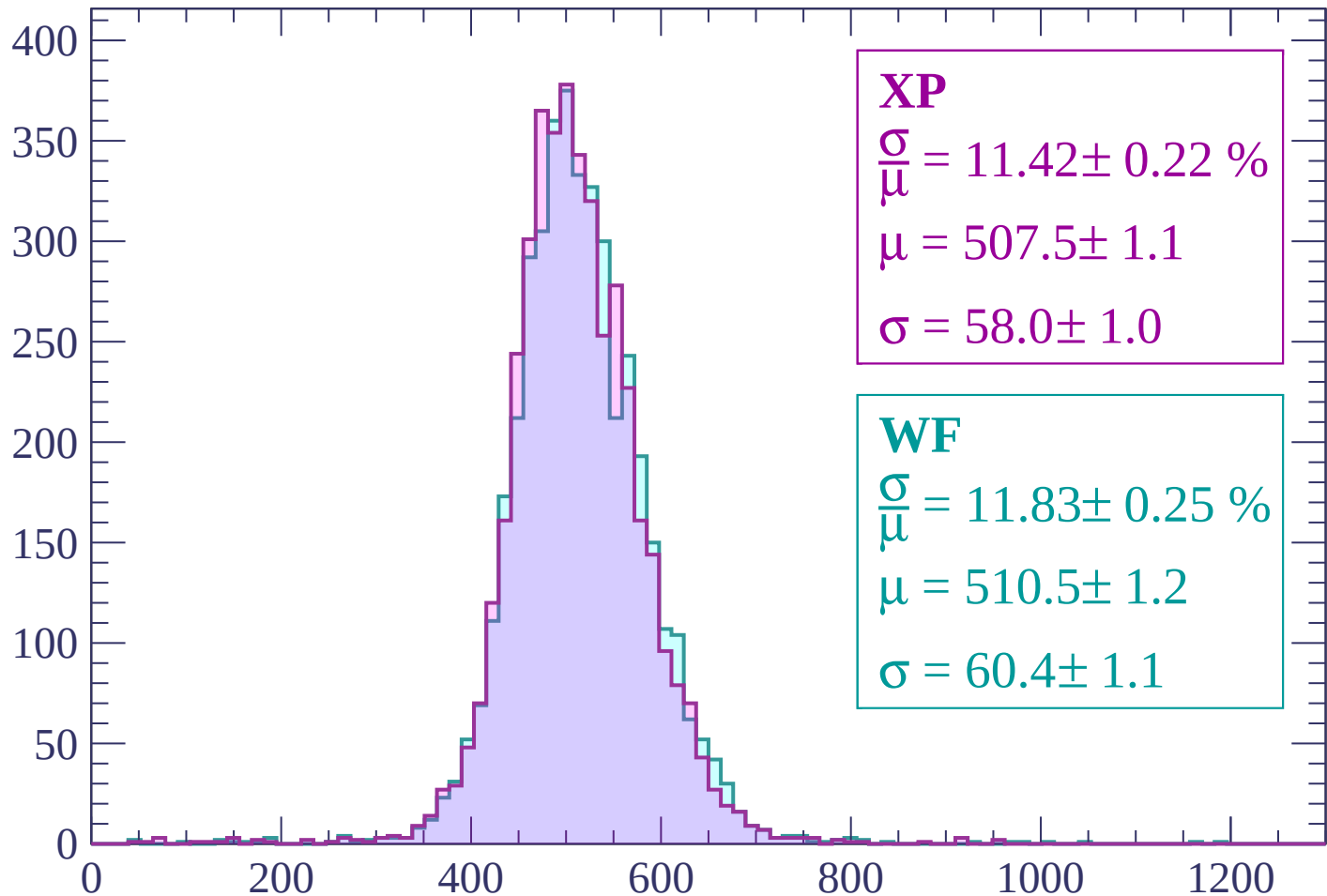
$$\mu = 511.8 \pm 1.1$$

$$\sigma = 61.4 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | -2520 < p < -2460

Count



XP

$$\frac{\sigma}{\mu} = 11.42 \pm 0.22 \%$$

$$\mu = 507.5 \pm 1.1$$

$$\sigma = 58.0 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 11.83 \pm 0.25 \%$$

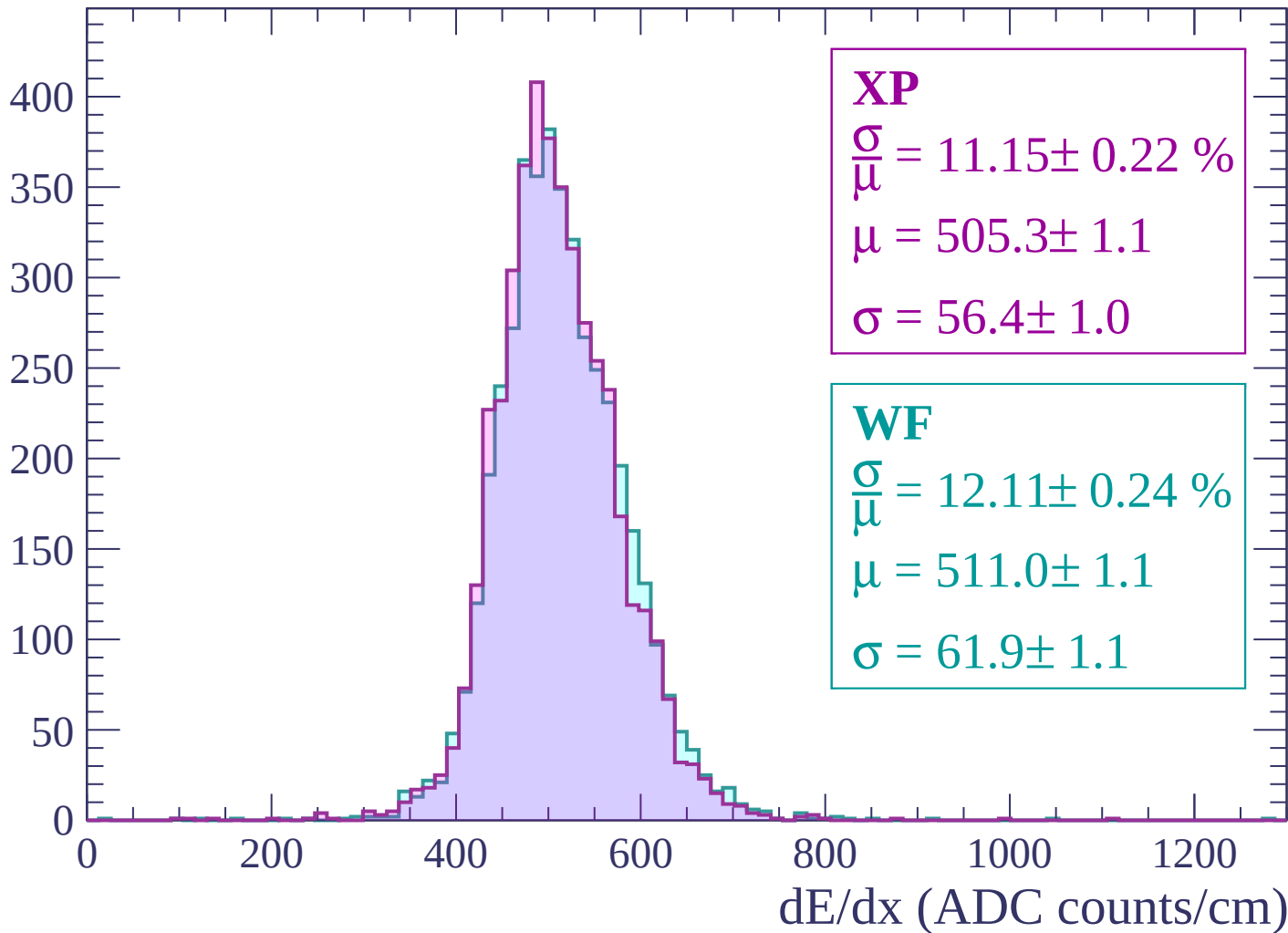
$$\mu = 510.5 \pm 1.2$$

$$\sigma = 60.4 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | $-2460 < p < -2400$

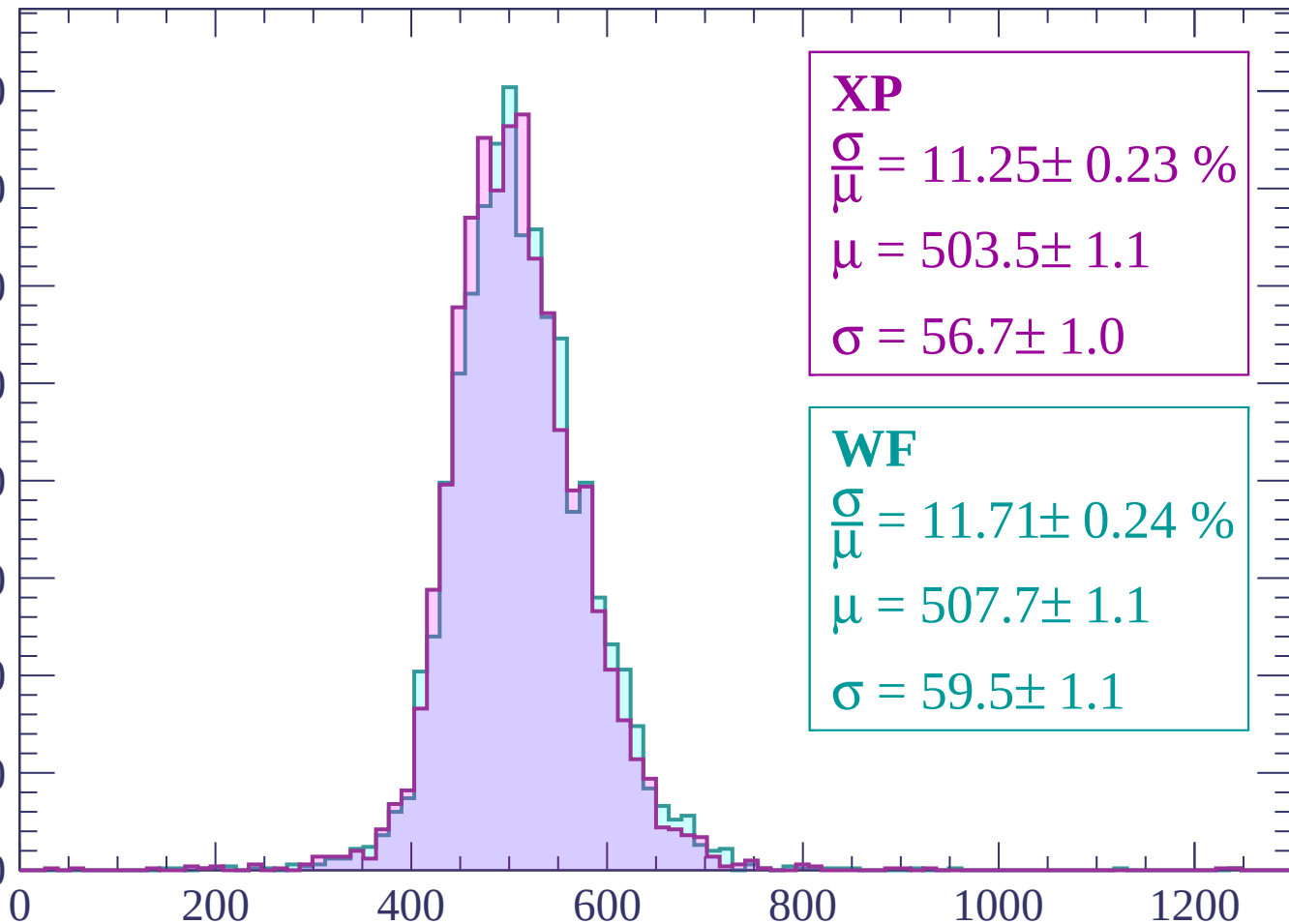
Count



Energy loss | -2400 < p < -2340

Count

400
350
300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 11.25 \pm 0.23 \%$$

$$\mu = 503.5 \pm 1.1$$

$$\sigma = 56.7 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 11.71 \pm 0.24 \%$$

$$\mu = 507.7 \pm 1.1$$

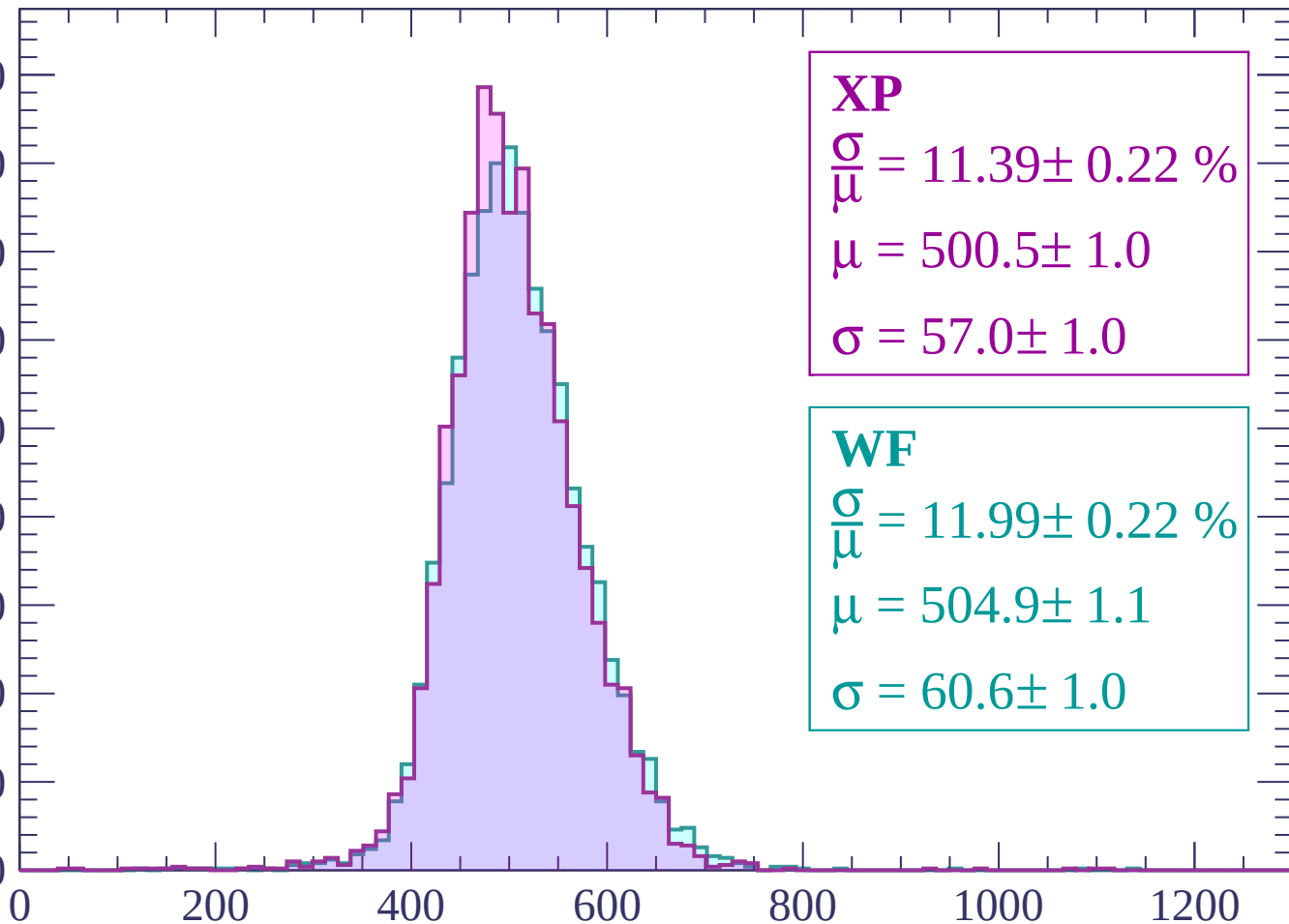
$$\sigma = 59.5 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | $-2340 < p < -2280$

Count

450
400
350
300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 11.39 \pm 0.22 \%$$

$$\mu = 500.5 \pm 1.0$$

$$\sigma = 57.0 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 11.99 \pm 0.22 \%$$

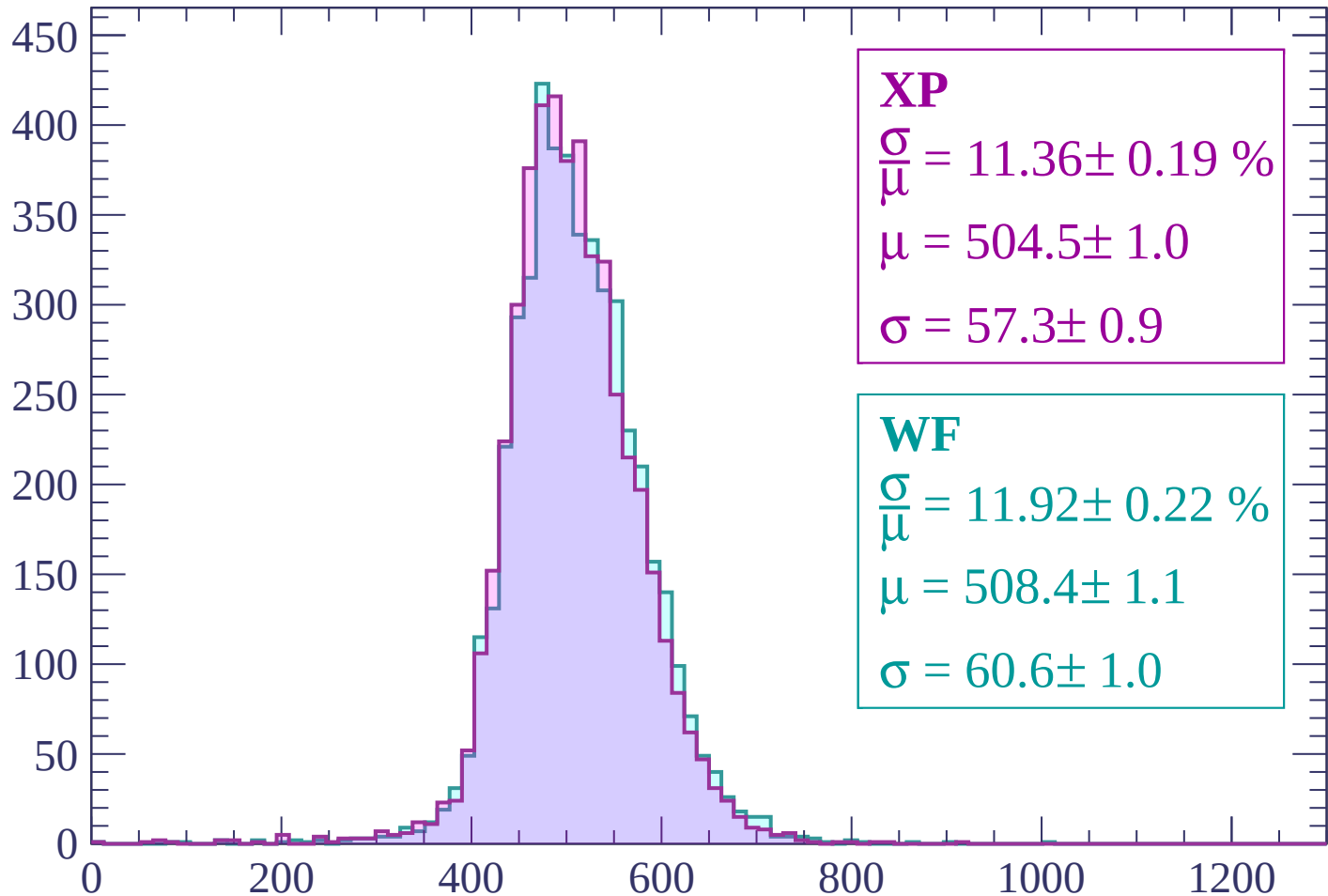
$$\mu = 504.9 \pm 1.1$$

$$\sigma = 60.6 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $-2280 < p < -2220$

Count



XP

$$\frac{\sigma}{\mu} = 11.36 \pm 0.19 \%$$

$$\mu = 504.5 \pm 1.0$$

$$\sigma = 57.3 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 11.92 \pm 0.22 \%$$

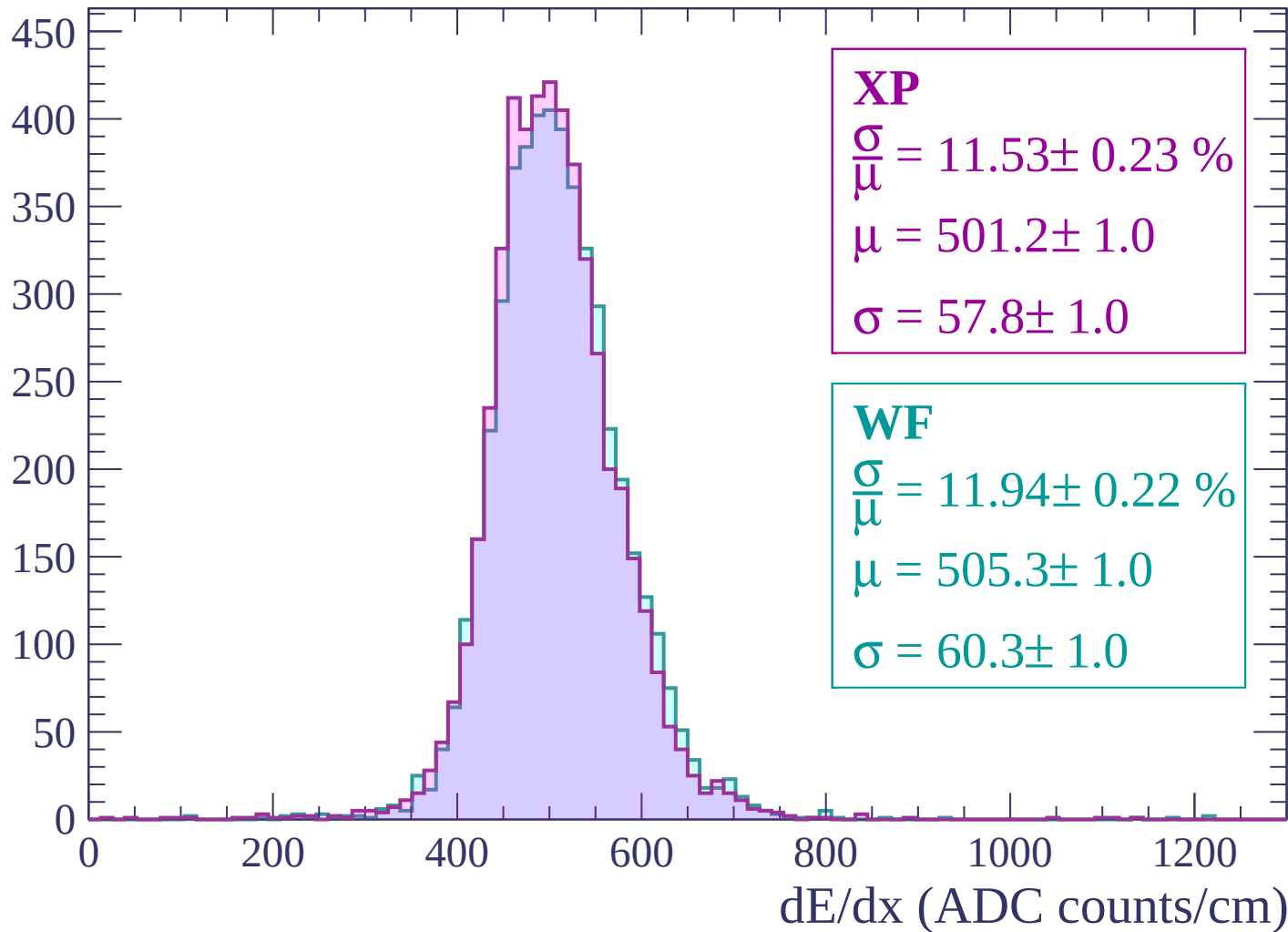
$$\mu = 508.4 \pm 1.1$$

$$\sigma = 60.6 \pm 1.0$$

dE/dx (ADC counts/cm)

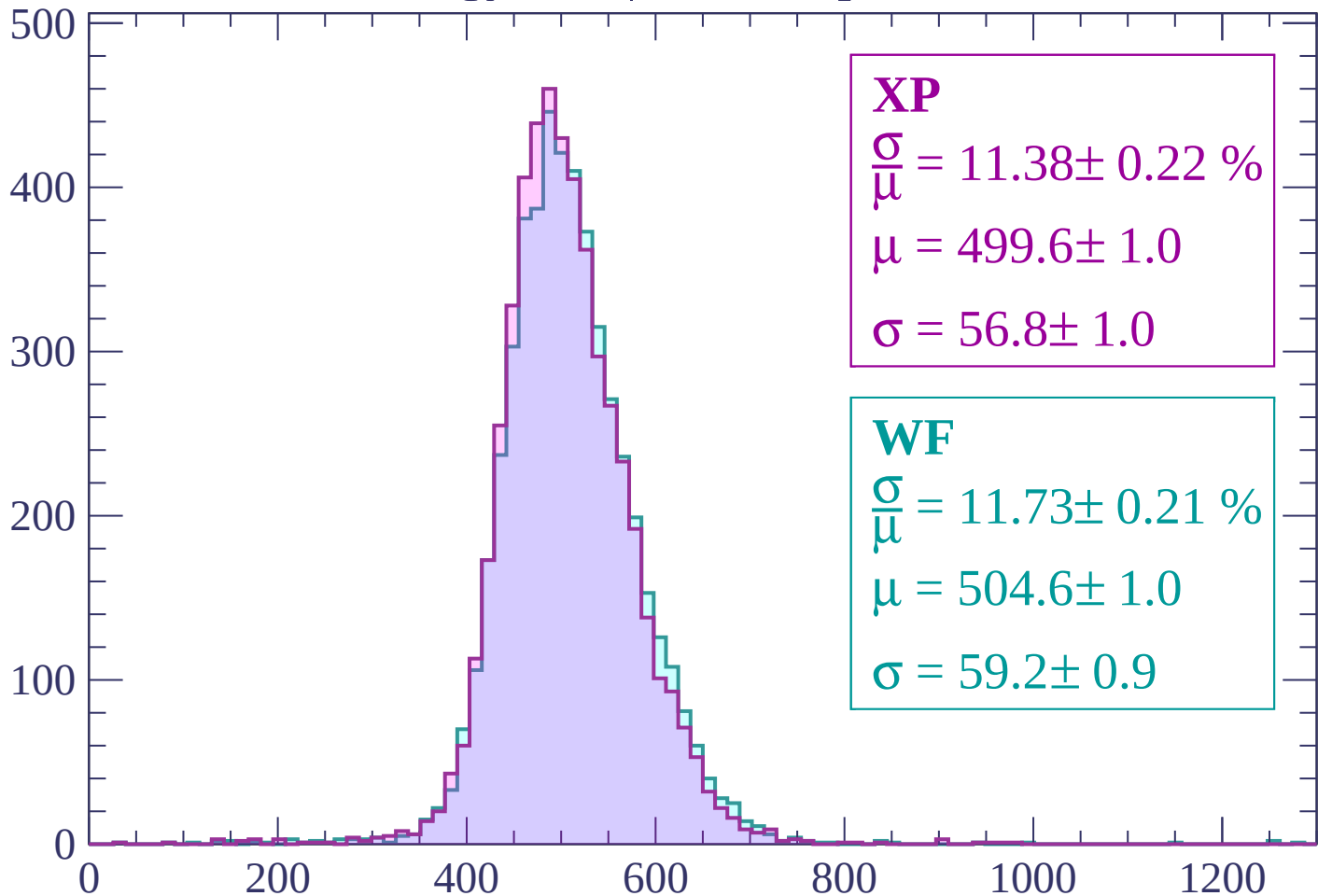
Energy loss | -2220 < p < -2160

Count



Energy loss | -2160 < p < -2100

Count



XP

$$\frac{\sigma}{\mu} = 11.38 \pm 0.22 \%$$

$$\mu = 499.6 \pm 1.0$$

$$\sigma = 56.8 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 11.73 \pm 0.21 \%$$

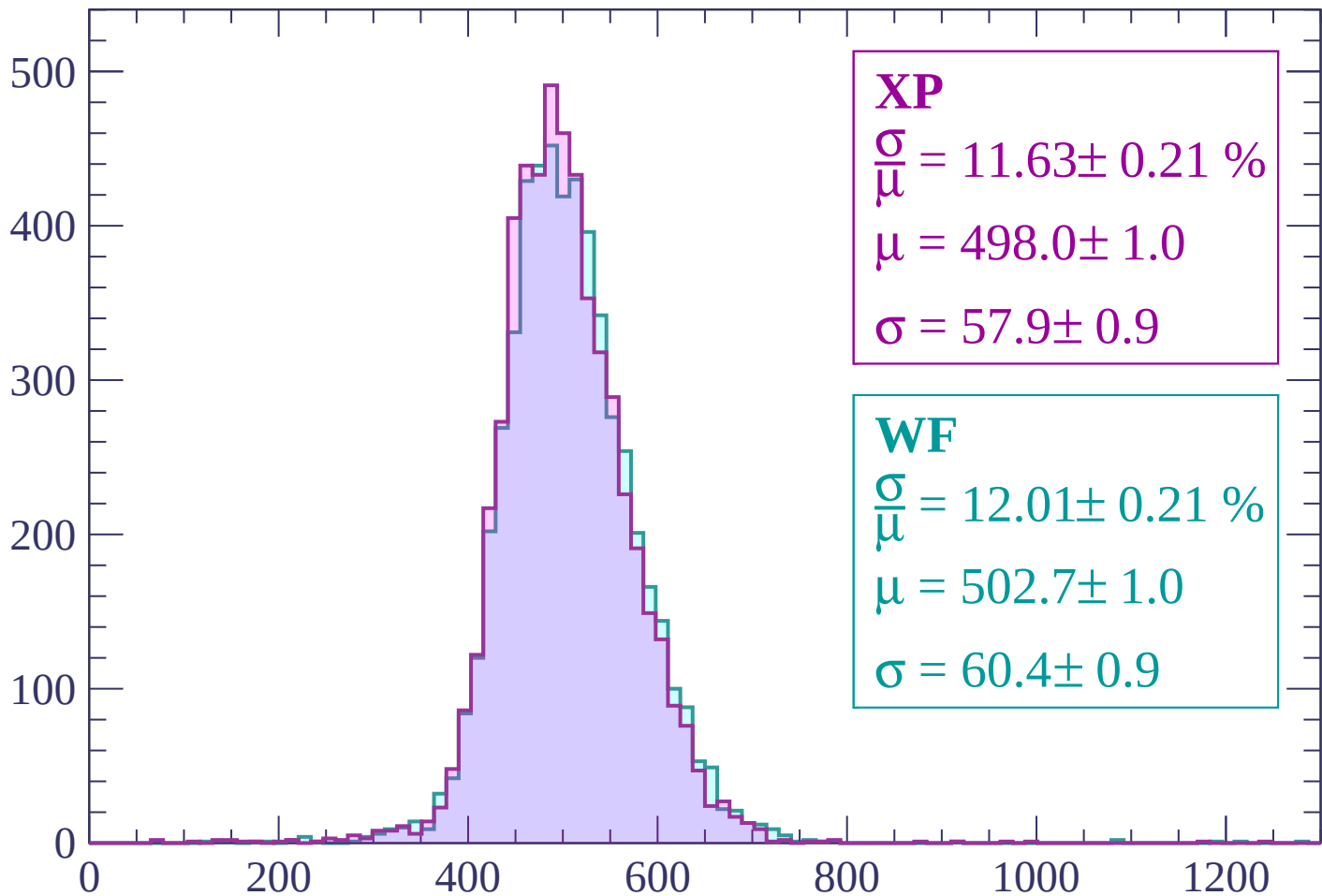
$$\mu = 504.6 \pm 1.0$$

$$\sigma = 59.2 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | -2100 < p < -2040

Count



XP

$$\frac{\sigma}{\mu} = 11.63 \pm 0.21 \%$$

$$\mu = 498.0 \pm 1.0$$

$$\sigma = 57.9 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 12.01 \pm 0.21 \%$$

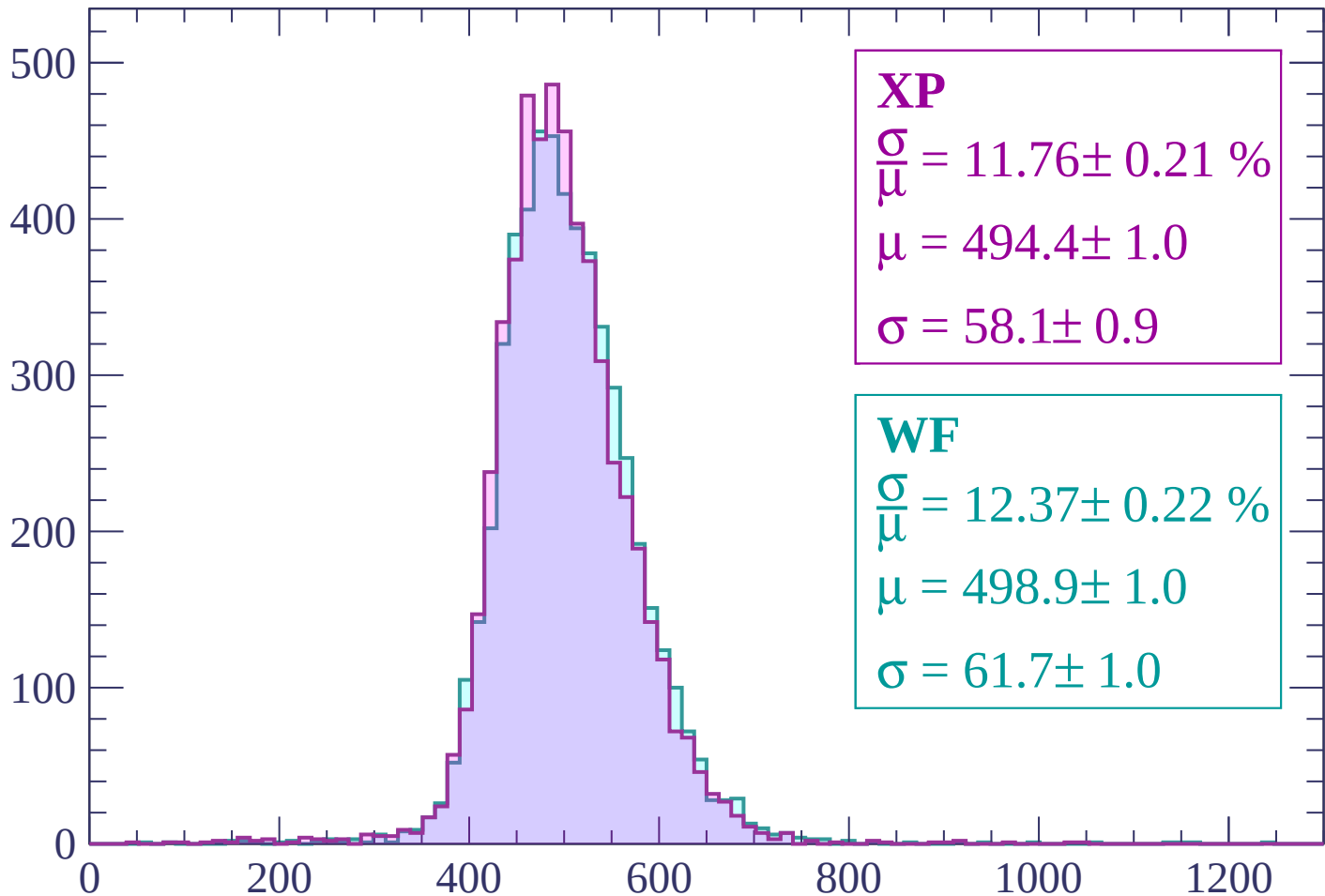
$$\mu = 502.7 \pm 1.0$$

$$\sigma = 60.4 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | -2040 < p < -1980

Count



XP

$$\frac{\sigma}{\mu} = 11.76 \pm 0.21 \%$$

$$\mu = 494.4 \pm 1.0$$

$$\sigma = 58.1 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 12.37 \pm 0.22 \%$$

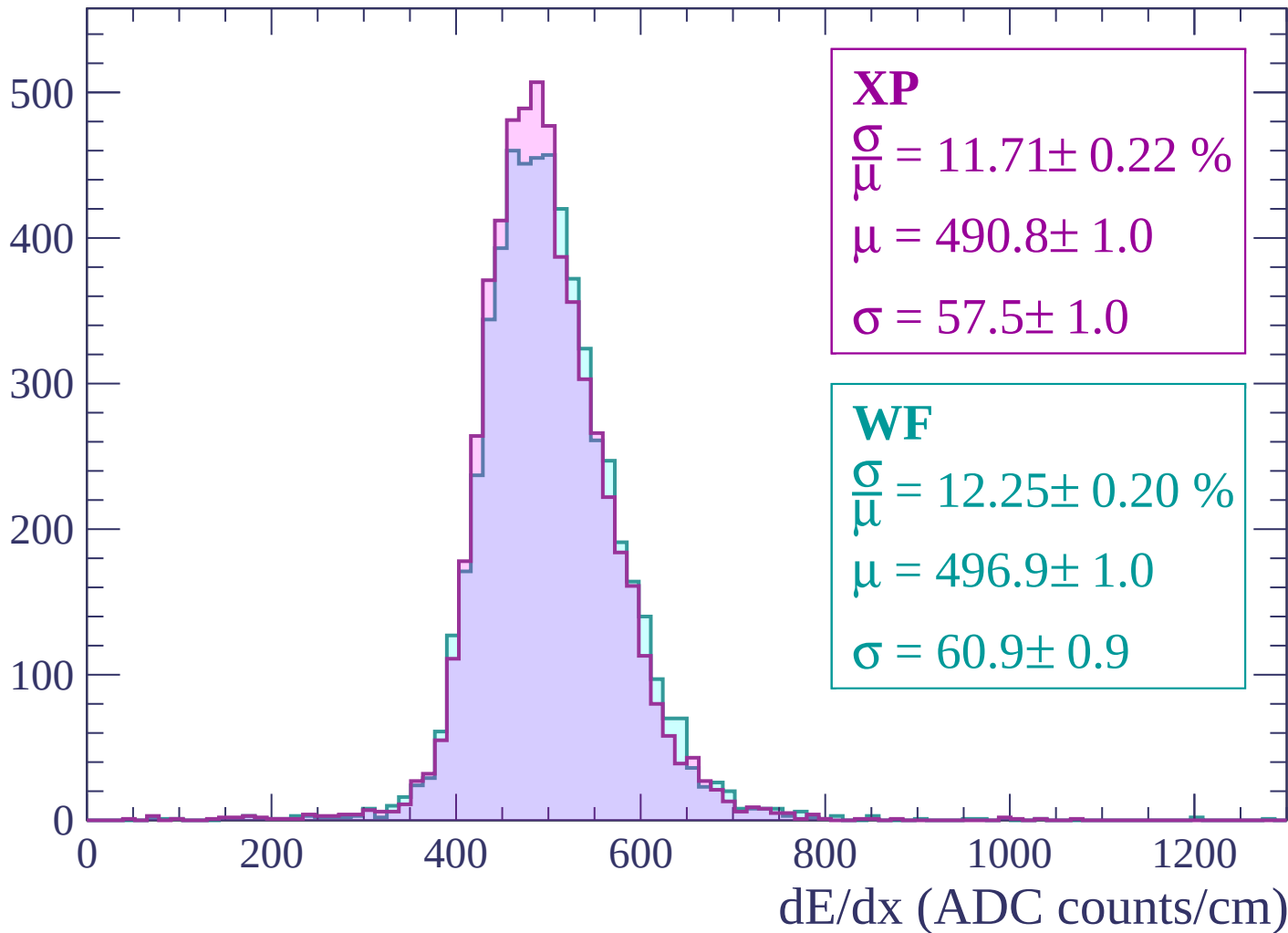
$$\mu = 498.9 \pm 1.0$$

$$\sigma = 61.7 \pm 1.0$$

dE/dx (ADC counts/cm)

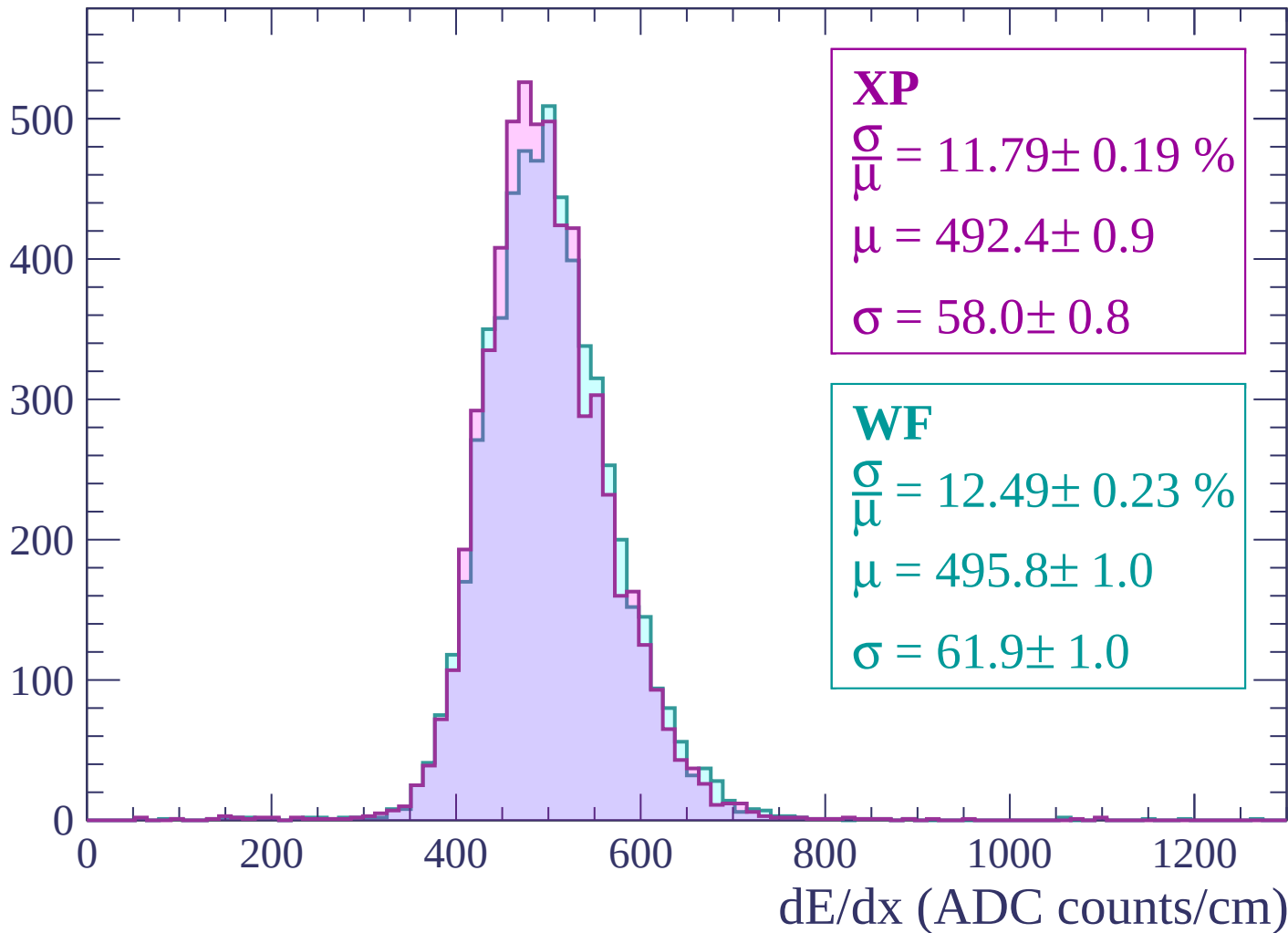
Energy loss | $-1980 < p < -1920$

Count



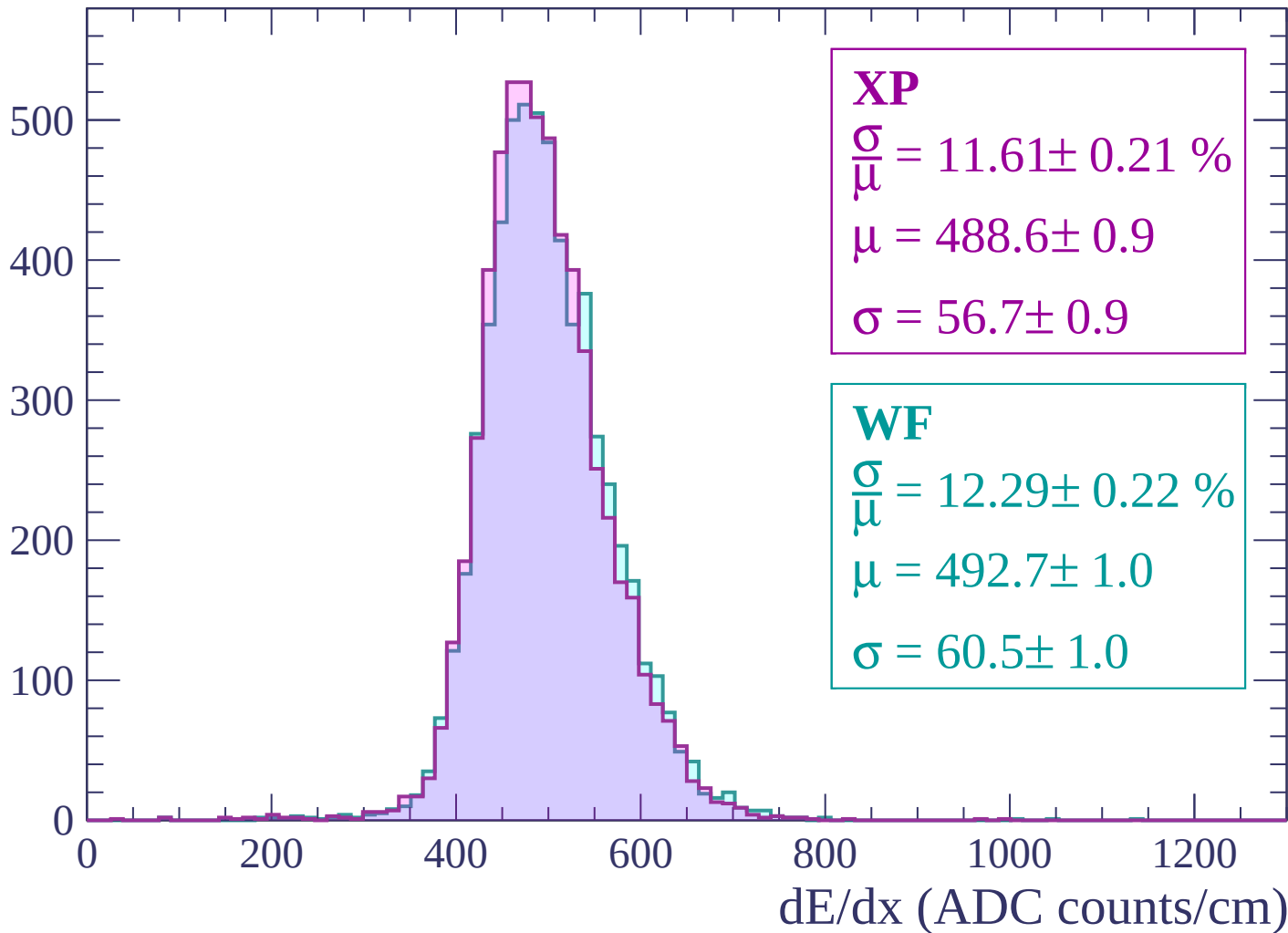
Energy loss | -1920 < p < -1860

Count



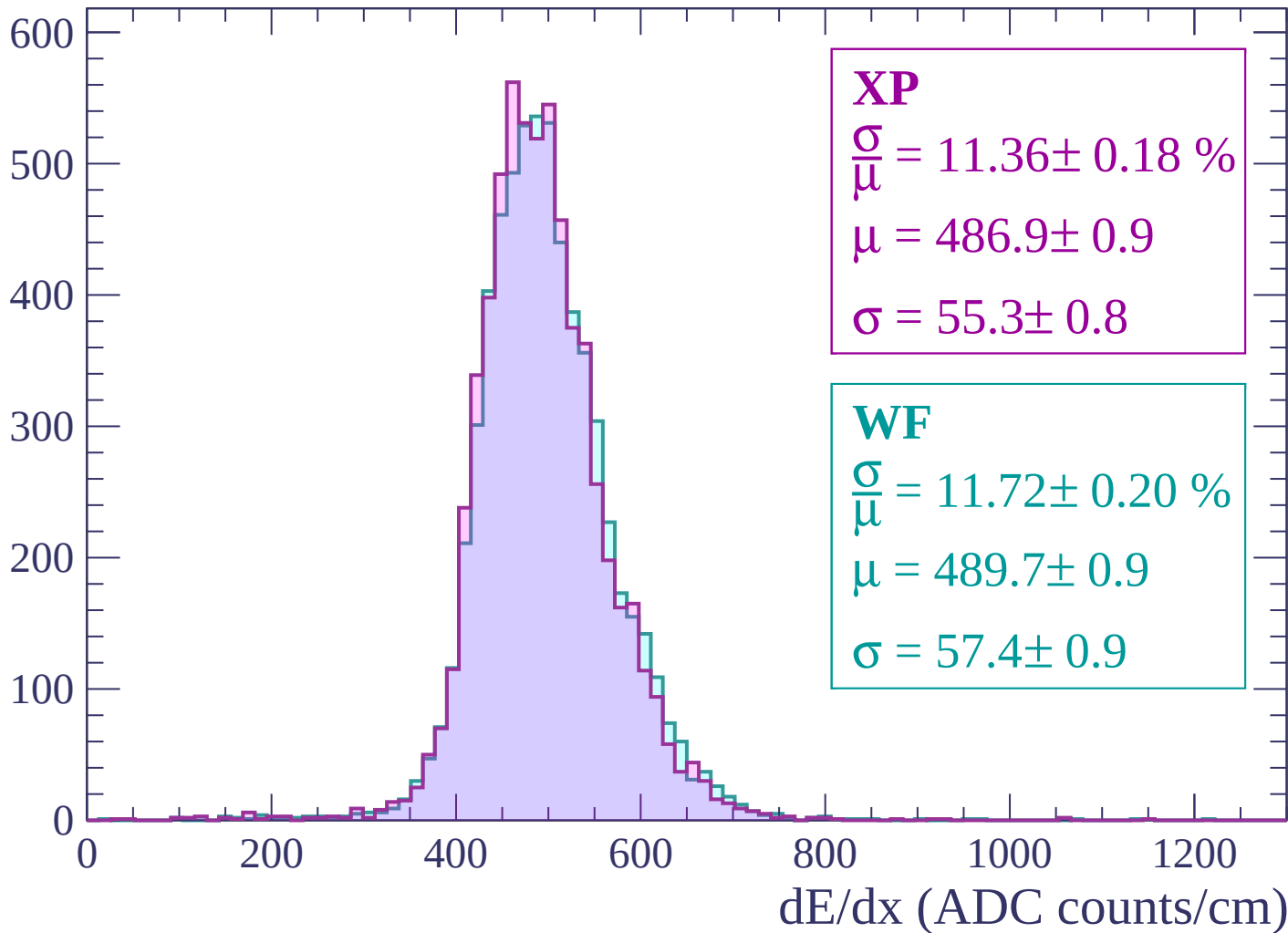
Energy loss | -1860 < p < -1800

Count



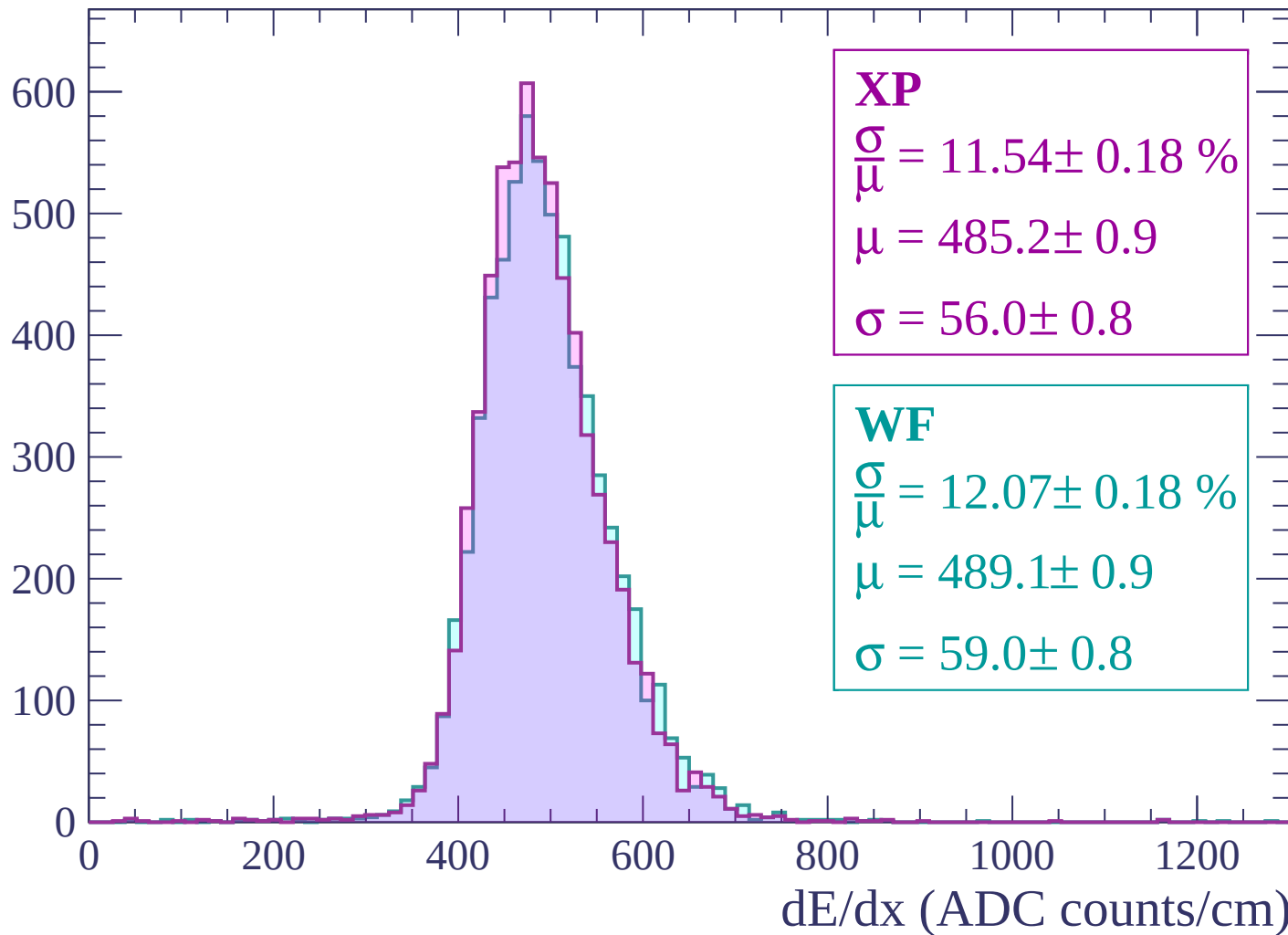
Energy loss | $-1800 < p < -1740$

Count



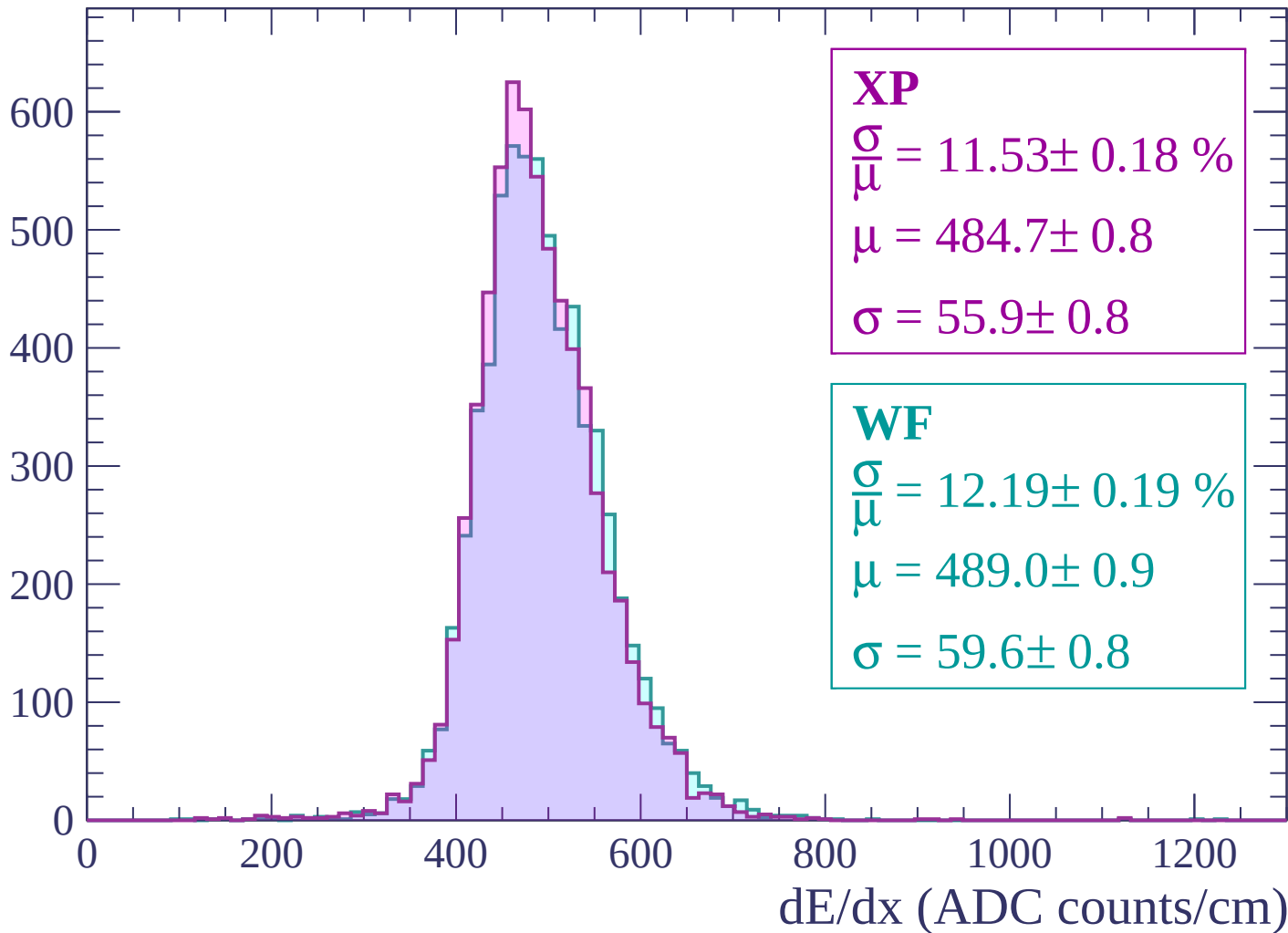
Energy loss | $-1740 < p < -1680$

Count



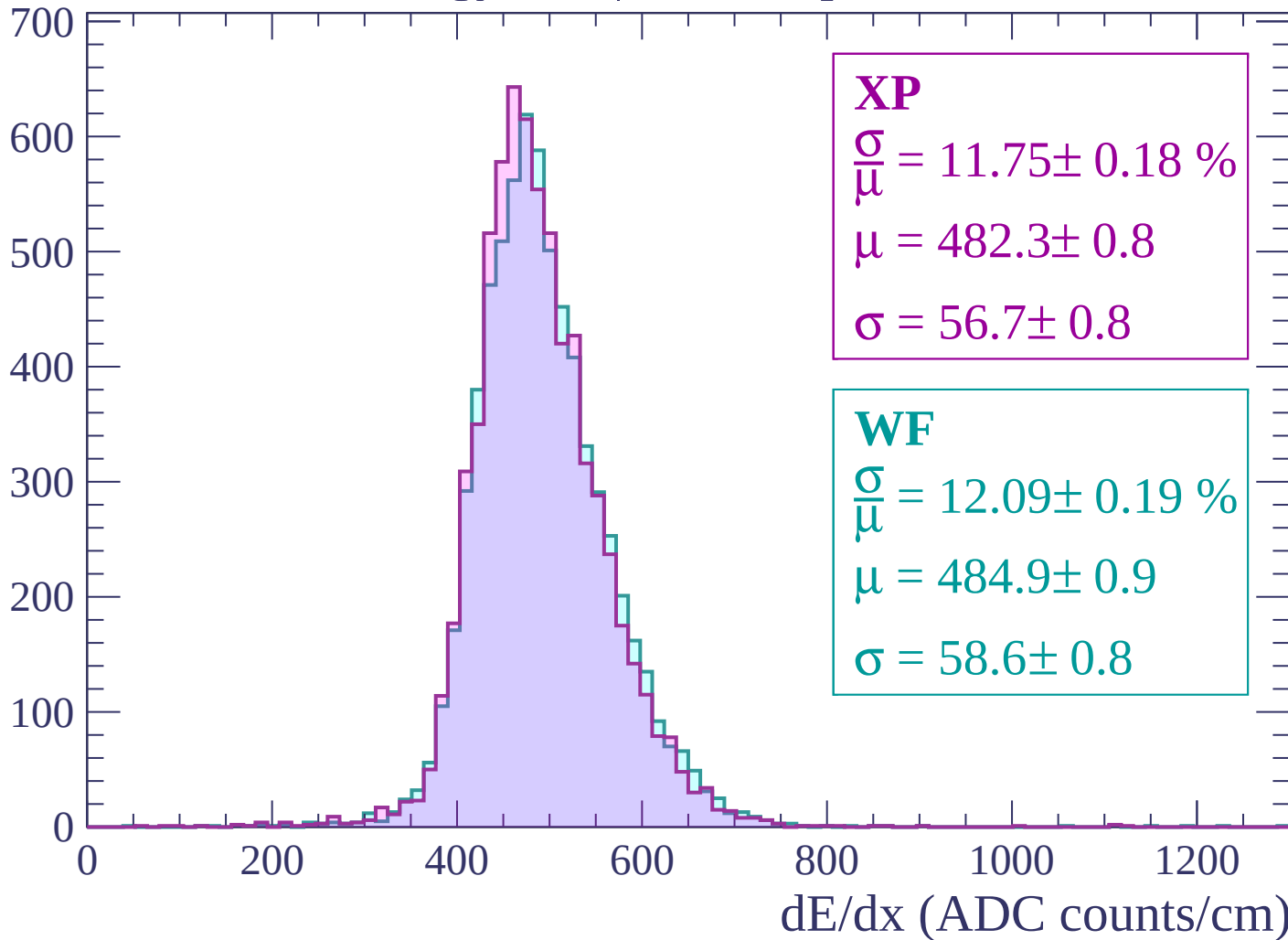
Energy loss | -1680 < p < -1620

Count



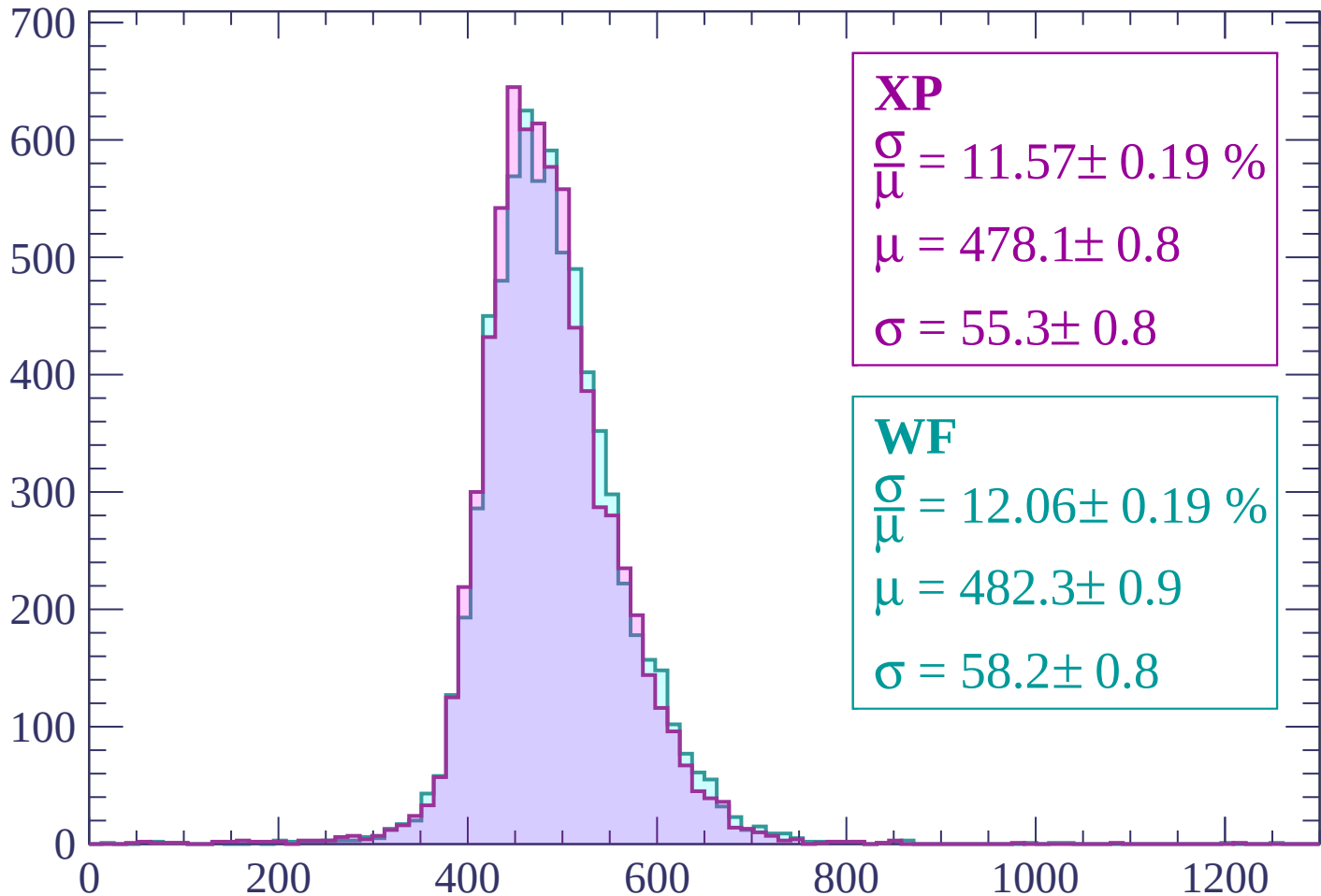
Energy loss | -1620 < p < -1560

Count



Energy loss | -1560 < p < -1500

Count



XP

$$\frac{\sigma}{\mu} = 11.57 \pm 0.19 \%$$

$$\mu = 478.1 \pm 0.8$$

$$\sigma = 55.3 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 12.06 \pm 0.19 \%$$

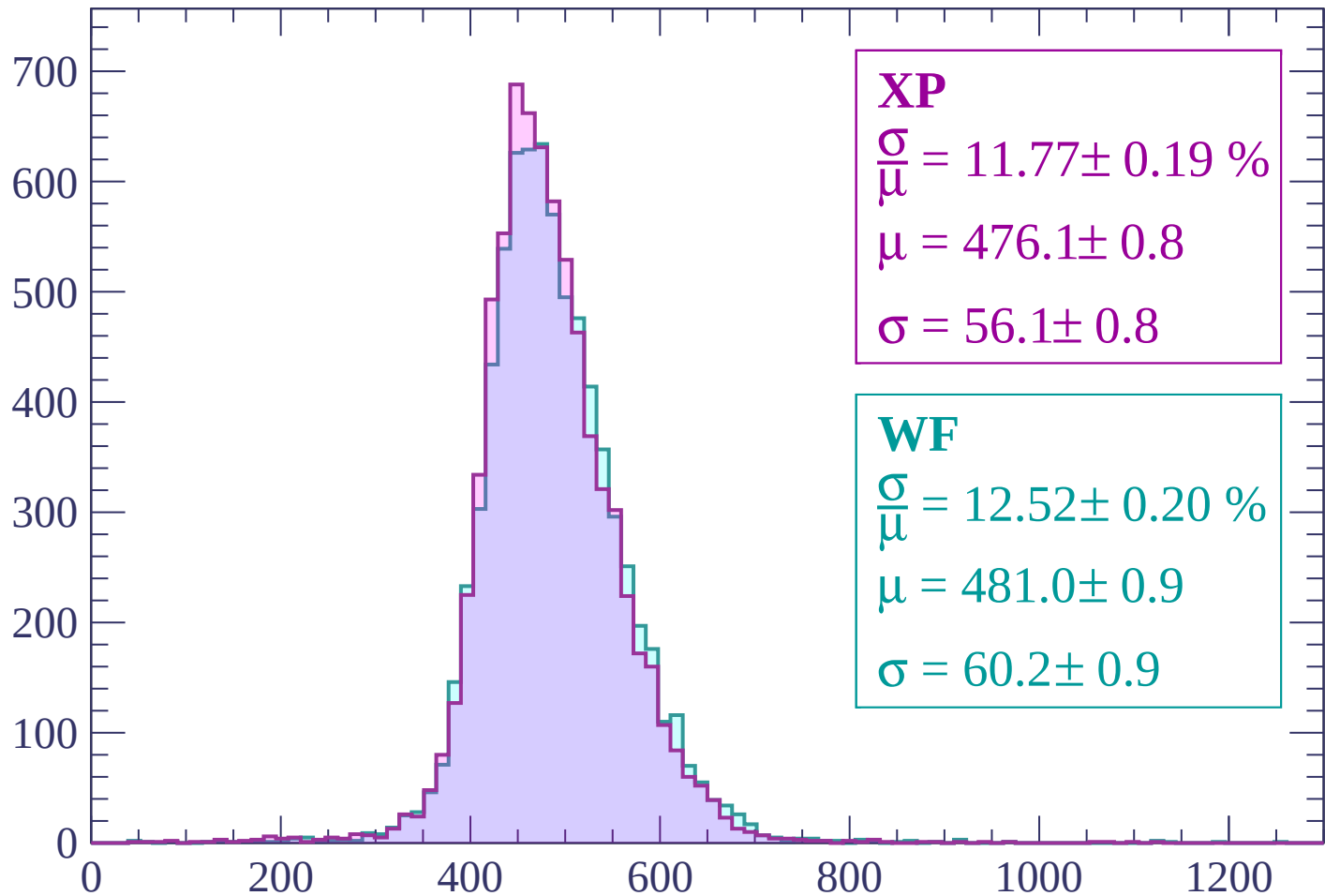
$$\mu = 482.3 \pm 0.9$$

$$\sigma = 58.2 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | -1500 < p < -1440

Count



XP

$$\frac{\sigma}{\mu} = 11.77 \pm 0.19 \%$$

$$\mu = 476.1 \pm 0.8$$

$$\sigma = 56.1 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 12.52 \pm 0.20 \%$$

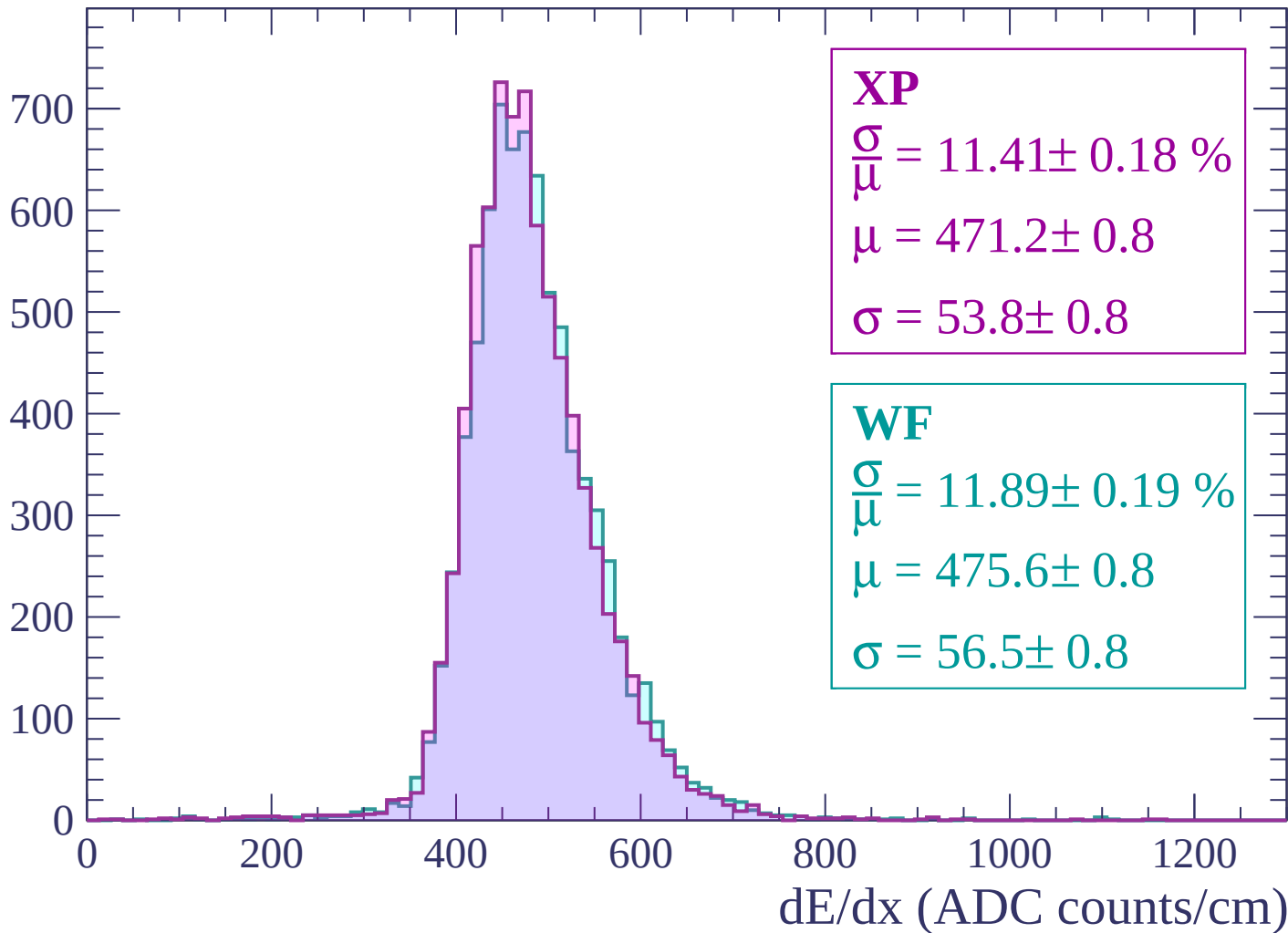
$$\mu = 481.0 \pm 0.9$$

$$\sigma = 60.2 \pm 0.9$$

dE/dx (ADC counts/cm)

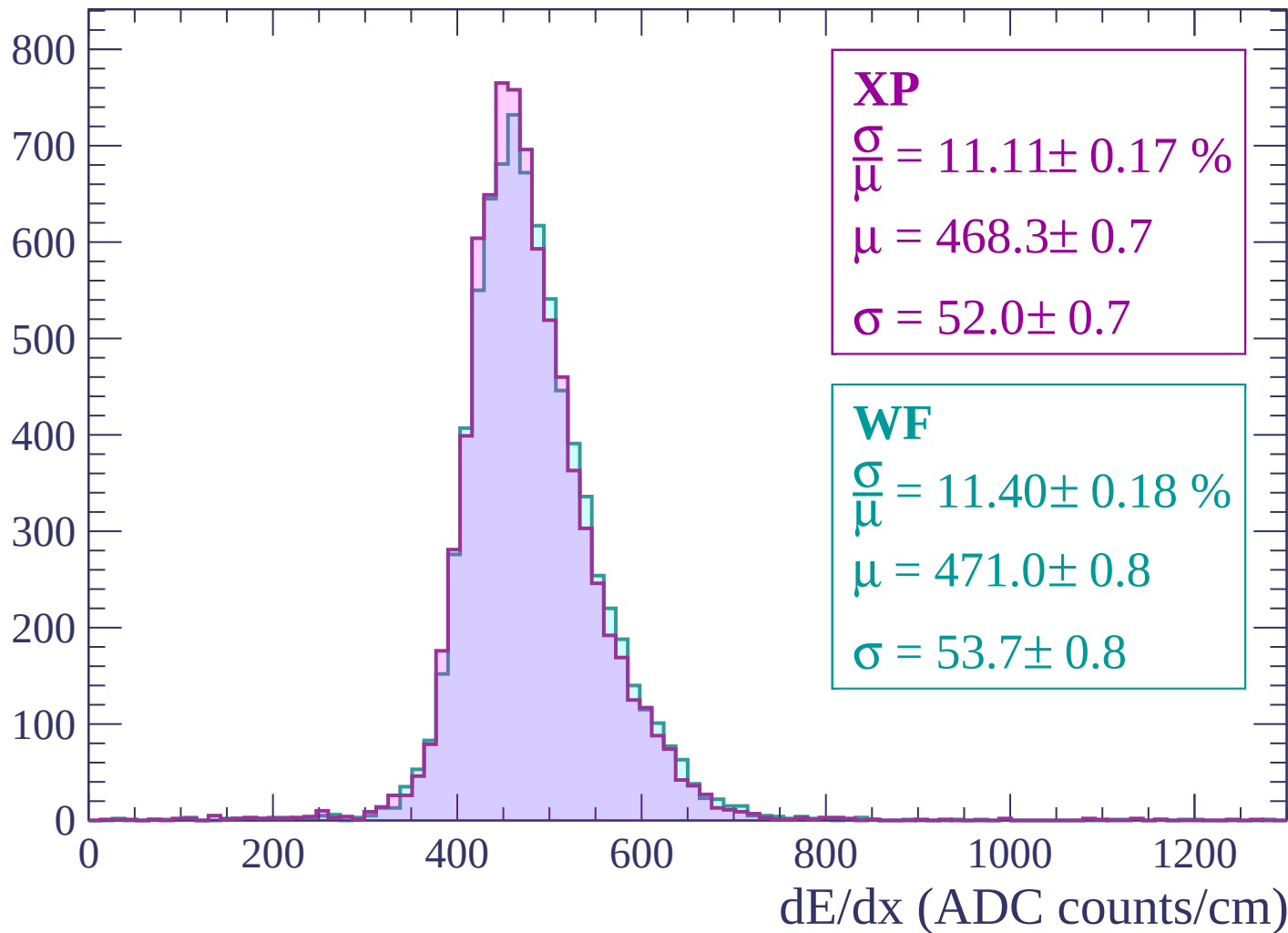
Energy loss | $-1440 < p < -1380$

Count



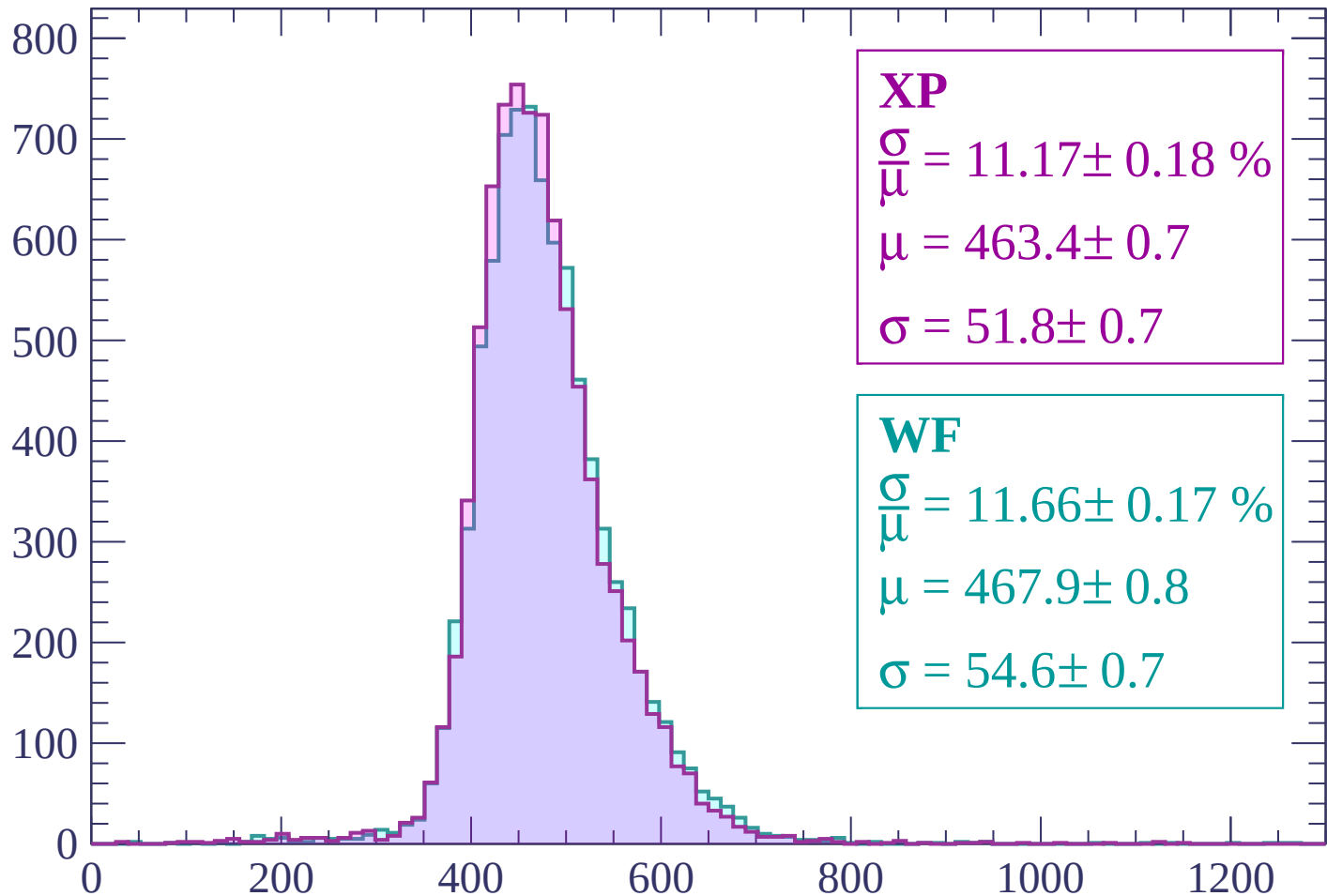
Energy loss | $-1380 < p < -1320$

Count



Energy loss | -1320 < p < -1260

Count



XP

$$\frac{\sigma}{\mu} = 11.17 \pm 0.18 \%$$

$$\mu = 463.4 \pm 0.7$$

$$\sigma = 51.8 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 11.66 \pm 0.17 \%$$

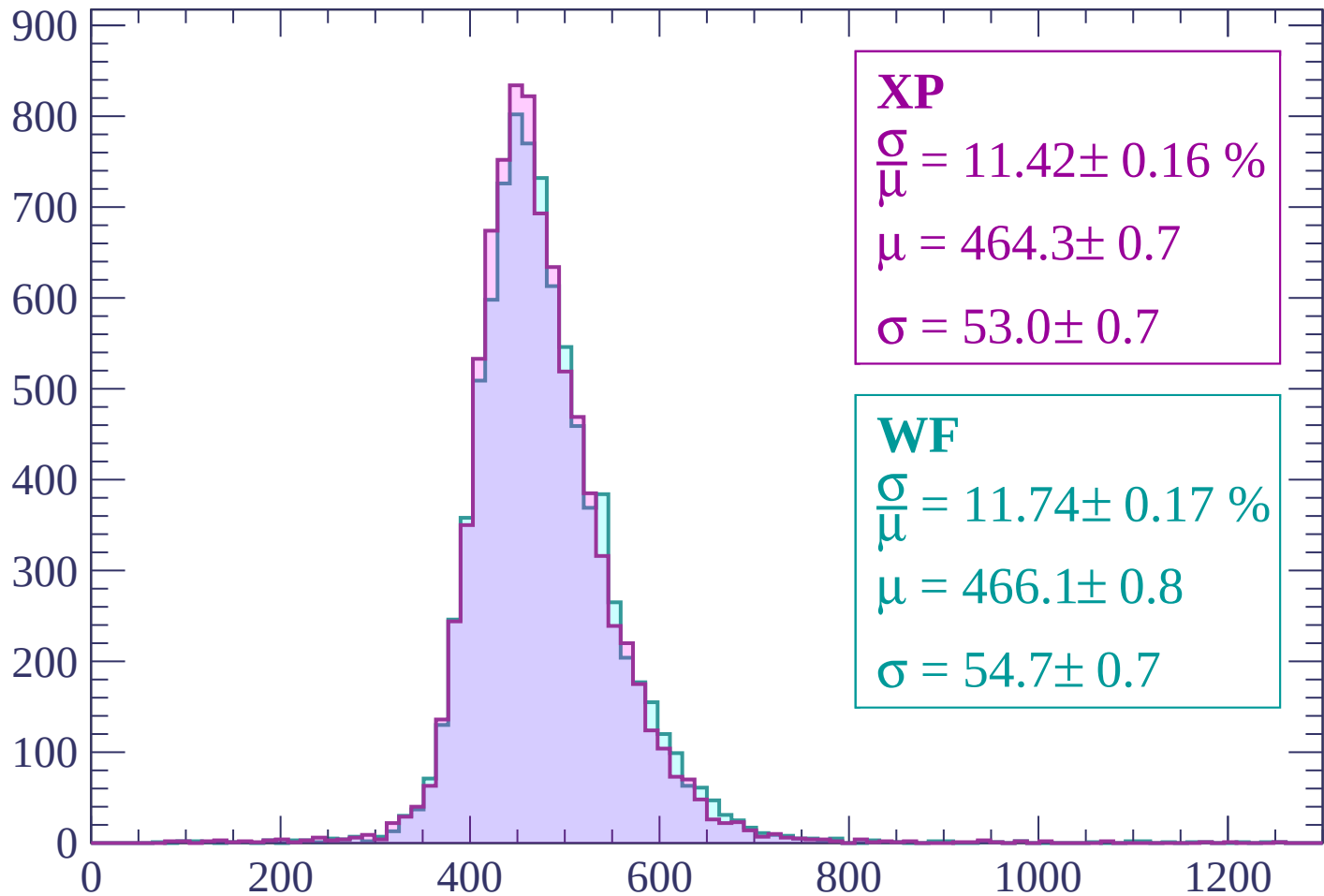
$$\mu = 467.9 \pm 0.8$$

$$\sigma = 54.6 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | -1260 < p < -1200

Count



XP

$$\frac{\sigma}{\mu} = 11.42 \pm 0.16 \%$$

$$\mu = 464.3 \pm 0.7$$

$$\sigma = 53.0 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 11.74 \pm 0.17 \%$$

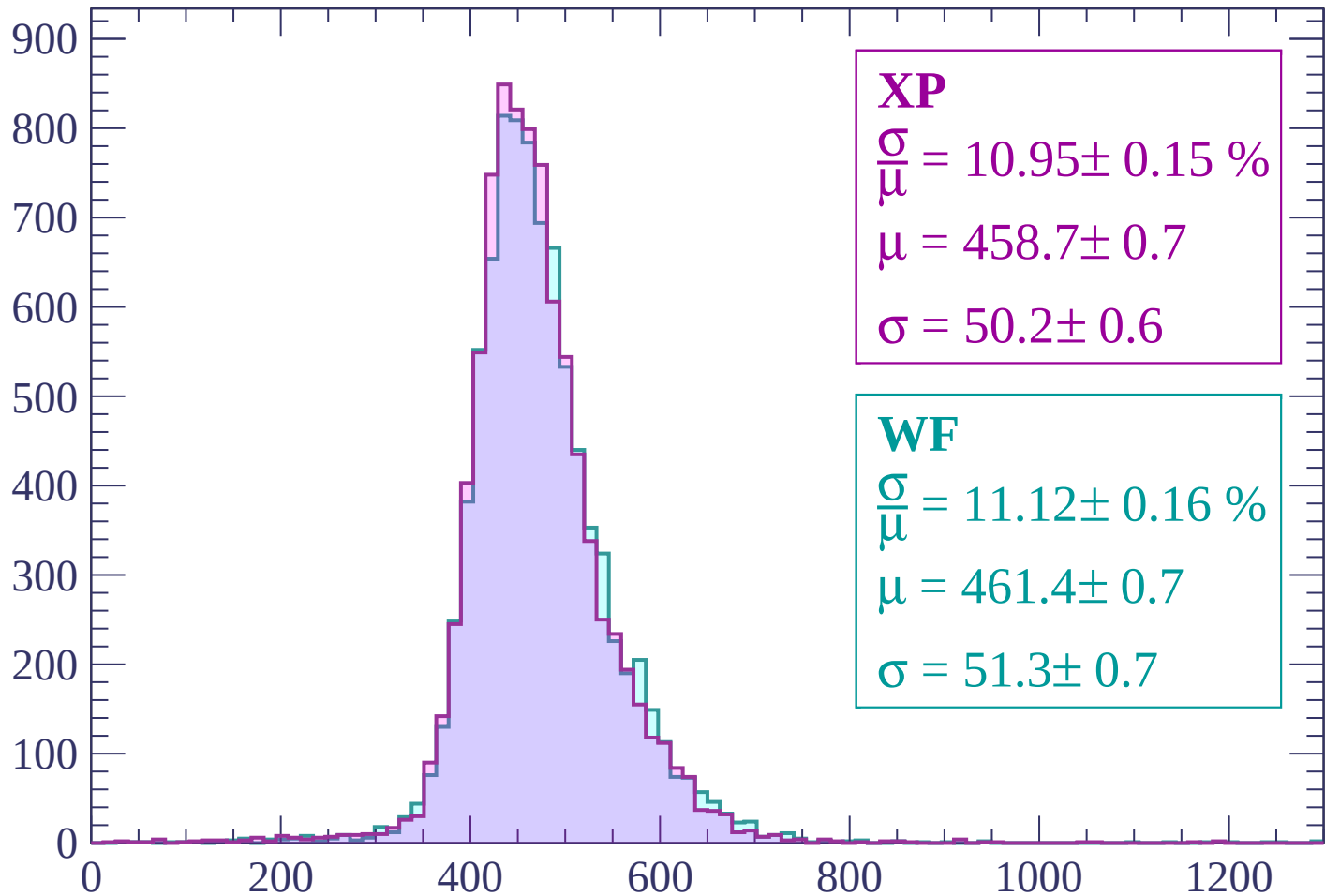
$$\mu = 466.1 \pm 0.8$$

$$\sigma = 54.7 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | -1200 < p < -1140

Count



XP

$$\frac{\sigma}{\mu} = 10.95 \pm 0.15 \%$$

$$\mu = 458.7 \pm 0.7$$

$$\sigma = 50.2 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 11.12 \pm 0.16 \%$$

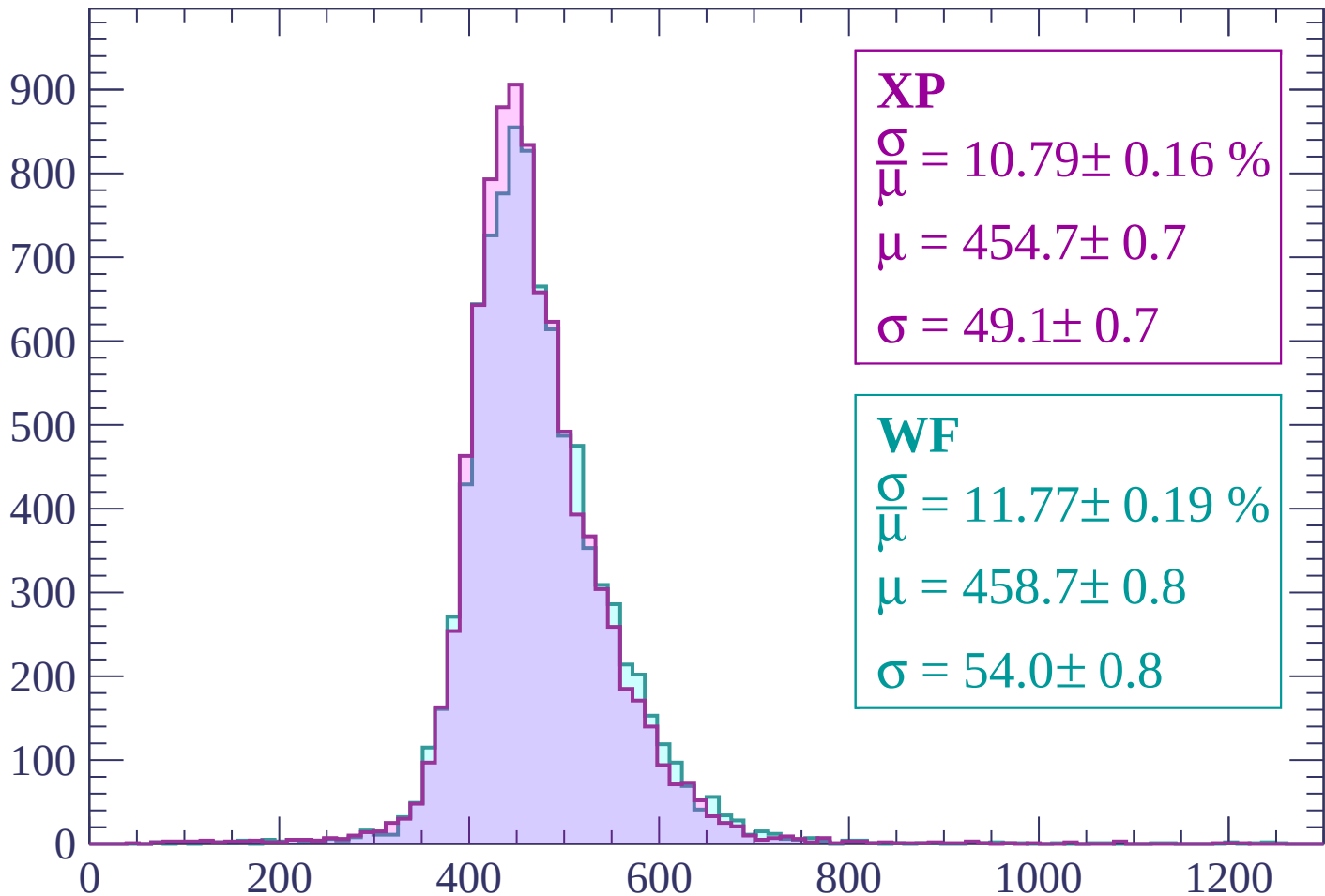
$$\mu = 461.4 \pm 0.7$$

$$\sigma = 51.3 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | -1140 < p < -1080

Count



XP

$$\frac{\sigma}{\mu} = 10.79 \pm 0.16 \%$$

$$\mu = 454.7 \pm 0.7$$

$$\sigma = 49.1 \pm 0.7$$

WF

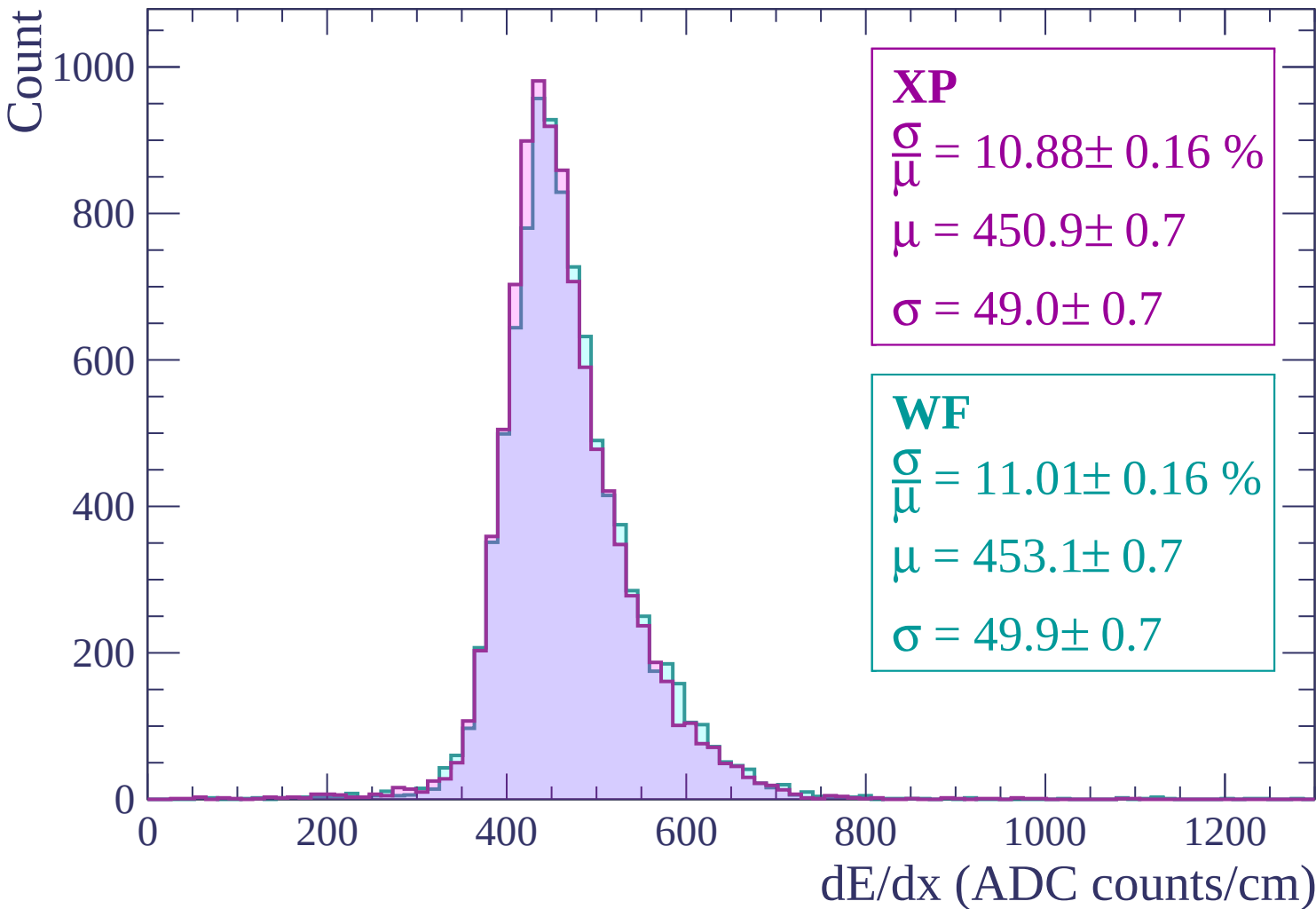
$$\frac{\sigma}{\mu} = 11.77 \pm 0.19 \%$$

$$\mu = 458.7 \pm 0.8$$

$$\sigma = 54.0 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | -1080 < p < -1020



Energy loss | $-1020 < p < -960$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

XP

$$\frac{\sigma}{\mu} = 10.47 \pm 0.15 \%$$

$$\mu = 445.5 \pm 0.6$$

$$\sigma = 46.7 \pm 0.6$$

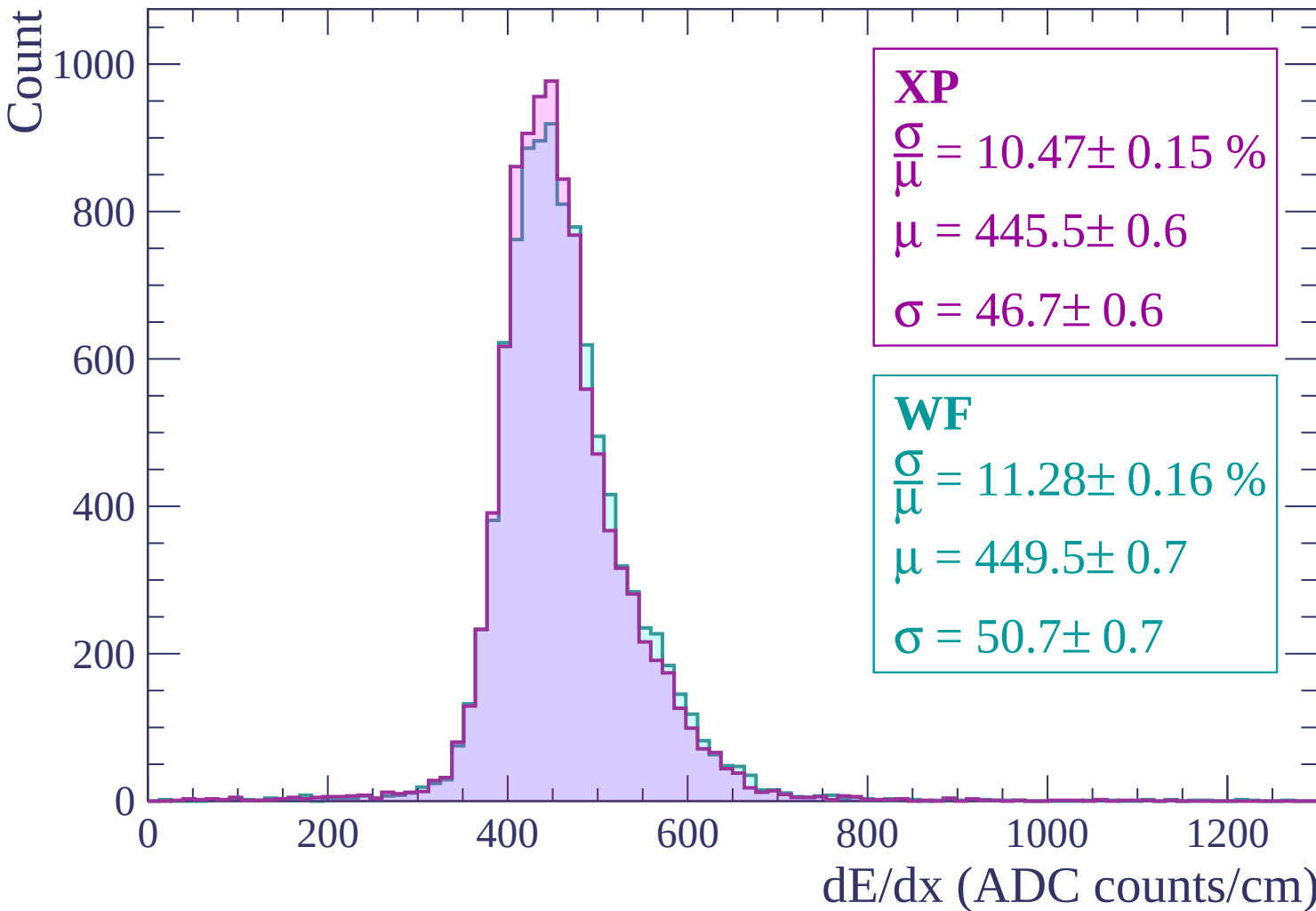
WF

$$\frac{\sigma}{\mu} = 11.28 \pm 0.16 \%$$

$$\mu = 449.5 \pm 0.7$$

$$\sigma = 50.7 \pm 0.7$$

dE/dx (ADC counts/cm)



Energy loss | $-960 < p < -900$

Count

1000

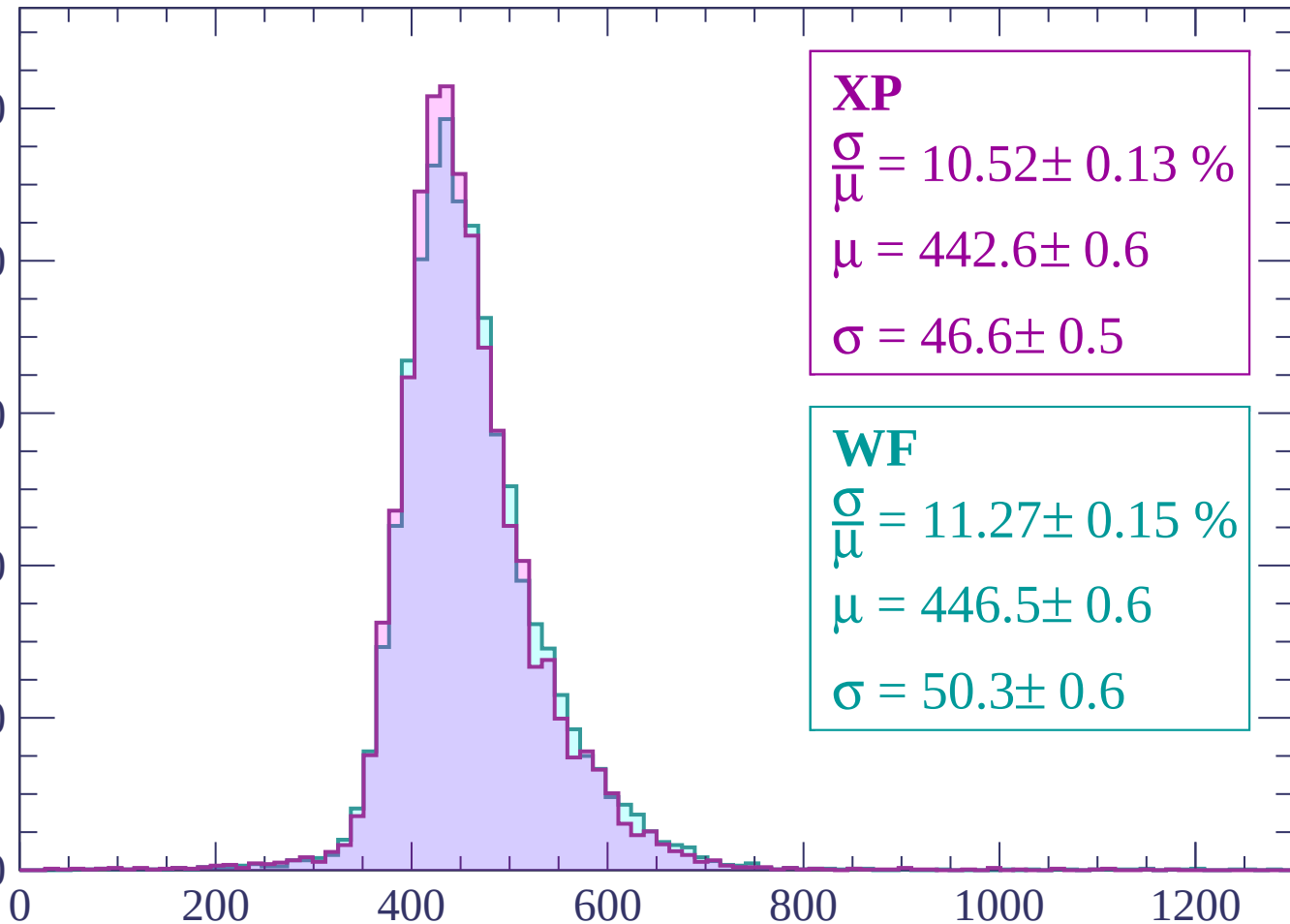
800

600

400

200

0



XP

$$\frac{\sigma}{\mu} = 10.52 \pm 0.13 \%$$

$$\mu = 442.6 \pm 0.6$$

$$\sigma = 46.6 \pm 0.5$$

WF

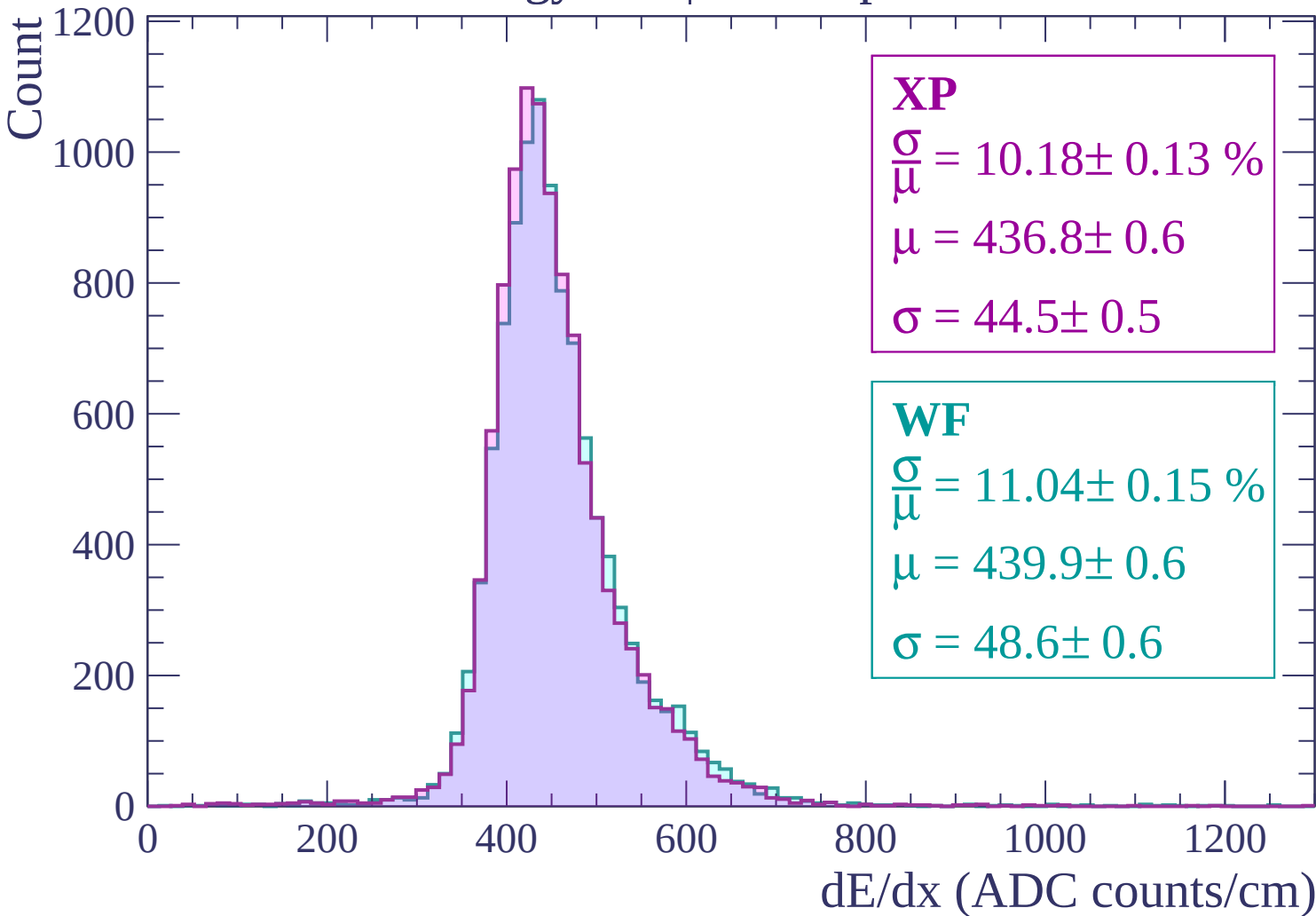
$$\frac{\sigma}{\mu} = 11.27 \pm 0.15 \%$$

$$\mu = 446.5 \pm 0.6$$

$$\sigma = 50.3 \pm 0.6$$

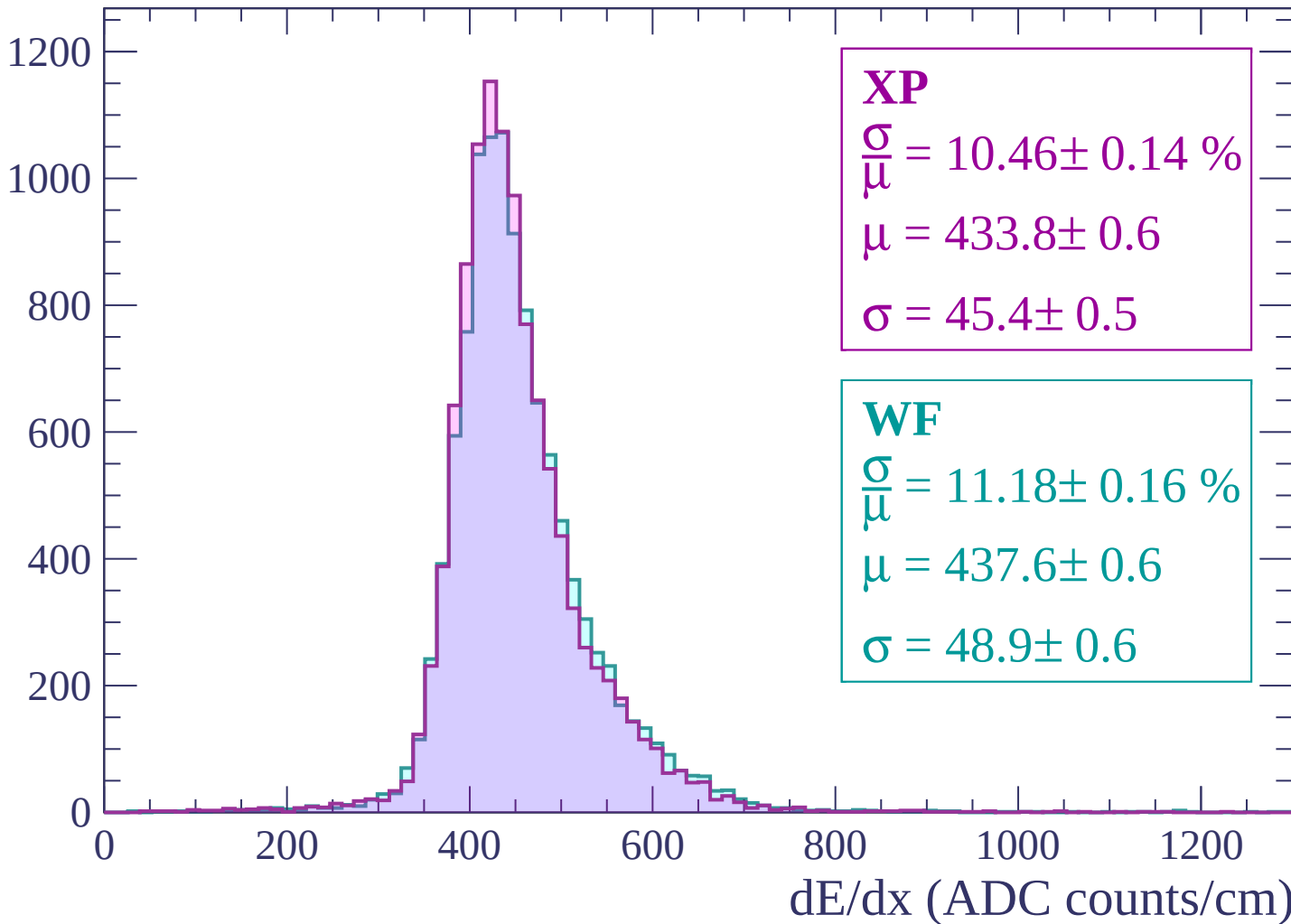
dE/dx (ADC counts/cm)

Energy loss | $-900 < p < -840$



Energy loss | $-840 < p < -780$

Count



Energy loss | $-780 < p < -720$

Count

1200
1000
800
600
400
200
0

XP

$$\frac{\sigma}{\mu} = 10.60 \pm 0.14 \%$$

$$\mu = 430.0 \pm 0.6$$

$$\sigma = 45.6 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 10.98 \pm 0.15 \%$$

$$\mu = 431.5 \pm 0.6$$

$$\sigma = 47.4 \pm 0.6$$

0 200 400 600 800 1000 1200

dE/dx (ADC counts/cm)

Energy loss | $-720 < p < -660$

Count

1200

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

XP

$$\frac{\sigma}{\mu} = 10.17 \pm 0.13 \%$$

$$\mu = 424.2 \pm 0.5$$

$$\sigma = 43.1 \pm 0.5$$

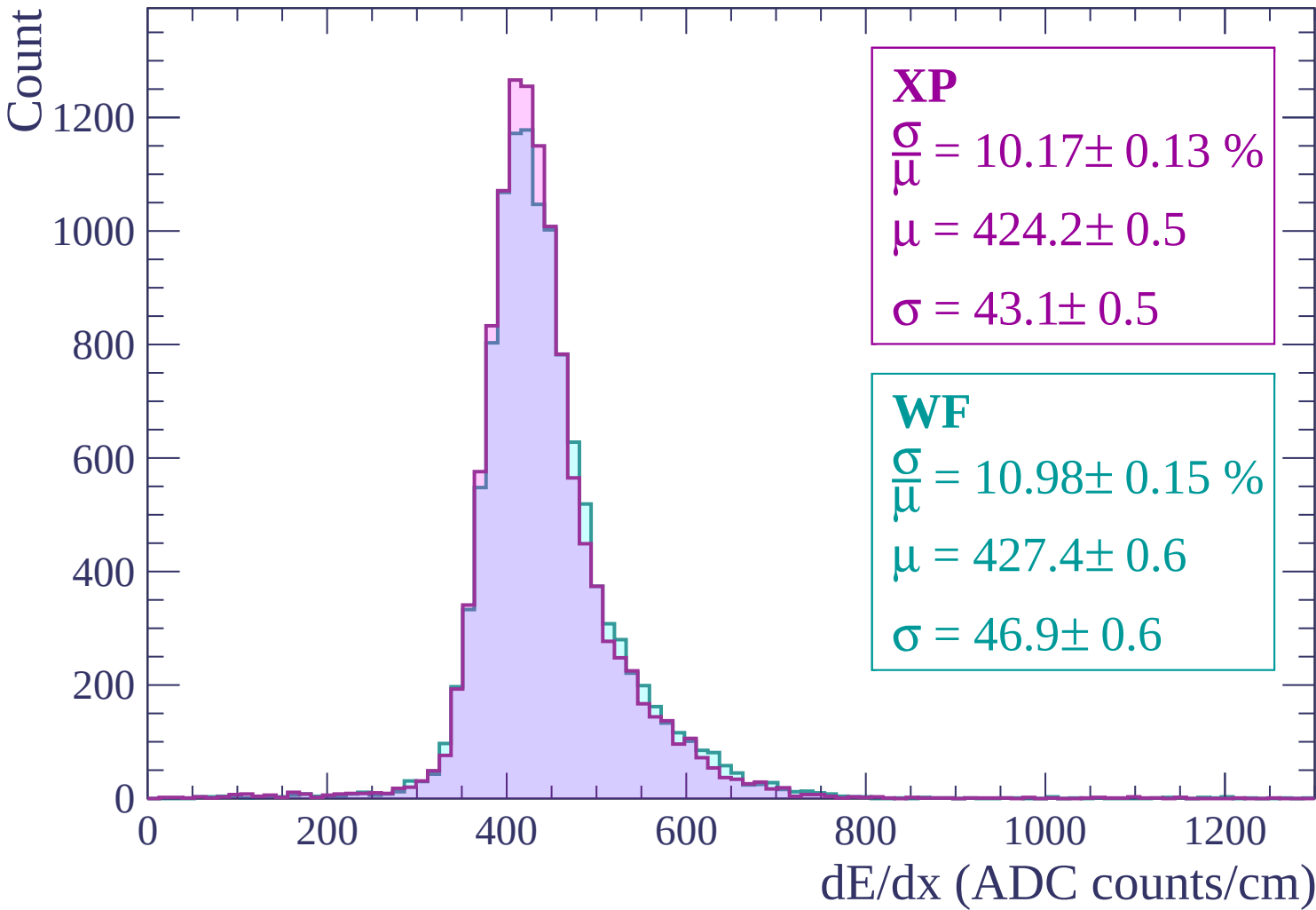
WF

$$\frac{\sigma}{\mu} = 10.98 \pm 0.15 \%$$

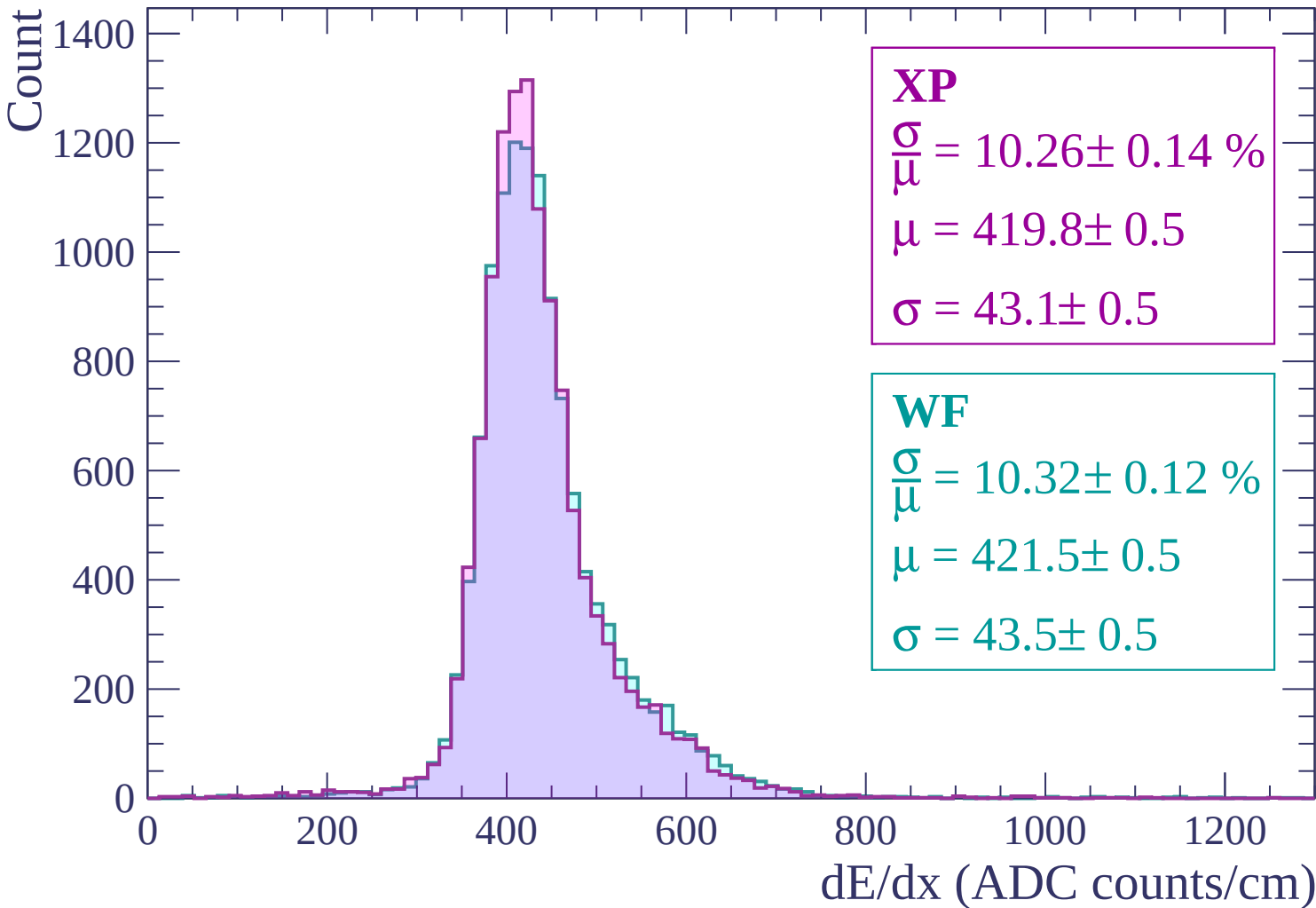
$$\mu = 427.4 \pm 0.6$$

$$\sigma = 46.9 \pm 0.6$$

dE/dx (ADC counts/cm)



Energy loss | $-660 < p < -600$



Energy loss | $-600 < p < -540$

Count

1400

1200

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

XP

$$\frac{\sigma}{\mu} = 9.96 \pm 0.14 \%$$

$$\mu = 414.5 \pm 0.5$$

$$\sigma = 41.3 \pm 0.5$$

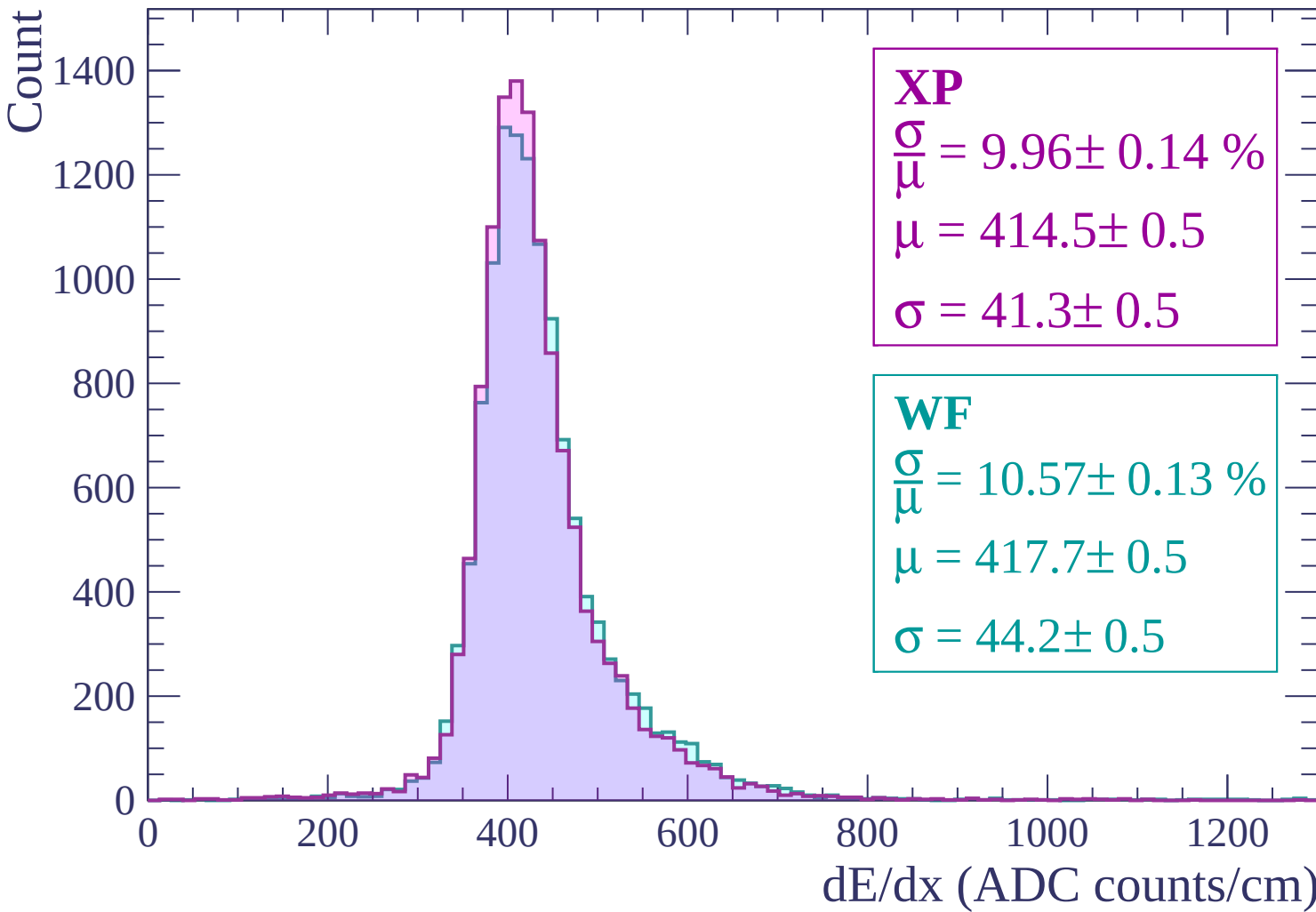
WF

$$\frac{\sigma}{\mu} = 10.57 \pm 0.13 \%$$

$$\mu = 417.7 \pm 0.5$$

$$\sigma = 44.2 \pm 0.5$$

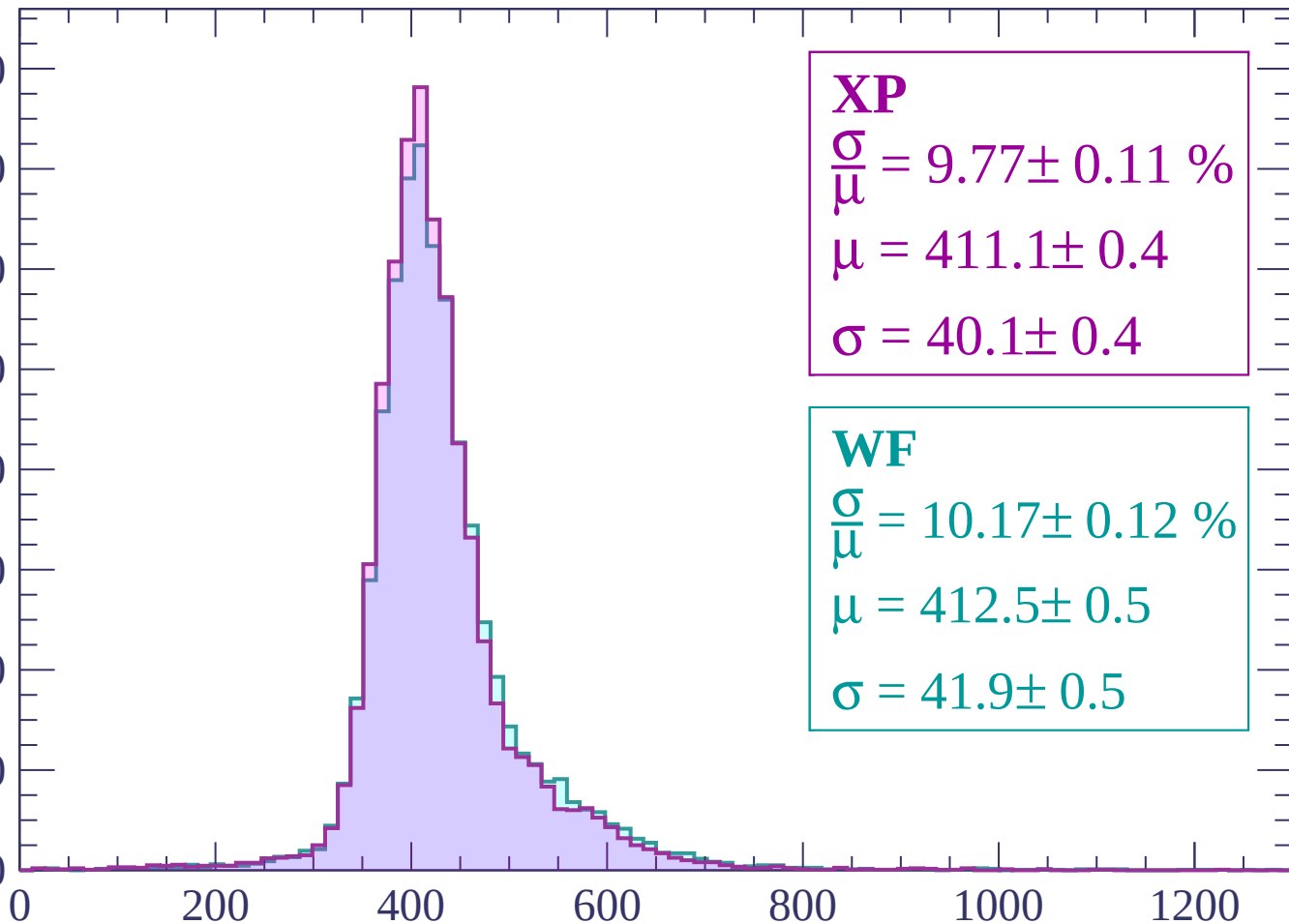
dE/dx (ADC counts/cm)



Energy loss | $-540 < p < -480$

Count

1600
1400
1200
1000
800
600
400
200
0



XP

$$\frac{\sigma}{\mu} = 9.77 \pm 0.11 \%$$

$$\mu = 411.1 \pm 0.4$$

$$\sigma = 40.1 \pm 0.4$$

WF

$$\frac{\sigma}{\mu} = 10.17 \pm 0.12 \%$$

$$\mu = 412.5 \pm 0.5$$

$$\sigma = 41.9 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $-480 < p < -420$

Count

1600
1400
1200
1000
800
600
400
200
0

XP

$$\frac{\sigma}{\mu} = 9.49 \pm 0.11 \%$$

$$\mu = 406.0 \pm 0.4$$

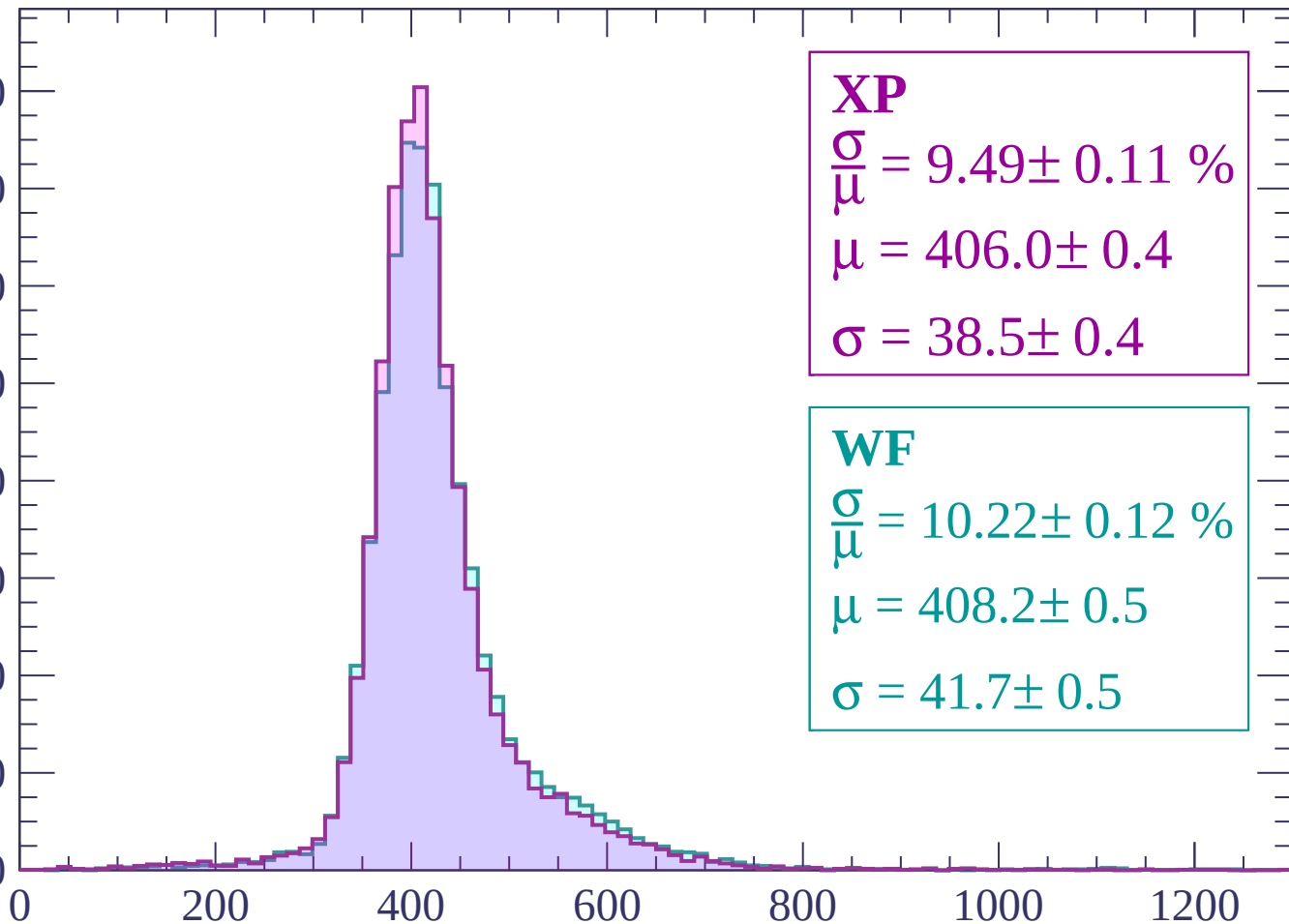
$$\sigma = 38.5 \pm 0.4$$

WF

$$\frac{\sigma}{\mu} = 10.22 \pm 0.12 \%$$

$$\mu = 408.2 \pm 0.5$$

$$\sigma = 41.7 \pm 0.5$$



dE/dx (ADC counts/cm)

Energy loss | $-420 < p < -360$

Count

1600
1400
1200
1000
800
600
400
200
0

XP

$$\frac{\sigma}{\mu} = 9.51 \pm 0.11 \%$$

$$\mu = 403.4 \pm 0.4$$

$$\sigma = 38.4 \pm 0.4$$

WF

$$\frac{\sigma}{\mu} = 9.86 \pm 0.12 \%$$

$$\mu = 404.0 \pm 0.4$$

$$\sigma = 39.9 \pm 0.4$$

0

200

400

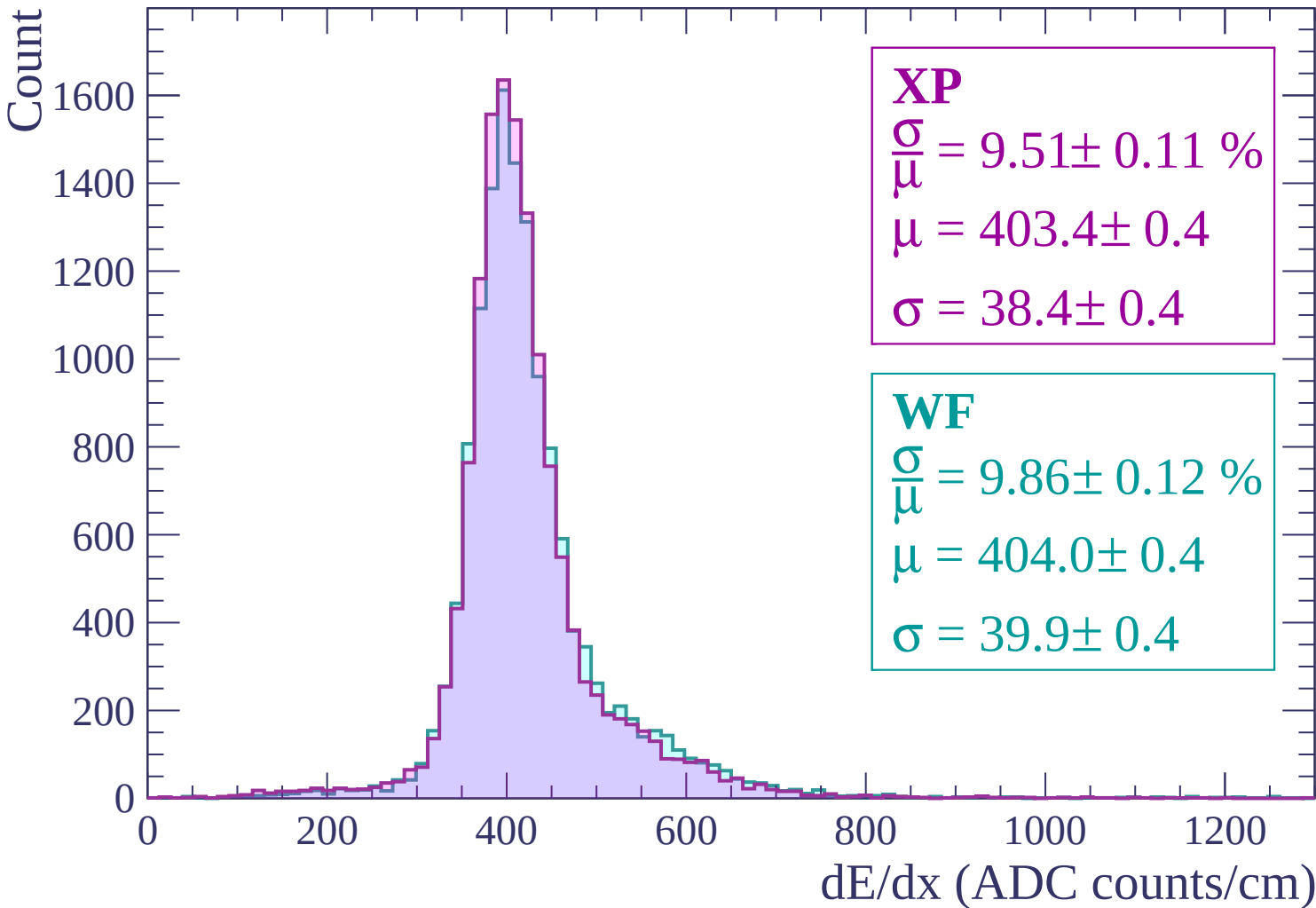
600

800

1000

1200

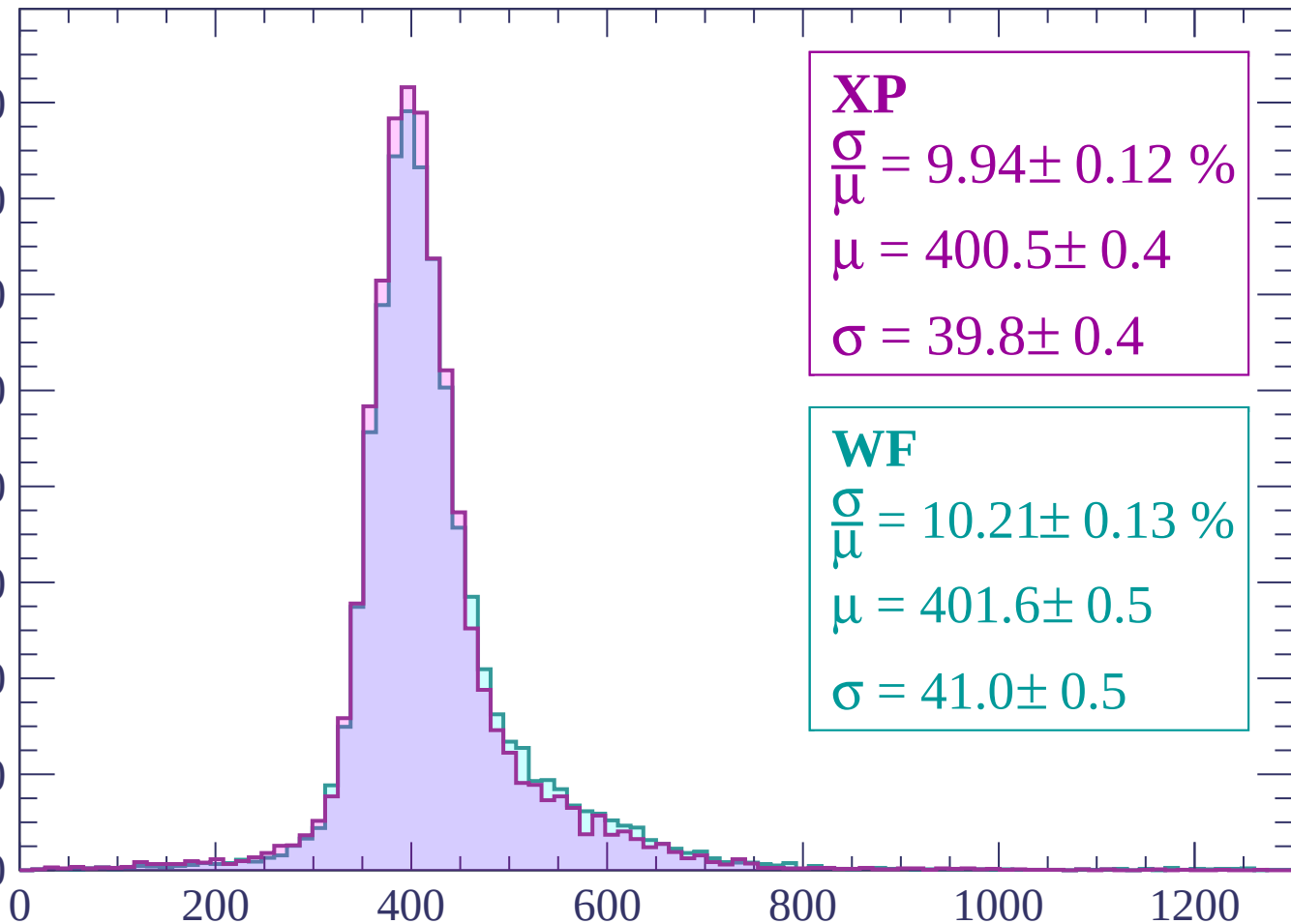
dE/dx (ADC counts/cm)



Energy loss | $-360 < p < -300$

Count

1600
1400
1200
1000
800
600
400
200
0



XP

$$\frac{\sigma}{\mu} = 9.94 \pm 0.12 \%$$

$$\mu = 400.5 \pm 0.4$$

$$\sigma = 39.8 \pm 0.4$$

WF

$$\frac{\sigma}{\mu} = 10.21 \pm 0.13 \%$$

$$\mu = 401.6 \pm 0.5$$

$$\sigma = 41.0 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $-300 < p < -240$

Count

1600
1400
1200
1000
800
600
400
200
0

XP

$$\frac{\sigma}{\mu} = 9.76 \pm 0.12 \%$$

$$\mu = 401.3 \pm 0.4$$

$$\sigma = 39.2 \pm 0.4$$

WF

$$\frac{\sigma}{\mu} = 10.02 \pm 0.13 \%$$

$$\mu = 403.0 \pm 0.5$$

$$\sigma = 40.4 \pm 0.5$$

0

200

400

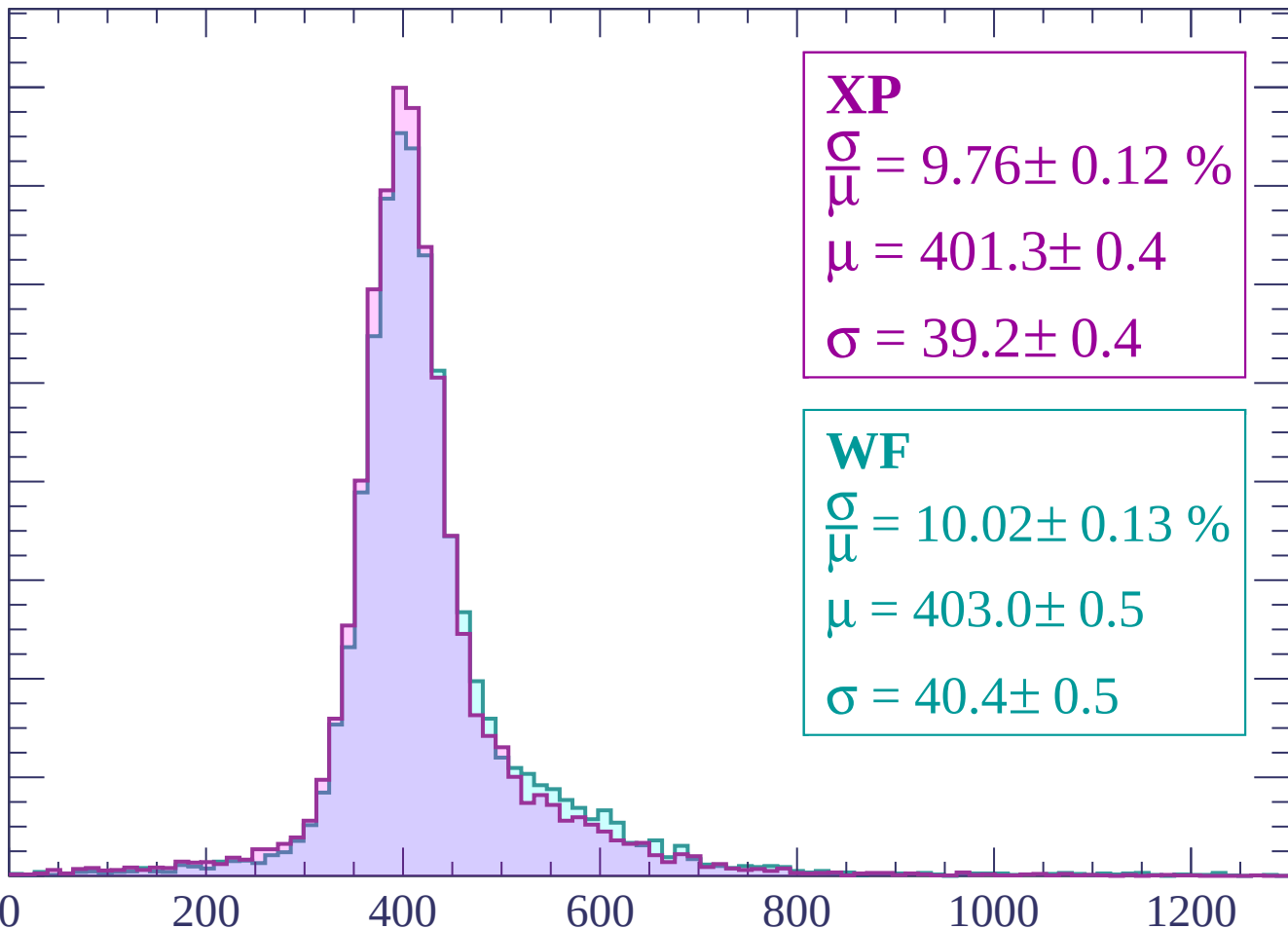
600

800

1000

1200

dE/dx (ADC counts/cm)



Energy loss | -240 < p < -180

Count

1200
1000
800
600
400
200
0

XP

$$\frac{\sigma}{\mu} = 11.37 \pm 0.16 \%$$

$$\mu = 412.3 \pm 0.5$$

$$\sigma = 46.9 \pm 0.6$$

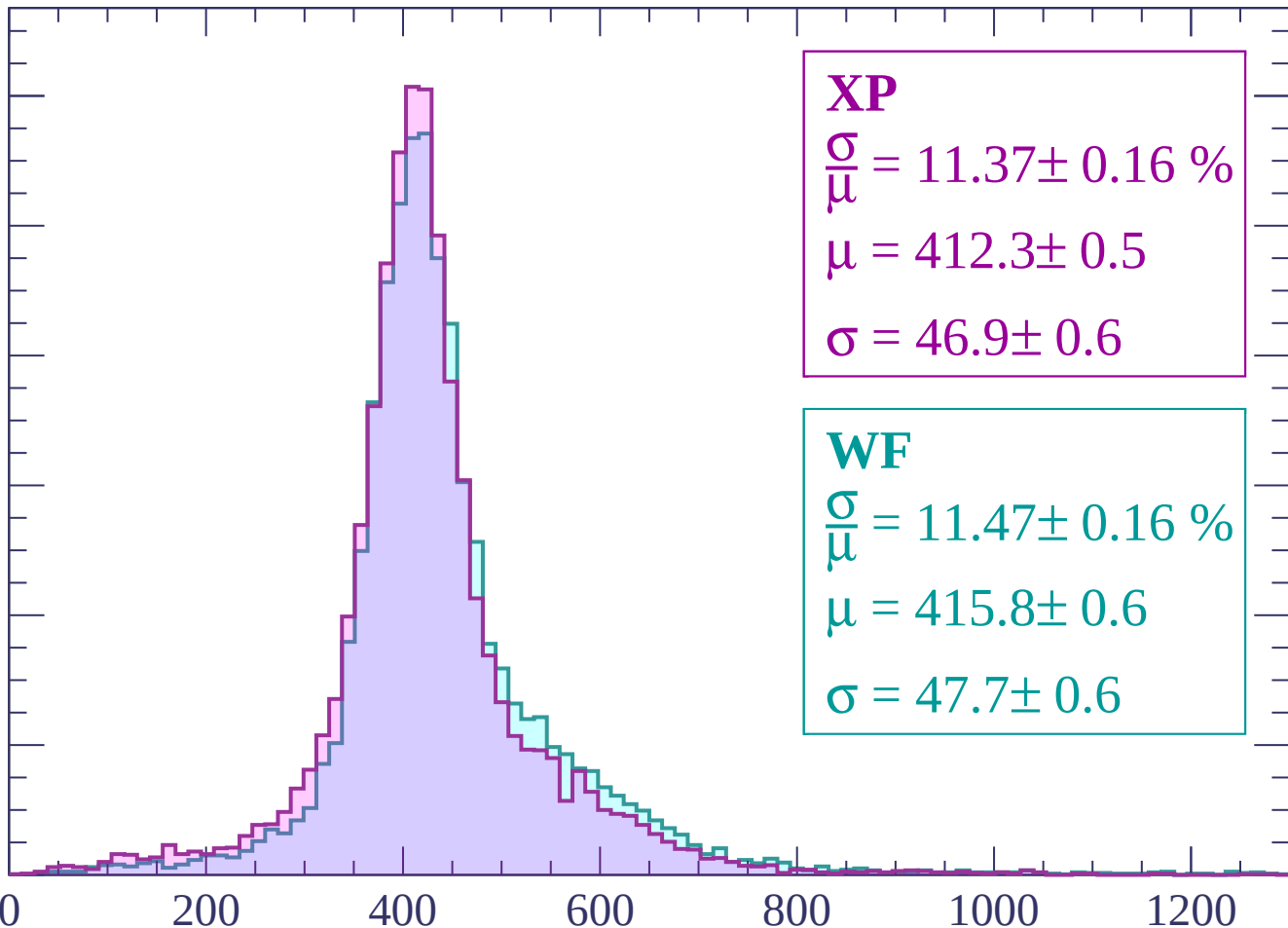
WF

$$\frac{\sigma}{\mu} = 11.47 \pm 0.16 \%$$

$$\mu = 415.8 \pm 0.6$$

$$\sigma = 47.7 \pm 0.6$$

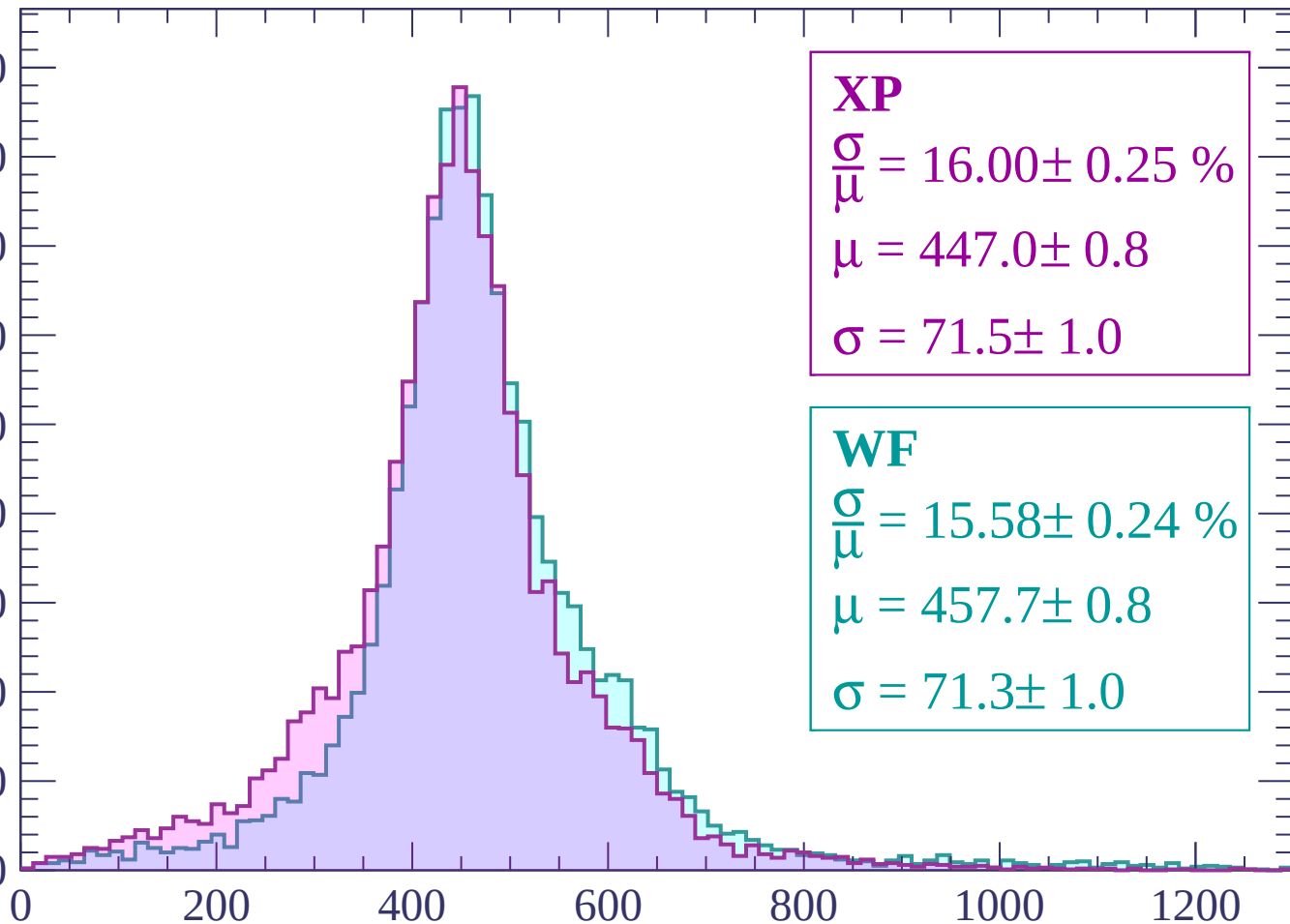
0 200 400 600 800 1000 1200
dE/dx (ADC counts/cm)



Energy loss | -180 < p < -120

Count

900
800
700
600
500
400
300
200
100
0



XP

$$\frac{\sigma}{\mu} = 16.00 \pm 0.25 \%$$

$$\mu = 447.0 \pm 0.8$$

$$\sigma = 71.5 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 15.58 \pm 0.24 \%$$

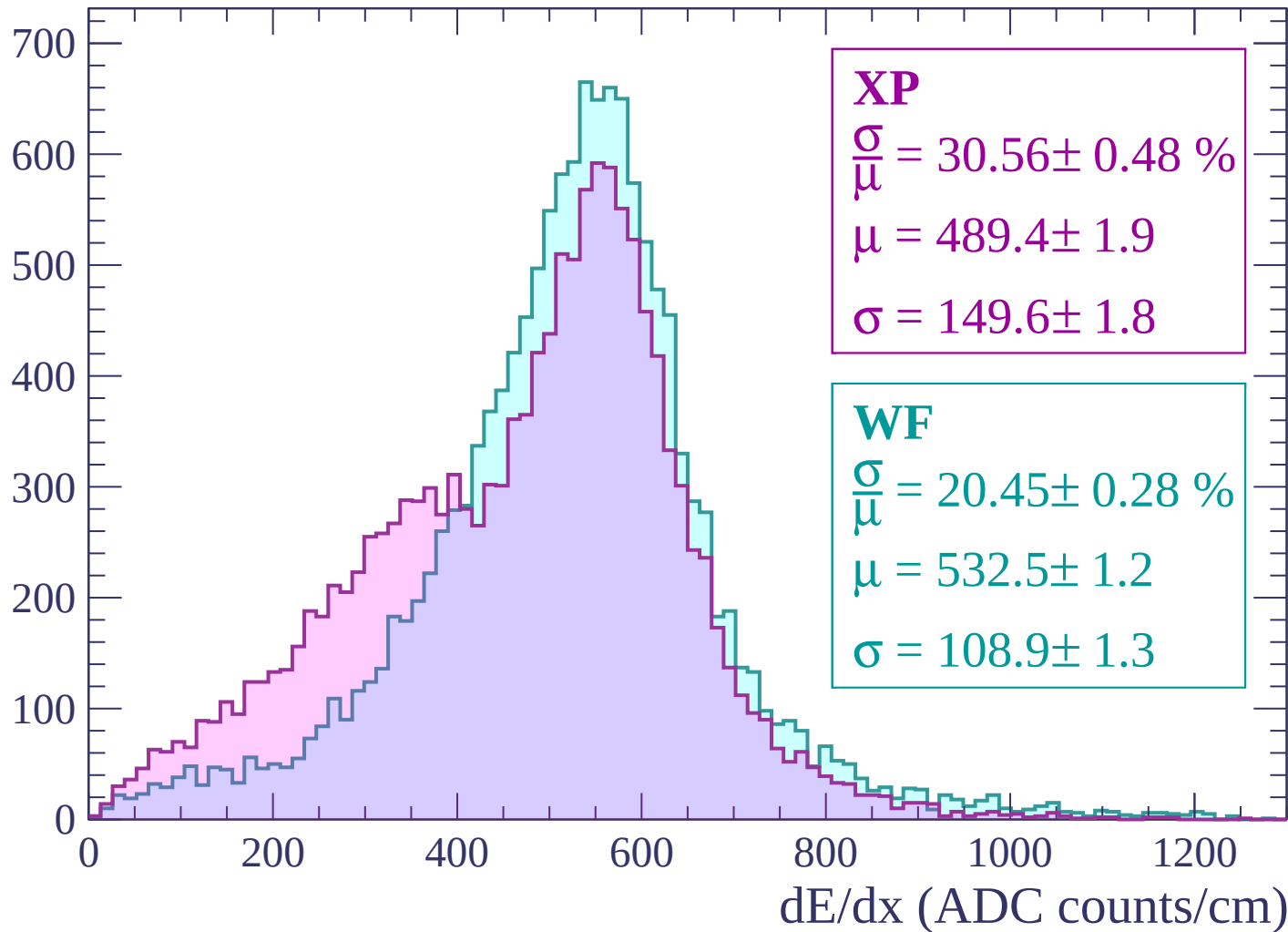
$$\mu = 457.7 \pm 0.8$$

$$\sigma = 71.3 \pm 1.0$$

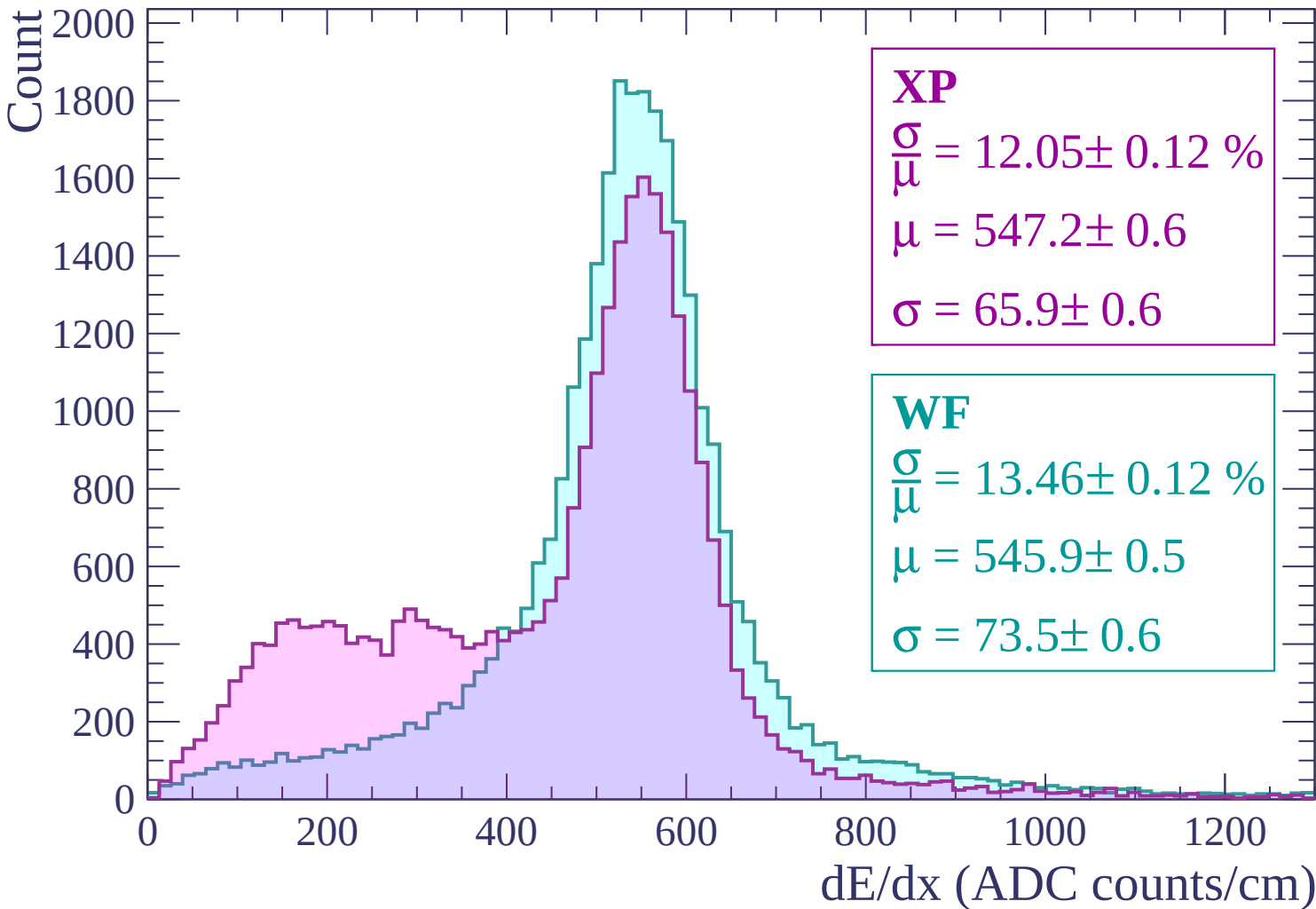
dE/dx (ADC counts/cm)

Energy loss | $-120 < p < -60$

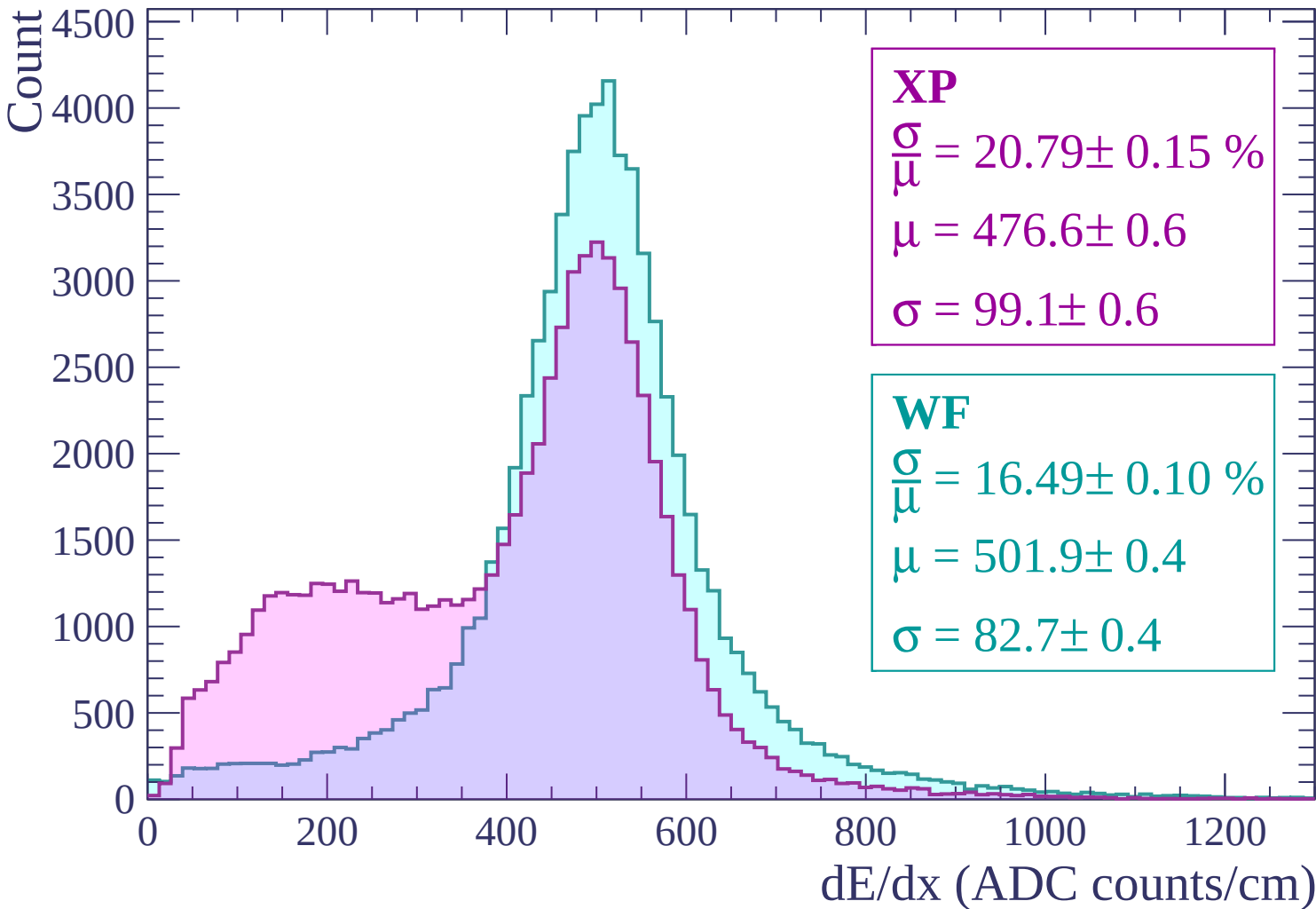
Count



Energy loss | $-60 < p < 0$

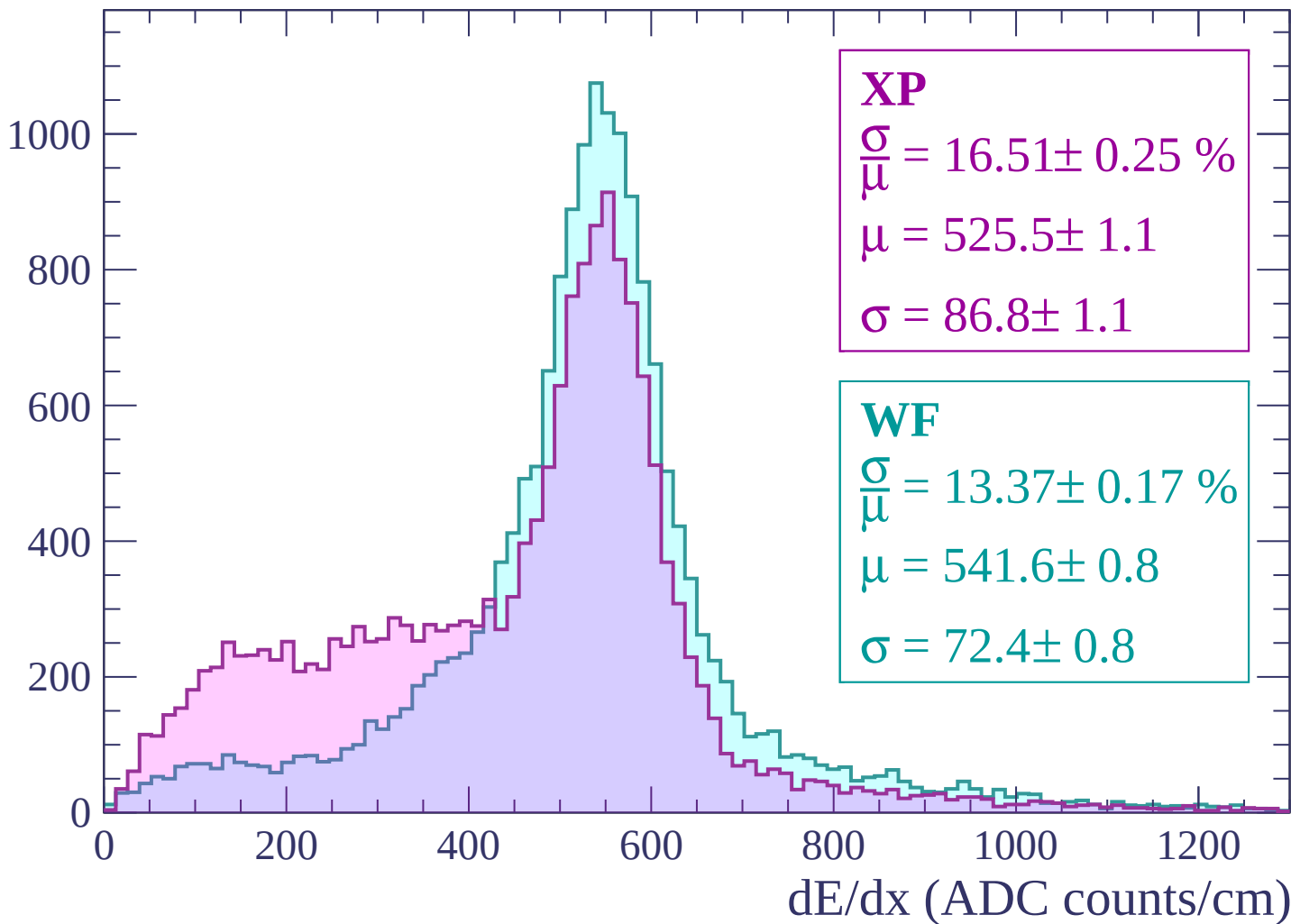


Energy loss | $0 < p < 60$



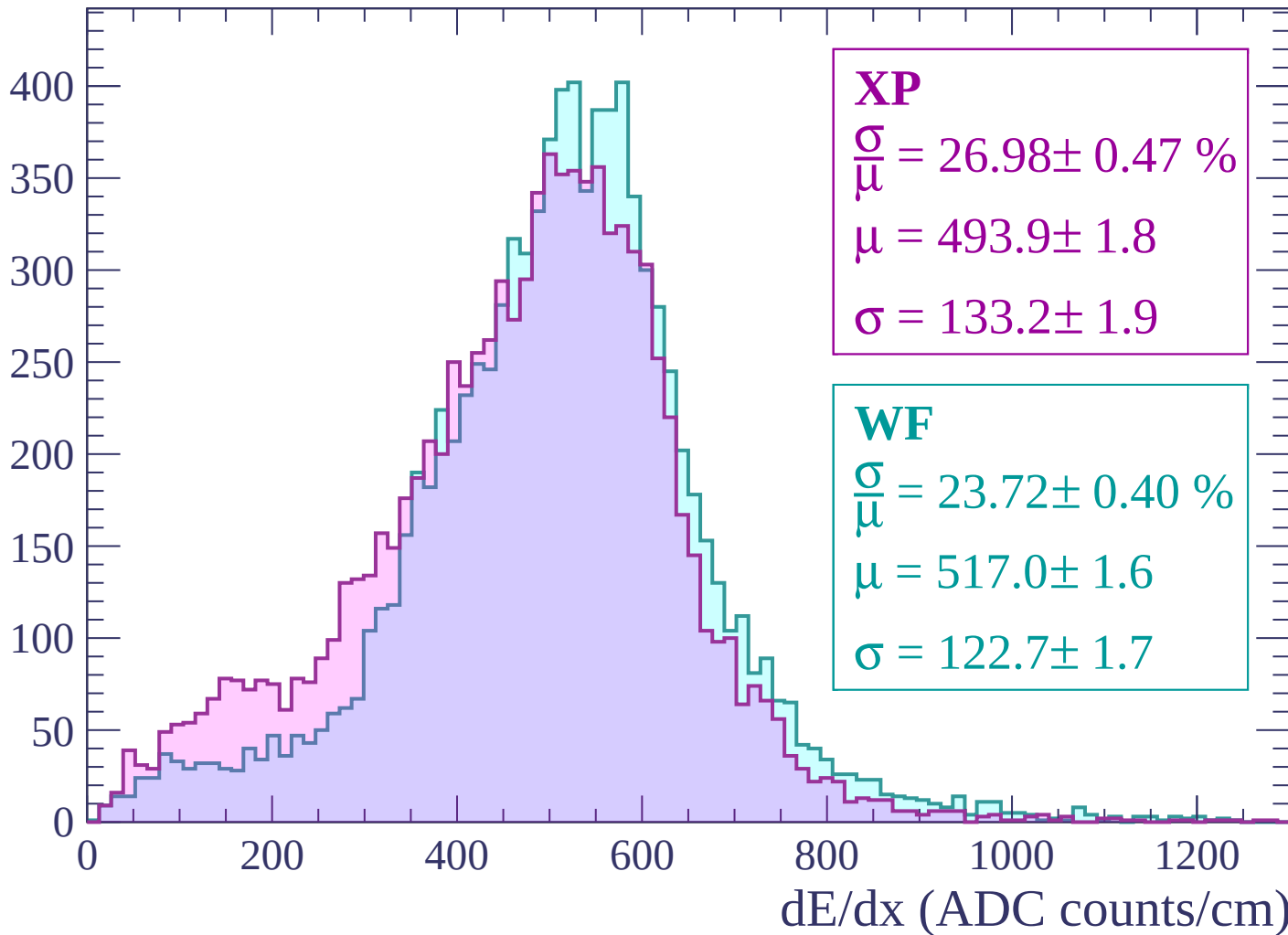
Energy loss | $60 < p < 120$

Count

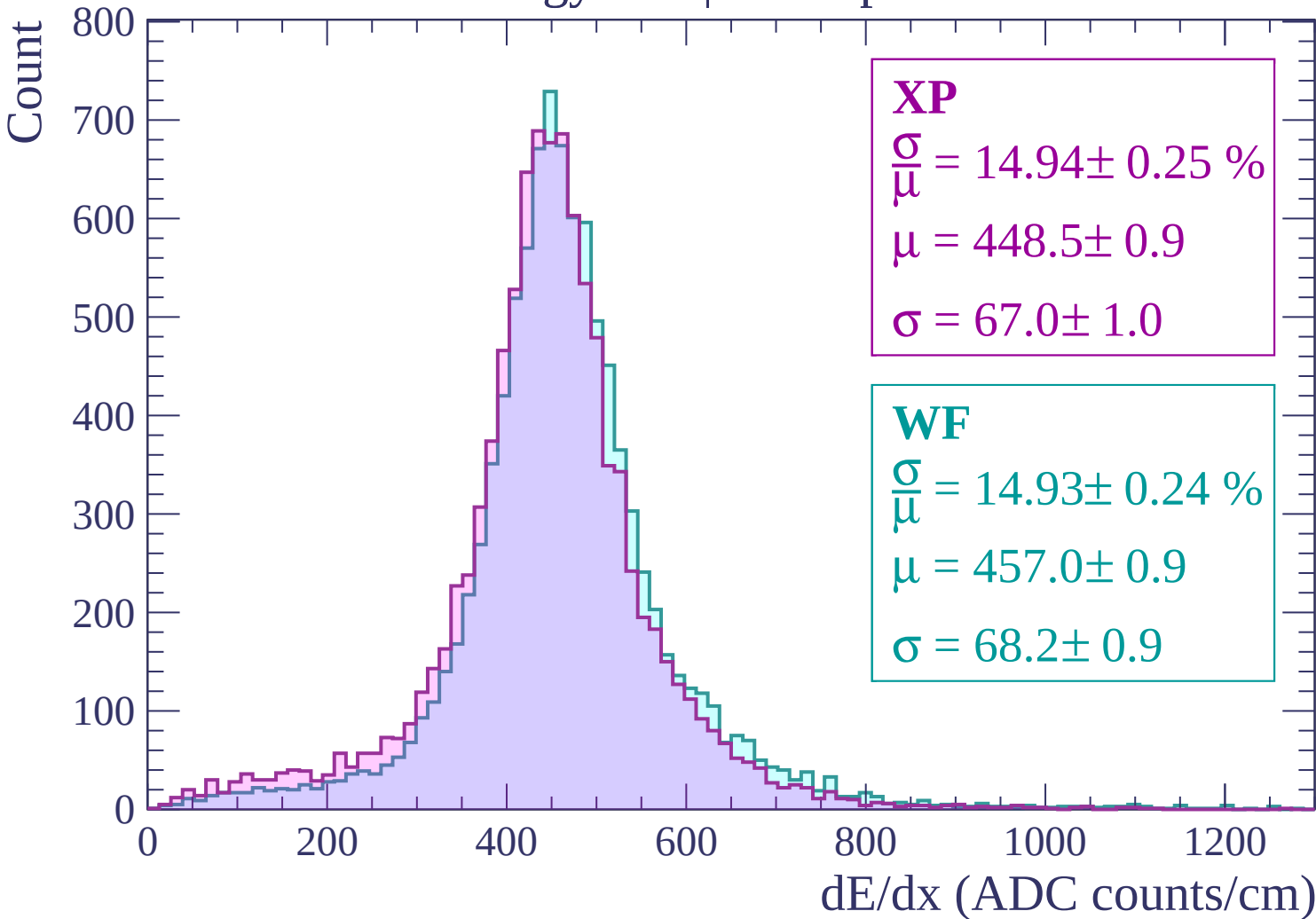


Energy loss | $120 < p < 180$

Count



Energy loss | $180 < p < 240$



Energy loss | $240 < p < 300$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 11.52 \pm 0.17 \%$$

$$\mu = 416.8 \pm 0.6$$

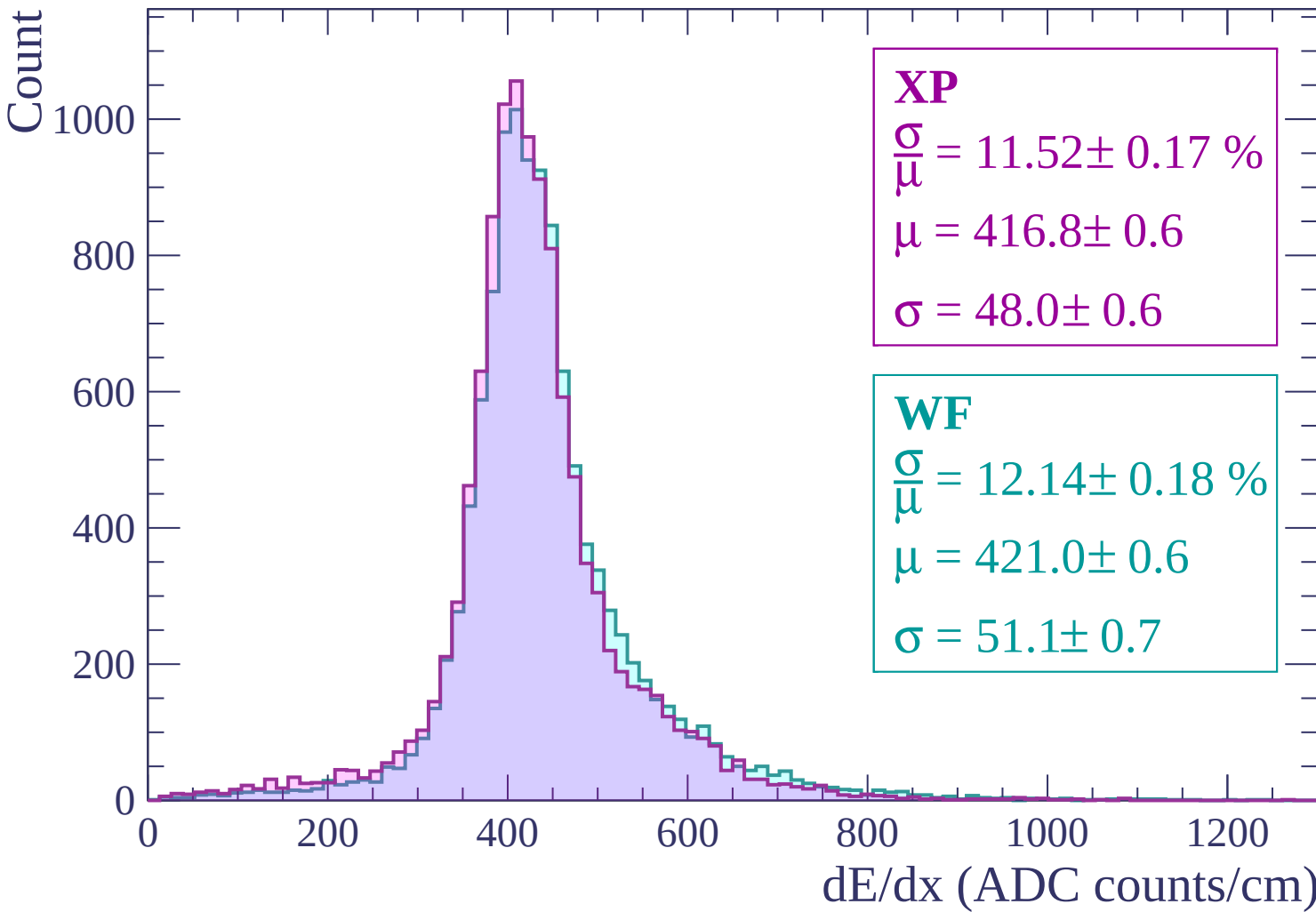
$$\sigma = 48.0 \pm 0.6$$

WF

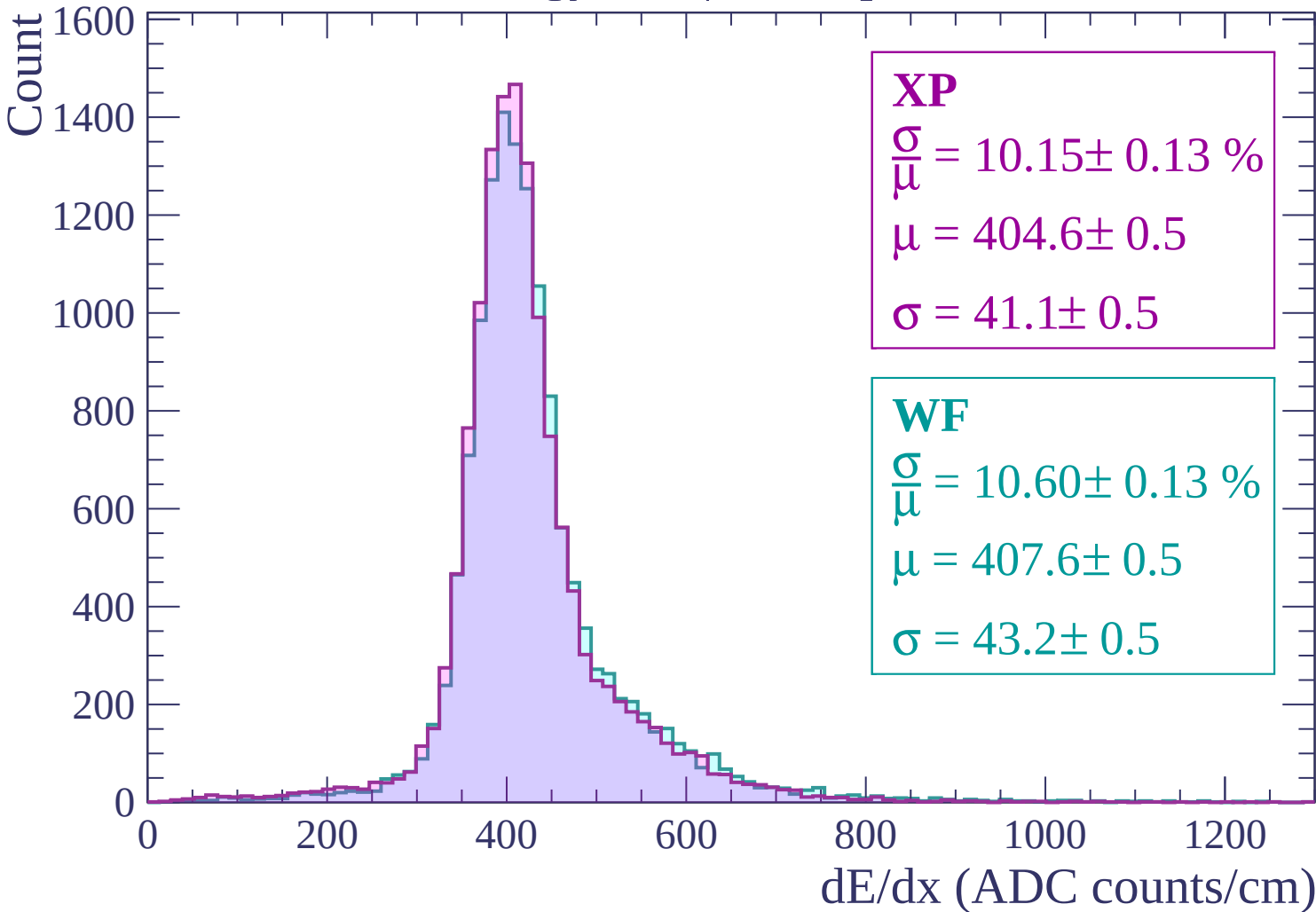
$$\frac{\sigma}{\mu} = 12.14 \pm 0.18 \%$$

$$\mu = 421.0 \pm 0.6$$

$$\sigma = 51.1 \pm 0.7$$



Energy loss | $300 < p < 360$



Energy loss | $360 < p < 420$

Count

1600
1400
1200
1000
800
600
400
200
0

XP

$$\frac{\sigma}{\mu} = 9.89 \pm 0.12 \%$$

$$\mu = 403.0 \pm 0.4$$

$$\sigma = 39.9 \pm 0.4$$

WF

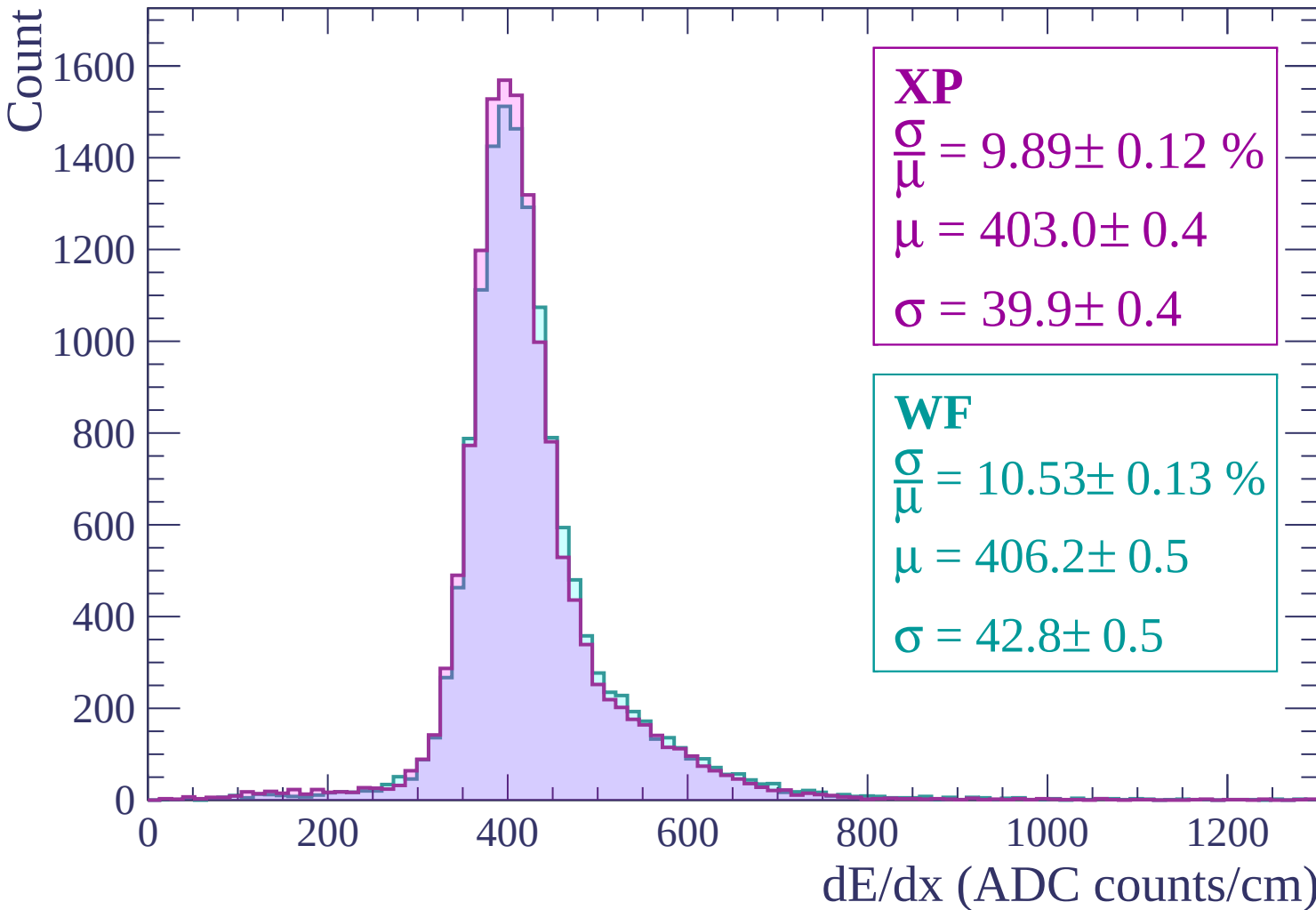
$$\frac{\sigma}{\mu} = 10.53 \pm 0.13 \%$$

$$\mu = 406.2 \pm 0.5$$

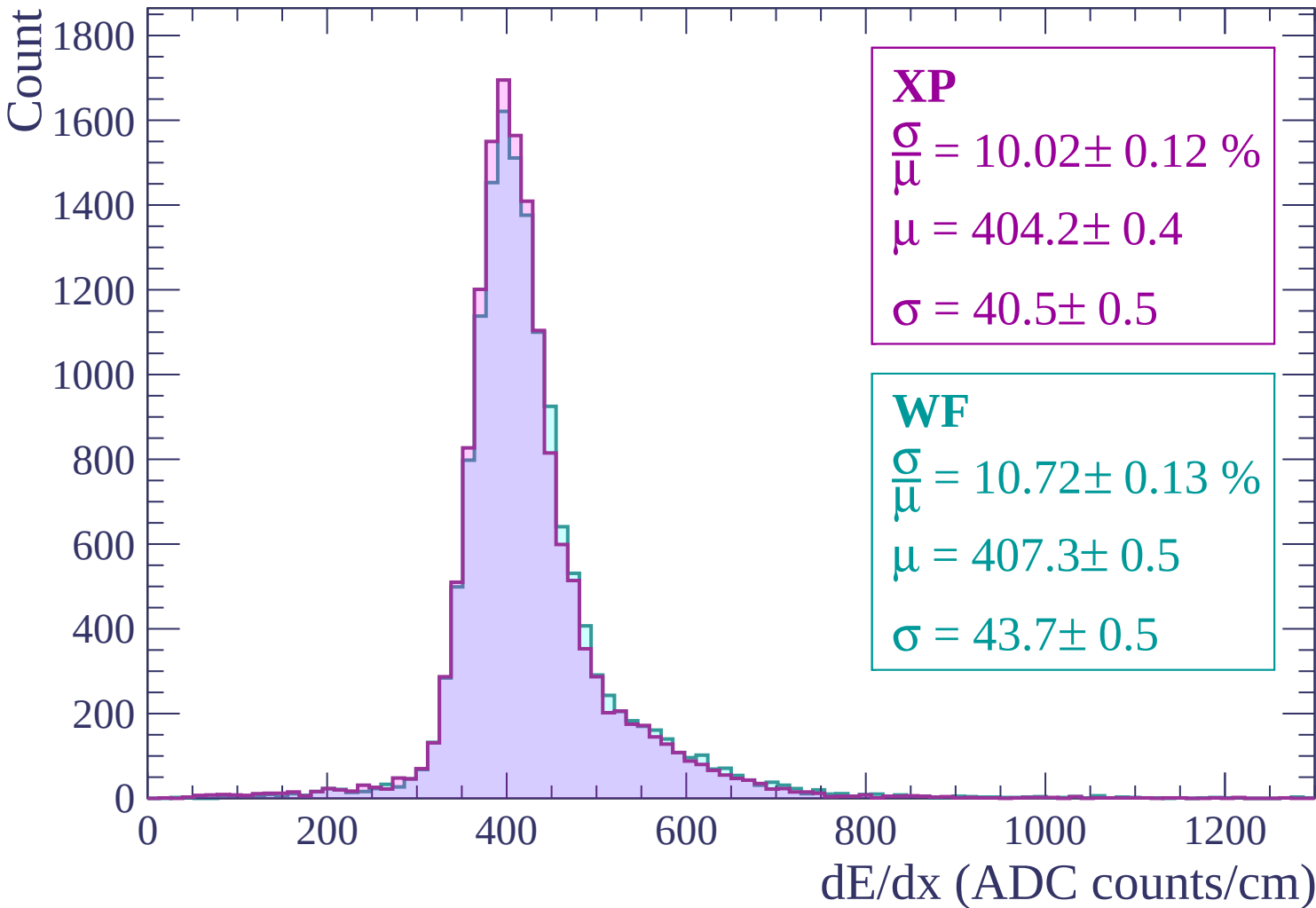
$$\sigma = 42.8 \pm 0.5$$

0 200 400 600 800 1000 1200

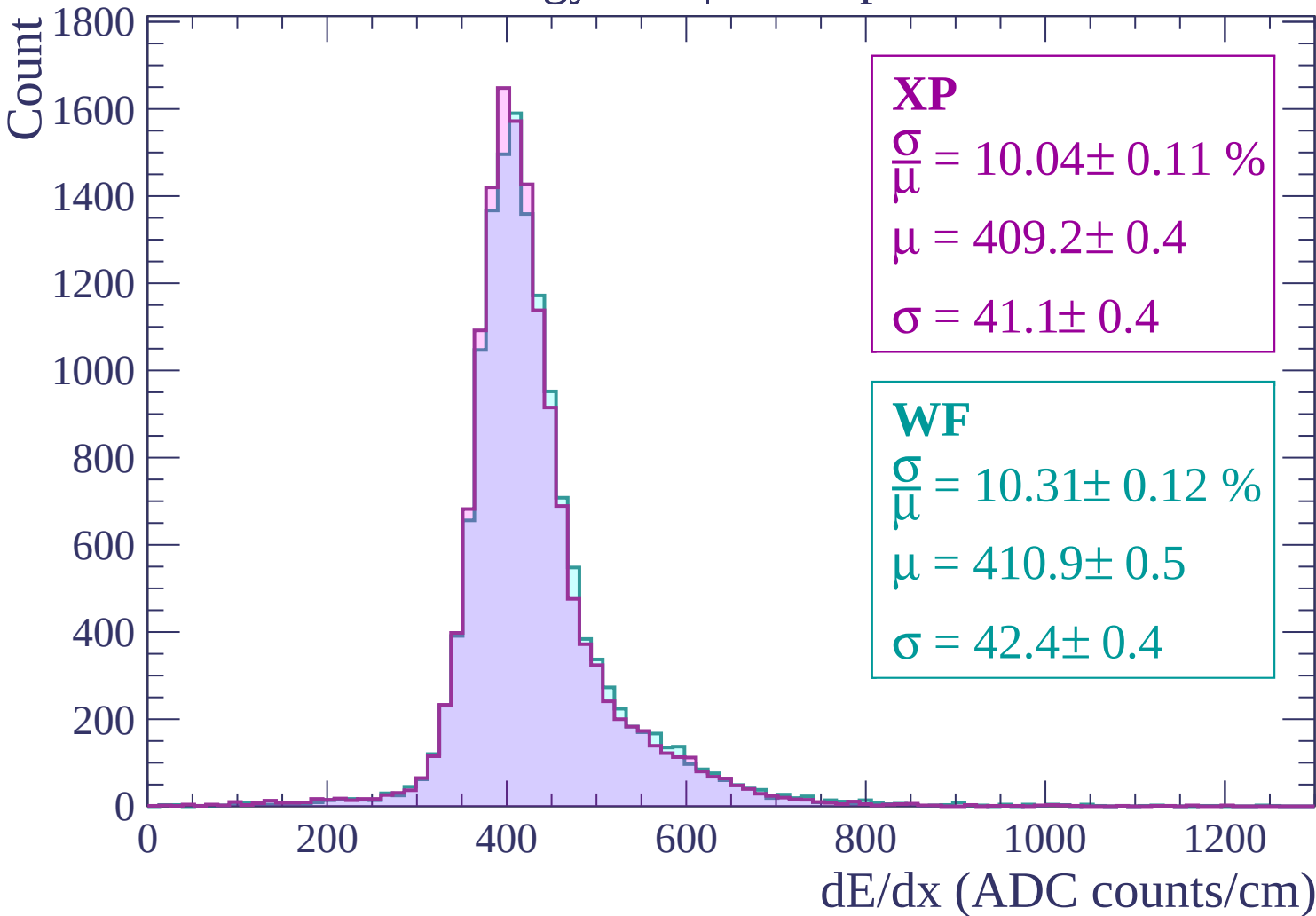
dE/dx (ADC counts/cm)



Energy loss | $420 < p < 480$



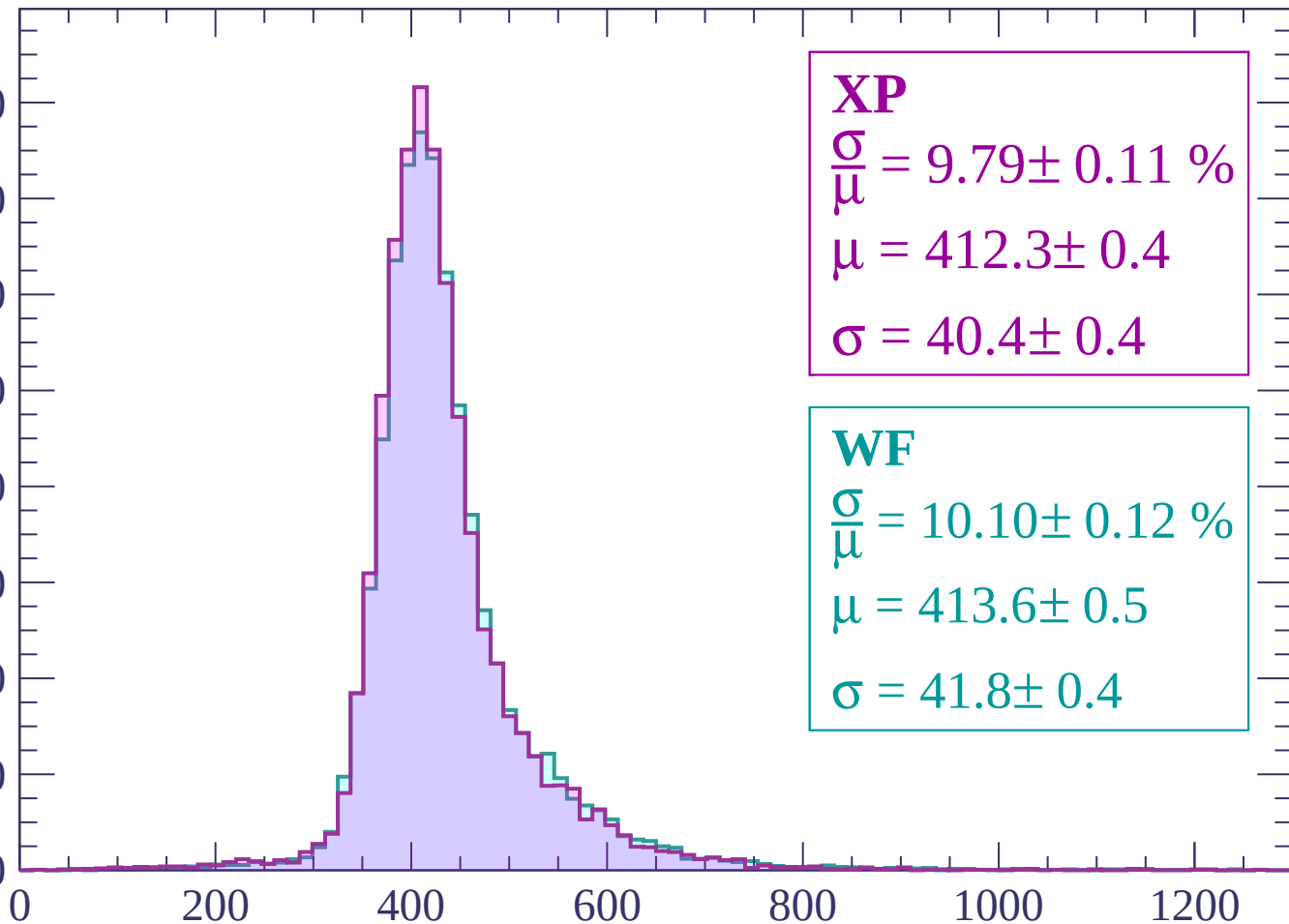
Energy loss | $480 < p < 540$



Energy loss | $540 < p < 600$

Count

1600
1400
1200
1000
800
600
400
200
0



XP

$$\frac{\sigma}{\mu} = 9.79 \pm 0.11 \%$$

$$\mu = 412.3 \pm 0.4$$

$$\sigma = 40.4 \pm 0.4$$

WF

$$\frac{\sigma}{\mu} = 10.10 \pm 0.12 \%$$

$$\mu = 413.6 \pm 0.5$$

$$\sigma = 41.8 \pm 0.4$$

dE/dx (ADC counts/cm)

Energy loss | $600 < p < 660$

Count

1600
1400
1200
1000
800
600
400
200
0

XP

$$\frac{\sigma}{\mu} = 9.69 \pm 0.12 \%$$

$$\mu = 416.3 \pm 0.4$$

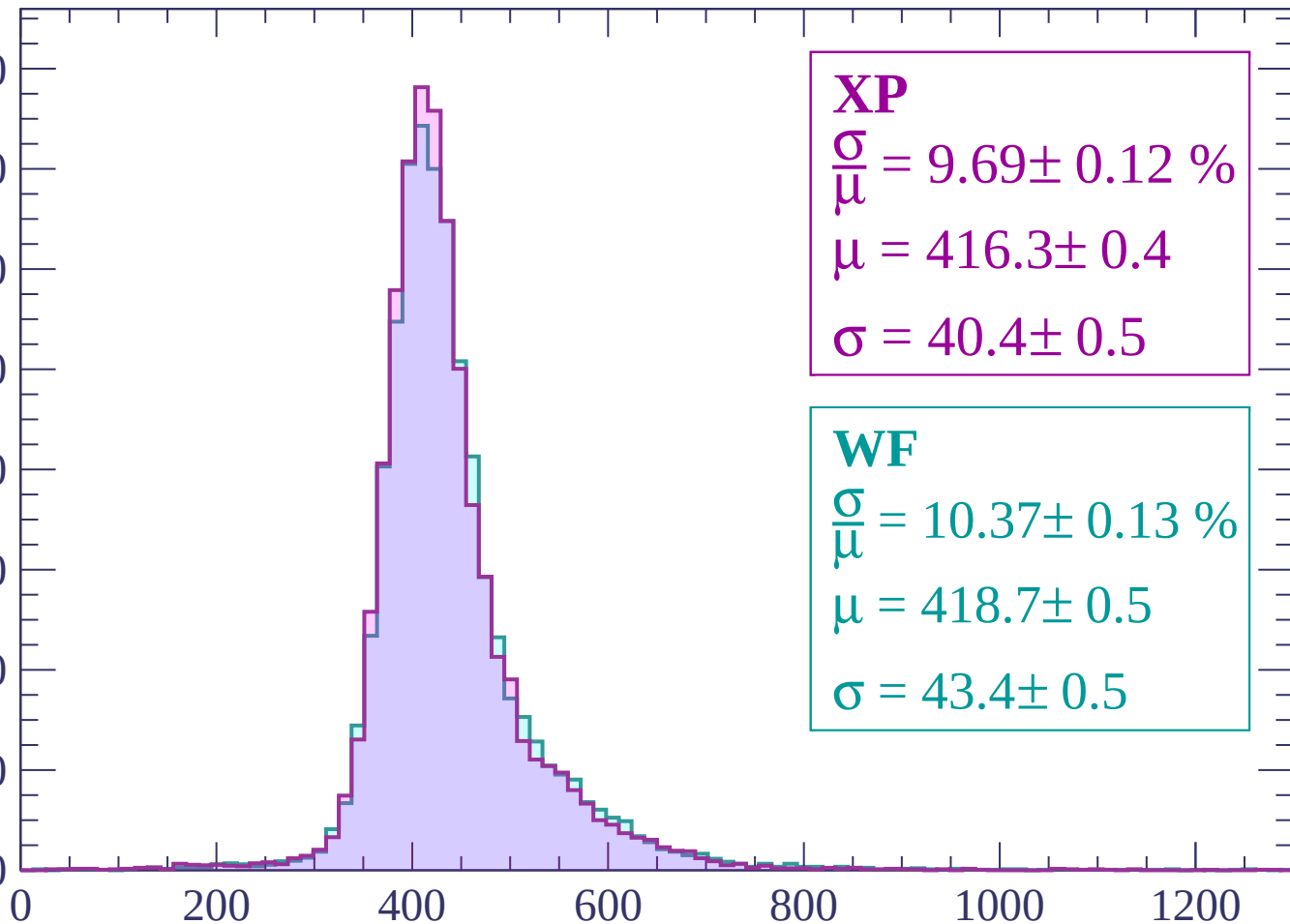
$$\sigma = 40.4 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 10.37 \pm 0.13 \%$$

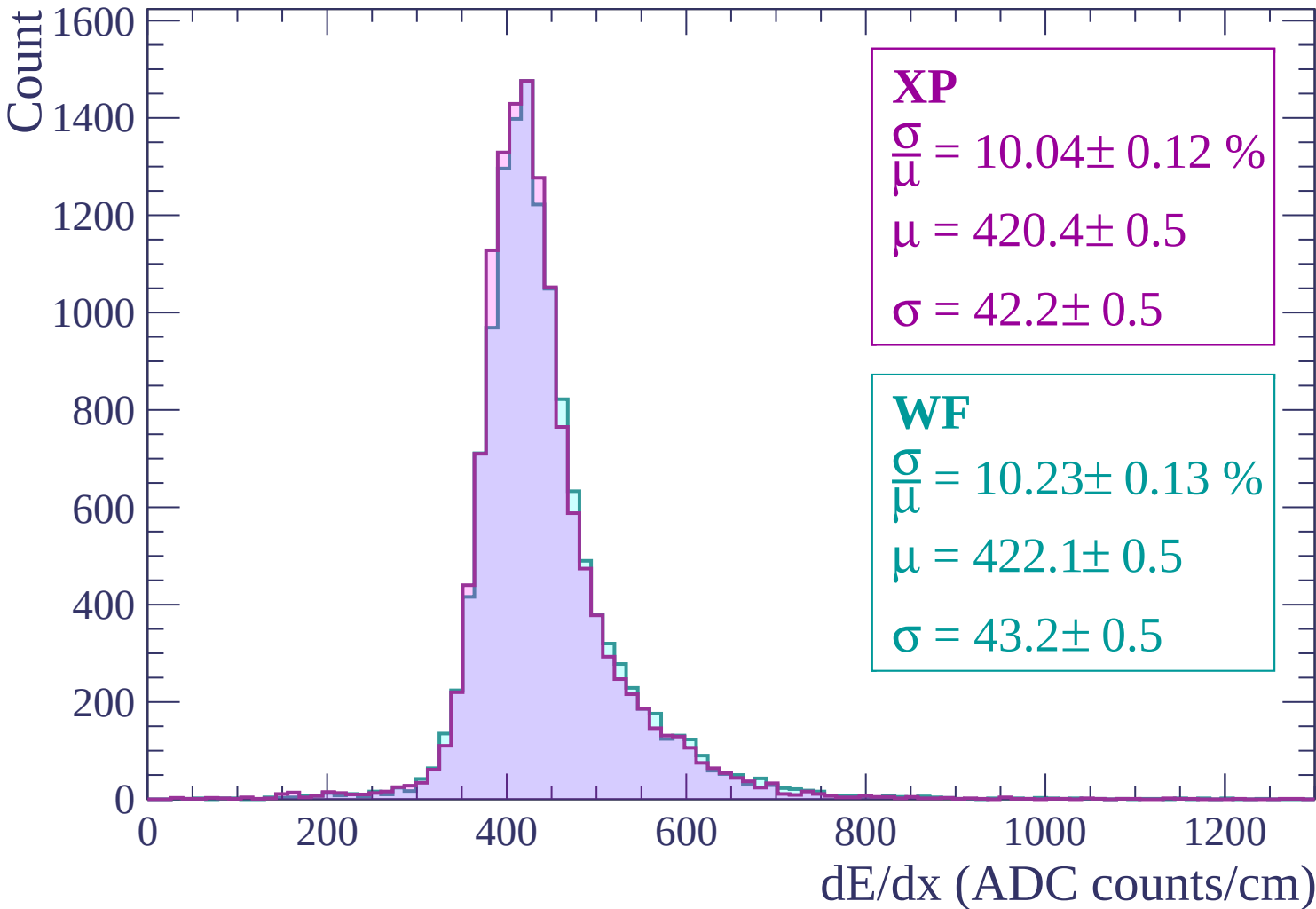
$$\mu = 418.7 \pm 0.5$$

$$\sigma = 43.4 \pm 0.5$$



dE/dx (ADC counts/cm)

Energy loss | $660 < p < 720$



Energy loss | $720 < p < 780$

Count

1400
1200
1000
800
600
400
200
0

XP

$$\frac{\sigma}{\mu} = 10.00 \pm 0.12 \%$$

$$\mu = 424.8 \pm 0.5$$

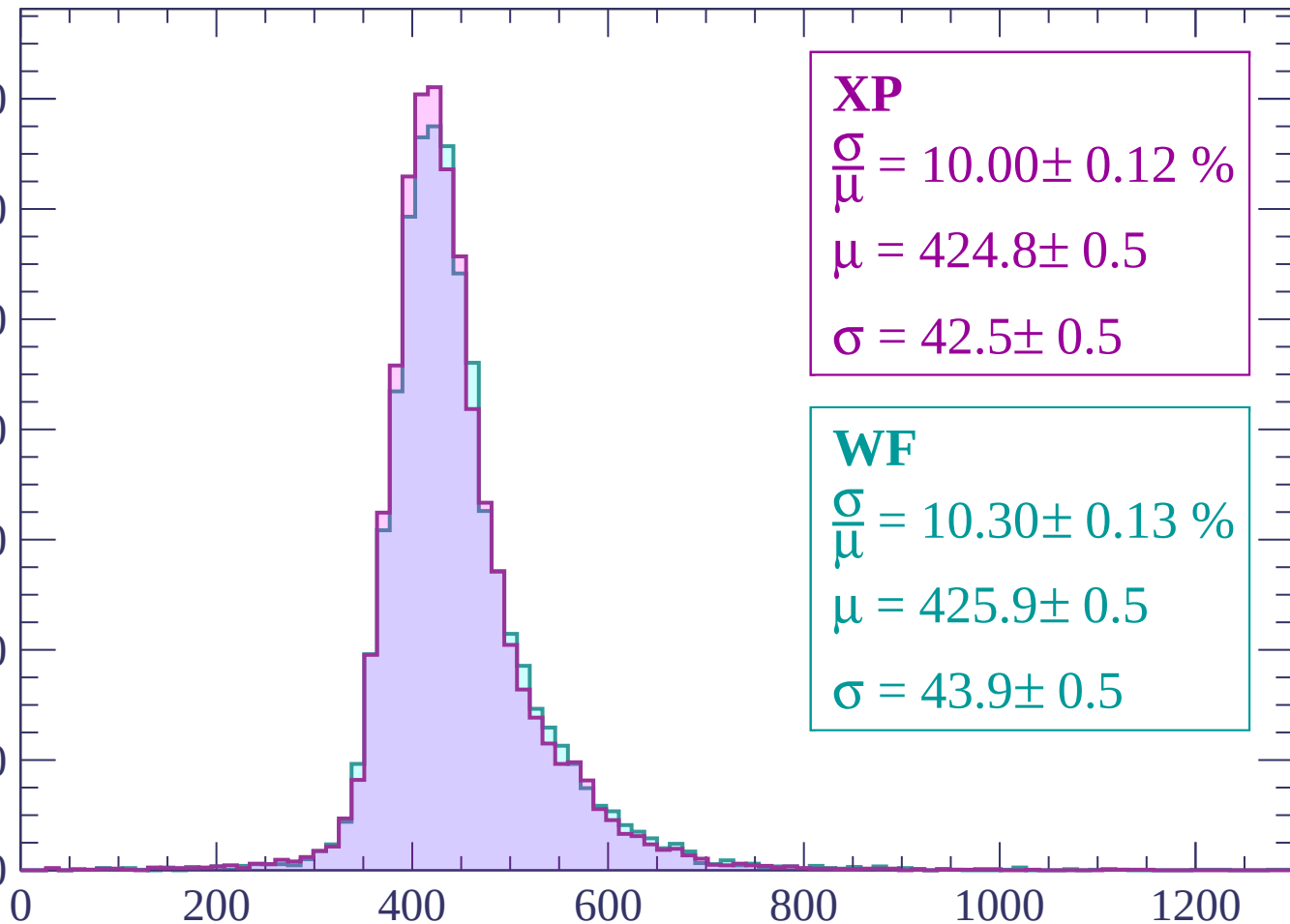
$$\sigma = 42.5 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 10.30 \pm 0.13 \%$$

$$\mu = 425.9 \pm 0.5$$

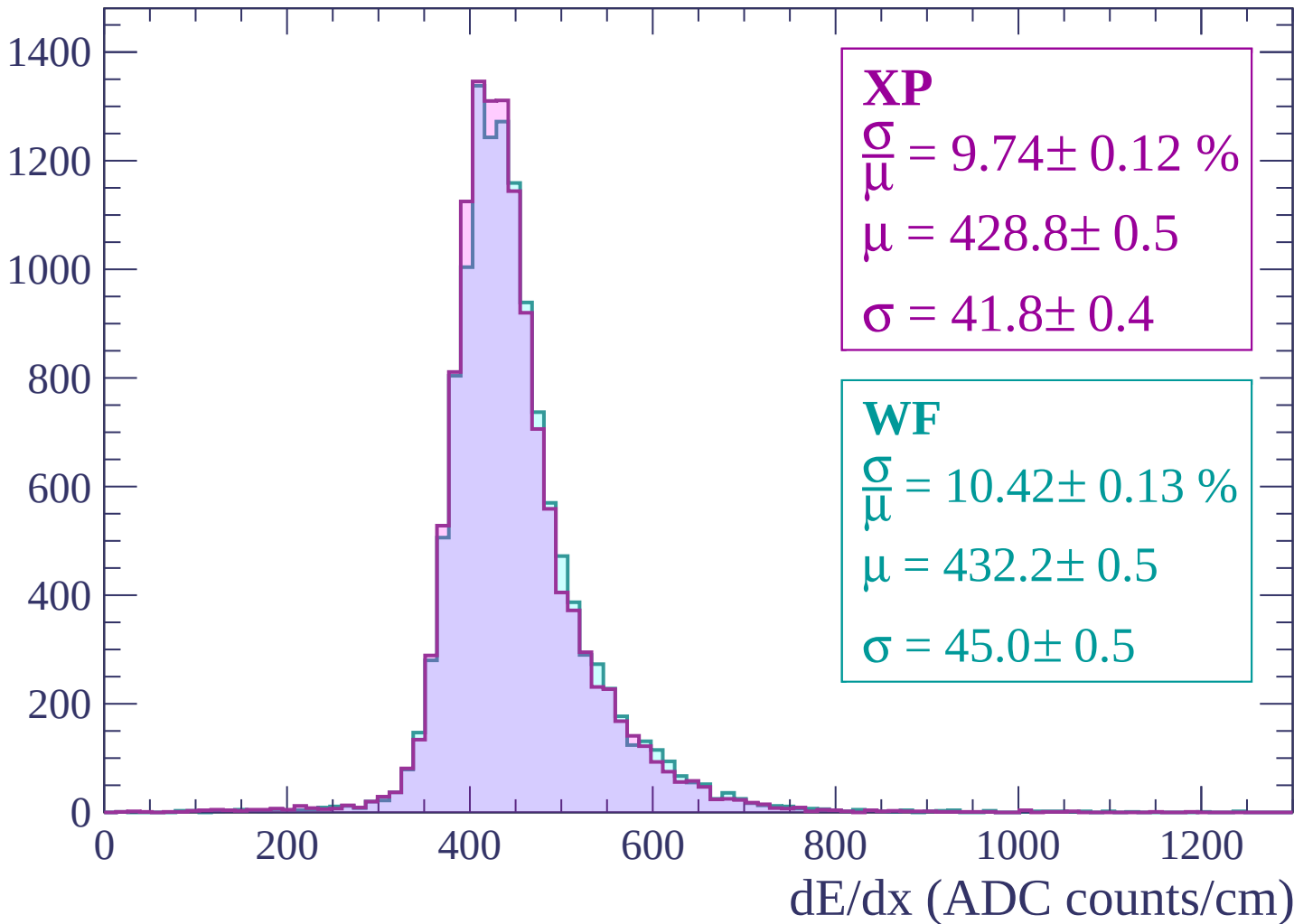
$$\sigma = 43.9 \pm 0.5$$



dE/dx (ADC counts/cm)

Energy loss | $780 < p < 840$

Count



Energy loss | $840 < p < 900$

Count

1400
1200
1000
800
600
400
200
0

XP

$$\frac{\sigma}{\mu} = 10.11 \pm 0.12 \%$$

$$\mu = 434.4 \pm 0.5$$

$$\sigma = 43.9 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 10.40 \pm 0.13 \%$$

$$\mu = 436.5 \pm 0.5$$

$$\sigma = 45.4 \pm 0.5$$

0

200

400

600

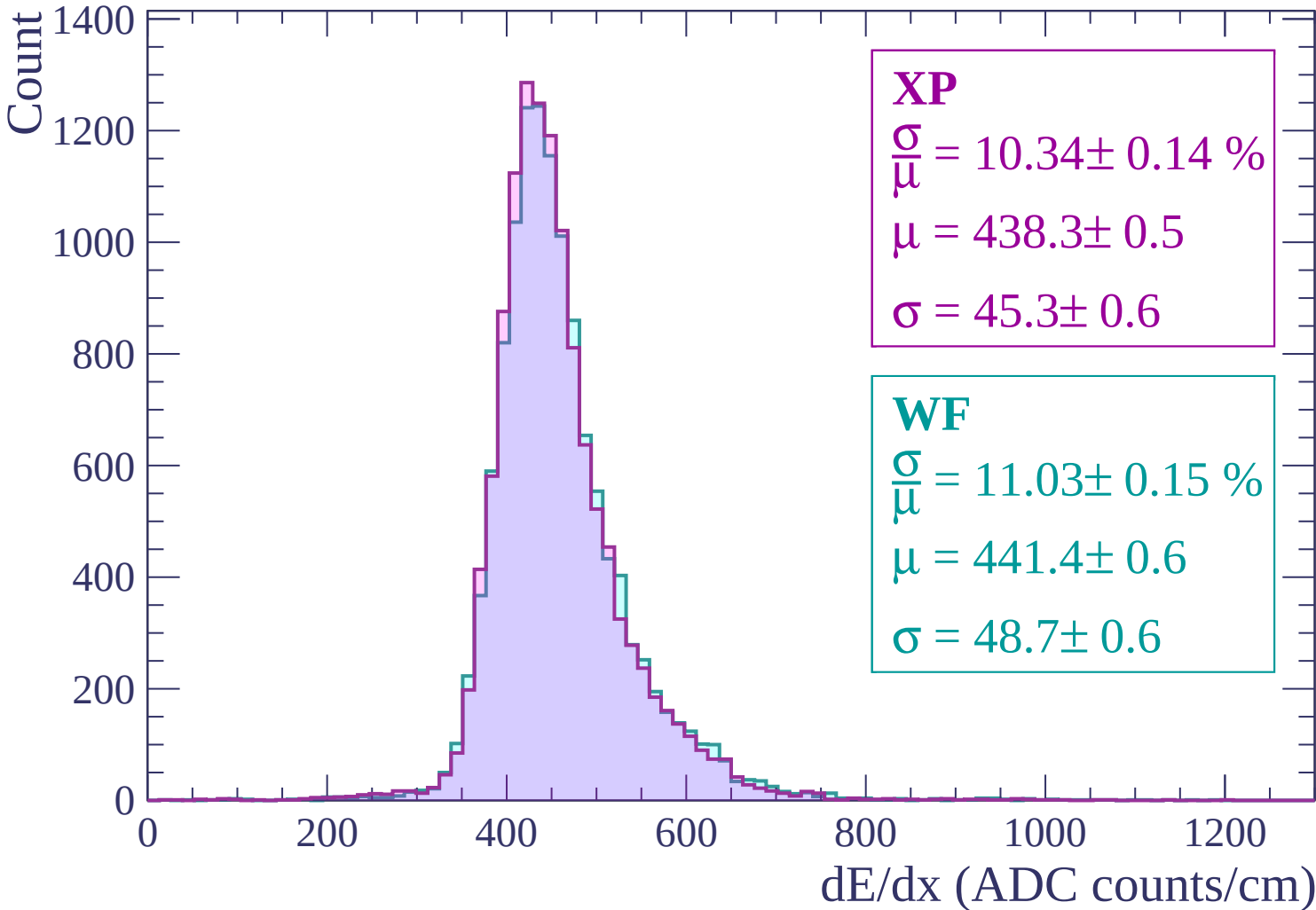
800

1000

1200

dE/dx (ADC counts/cm)

Energy loss | $900 < p < 960$



Energy loss | $960 < p < 1020$

Count

1200

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

XP

$$\frac{\sigma}{\mu} = 10.46 \pm 0.14 \%$$

$$\mu = 443.0 \pm 0.6$$

$$\sigma = 46.3 \pm 0.6$$

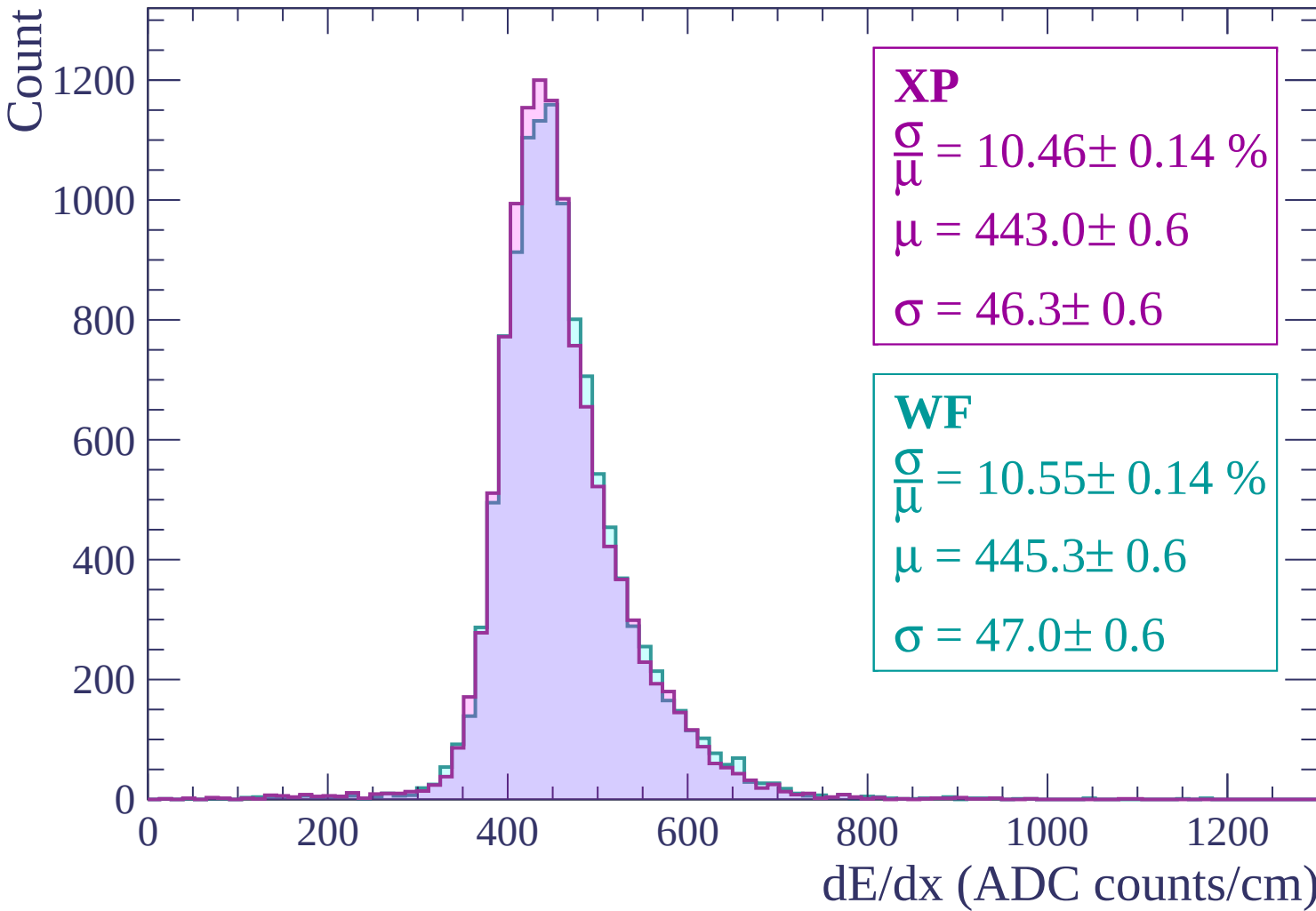
WF

$$\frac{\sigma}{\mu} = 10.55 \pm 0.14 \%$$

$$\mu = 445.3 \pm 0.6$$

$$\sigma = 47.0 \pm 0.6$$

dE/dx (ADC counts/cm)



Energy loss | $1020 < p < 1080$

Count

1200

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

XP

$$\frac{\sigma}{\mu} = 10.18 \pm 0.13 \%$$

$$\mu = 446.0 \pm 0.6$$

$$\sigma = 45.4 \pm 0.5$$

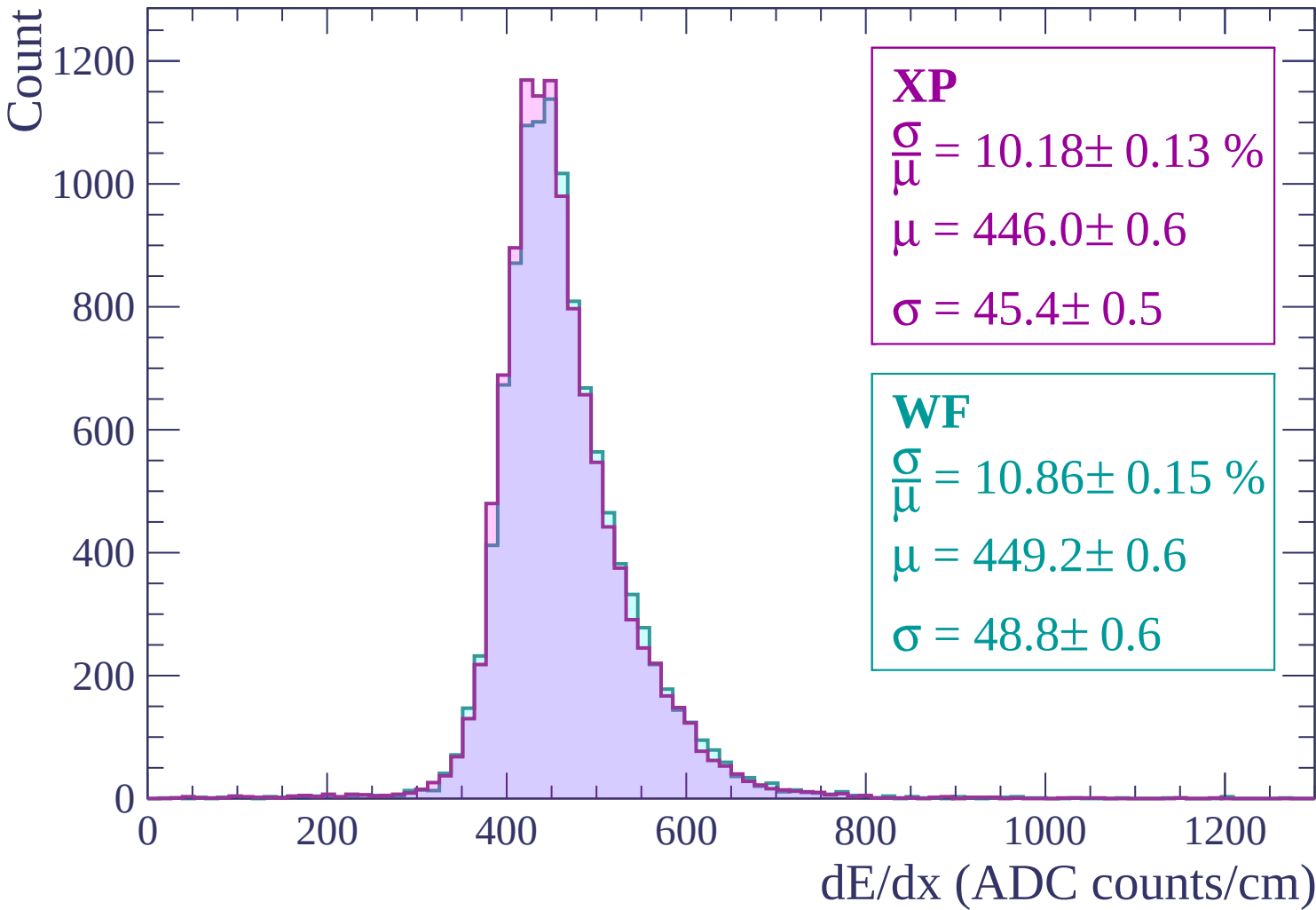
WF

$$\frac{\sigma}{\mu} = 10.86 \pm 0.15 \%$$

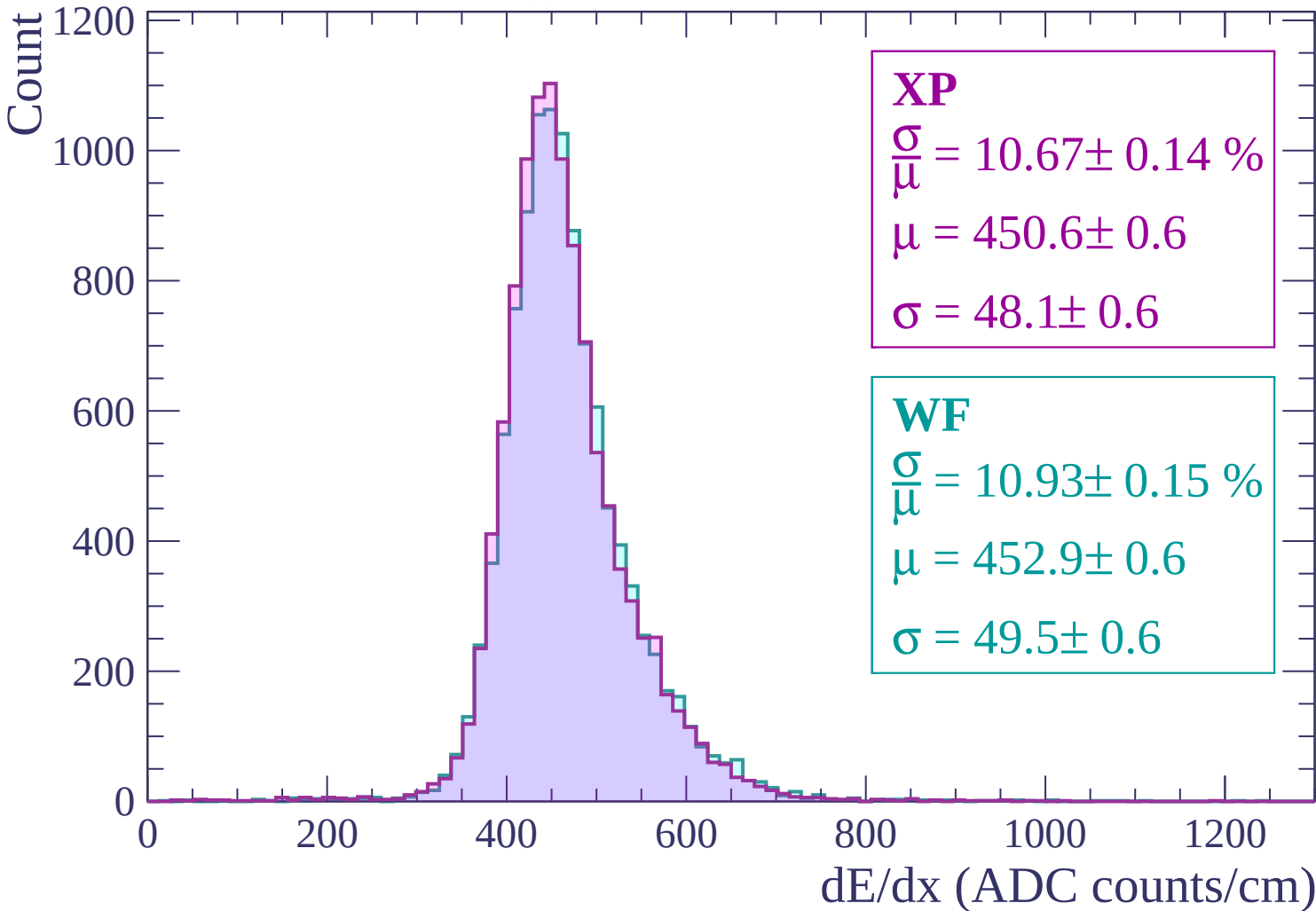
$$\mu = 449.2 \pm 0.6$$

$$\sigma = 48.8 \pm 0.6$$

dE/dx (ADC counts/cm)



Energy loss | $1080 < p < 1140$



Energy loss | $1140 < p < 1200$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

XP

$$\frac{\sigma}{\mu} = 10.43 \pm 0.13 \%$$

$$\mu = 453.4 \pm 0.6$$

$$\sigma = 47.3 \pm 0.5$$

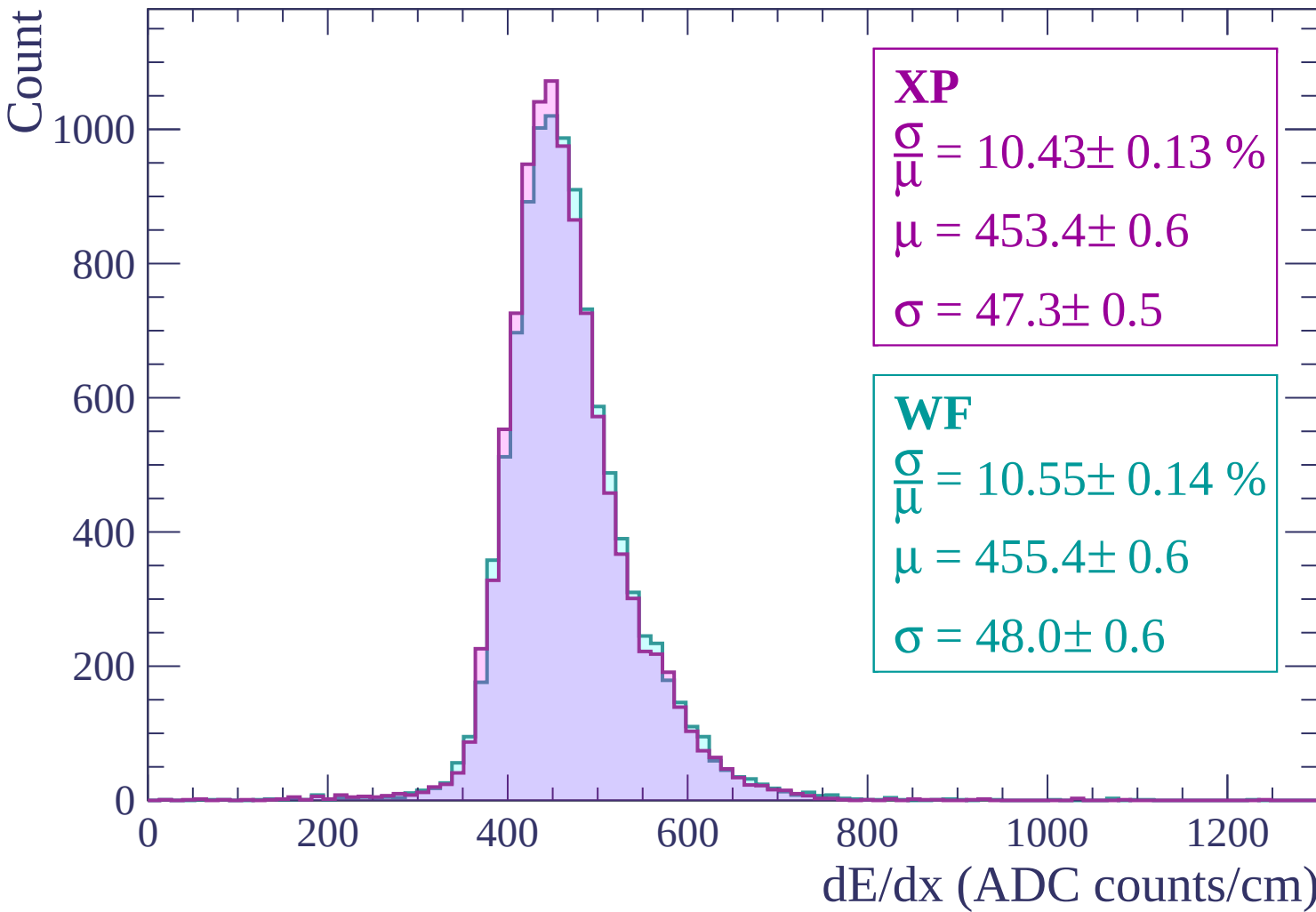
WF

$$\frac{\sigma}{\mu} = 10.55 \pm 0.14 \%$$

$$\mu = 455.4 \pm 0.6$$

$$\sigma = 48.0 \pm 0.6$$

dE/dx (ADC counts/cm)



Energy loss | $1200 < p < 1260$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 11.05 \pm 0.15 \%$$

$$\mu = 458.4 \pm 0.6$$

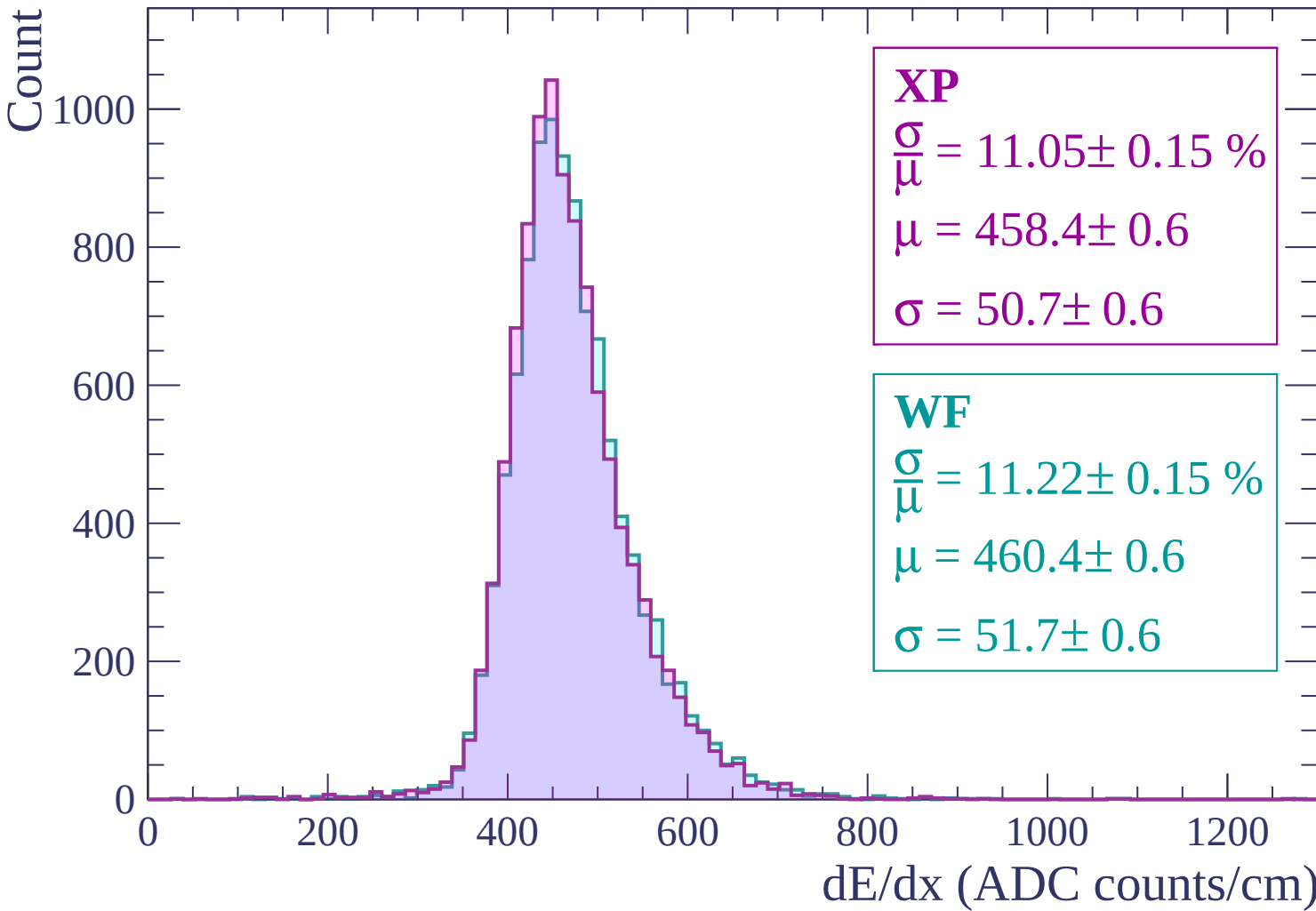
$$\sigma = 50.7 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 11.22 \pm 0.15 \%$$

$$\mu = 460.4 \pm 0.6$$

$$\sigma = 51.7 \pm 0.6$$



Energy loss | $1260 < p < 1320$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

XP

$$\frac{\sigma}{\mu} = 11.02 \pm 0.16 \%$$

$$\mu = 461.7 \pm 0.7$$

$$\sigma = 50.9 \pm 0.7$$

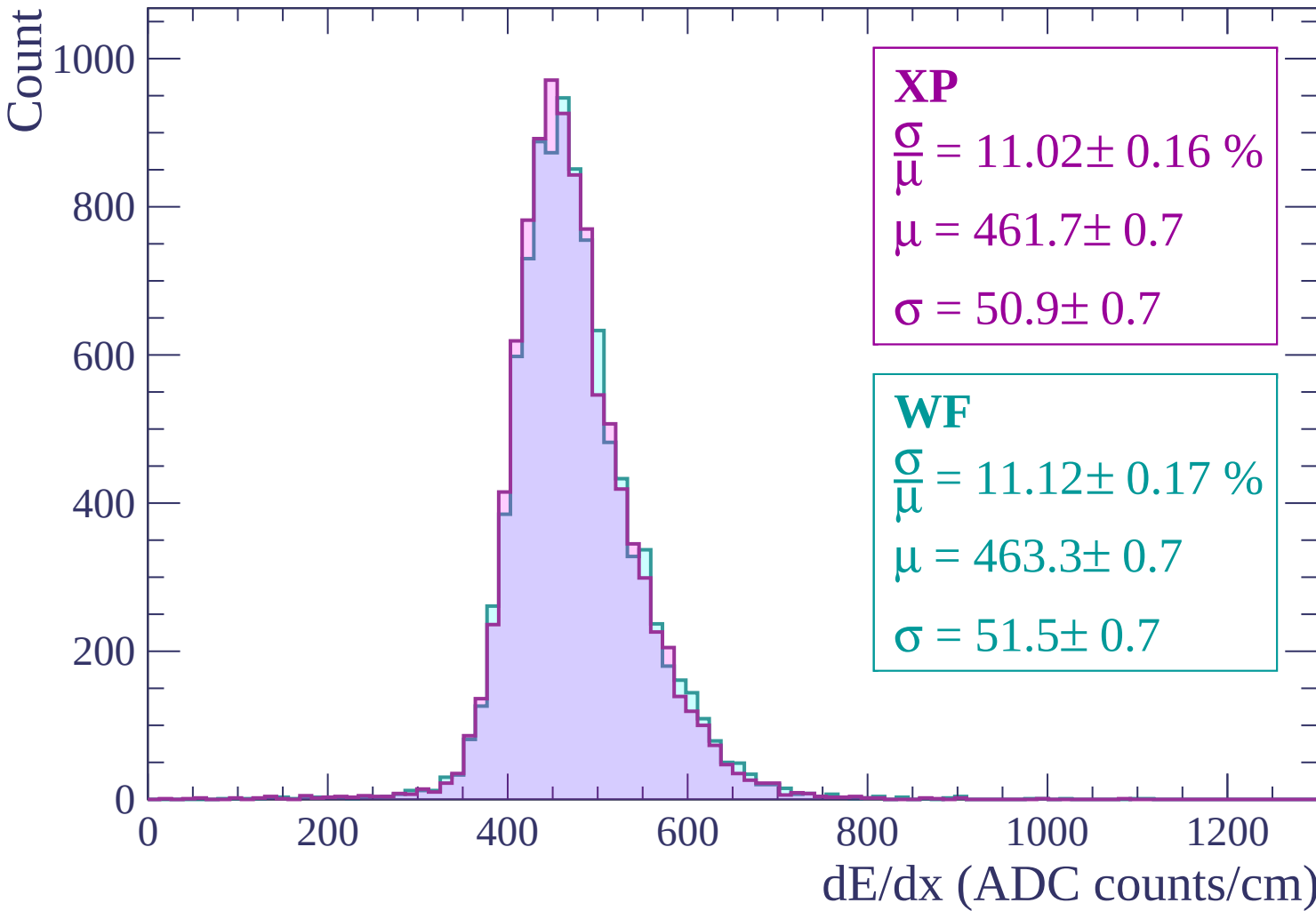
WF

$$\frac{\sigma}{\mu} = 11.12 \pm 0.17 \%$$

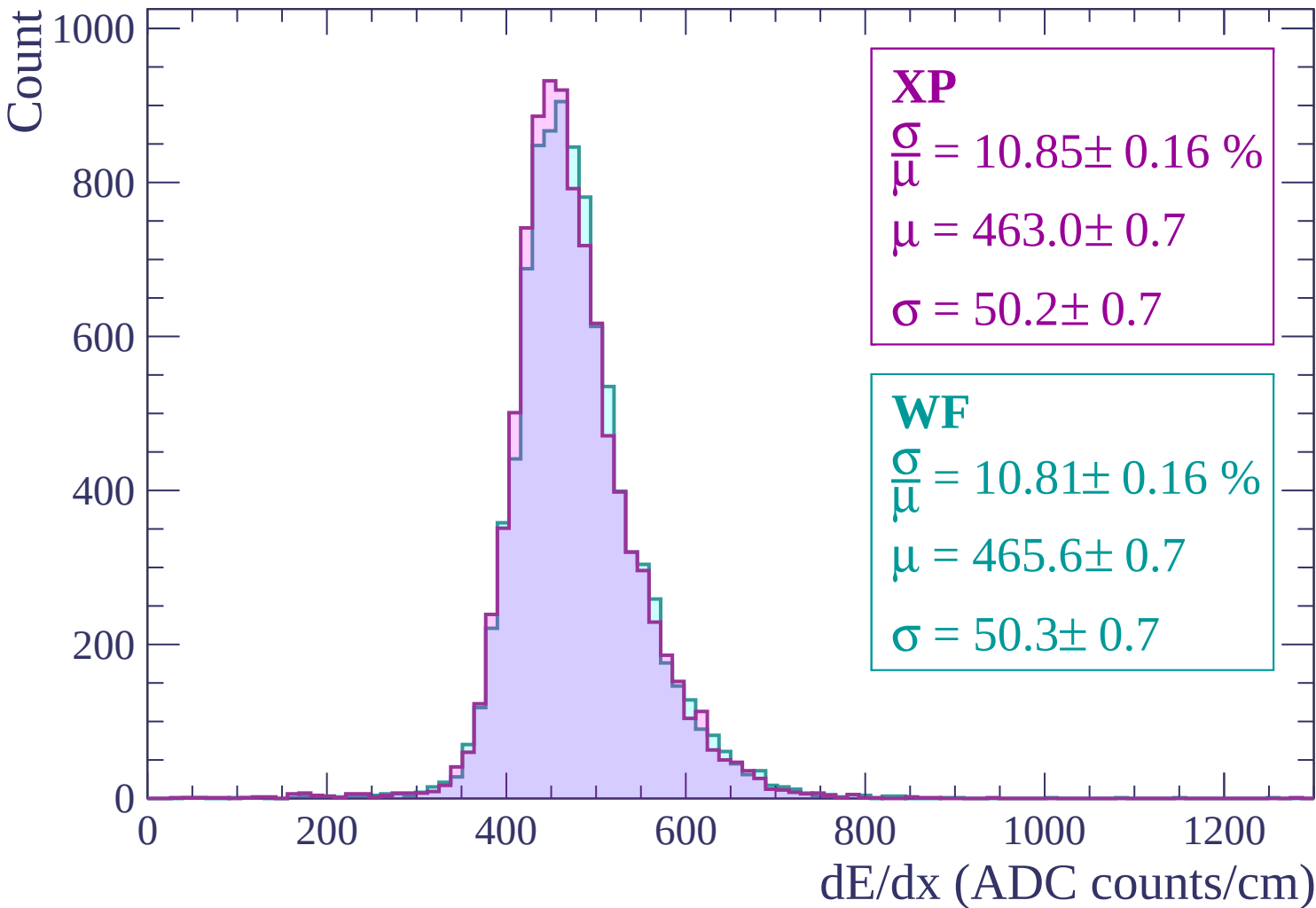
$$\mu = 463.3 \pm 0.7$$

$$\sigma = 51.5 \pm 0.7$$

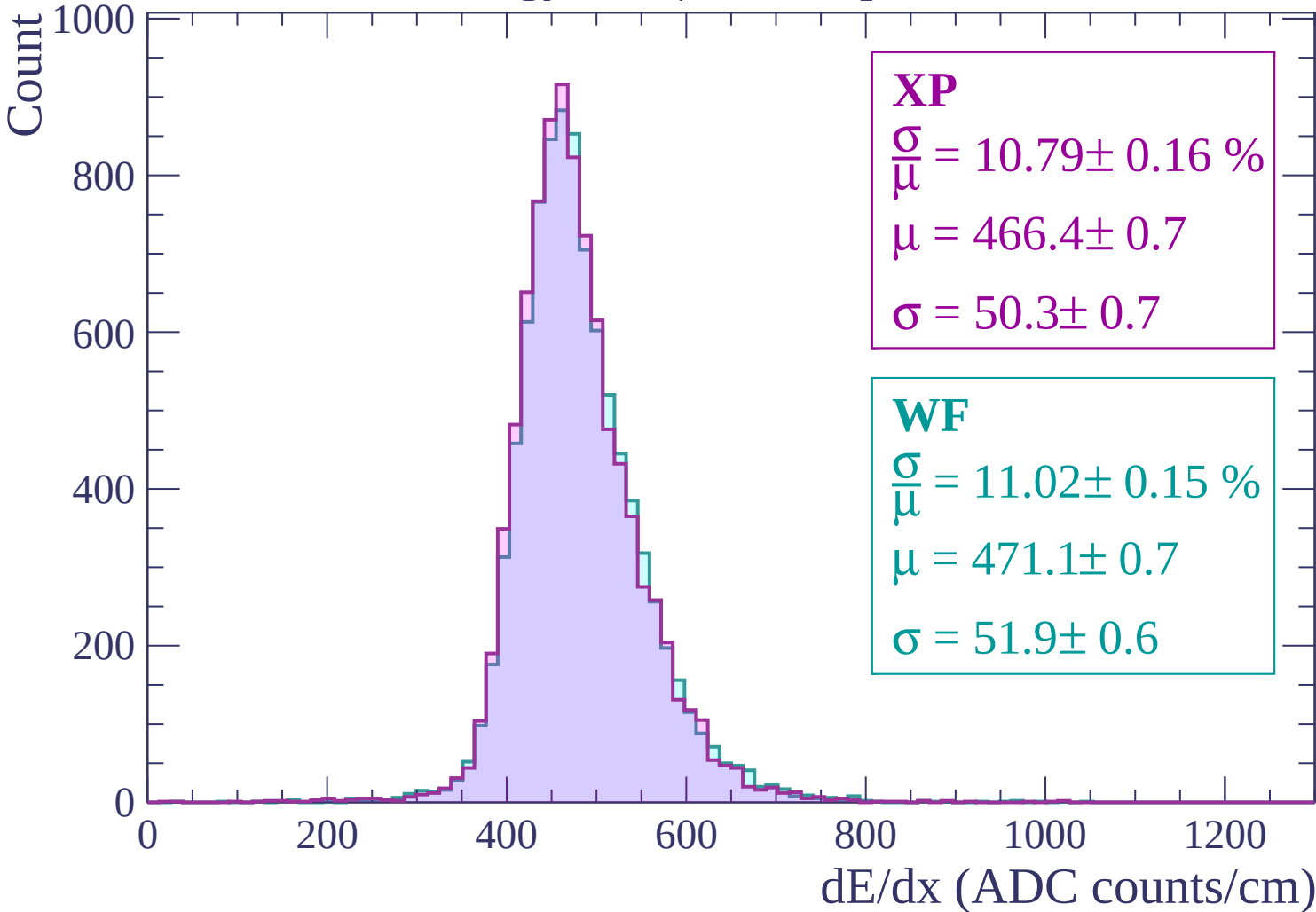
dE/dx (ADC counts/cm)



Energy loss | $1320 < p < 1380$

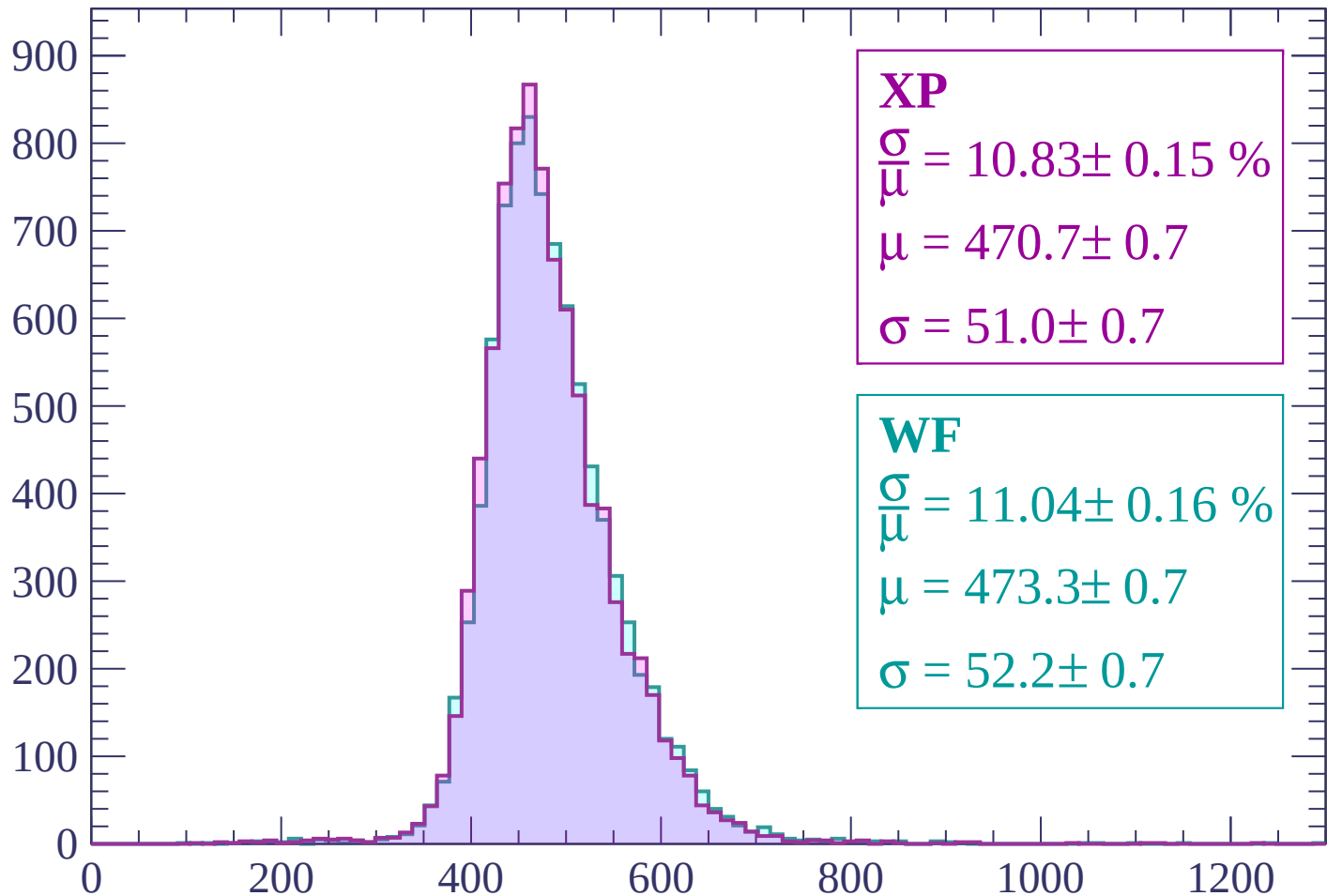


Energy loss | $1380 < p < 1440$



Energy loss | $1440 < p < 1500$

Count



XP

$$\frac{\sigma}{\mu} = 10.83 \pm 0.15 \%$$

$$\mu = 470.7 \pm 0.7$$

$$\sigma = 51.0 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 11.04 \pm 0.16 \%$$

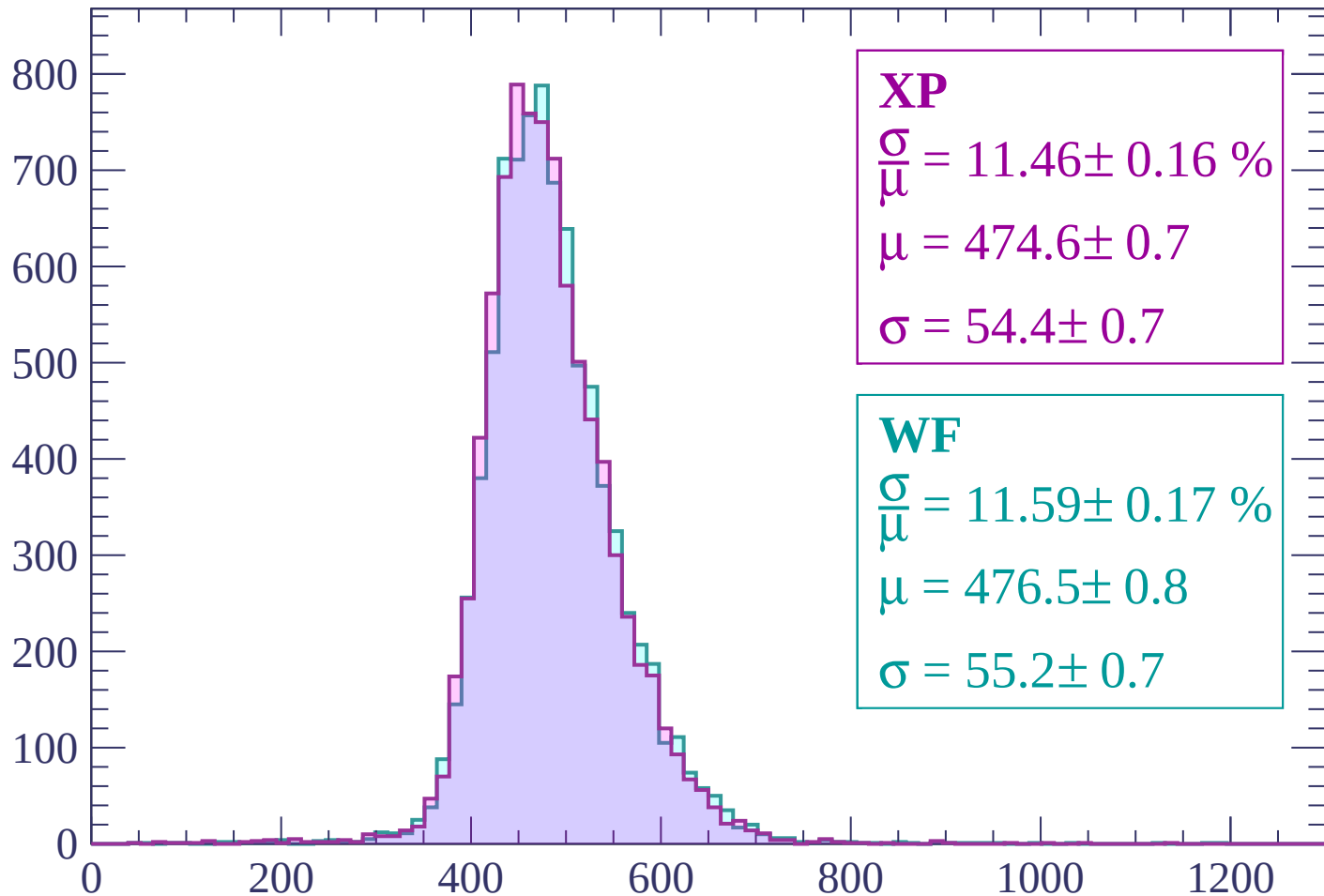
$$\mu = 473.3 \pm 0.7$$

$$\sigma = 52.2 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1500 < p < 1560$

Count



XP

$$\frac{\sigma}{\mu} = 11.46 \pm 0.16 \%$$

$$\mu = 474.6 \pm 0.7$$

$$\sigma = 54.4 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 11.59 \pm 0.17 \%$$

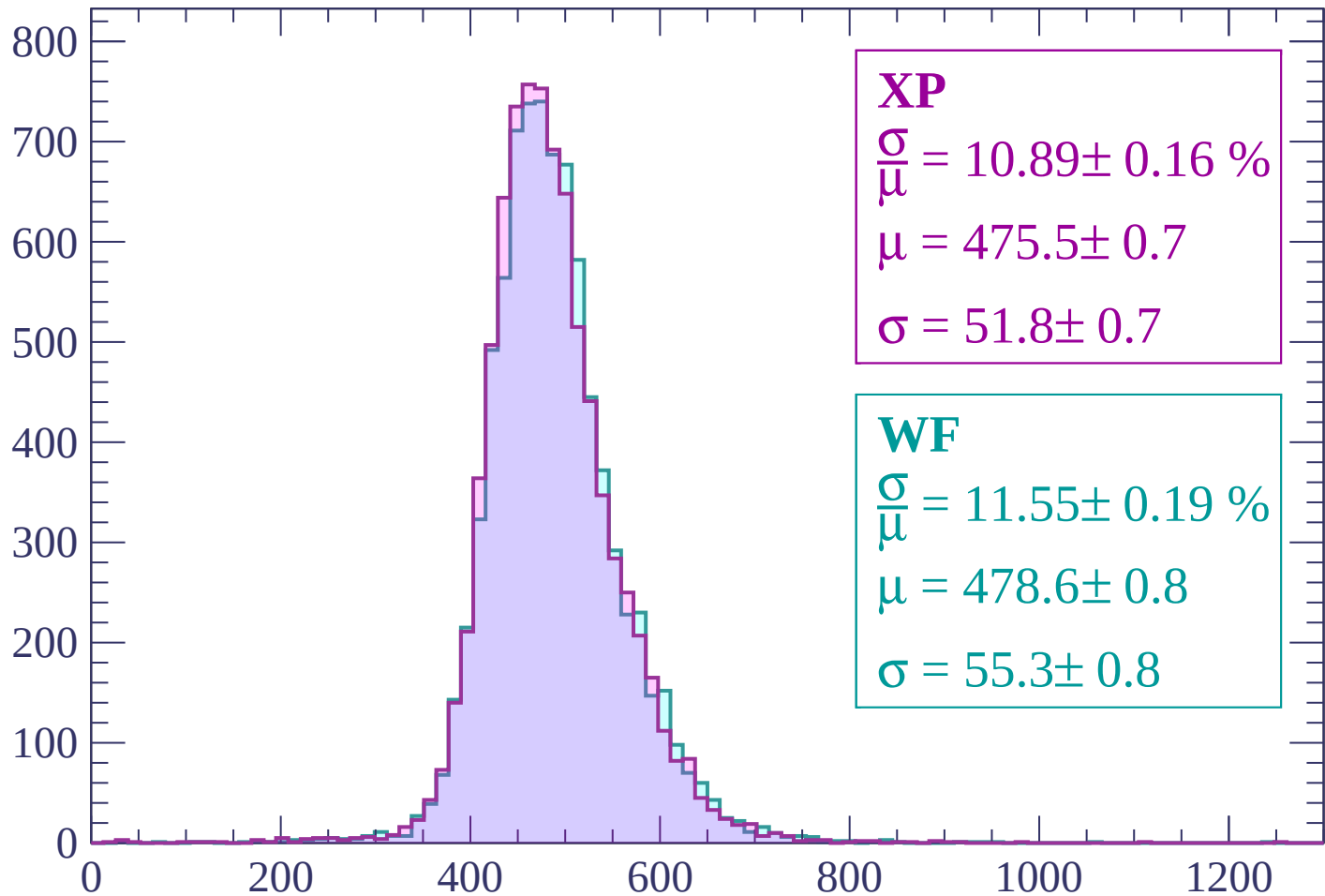
$$\mu = 476.5 \pm 0.8$$

$$\sigma = 55.2 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1560 < p < 1620$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 10.89 \pm 0.16 \%$$

$$\mu = 475.5 \pm 0.7$$

$$\sigma = 51.8 \pm 0.7$$

WF

$$\frac{\sigma}{\bar{\mu}} = 11.55 \pm 0.19 \%$$

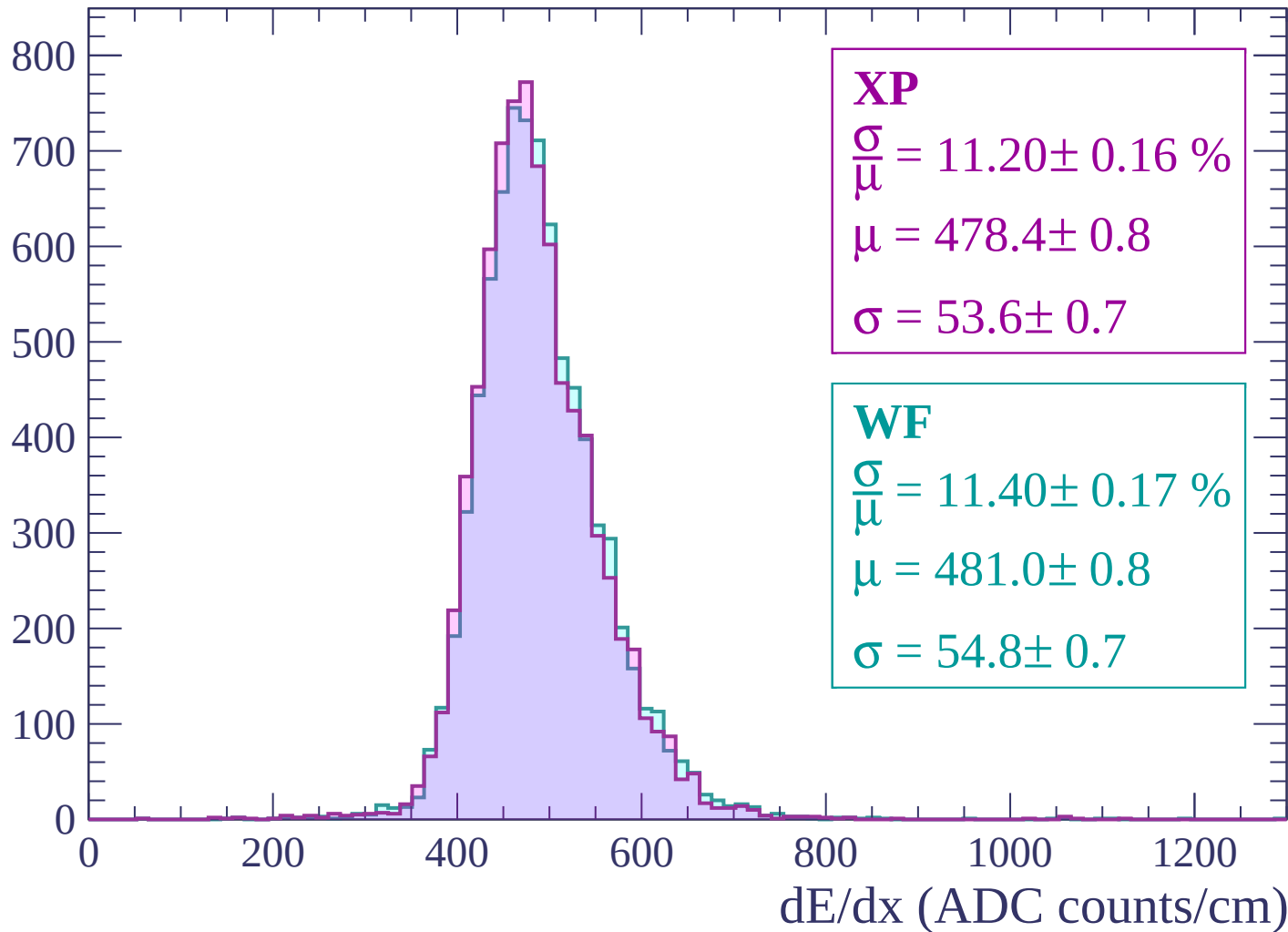
$$\mu = 478.6 \pm 0.8$$

$$\sigma = 55.3 \pm 0.8$$

dE/dx (ADC counts/cm)

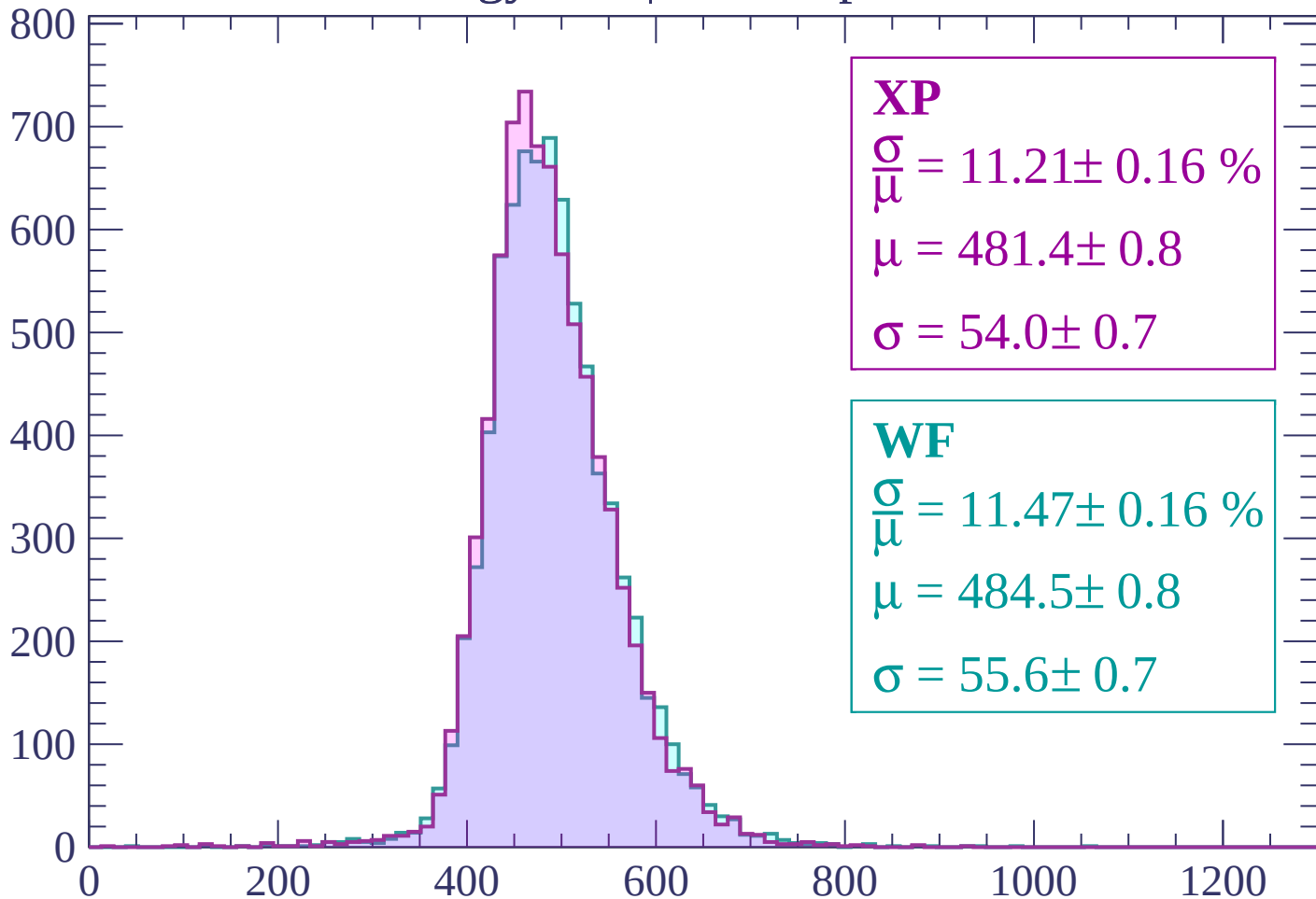
Energy loss | $1620 < p < 1680$

Count



Energy loss | $1680 < p < 1740$

Count



XP

$$\frac{\sigma}{\mu} = 11.21 \pm 0.16 \%$$

$$\mu = 481.4 \pm 0.8$$

$$\sigma = 54.0 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 11.47 \pm 0.16 \%$$

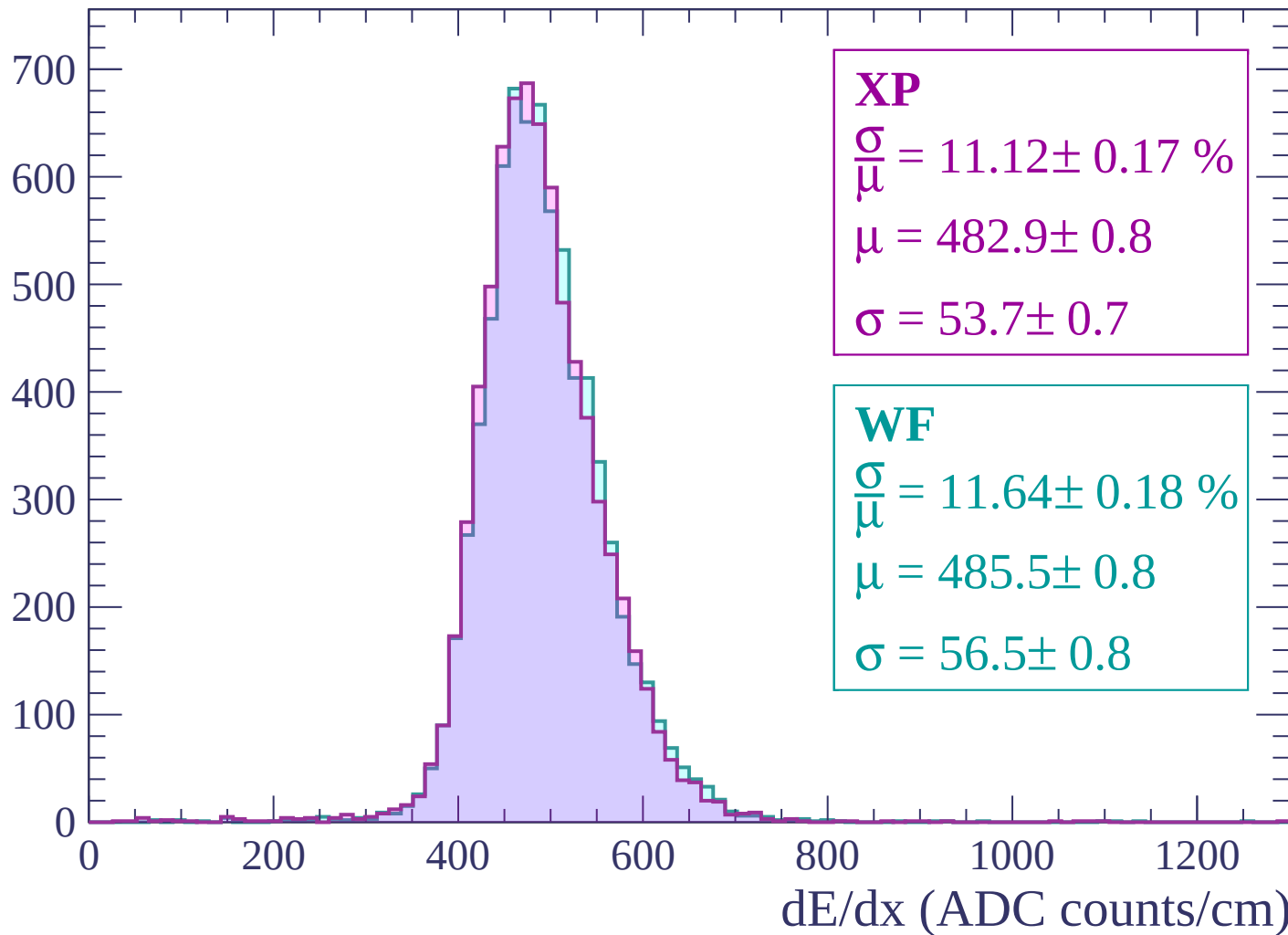
$$\mu = 484.5 \pm 0.8$$

$$\sigma = 55.6 \pm 0.7$$

dE/dx (ADC counts/cm)

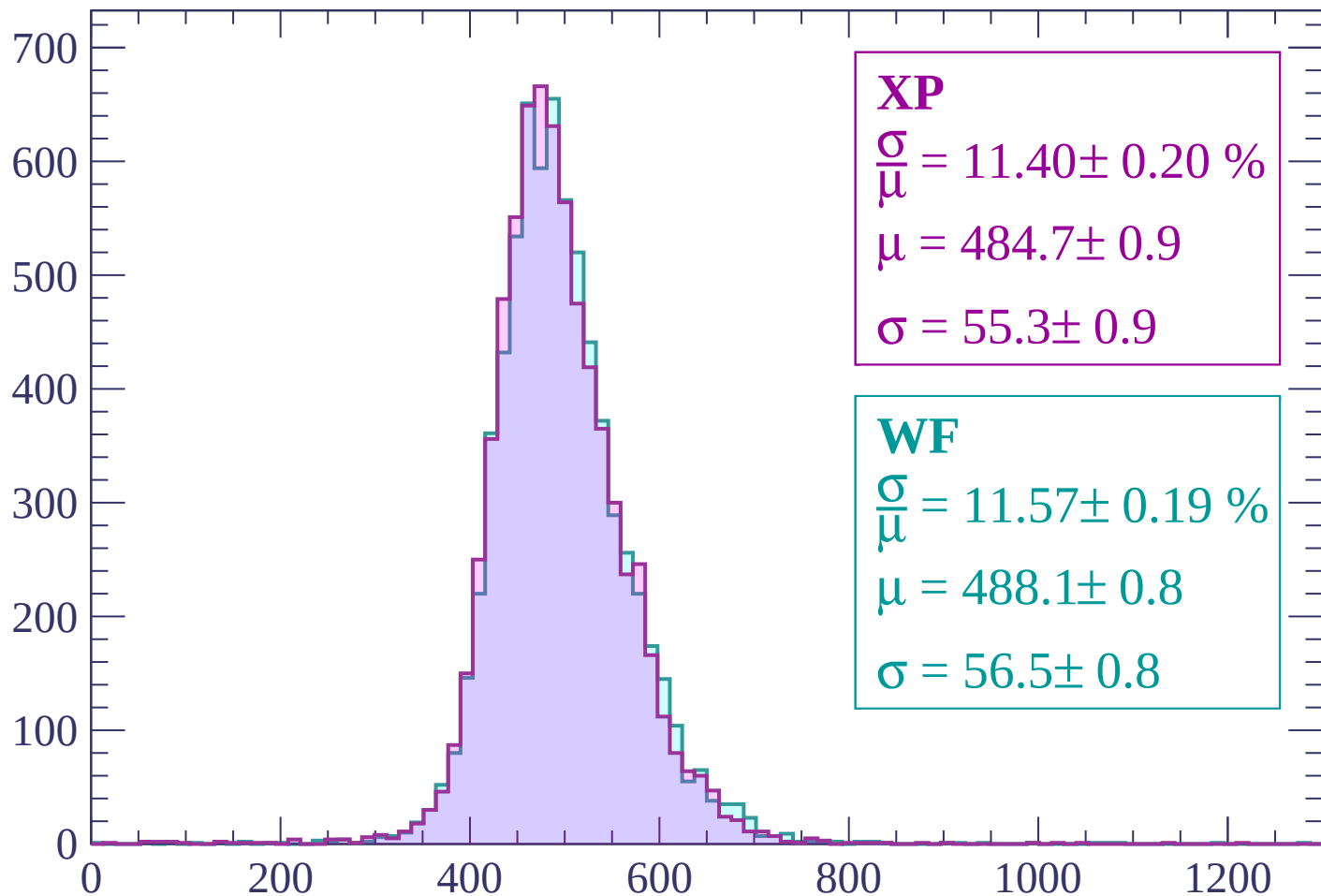
Energy loss | $1740 < p < 1800$

Count



Energy loss | $1800 < p < 1860$

Count



XP

$$\frac{\sigma}{\mu} = 11.40 \pm 0.20 \%$$

$$\mu = 484.7 \pm 0.9$$

$$\sigma = 55.3 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 11.57 \pm 0.19 \%$$

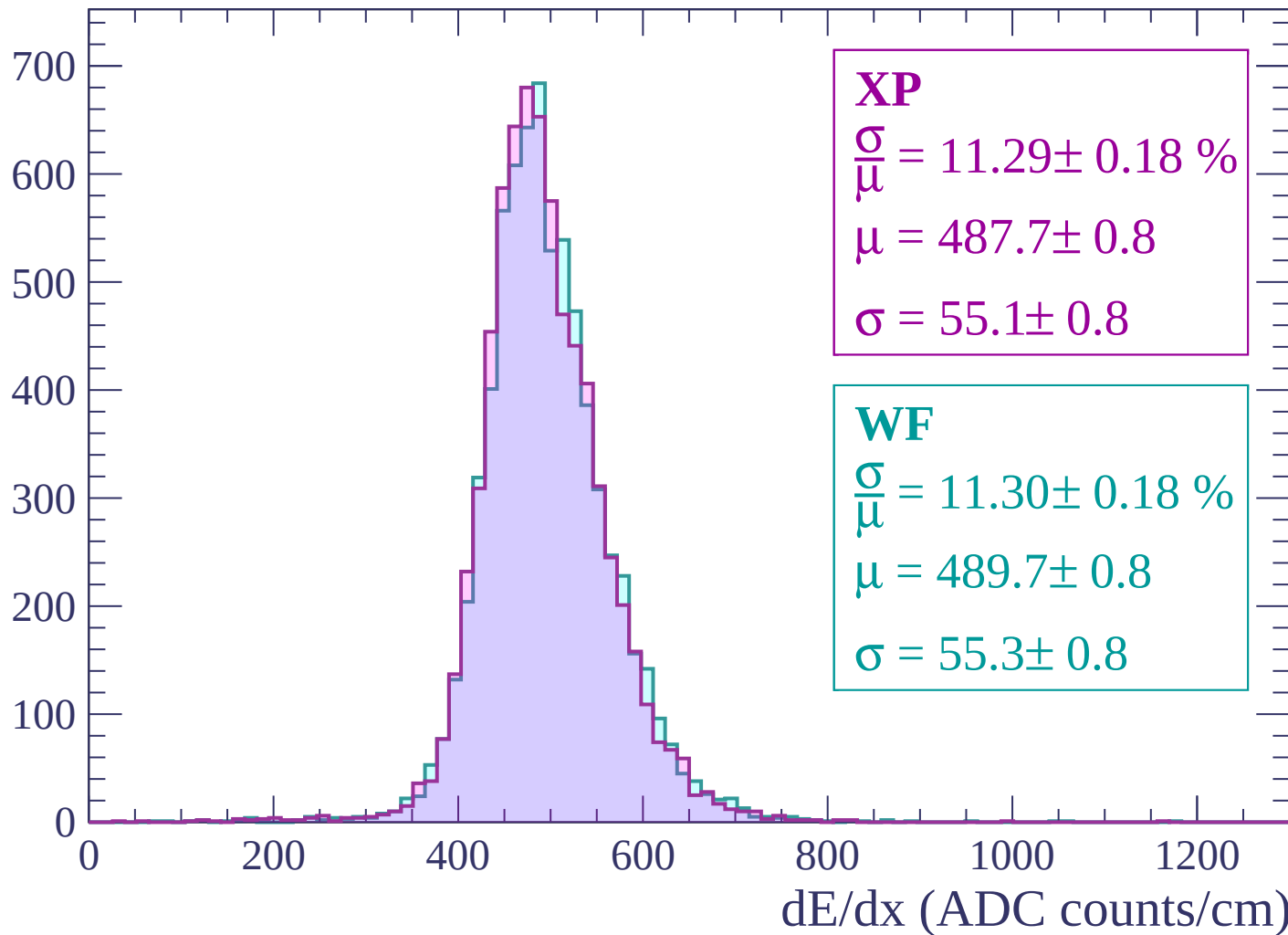
$$\mu = 488.1 \pm 0.8$$

$$\sigma = 56.5 \pm 0.8$$

dE/dx (ADC counts/cm)

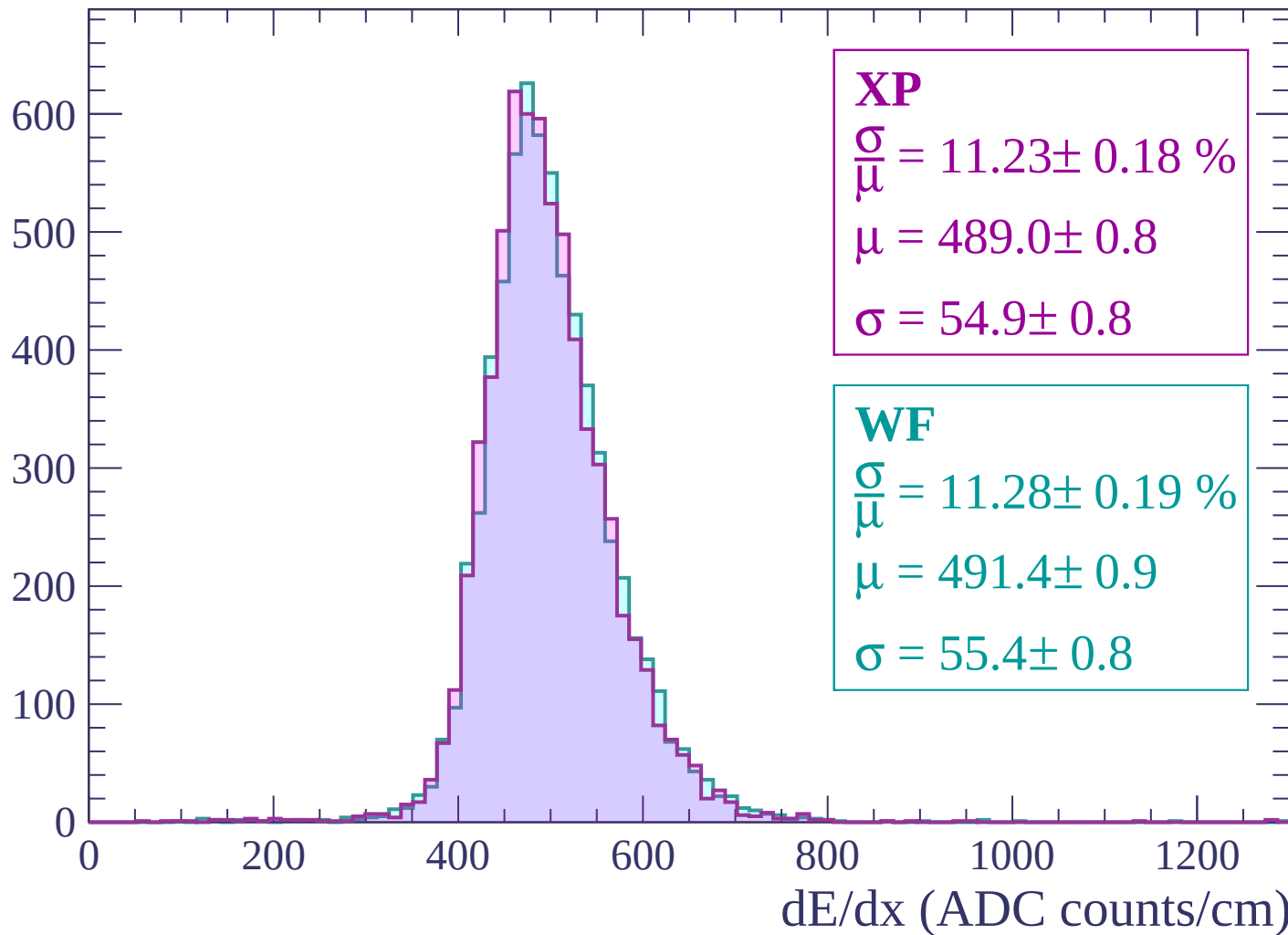
Energy loss | $1860 < p < 1920$

Count



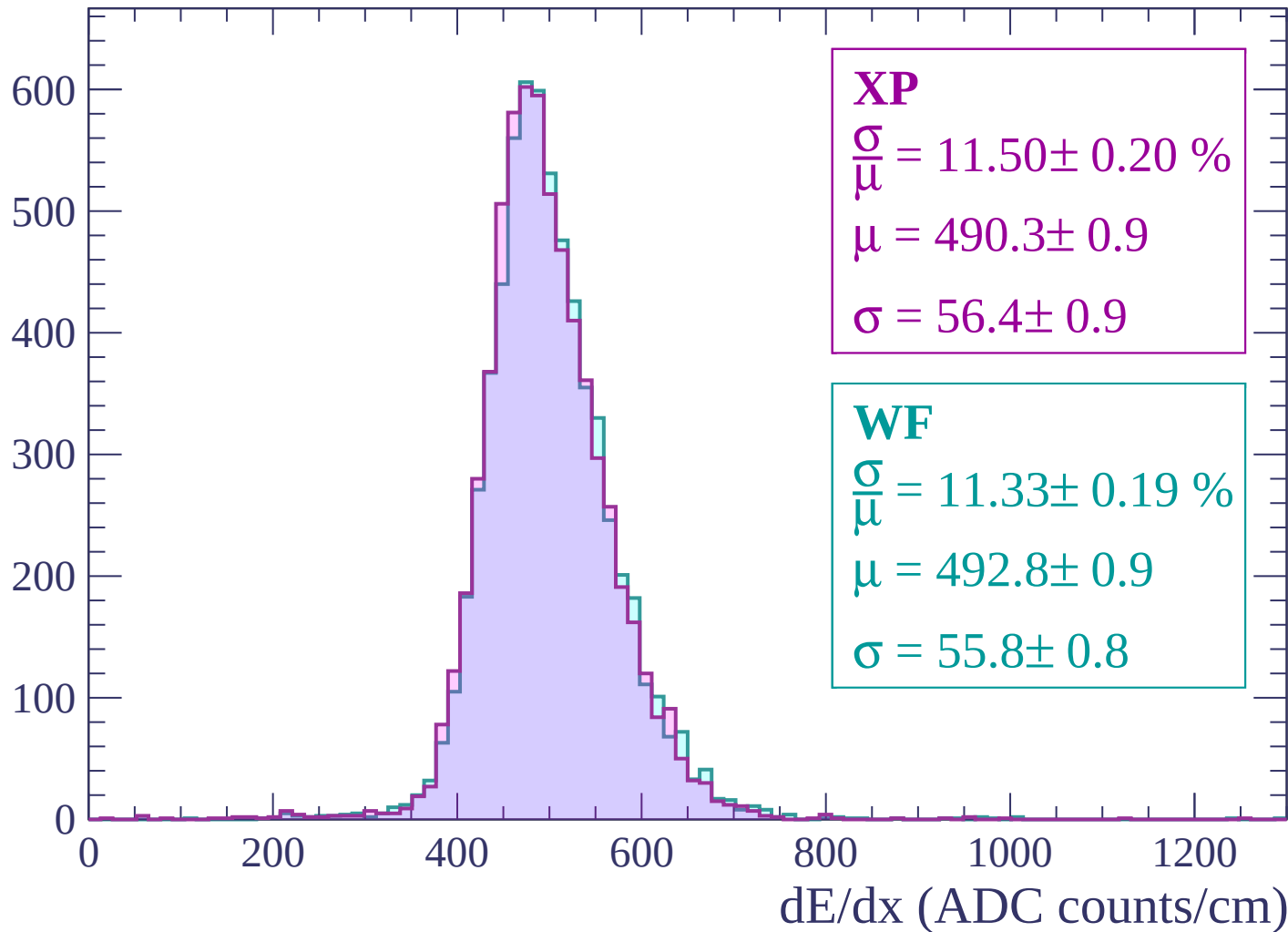
Energy loss | $1920 < p < 1980$

Count



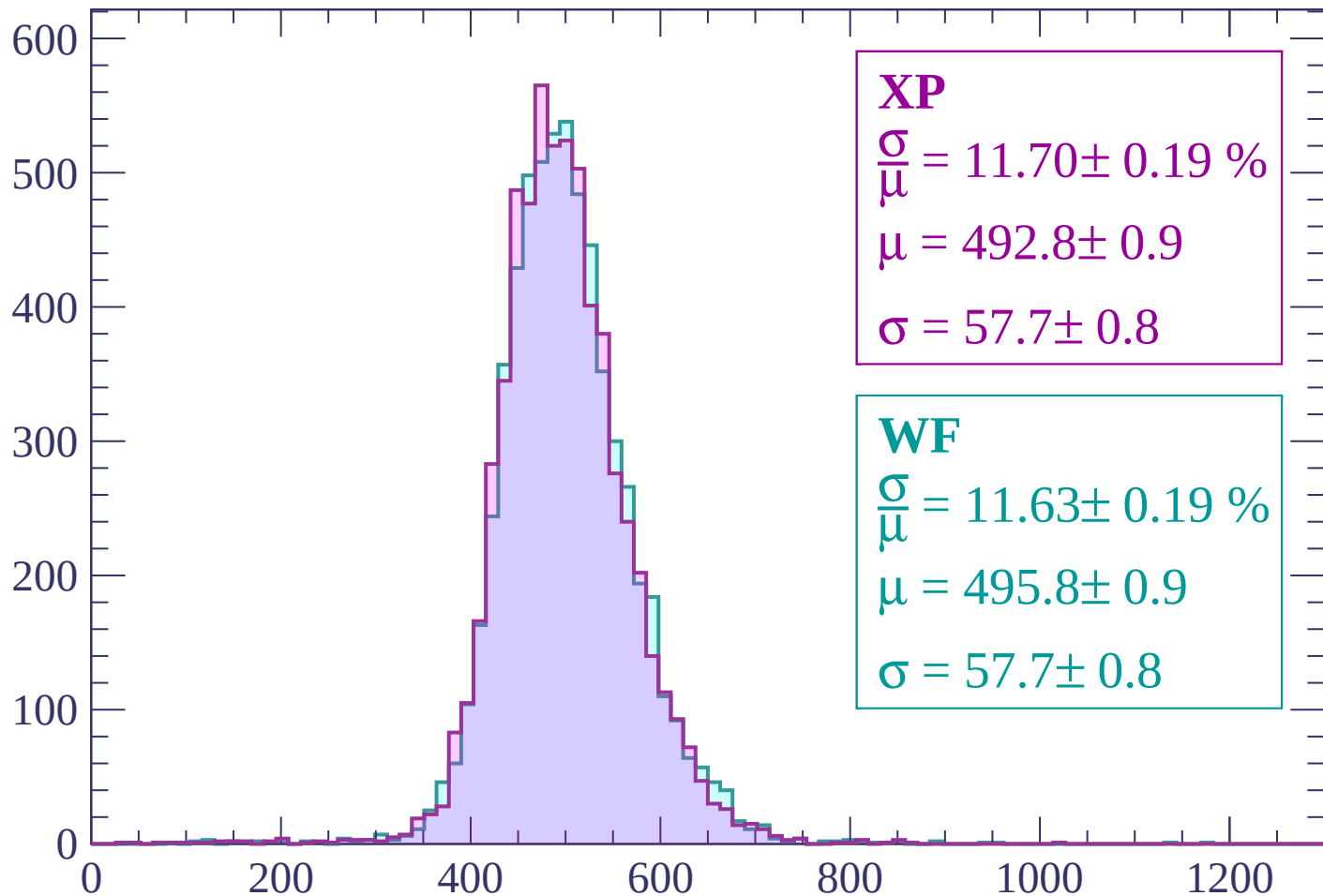
Energy loss | $1980 < p < 2040$

Count



Energy loss | $2040 < p < 2100$

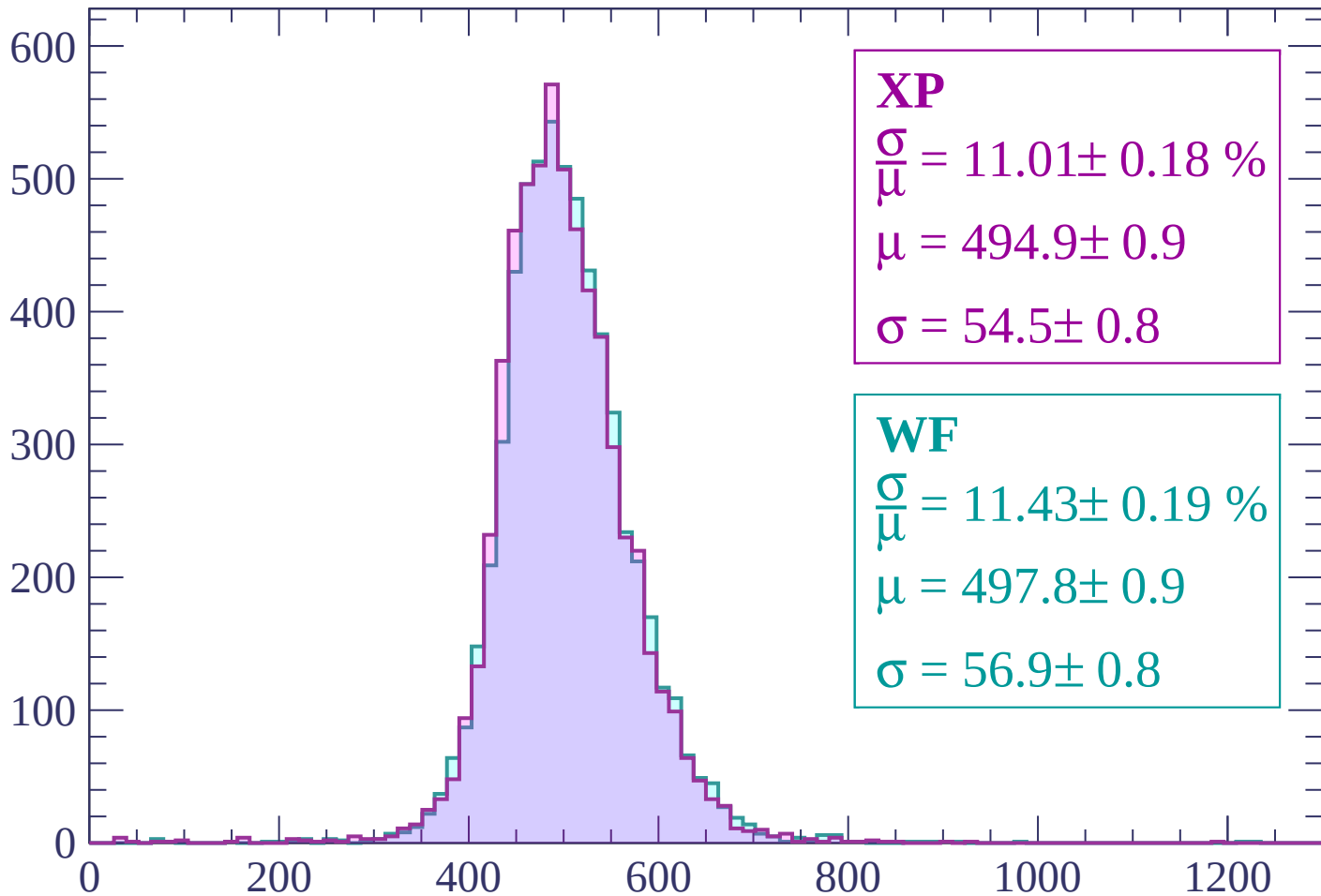
Count



dE/dx (ADC counts/cm)

Energy loss | $2100 < p < 2160$

Count



XP

$$\frac{\sigma}{\mu} = 11.01 \pm 0.18 \%$$

$$\mu = 494.9 \pm 0.9$$

$$\sigma = 54.5 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 11.43 \pm 0.19 \%$$

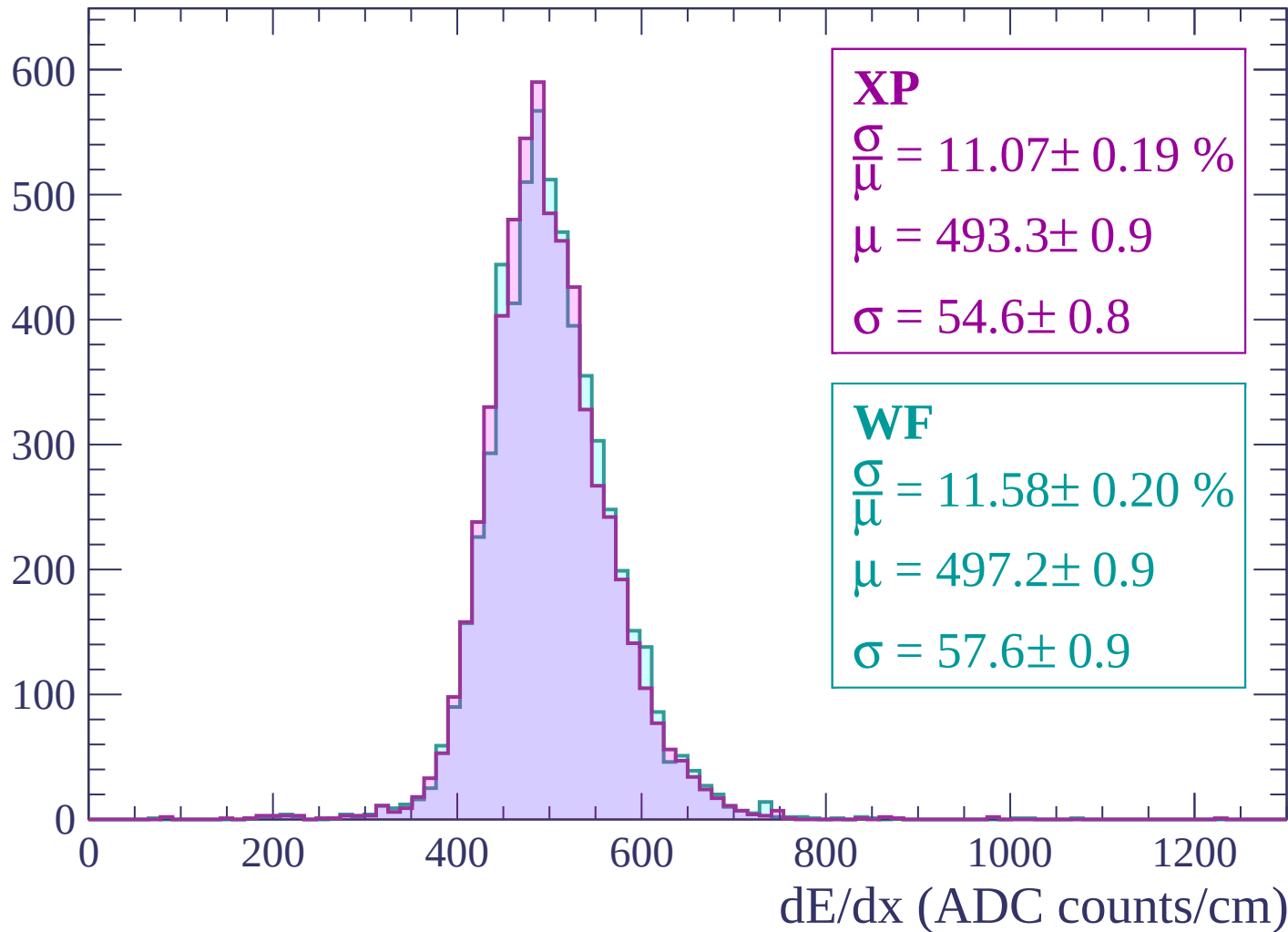
$$\mu = 497.8 \pm 0.9$$

$$\sigma = 56.9 \pm 0.8$$

dE/dx (ADC counts/cm)

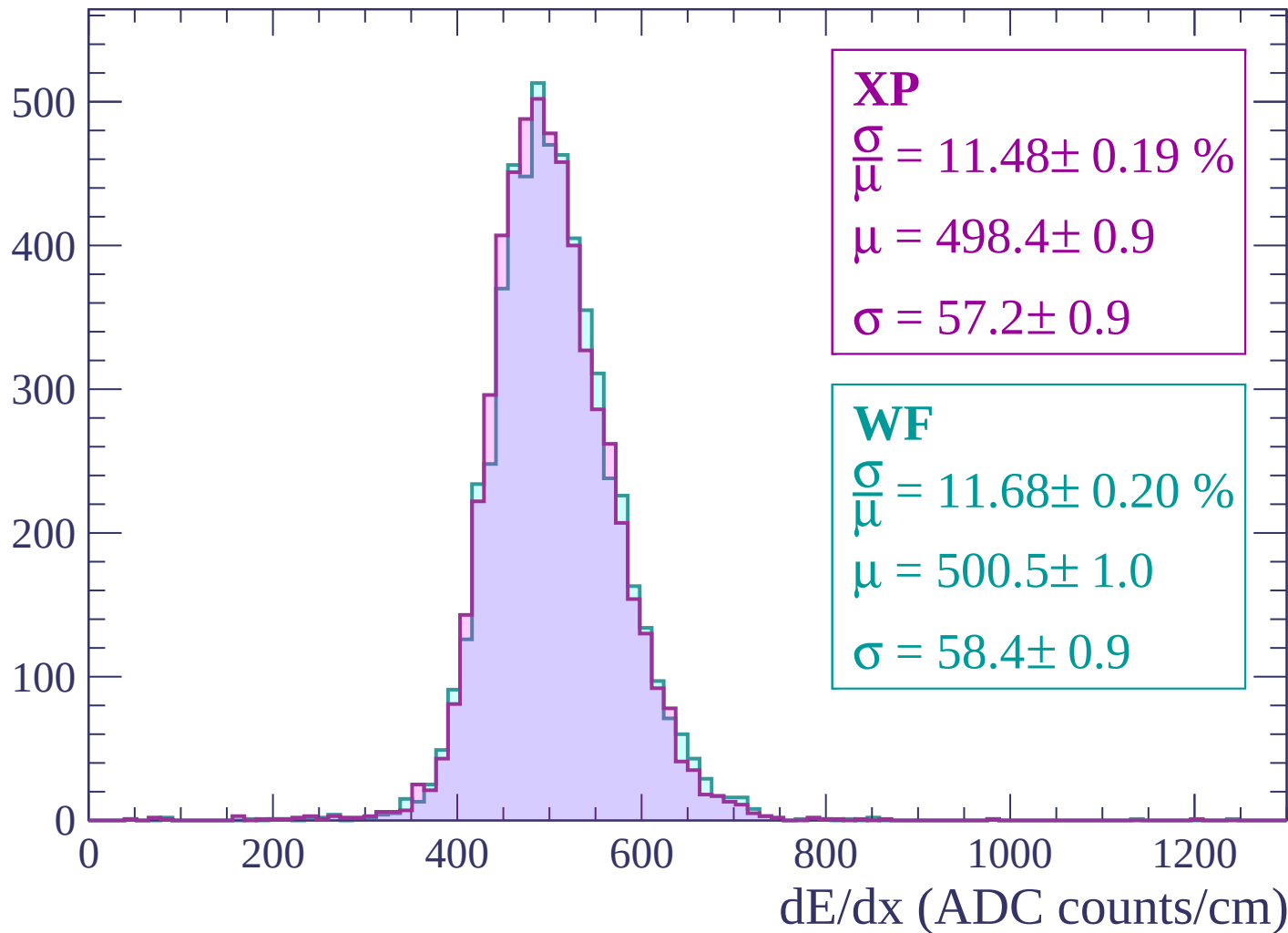
Energy loss | $2160 < p < 2220$

Count



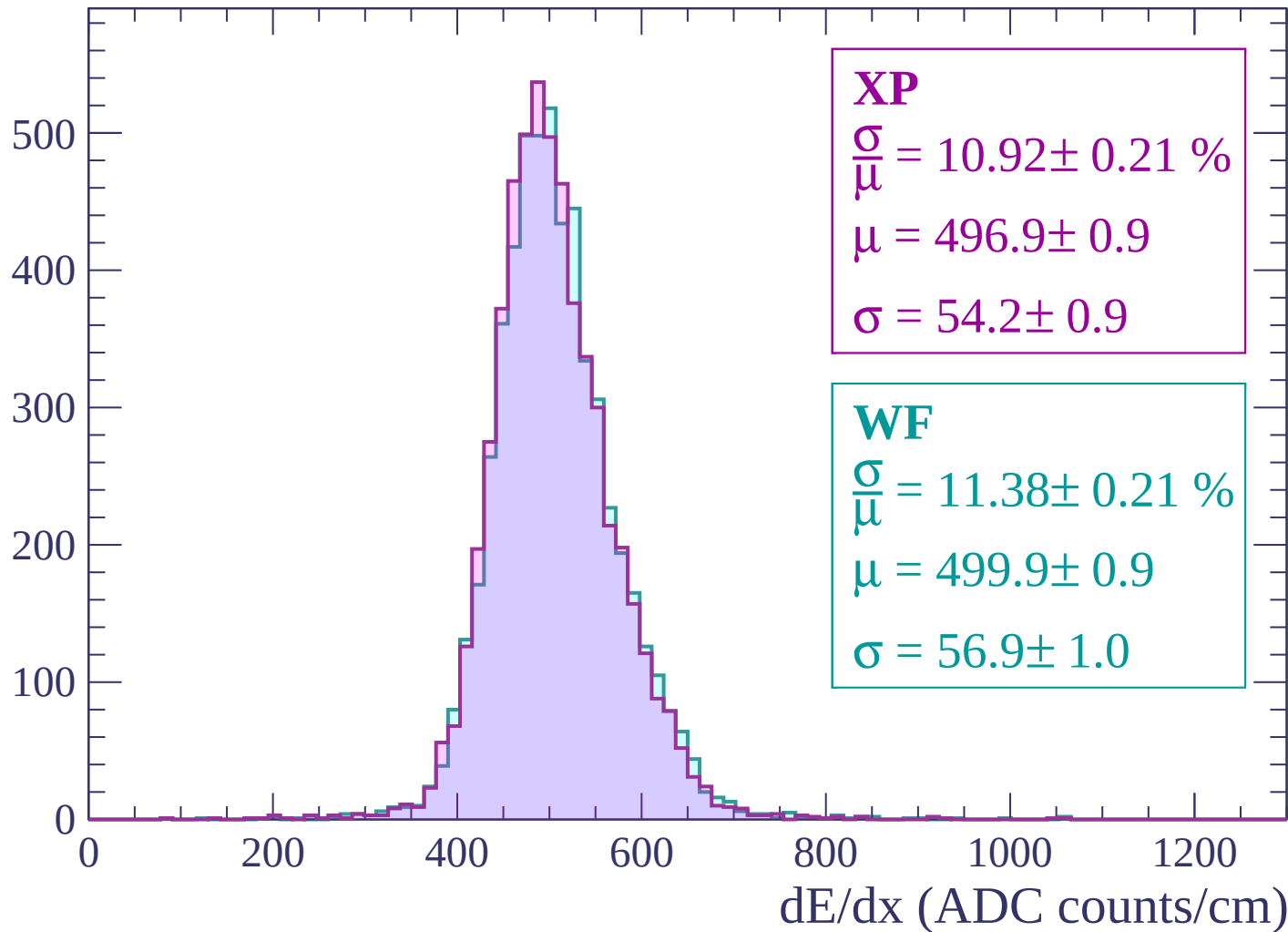
Energy loss | $2220 < p < 2280$

Count



Energy loss | $2280 < p < 2340$

Count



Energy loss | $2340 < p < 2400$

Count

500

400

300

200

100

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 11.14 \pm 0.21 \%$$

$$\mu = 500.7 \pm 0.9$$

$$\sigma = 55.8 \pm 0.9$$

WF

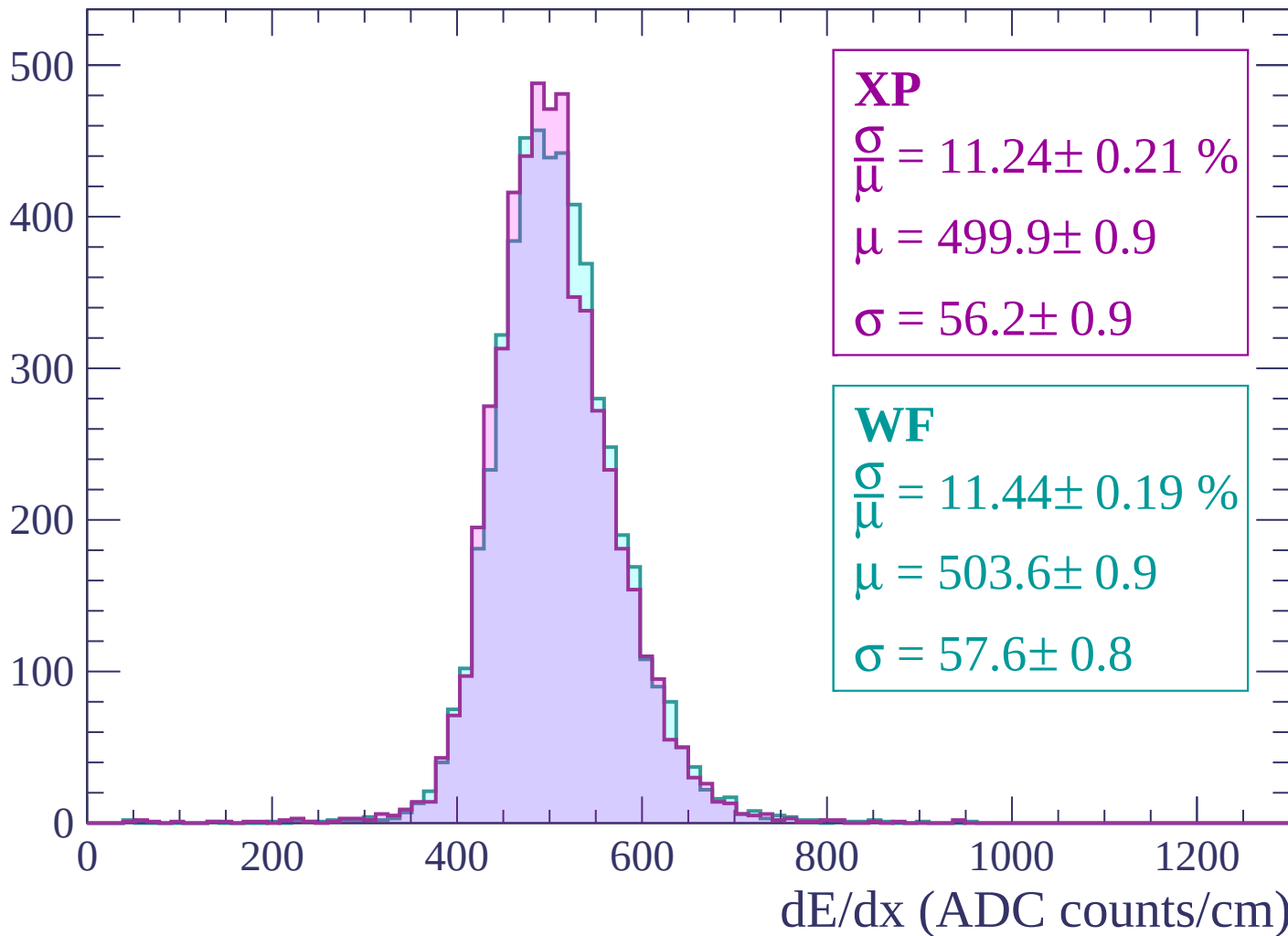
$$\frac{\sigma}{\mu} = 11.22 \pm 0.21 \%$$

$$\mu = 503.1 \pm 1.0$$

$$\sigma = 56.4 \pm 0.9$$

Energy loss | $2400 < p < 2460$

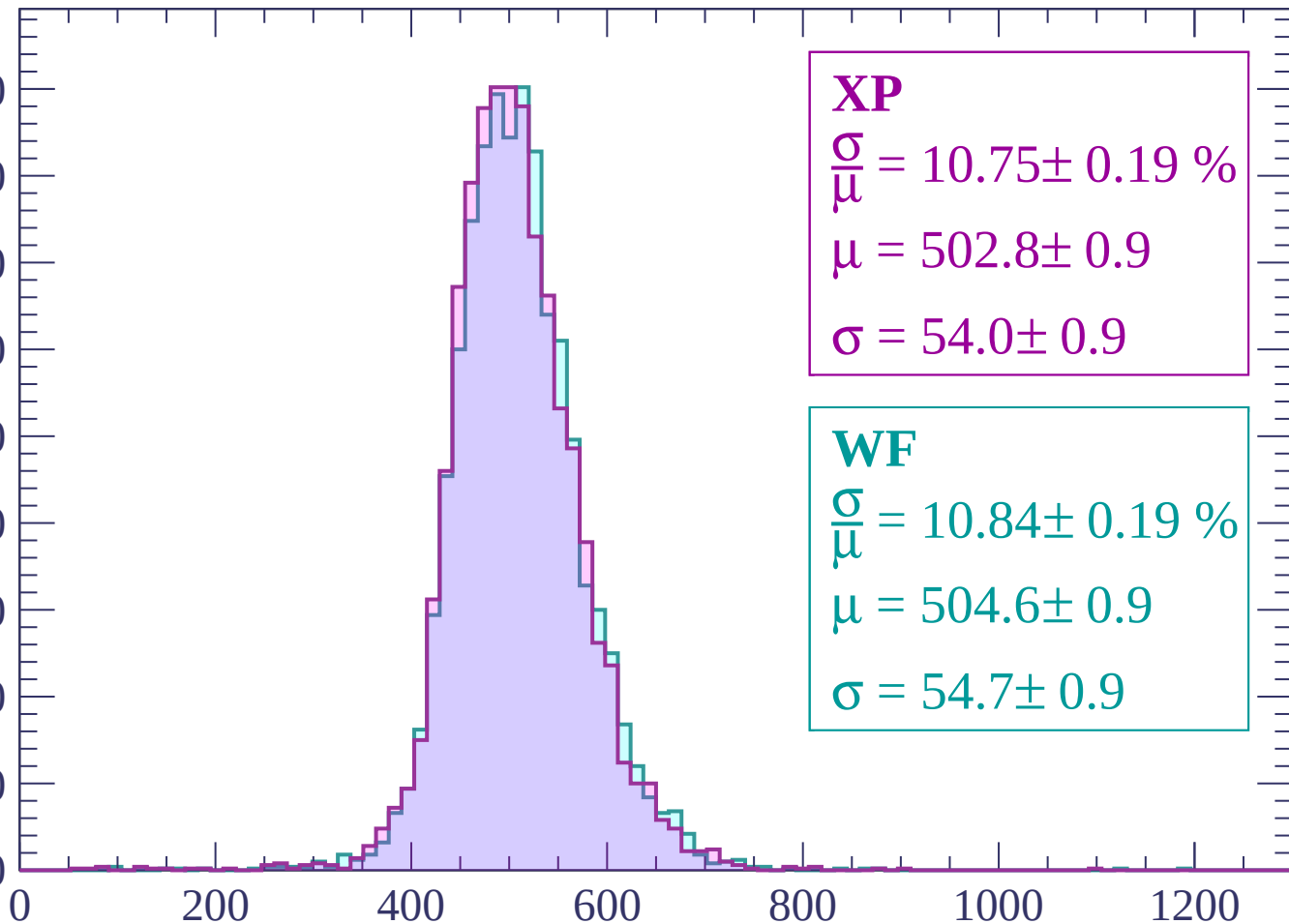
Count



Energy loss | $2460 < p < 2520$

Count

450
400
350
300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 10.75 \pm 0.19 \%$$

$$\mu = 502.8 \pm 0.9$$

$$\sigma = 54.0 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 10.84 \pm 0.19 \%$$

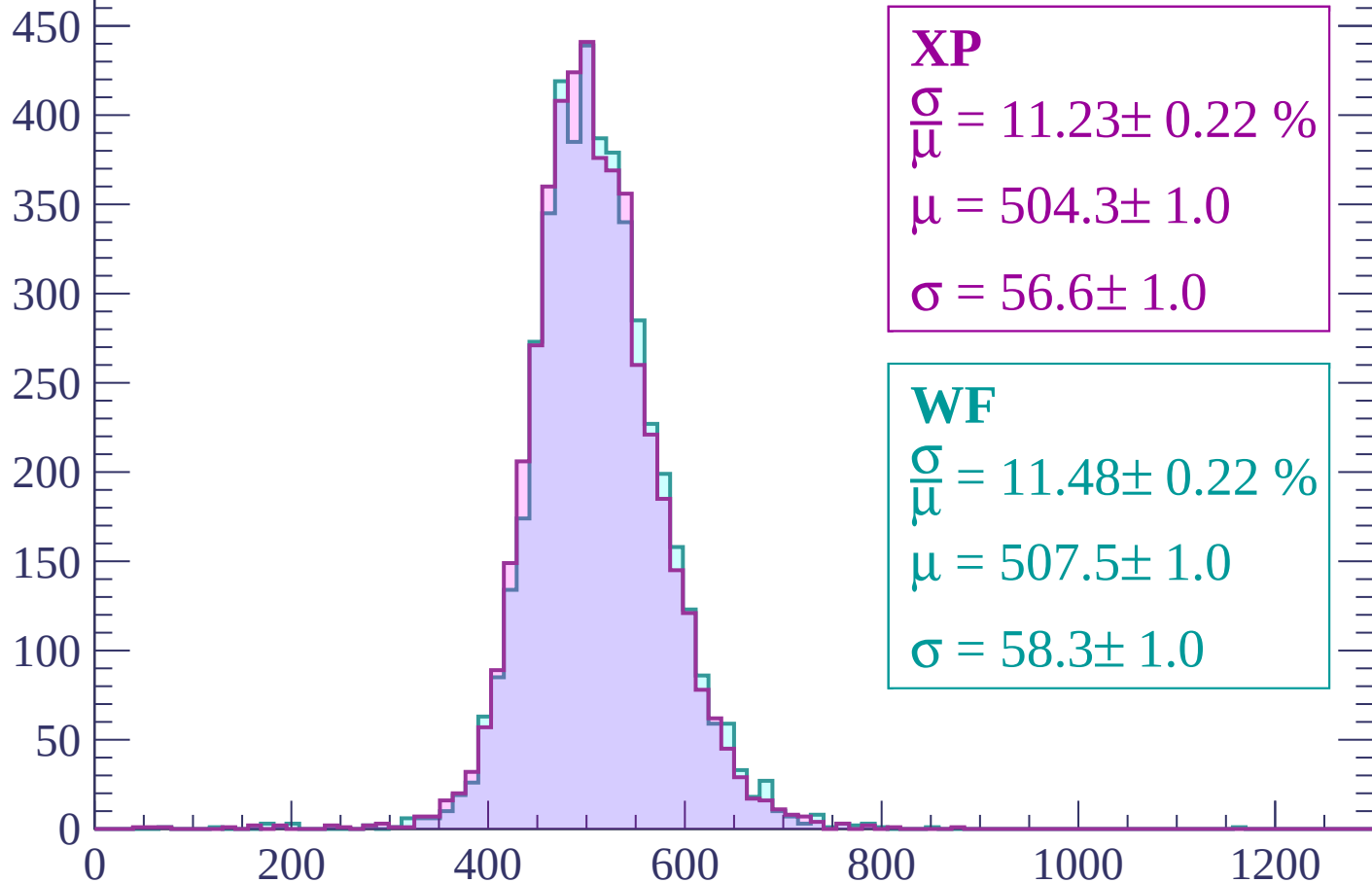
$$\mu = 504.6 \pm 0.9$$

$$\sigma = 54.7 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $2520 < p < 2580$

Count



XP

$$\frac{\sigma}{\mu} = 11.23 \pm 0.22 \%$$

$$\mu = 504.3 \pm 1.0$$

$$\sigma = 56.6 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 11.48 \pm 0.22 \%$$

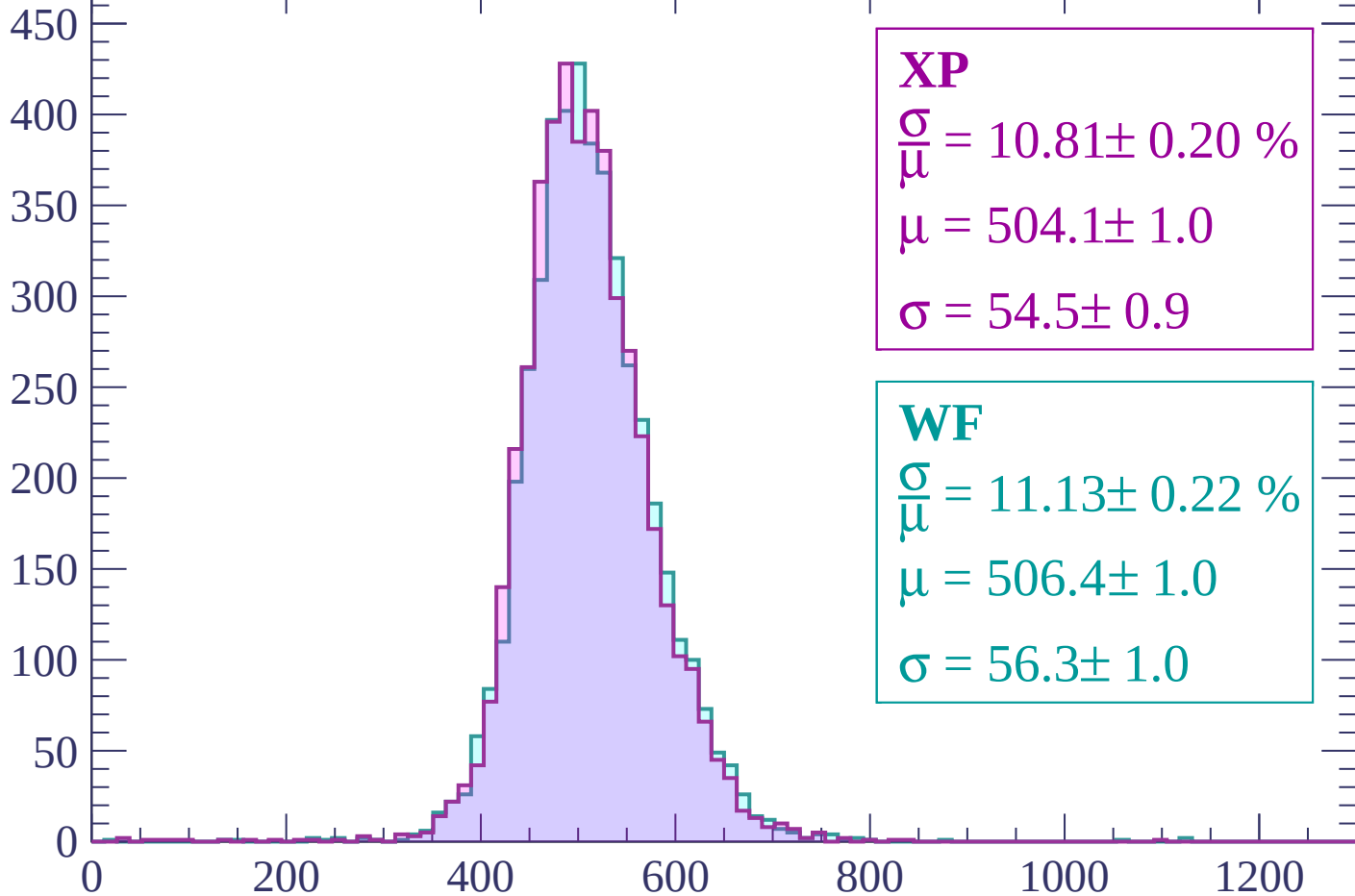
$$\mu = 507.5 \pm 1.0$$

$$\sigma = 58.3 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $2580 < p < 2640$

Count



XP

$$\frac{\sigma}{\mu} = 10.81 \pm 0.20 \%$$

$$\mu = 504.1 \pm 1.0$$

$$\sigma = 54.5 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 11.13 \pm 0.22 \%$$

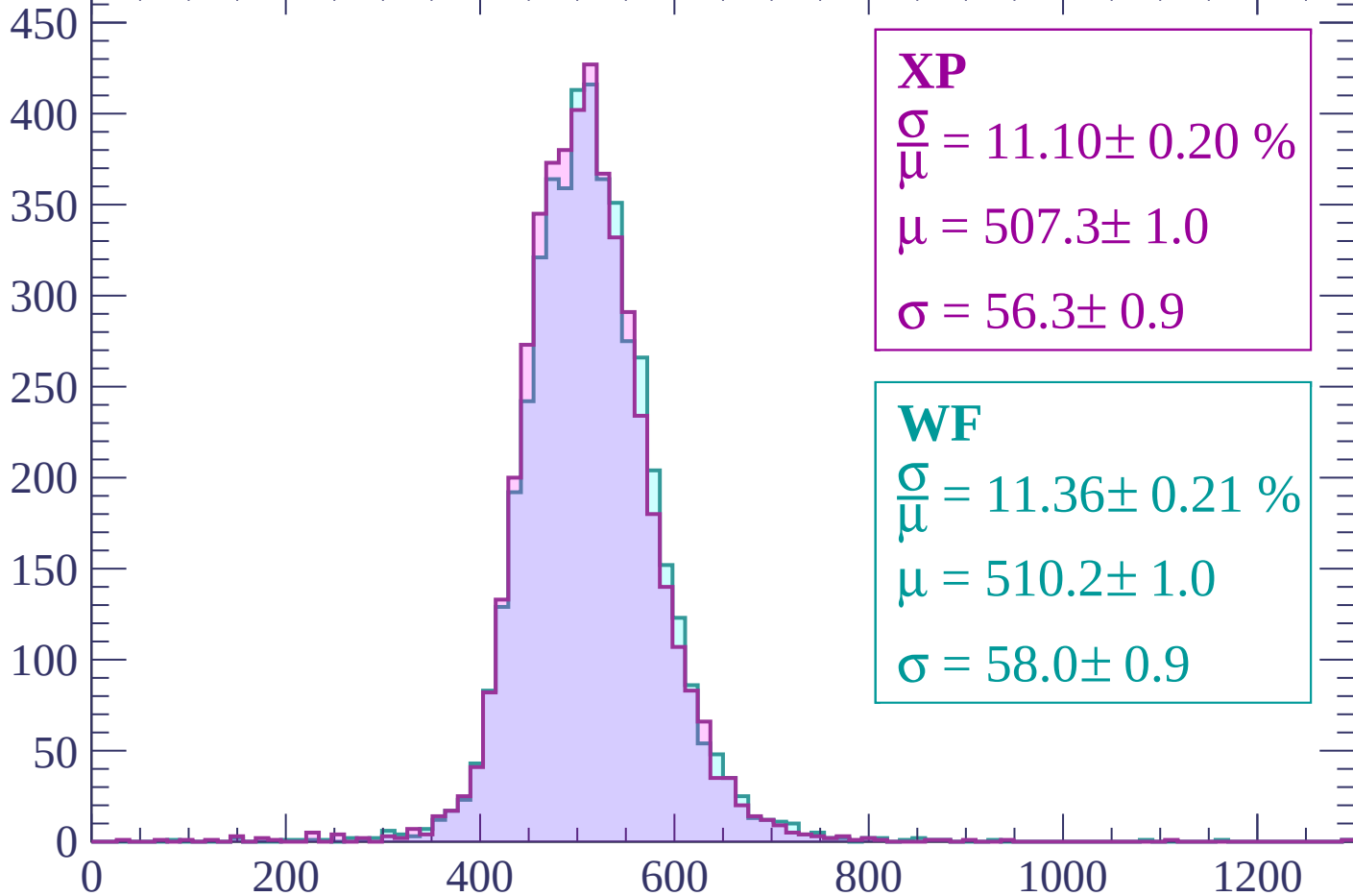
$$\mu = 506.4 \pm 1.0$$

$$\sigma = 56.3 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $2640 < p < 2700$

Count



XP

$$\frac{\sigma}{\mu} = 11.10 \pm 0.20 \%$$

$$\mu = 507.3 \pm 1.0$$

$$\sigma = 56.3 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 11.36 \pm 0.21 \%$$

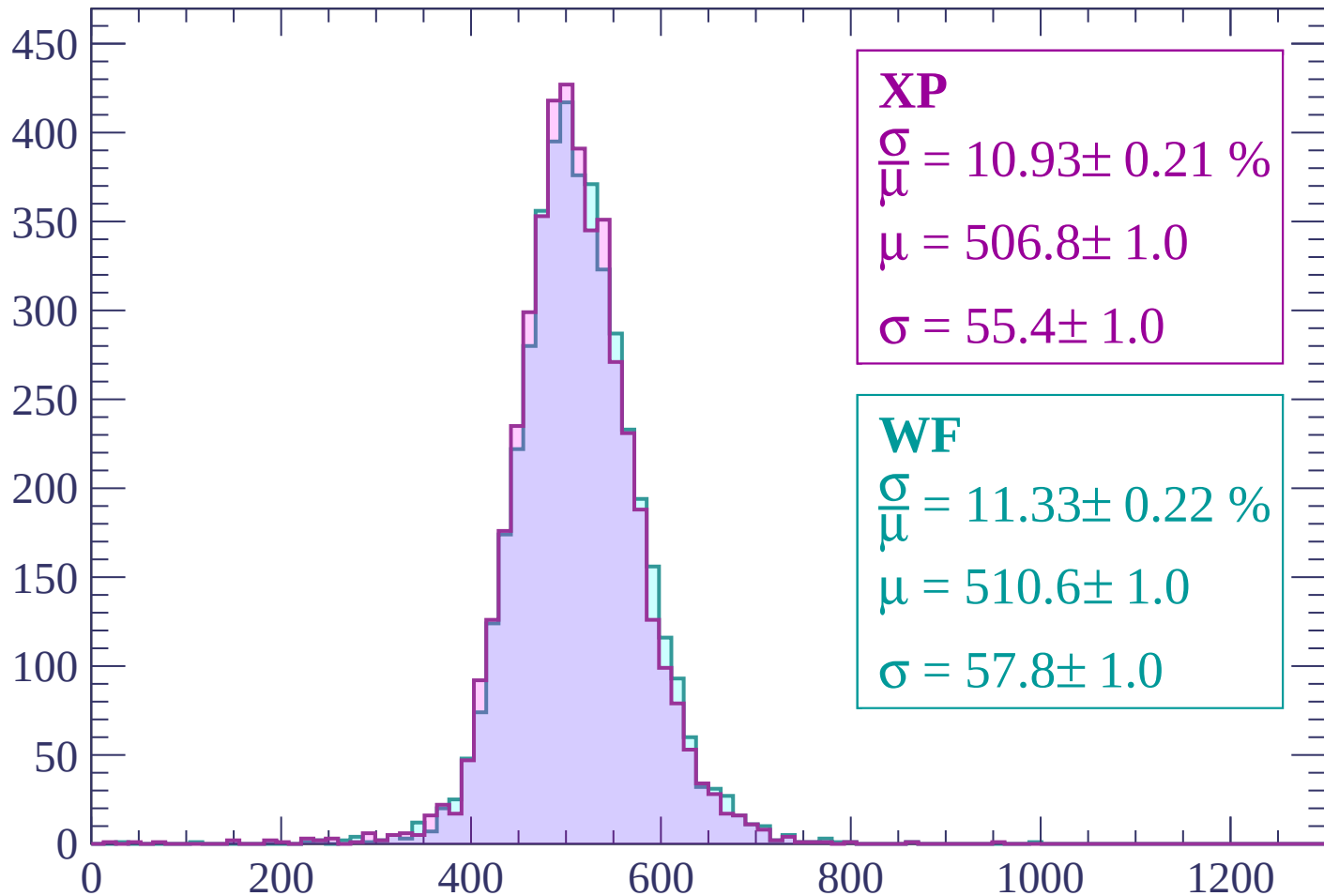
$$\mu = 510.2 \pm 1.0$$

$$\sigma = 58.0 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $2700 < p < 2760$

Count



XP

$$\frac{\sigma}{\mu} = 10.93 \pm 0.21 \%$$

$$\mu = 506.8 \pm 1.0$$

$$\sigma = 55.4 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 11.33 \pm 0.22 \%$$

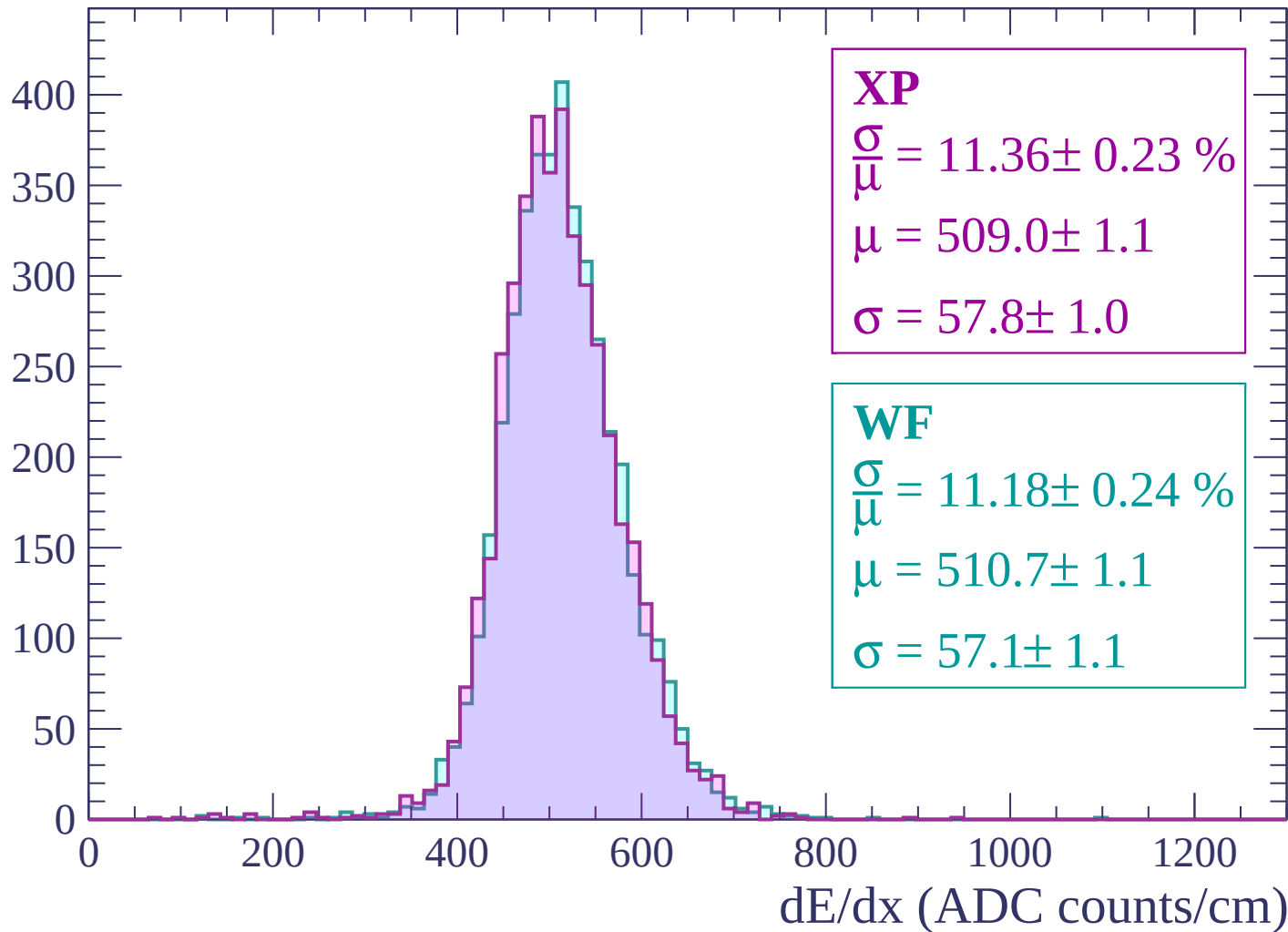
$$\mu = 510.6 \pm 1.0$$

$$\sigma = 57.8 \pm 1.0$$

dE/dx (ADC counts/cm)

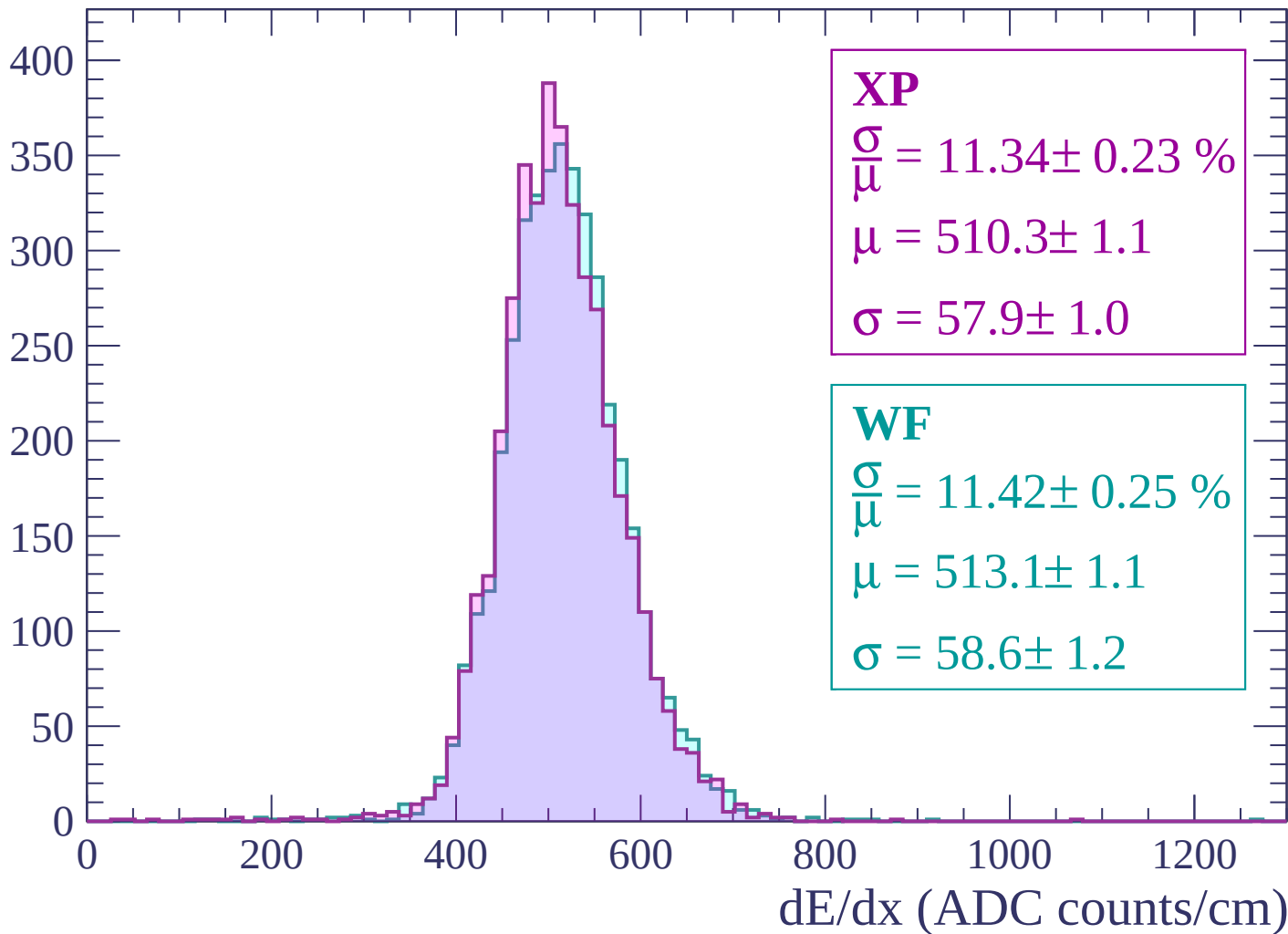
Energy loss | $2760 < p < 2820$

Count



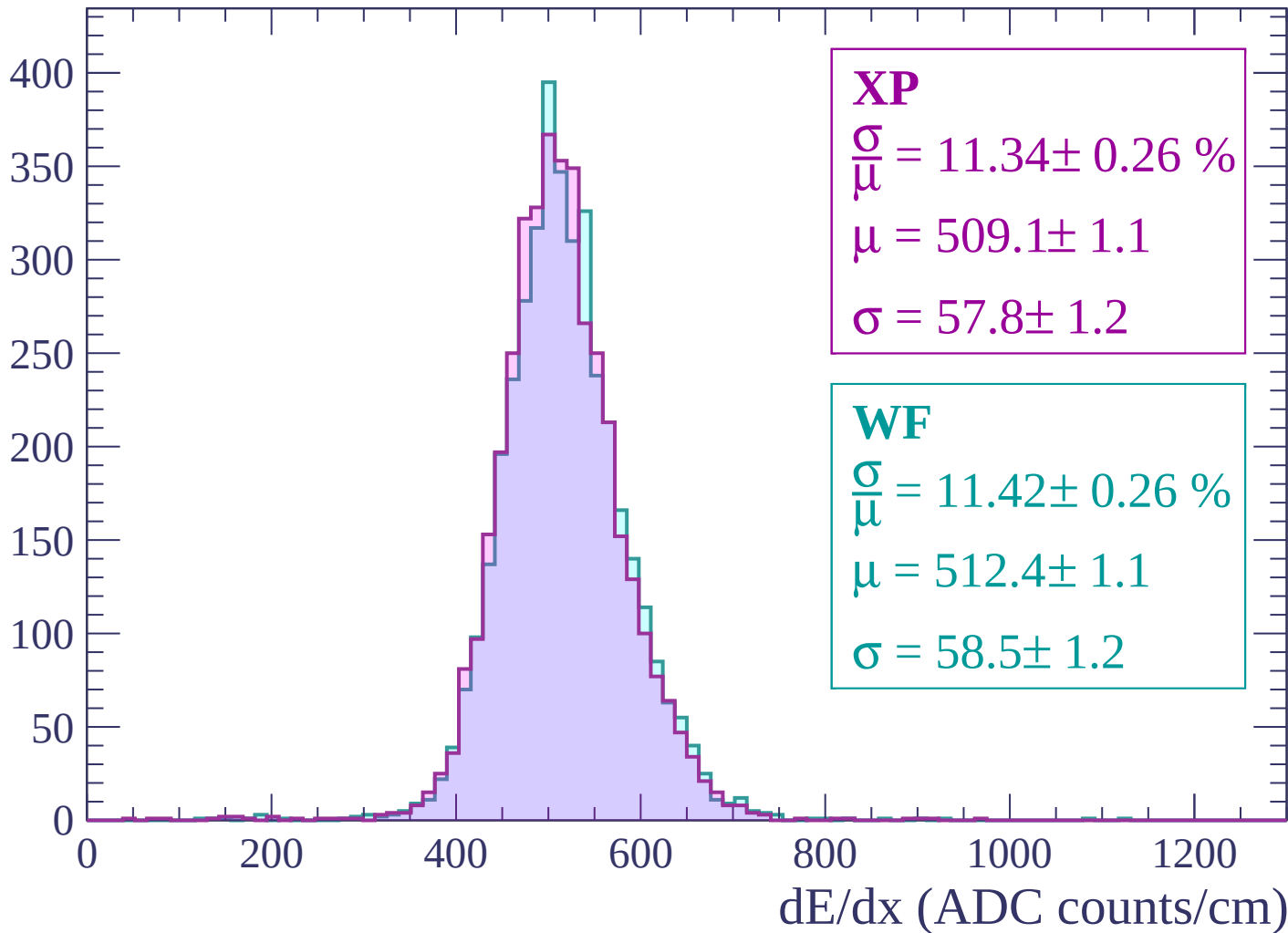
Energy loss | $2820 < p < 2880$

Count



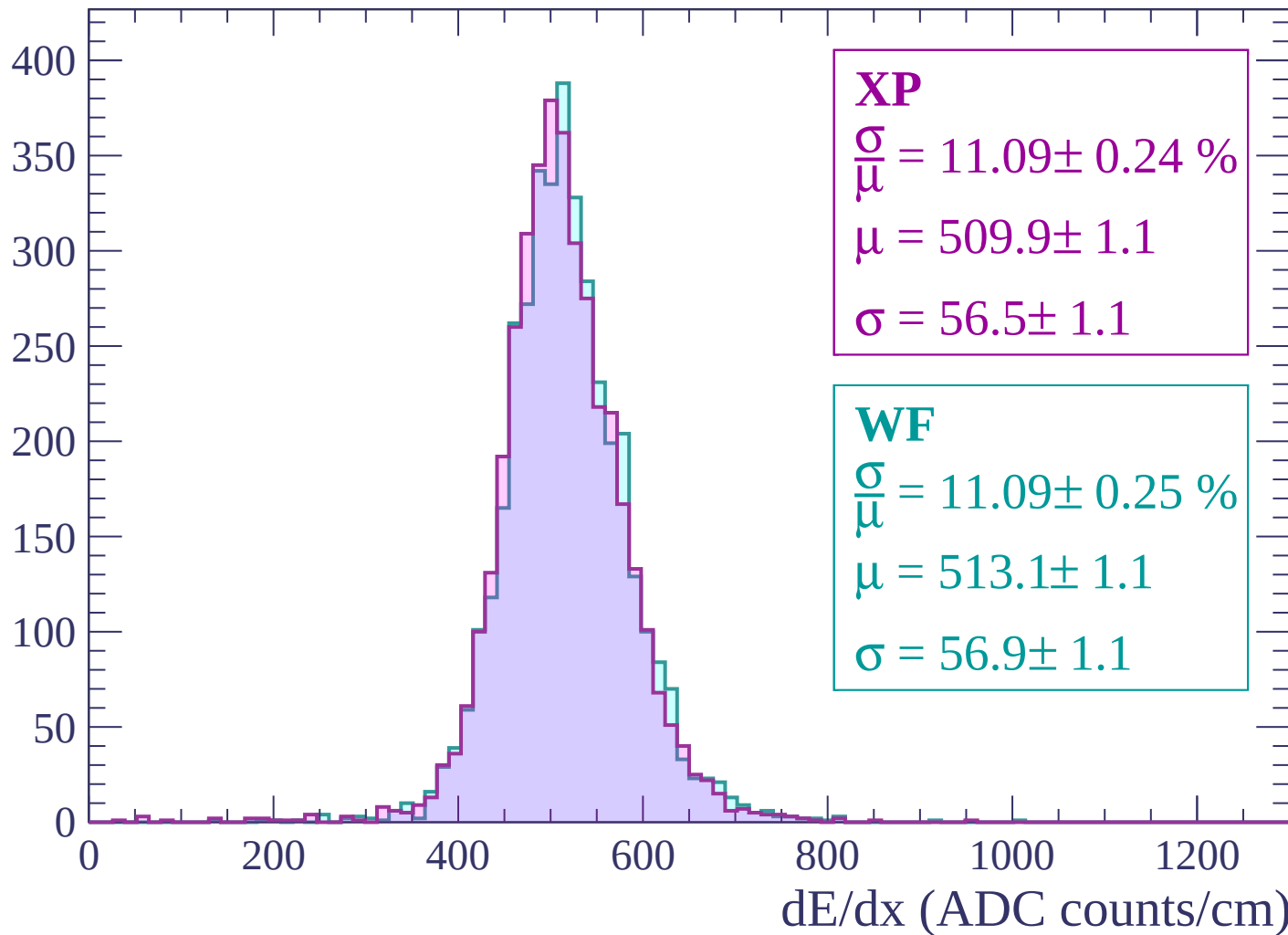
Energy loss | $2880 < p < 2940$

Count



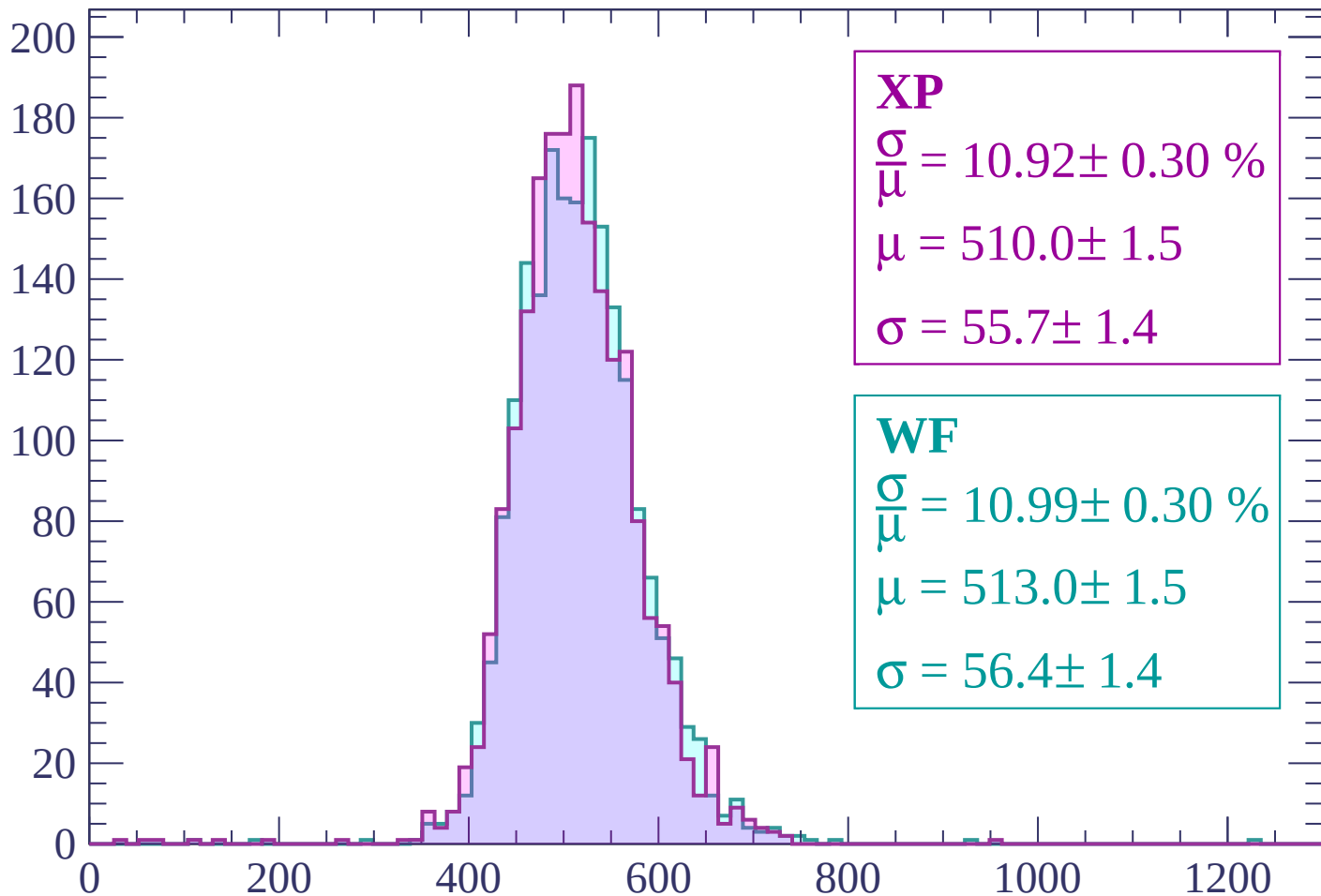
Energy loss | $2940 < p < 3000$

Count



Energy loss | $3000 < p < 3060$

Count



Energy loss | $-90 < \theta < -88$

Count

1.0

0.8

0.6

0.4

0.2

0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 4.16 \pm 21.99 \%$$

$$\mu = 6.2 \pm 2.4$$

$$\sigma = 0.3 \pm 1.3$$

WF

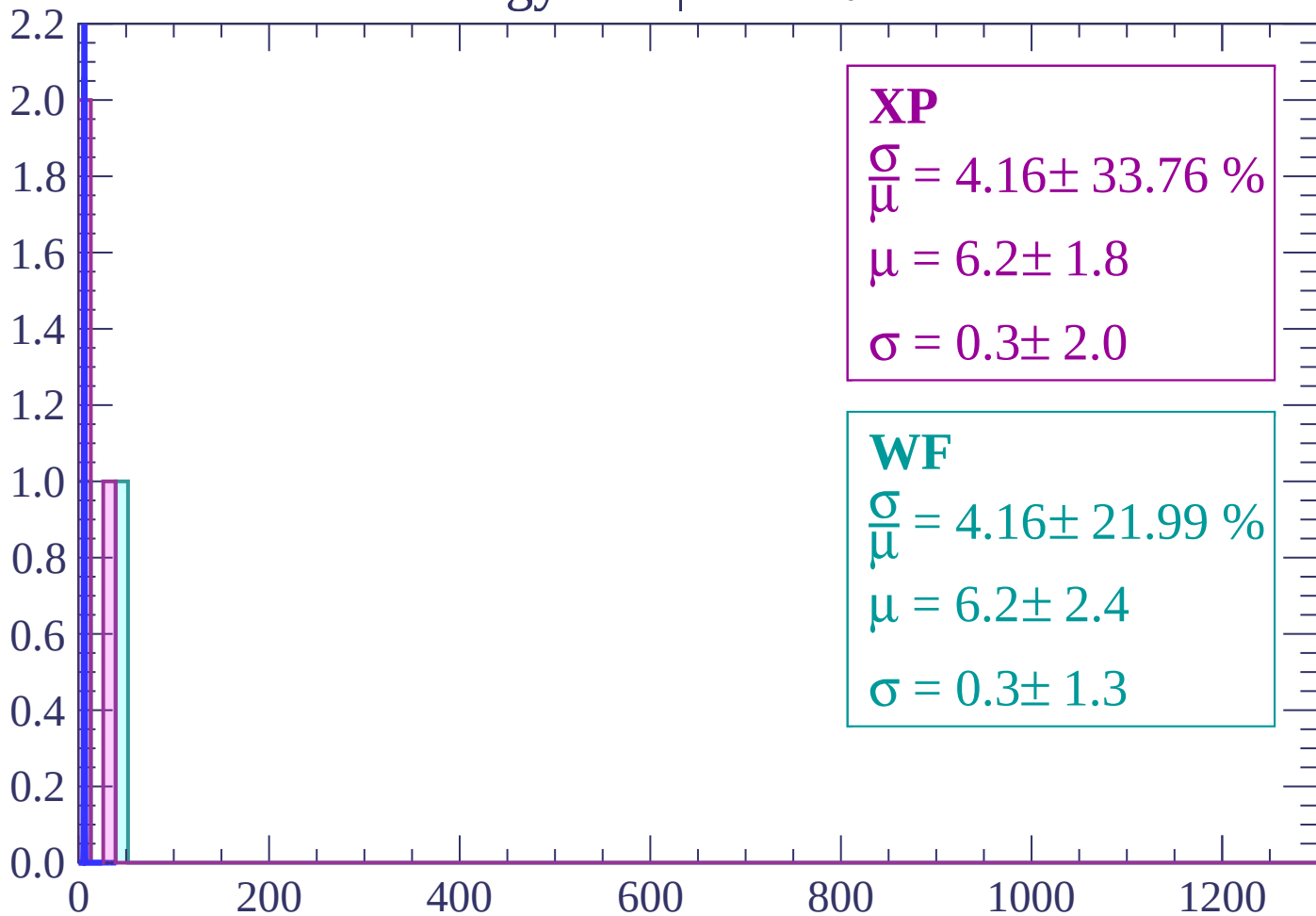
$$\frac{\sigma}{\mu} = 4.16 \pm 21.99 \%$$

$$\mu = 6.2 \pm 2.4$$

$$\sigma = 0.3 \pm 1.3$$

Energy loss | $-88 < \theta < -86$

Count



XP

$$\frac{\sigma}{\mu} = 4.16 \pm 33.76 \%$$

$$\mu = 6.2 \pm 1.8$$

$$\sigma = 0.3 \pm 2.0$$

WF

$$\frac{\sigma}{\mu} = 4.16 \pm 21.99 \%$$

$$\mu = 6.2 \pm 2.4$$

$$\sigma = 0.3 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $-86 < \theta < -84$

Count

1.0

0.8

0.6

0.4

0.2

0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 4.16 \pm 21.99 \%$$

$$\mu = 6.2 \pm 2.4$$

$$\sigma = 0.3 \pm 1.3$$

WF

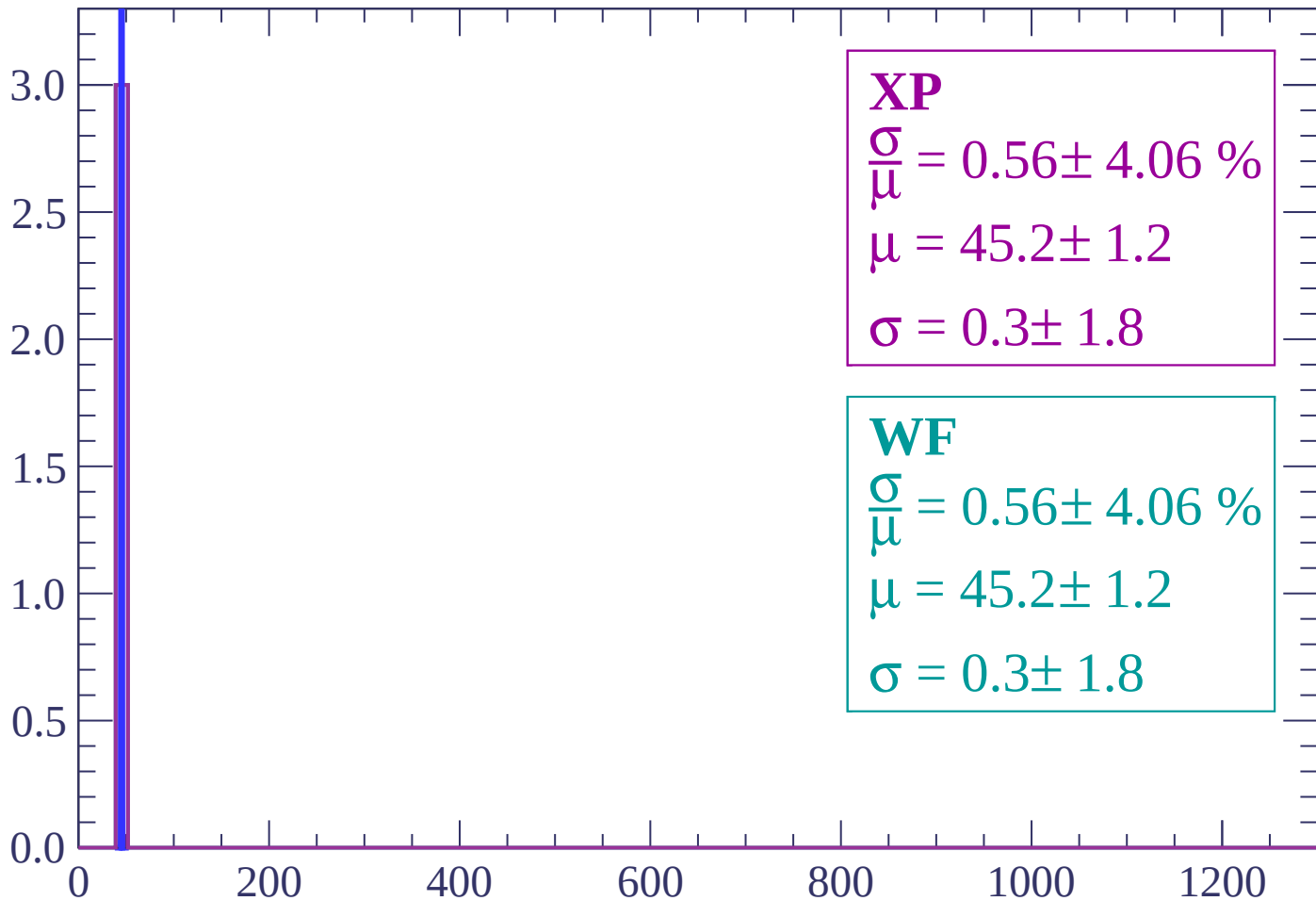
$$\frac{\sigma}{\mu} = 66.67 \pm 706.75 \%$$

$$\mu = 19.5 \pm 47.2$$

$$\sigma = 13.0 \pm 106.3$$

Energy loss | $-84 < \theta < -82$

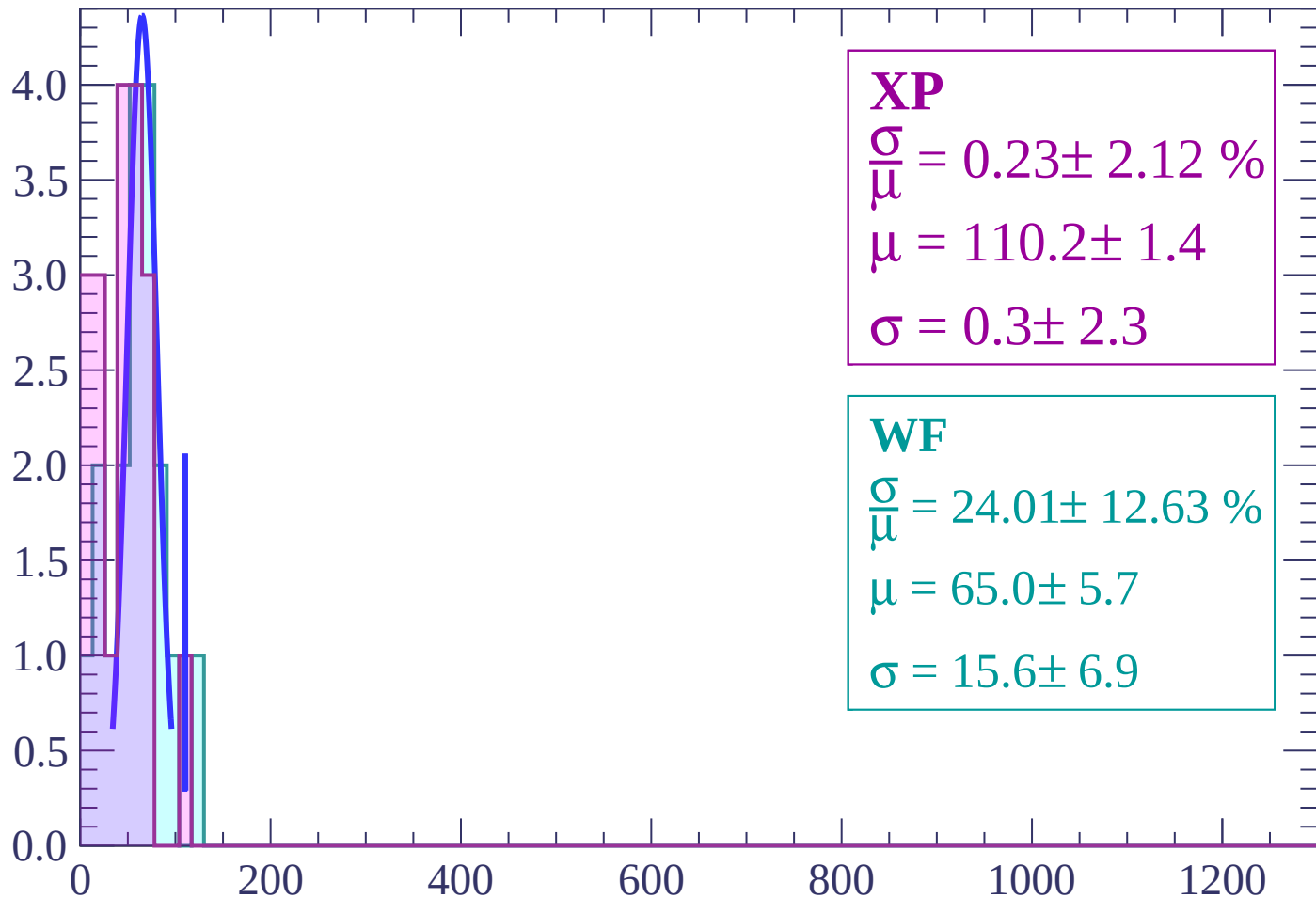
Count



dE/dx (ADC counts/cm)

Energy loss | $-82 < \theta < -80$

Count



XP

$$\frac{\sigma}{\mu} = 0.23 \pm 2.12 \%$$

$$\mu = 110.2 \pm 1.4$$

$$\sigma = 0.3 \pm 2.3$$

WF

$$\frac{\sigma}{\mu} = 24.01 \pm 12.63 \%$$

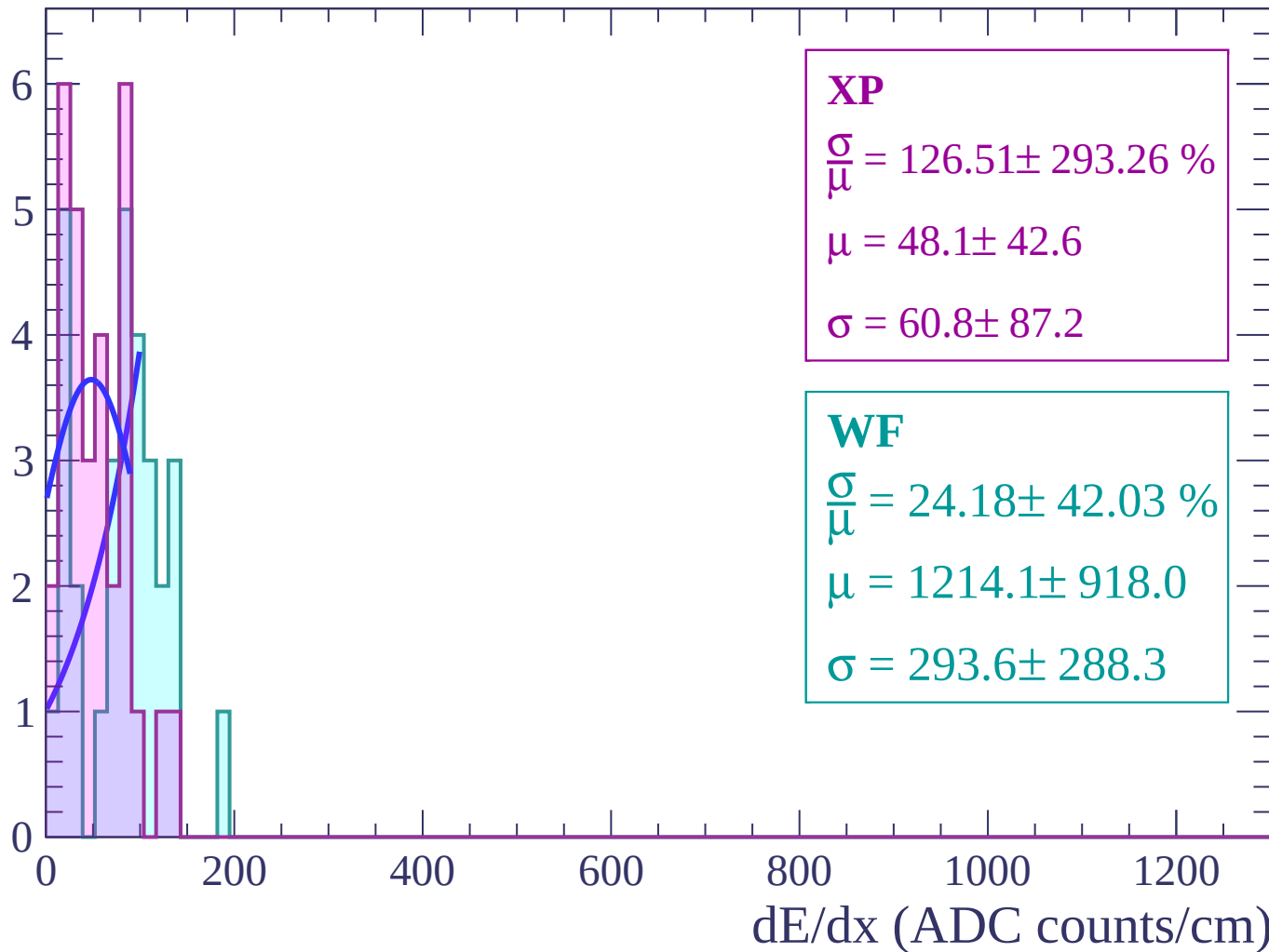
$$\mu = 65.0 \pm 5.7$$

$$\sigma = 15.6 \pm 6.9$$

dE/dx (ADC counts/cm)

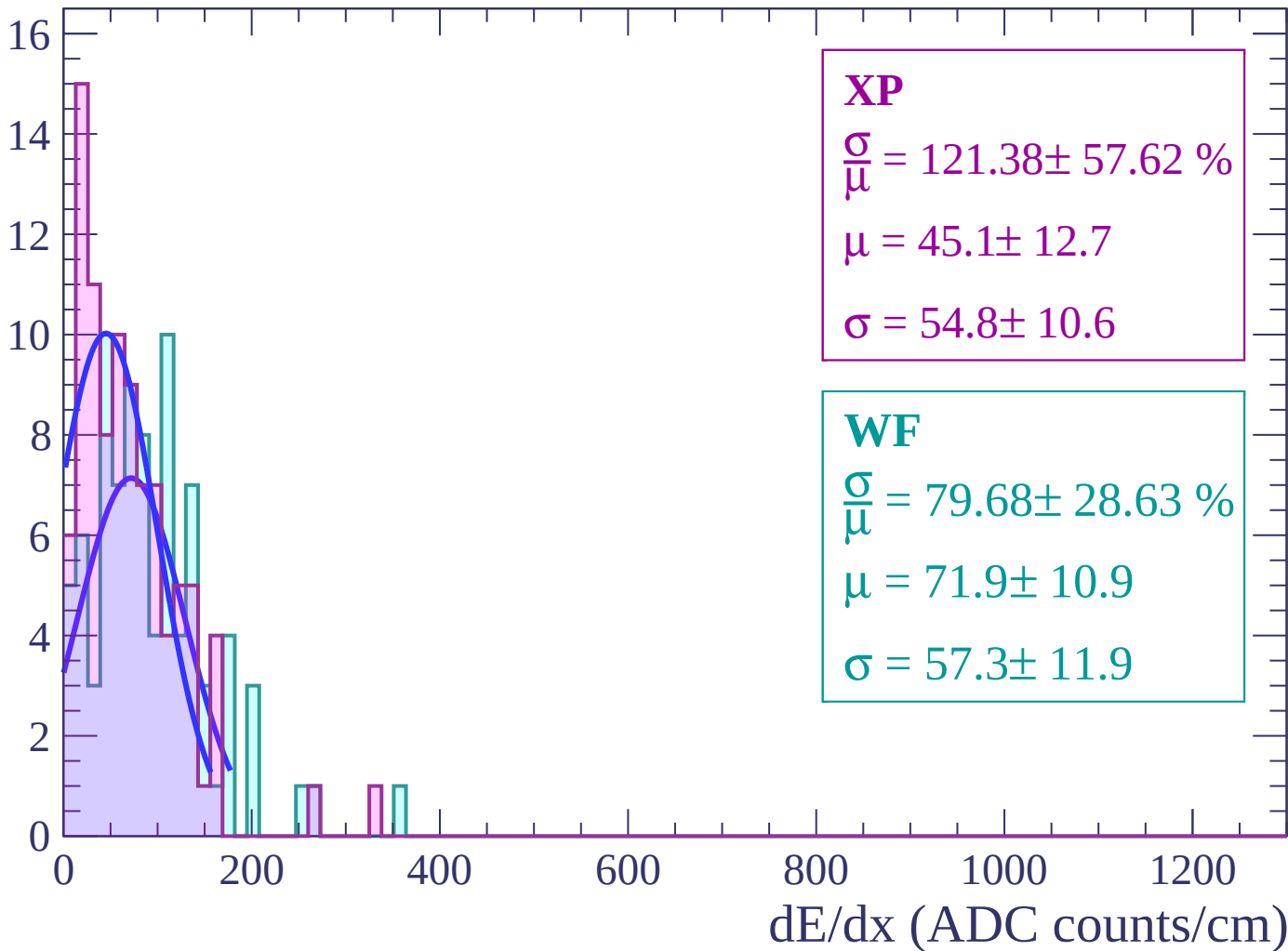
Energy loss | $-80 < \theta < -78$

Count



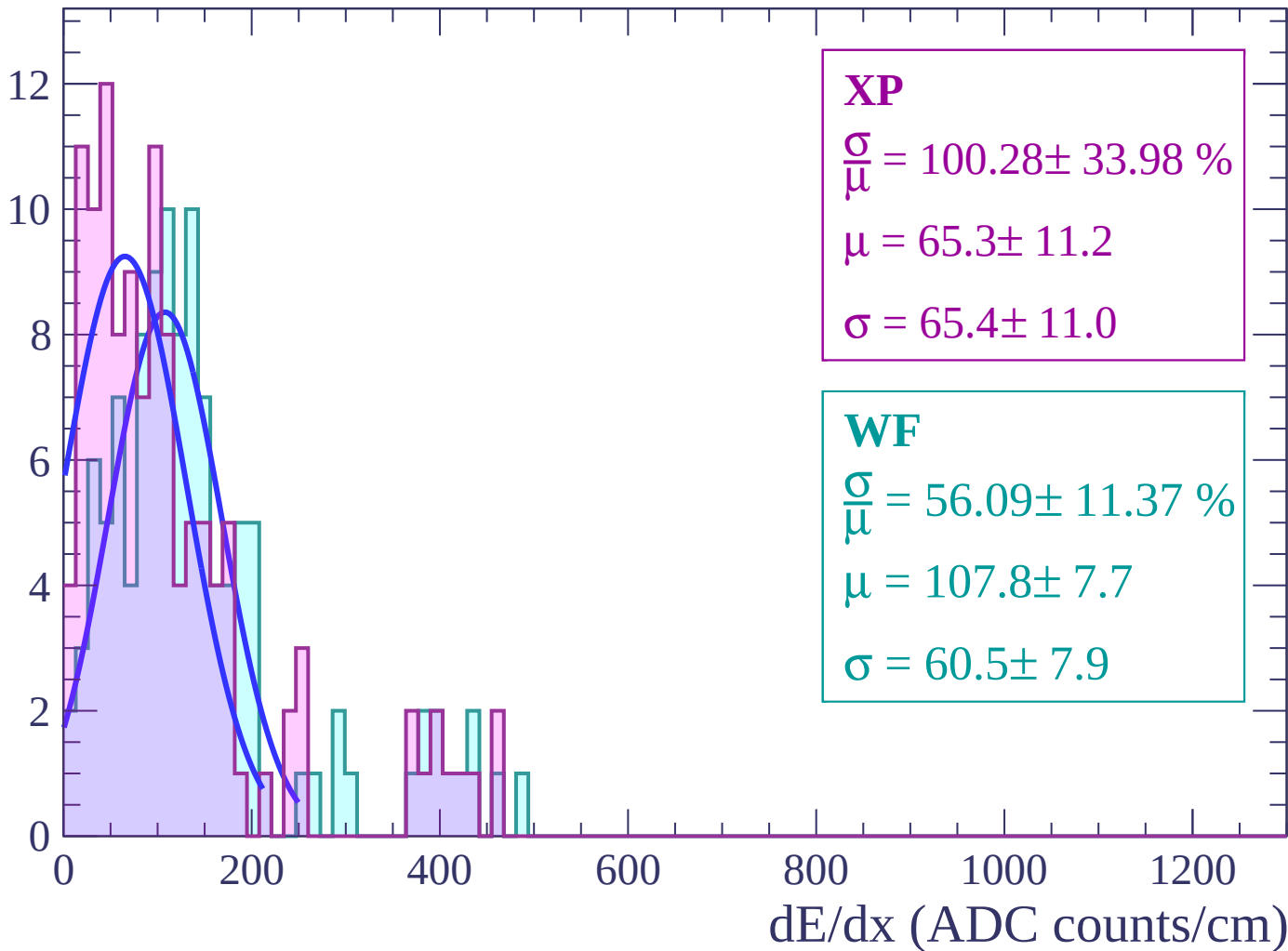
Energy loss | $-78 < \theta < -76$

Count



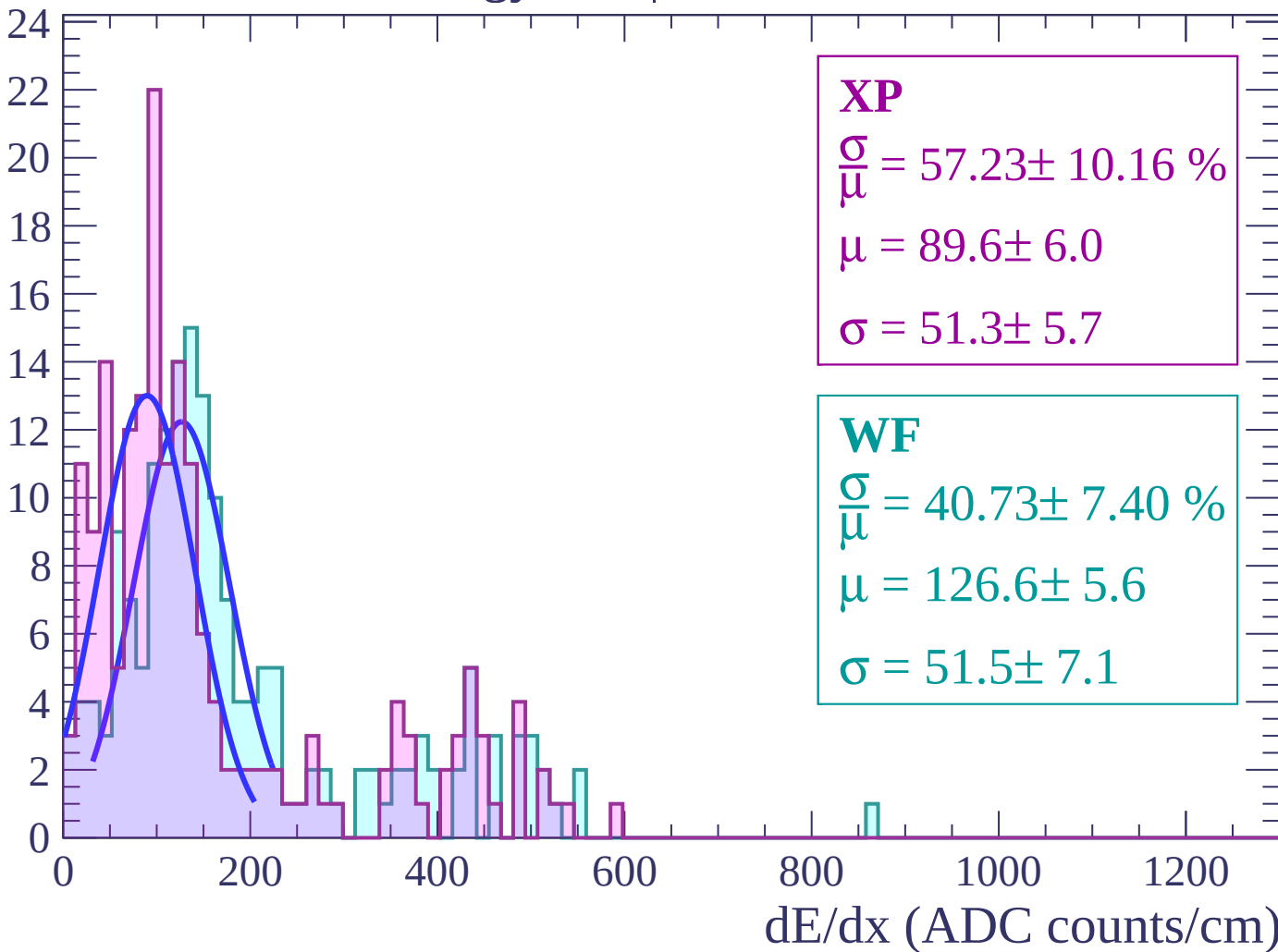
Energy loss | $-76 < \theta < -74$

Count



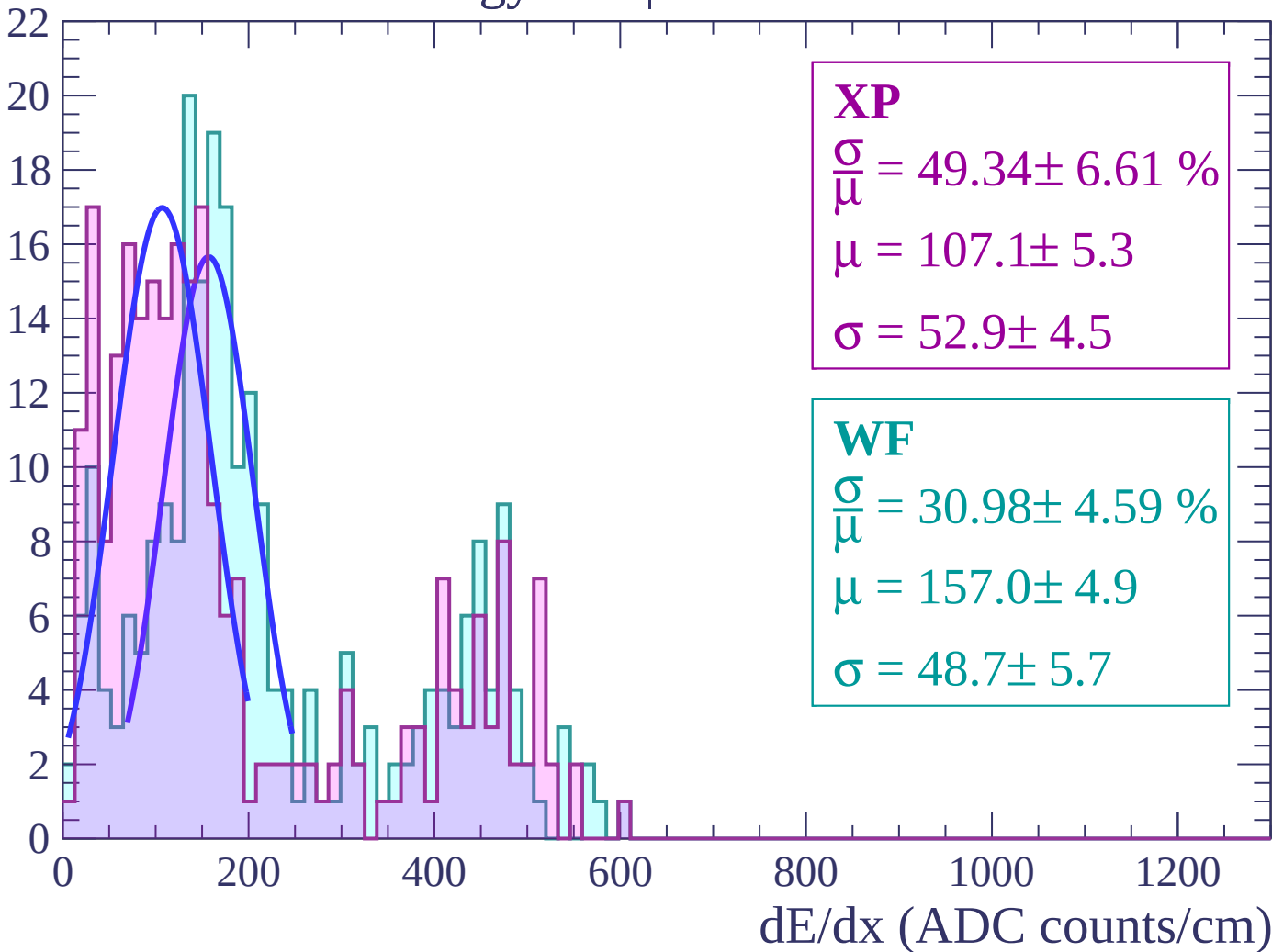
Energy loss | $-74 < \theta < -72$

Count



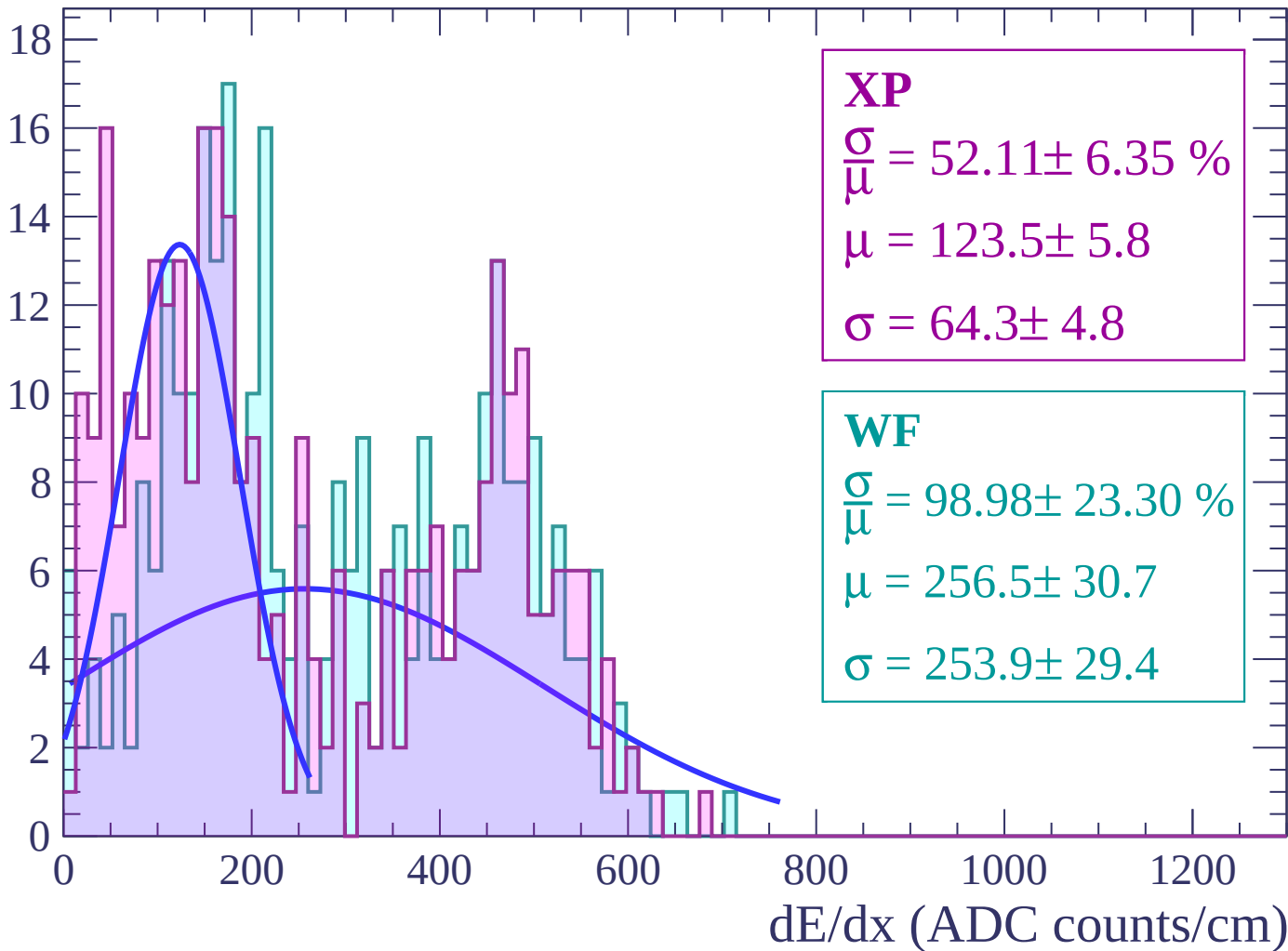
Energy loss | $-72 < \theta < -70$

Count



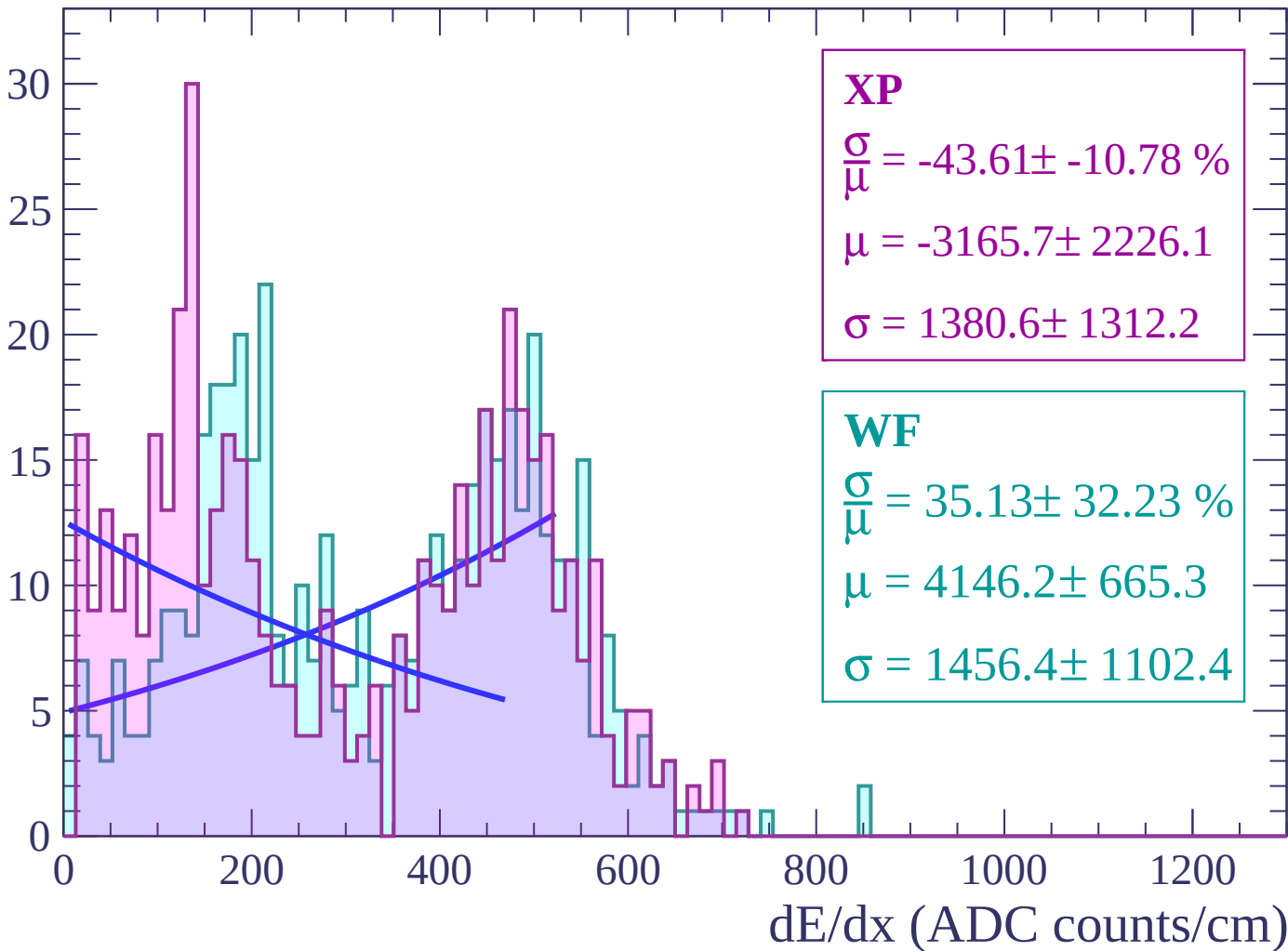
Energy loss | $-70 < \theta < -68$

Count



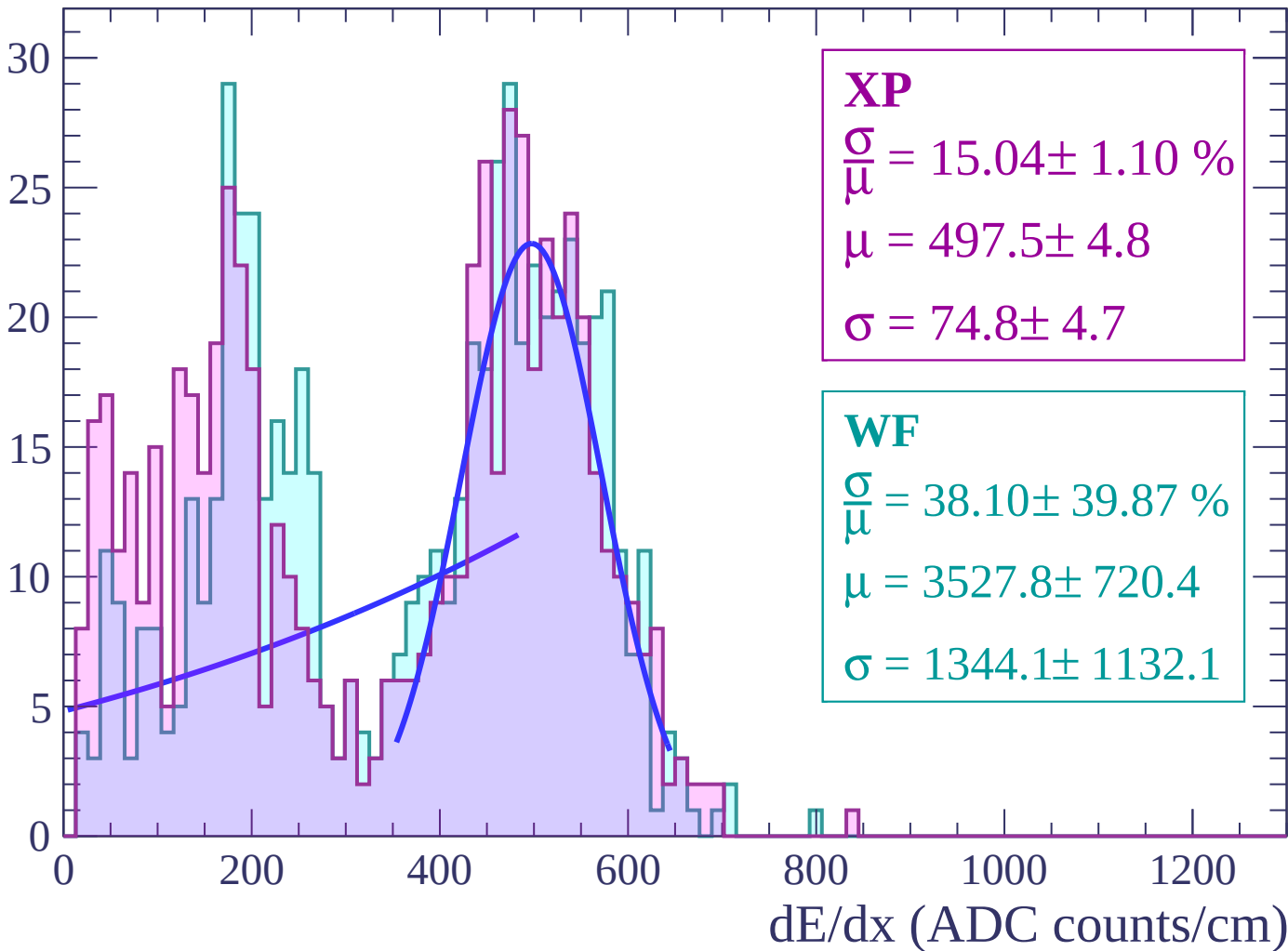
Energy loss | $-68 < \theta < -66$

Count



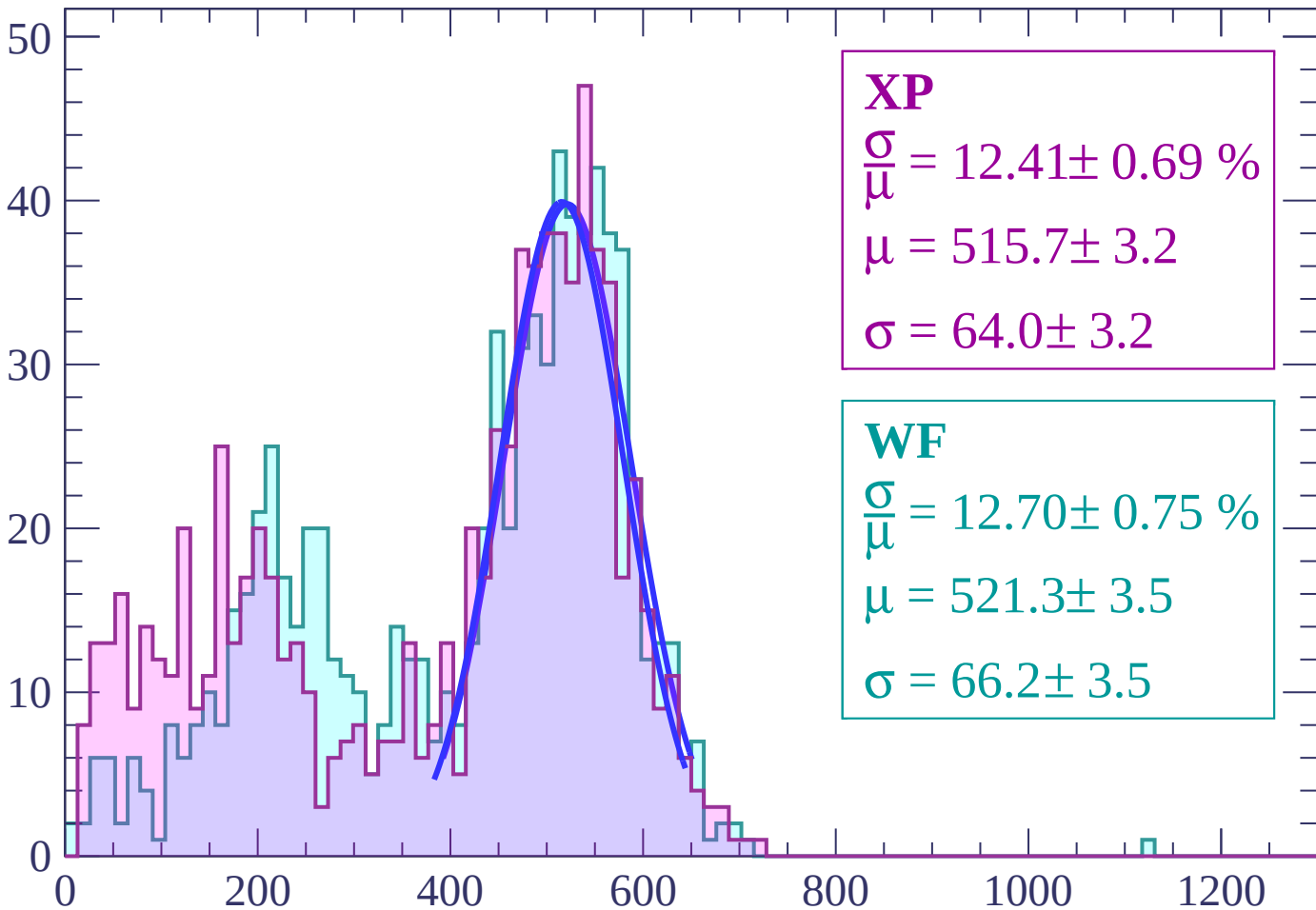
Energy loss | $-66 < \theta < -64$

Count



Energy loss | $-64 < \theta < -62$

Count



XP

$$\frac{\sigma}{\mu} = 12.41 \pm 0.69 \%$$

$$\mu = 515.7 \pm 3.2$$

$$\sigma = 64.0 \pm 3.2$$

WF

$$\frac{\sigma}{\mu} = 12.70 \pm 0.75 \%$$

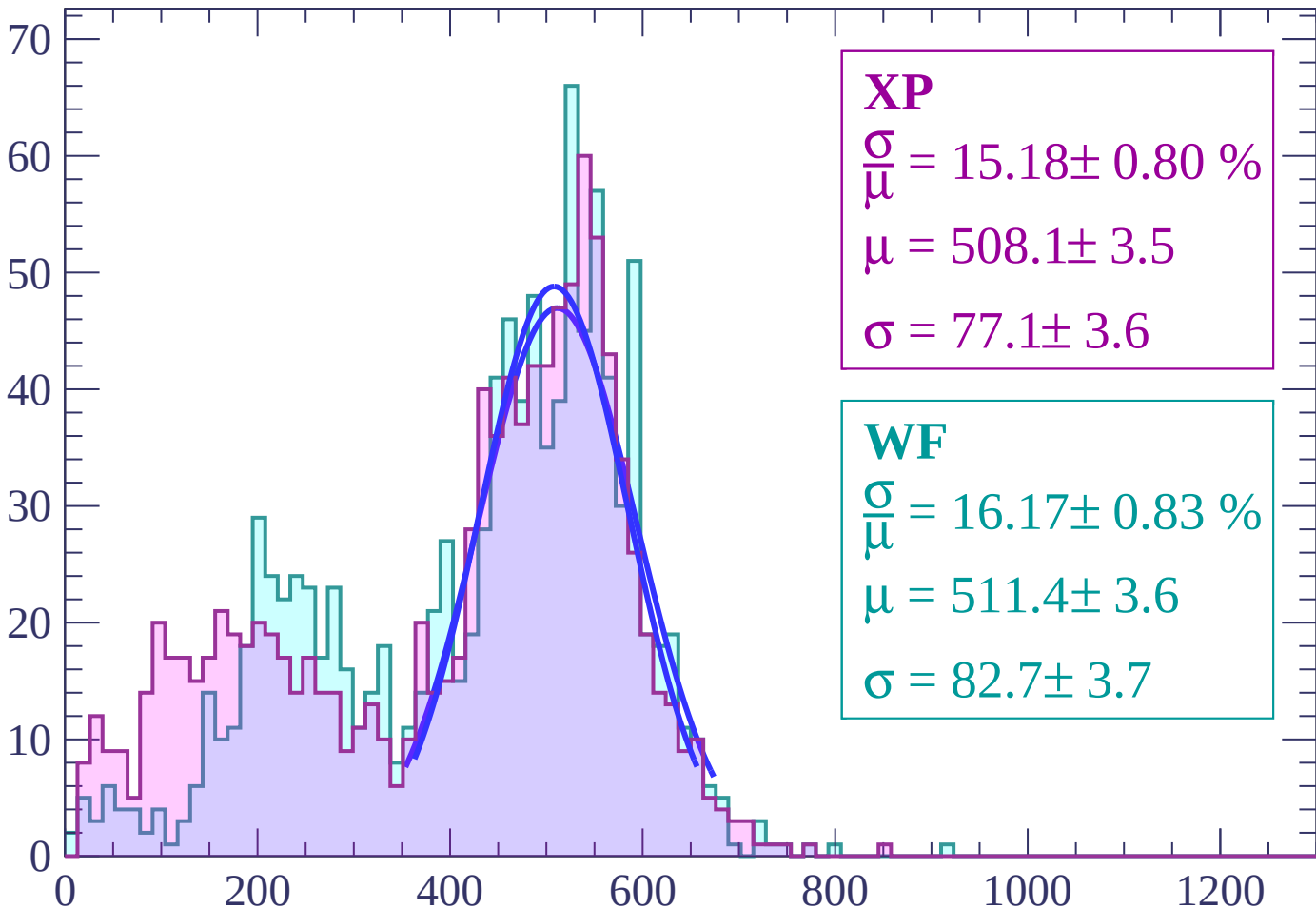
$$\mu = 521.3 \pm 3.5$$

$$\sigma = 66.2 \pm 3.5$$

dE/dx (ADC counts/cm)

Energy loss | $-62 < \theta < -60$

Count



XP

$$\frac{\sigma}{\mu} = 15.18 \pm 0.80 \%$$

$$\mu = 508.1 \pm 3.5$$

$$\sigma = 77.1 \pm 3.6$$

WF

$$\frac{\sigma}{\mu} = 16.17 \pm 0.83 \%$$

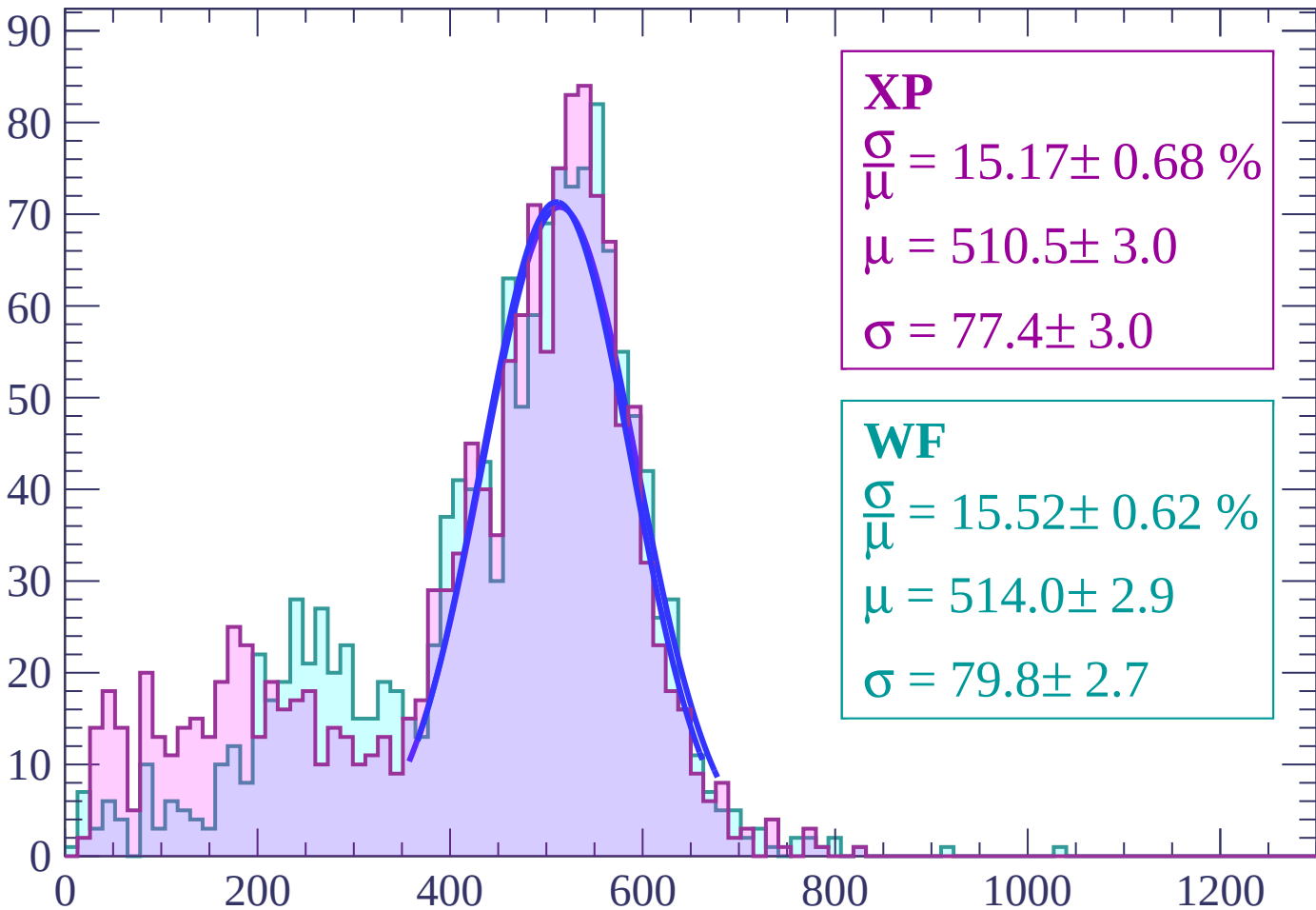
$$\mu = 511.4 \pm 3.6$$

$$\sigma = 82.7 \pm 3.7$$

dE/dx (ADC counts/cm)

Energy loss | $-60 < \theta < -58$

Count



XP

$$\frac{\sigma}{\mu} = 15.17 \pm 0.68 \%$$

$$\mu = 510.5 \pm 3.0$$

$$\sigma = 77.4 \pm 3.0$$

WF

$$\frac{\sigma}{\mu} = 15.52 \pm 0.62 \%$$

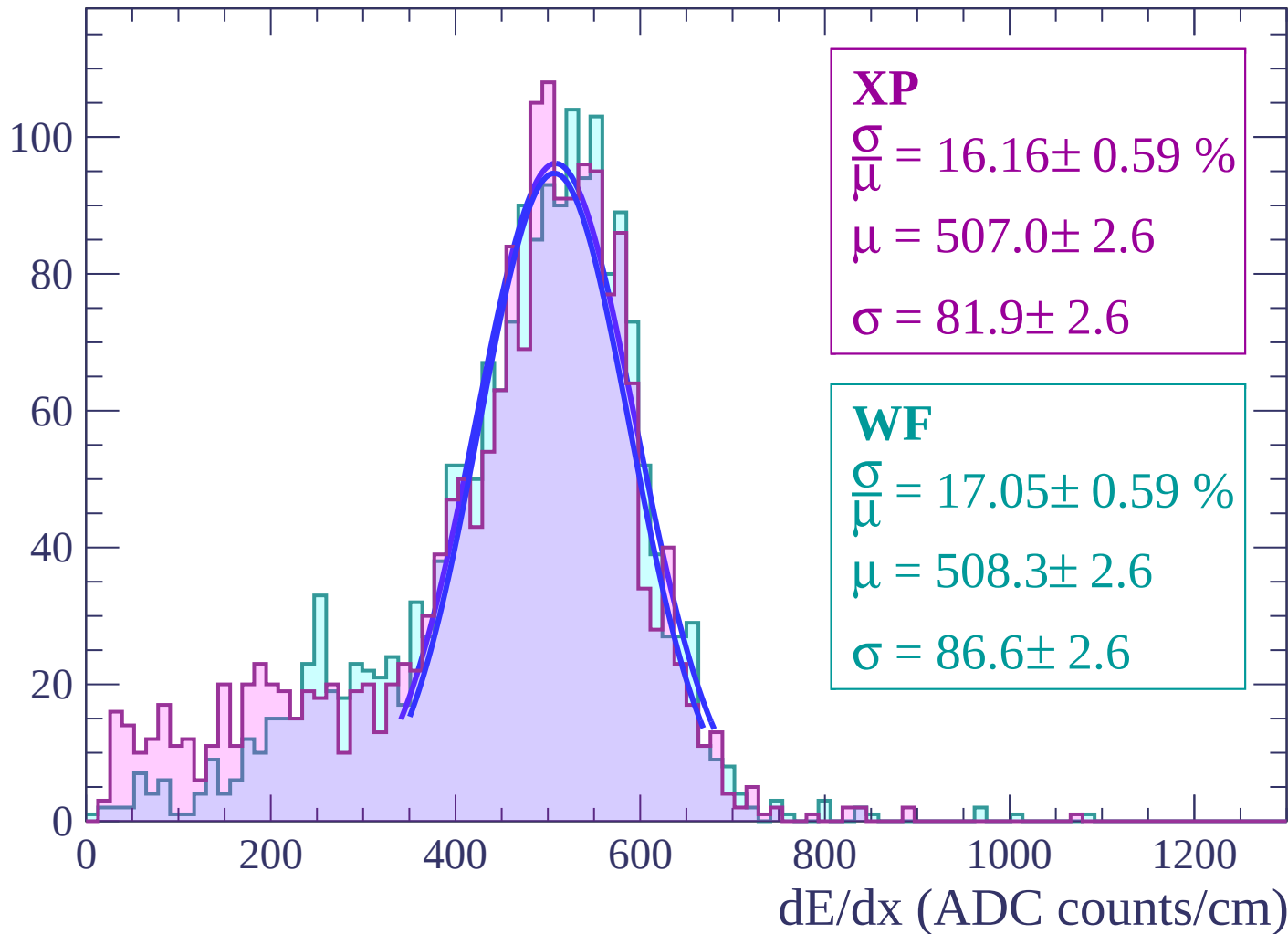
$$\mu = 514.0 \pm 2.9$$

$$\sigma = 79.8 \pm 2.7$$

dE/dx (ADC counts/cm)

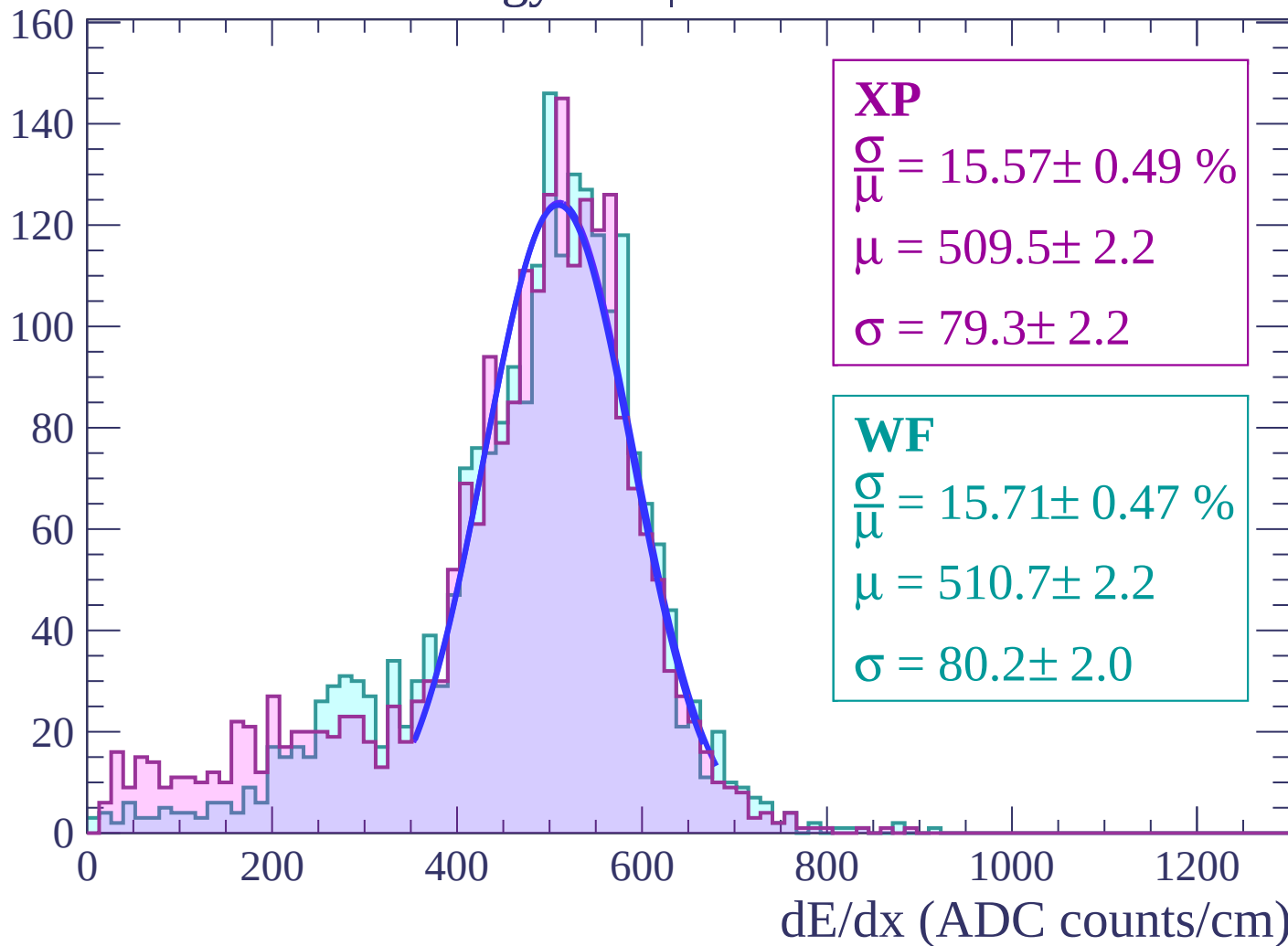
Energy loss | $-58 < \theta < -56$

Count



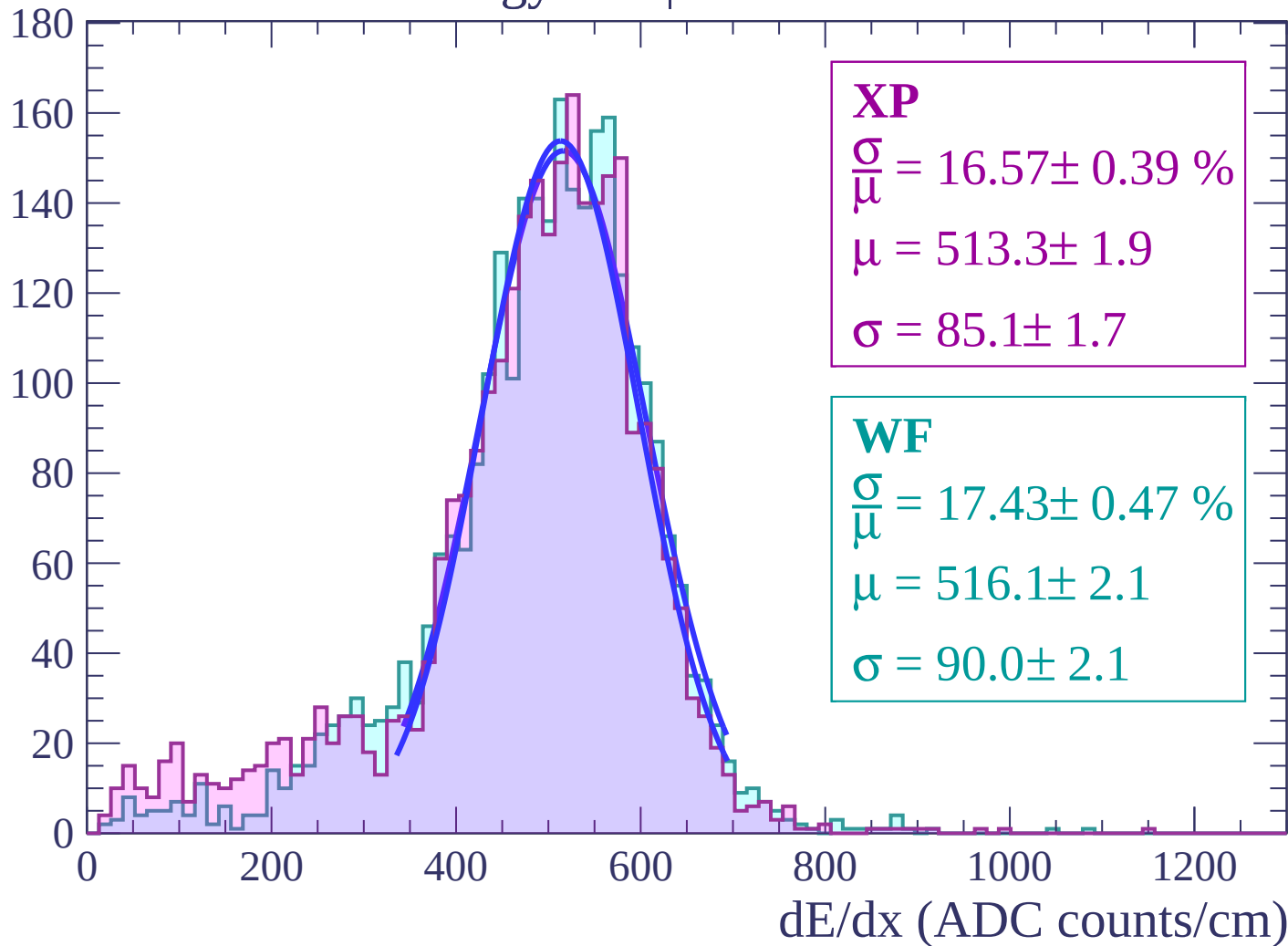
Energy loss | $-56 < \theta < -54$

Count



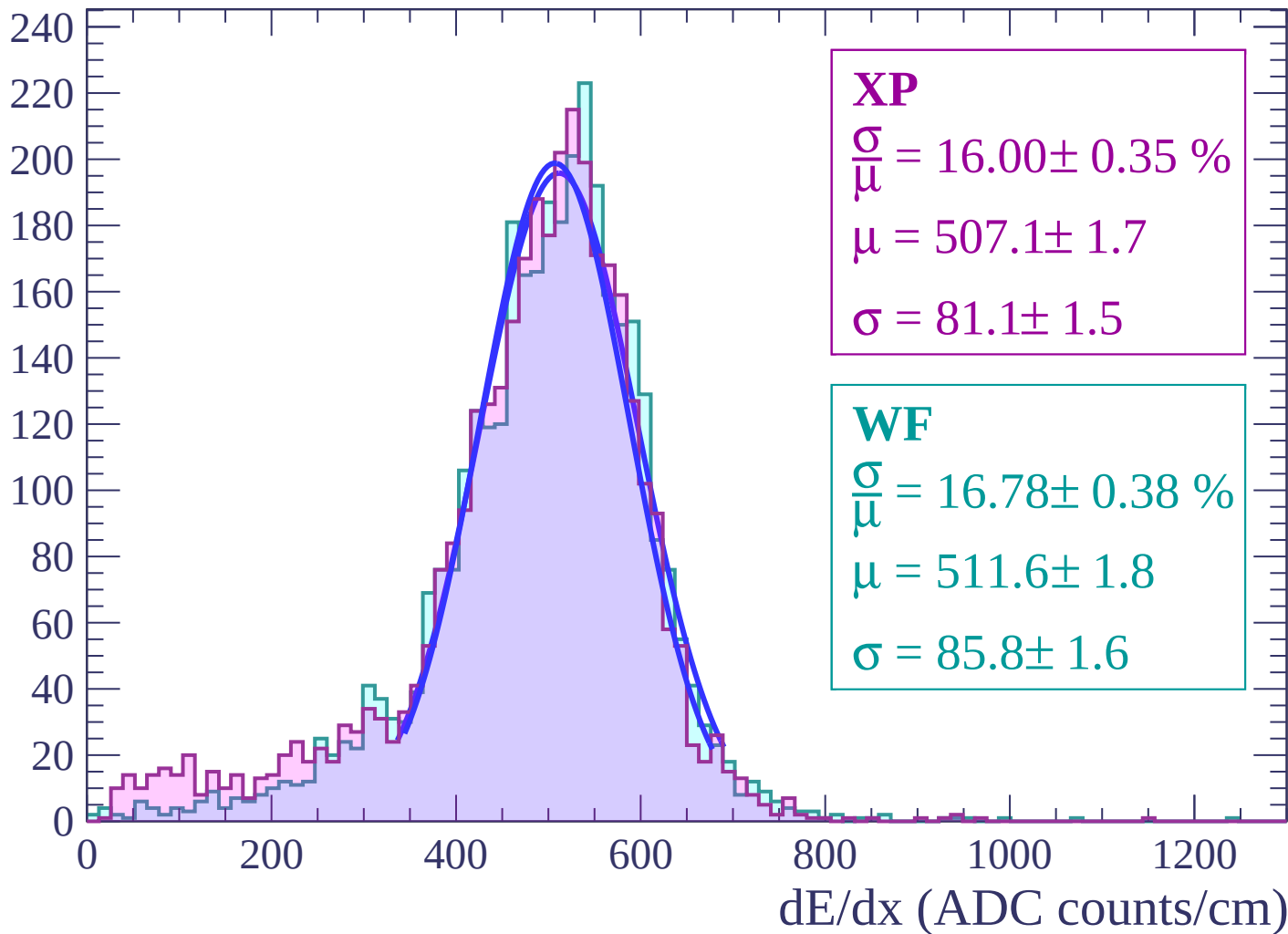
Energy loss | $-54 < \theta < -52$

Count



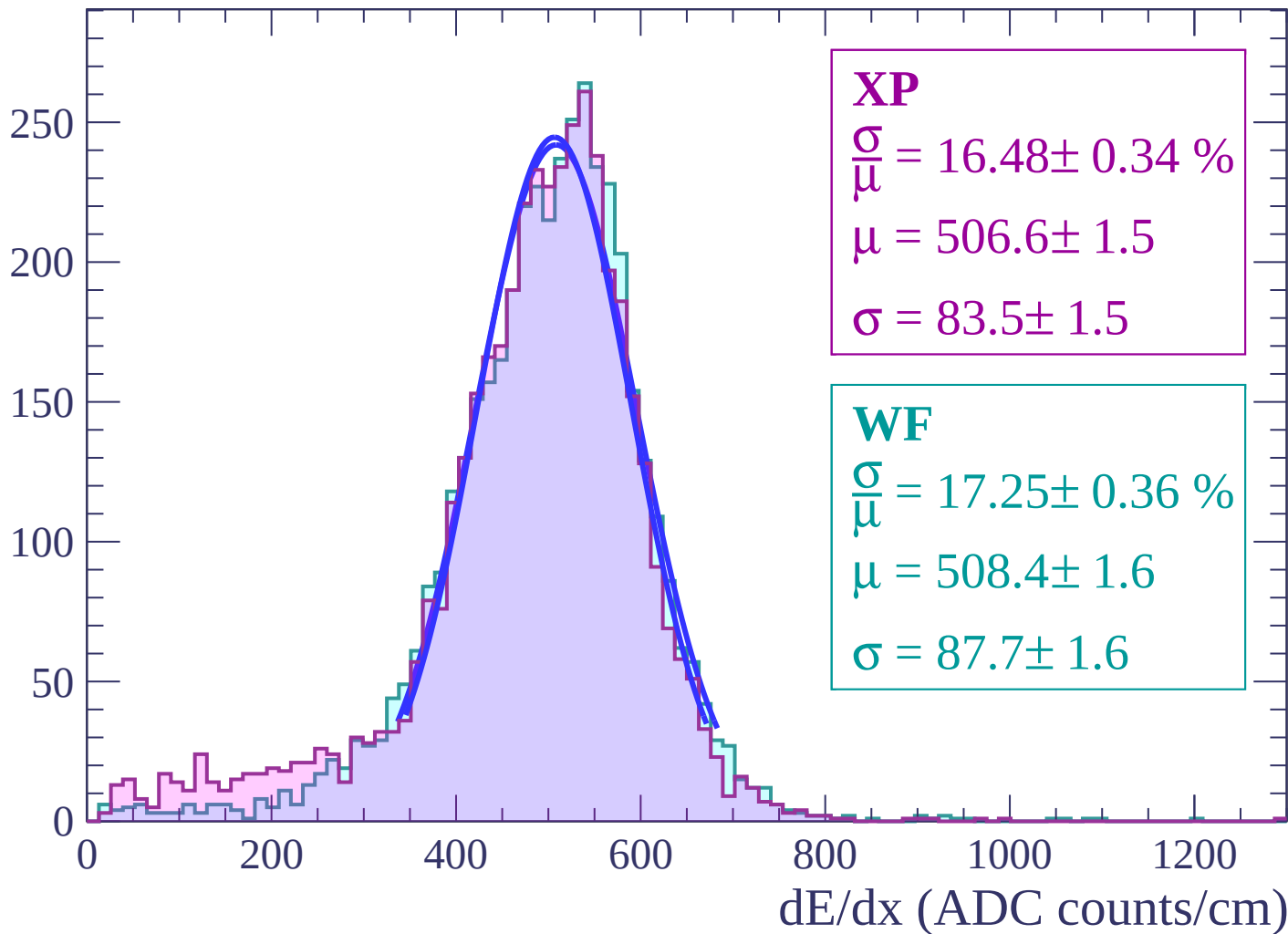
Energy loss | $-52 < \theta < -50$

Count



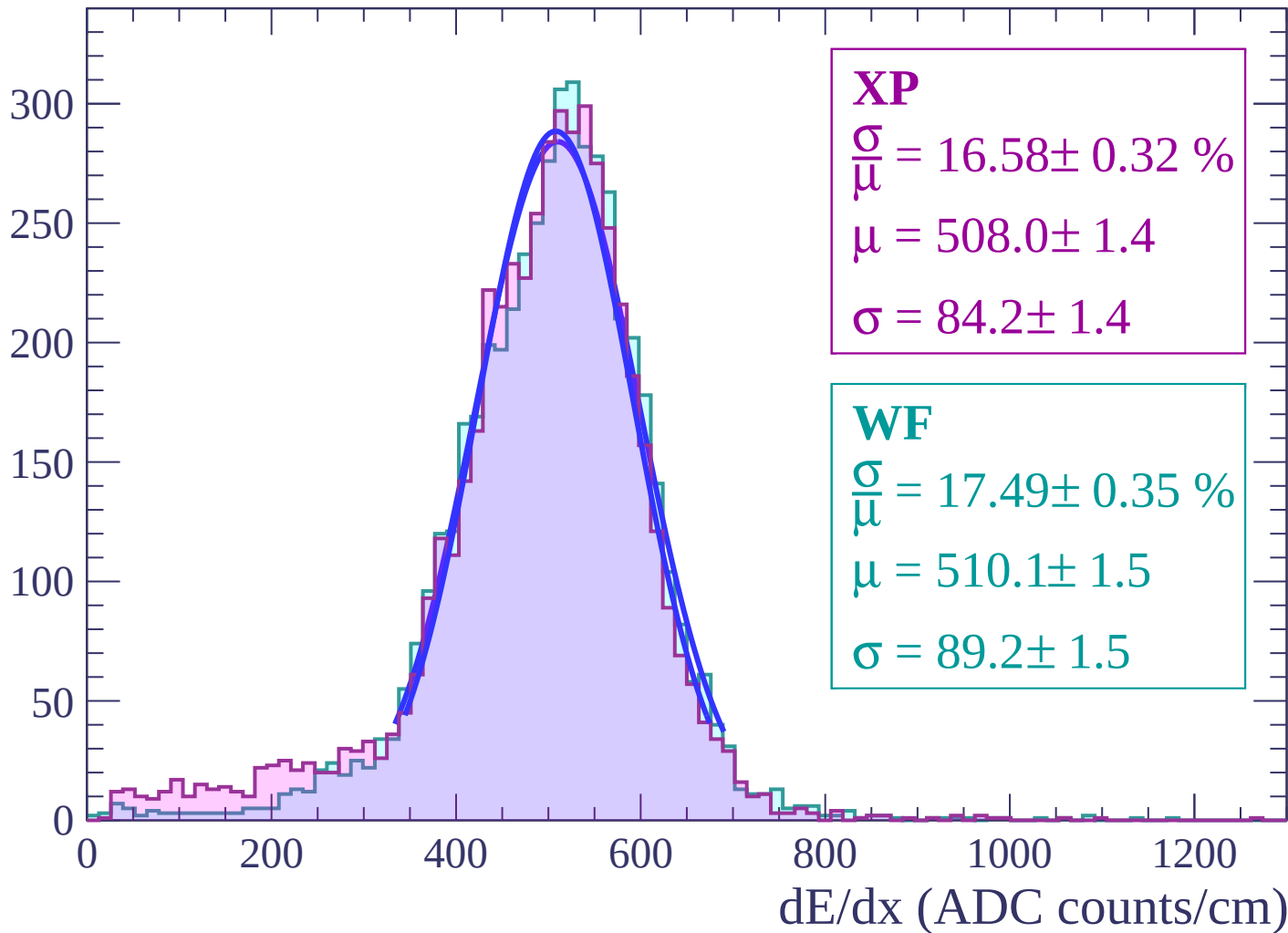
Energy loss | $-50 < \theta < -48$

Count



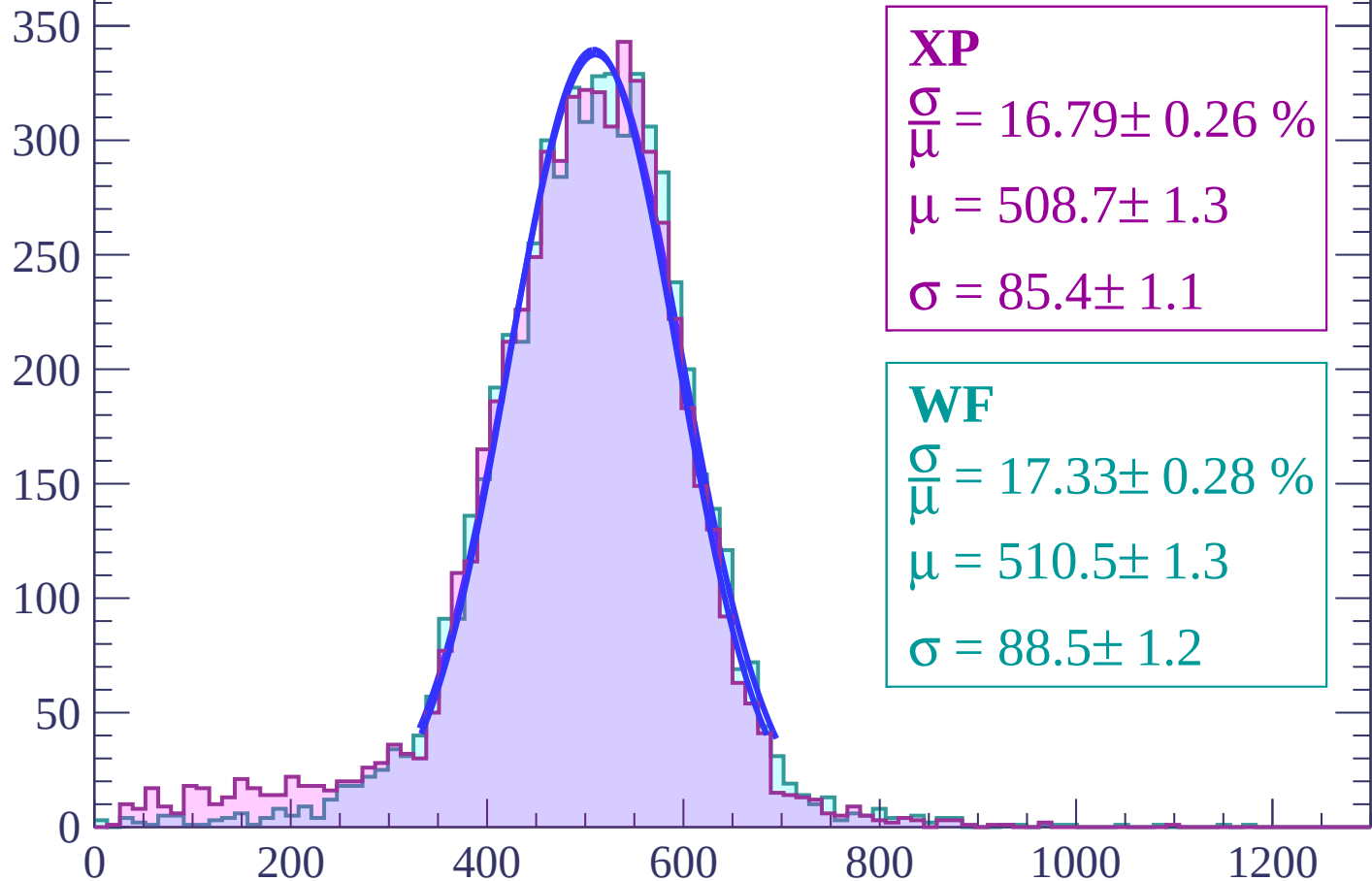
Energy loss | $-48 < \theta < -46$

Count



Energy loss | $-46 < \theta < -44$

Count



XP

$$\frac{\sigma}{\mu} = 16.79 \pm 0.26 \%$$

$$\mu = 508.7 \pm 1.3$$

$$\sigma = 85.4 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 17.33 \pm 0.28 \%$$

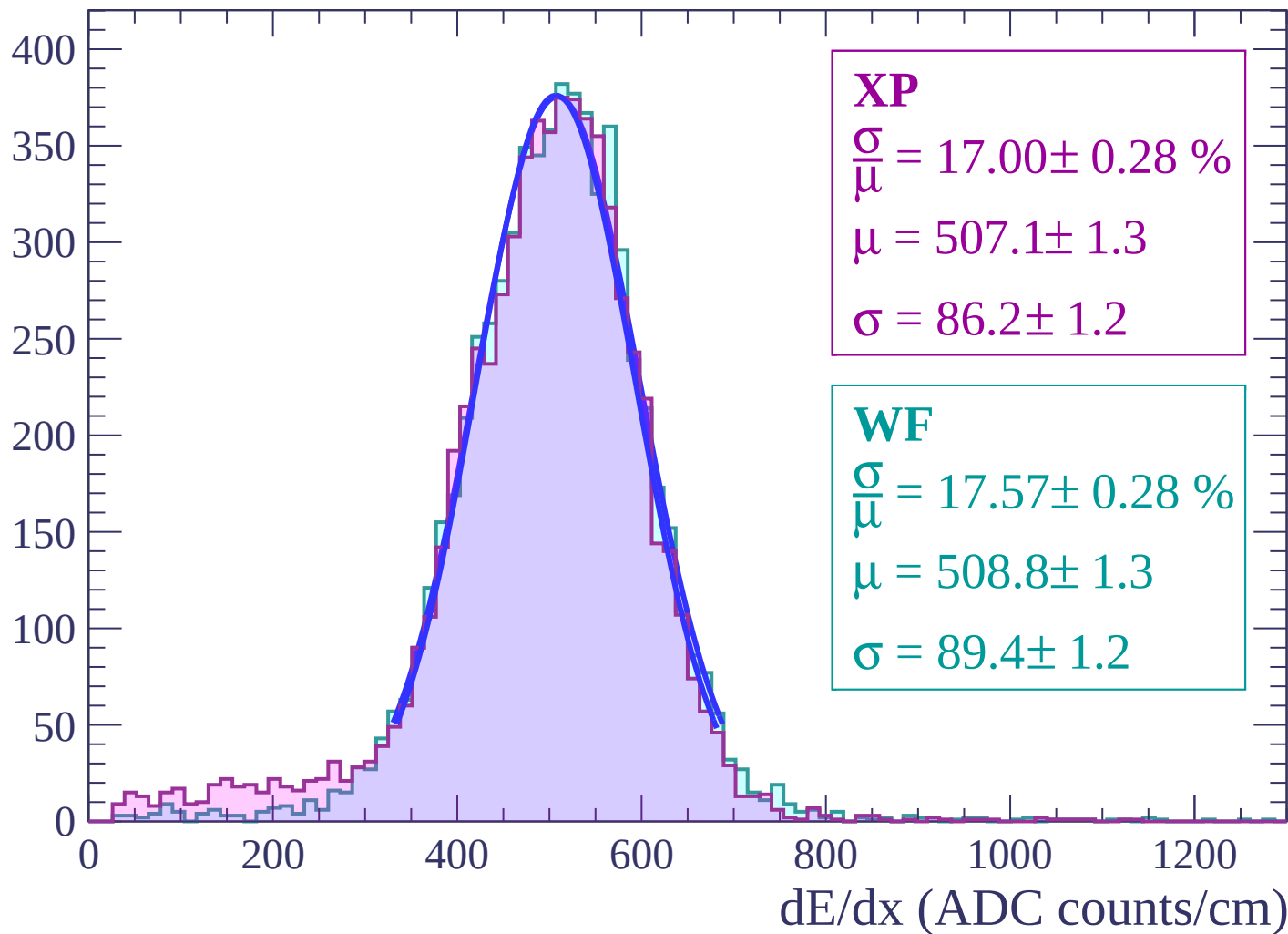
$$\mu = 510.5 \pm 1.3$$

$$\sigma = 88.5 \pm 1.2$$

dE/dx (ADC counts/cm)

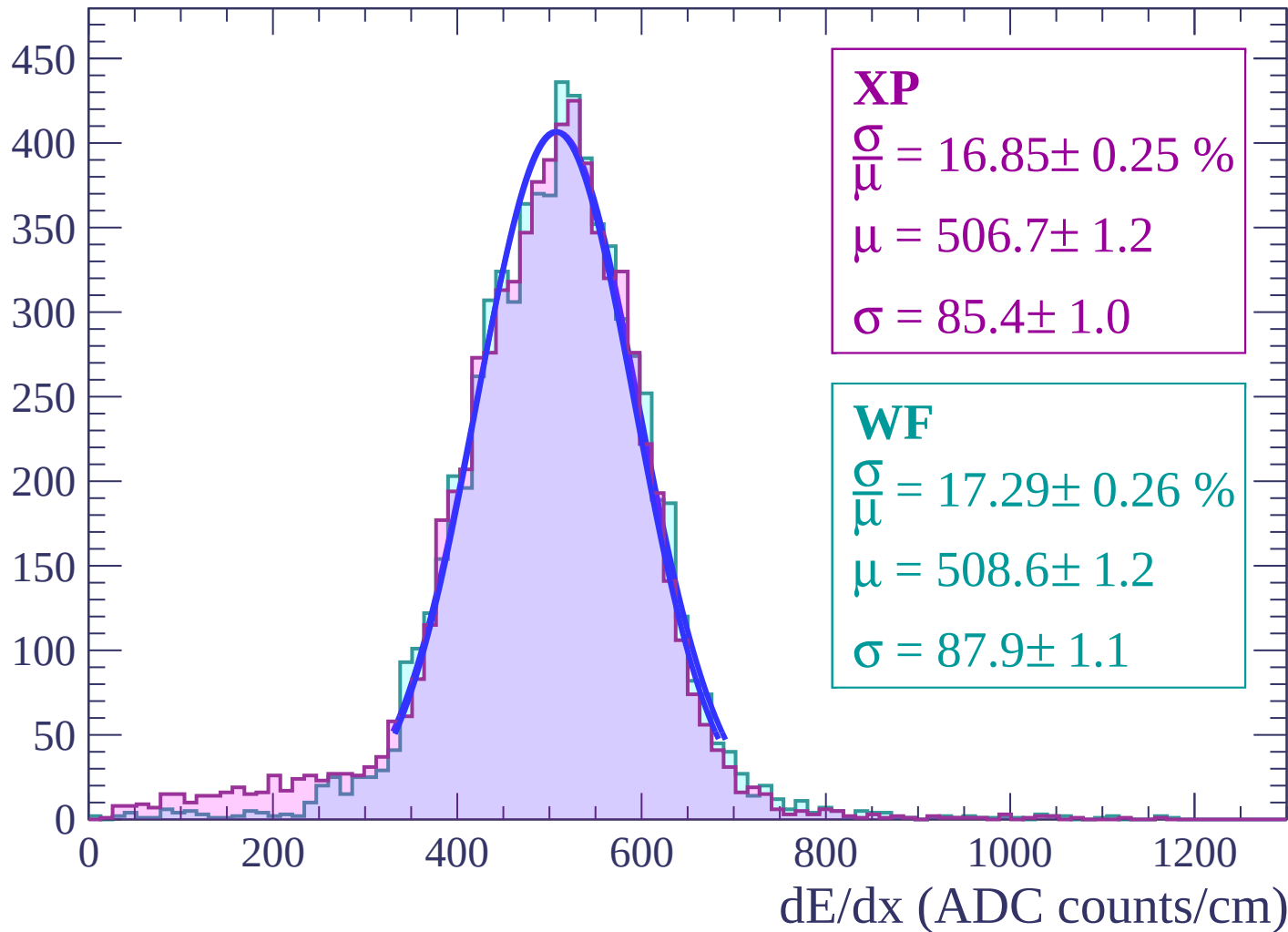
Energy loss | $-44 < \theta < -42$

Count



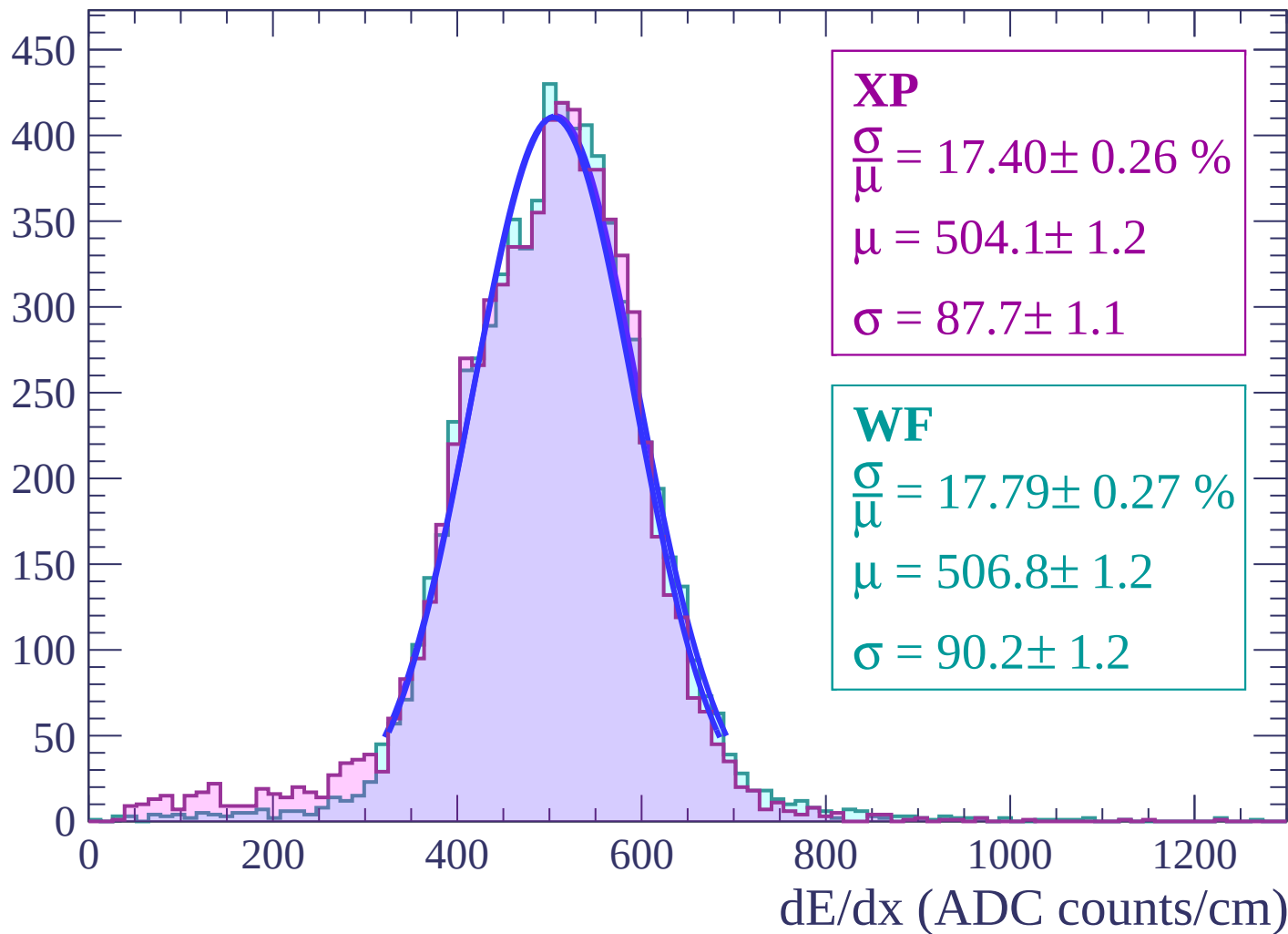
Energy loss | $-42 < \theta < -40$

Count



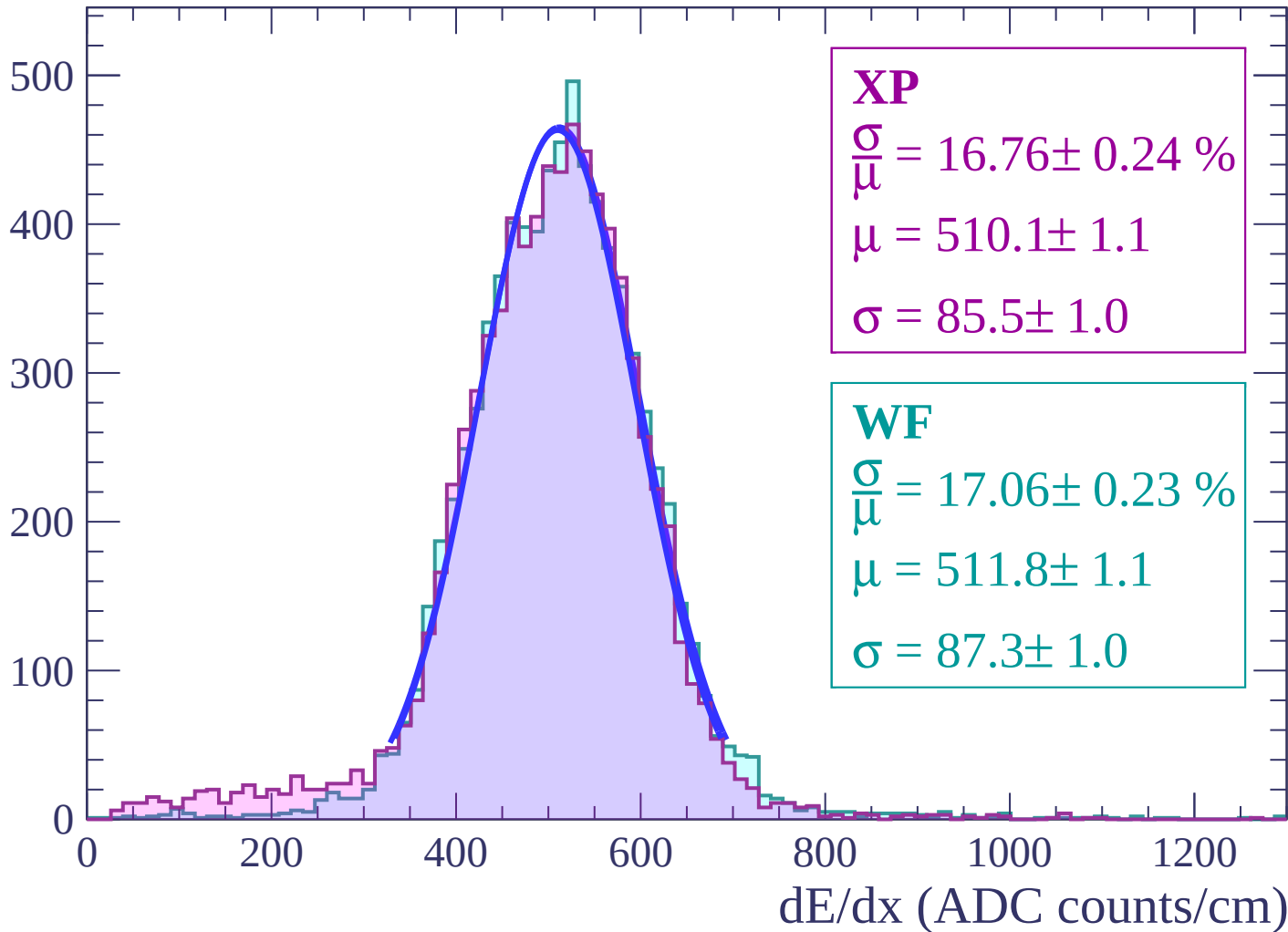
Energy loss | $-40 < \theta < -38$

Count



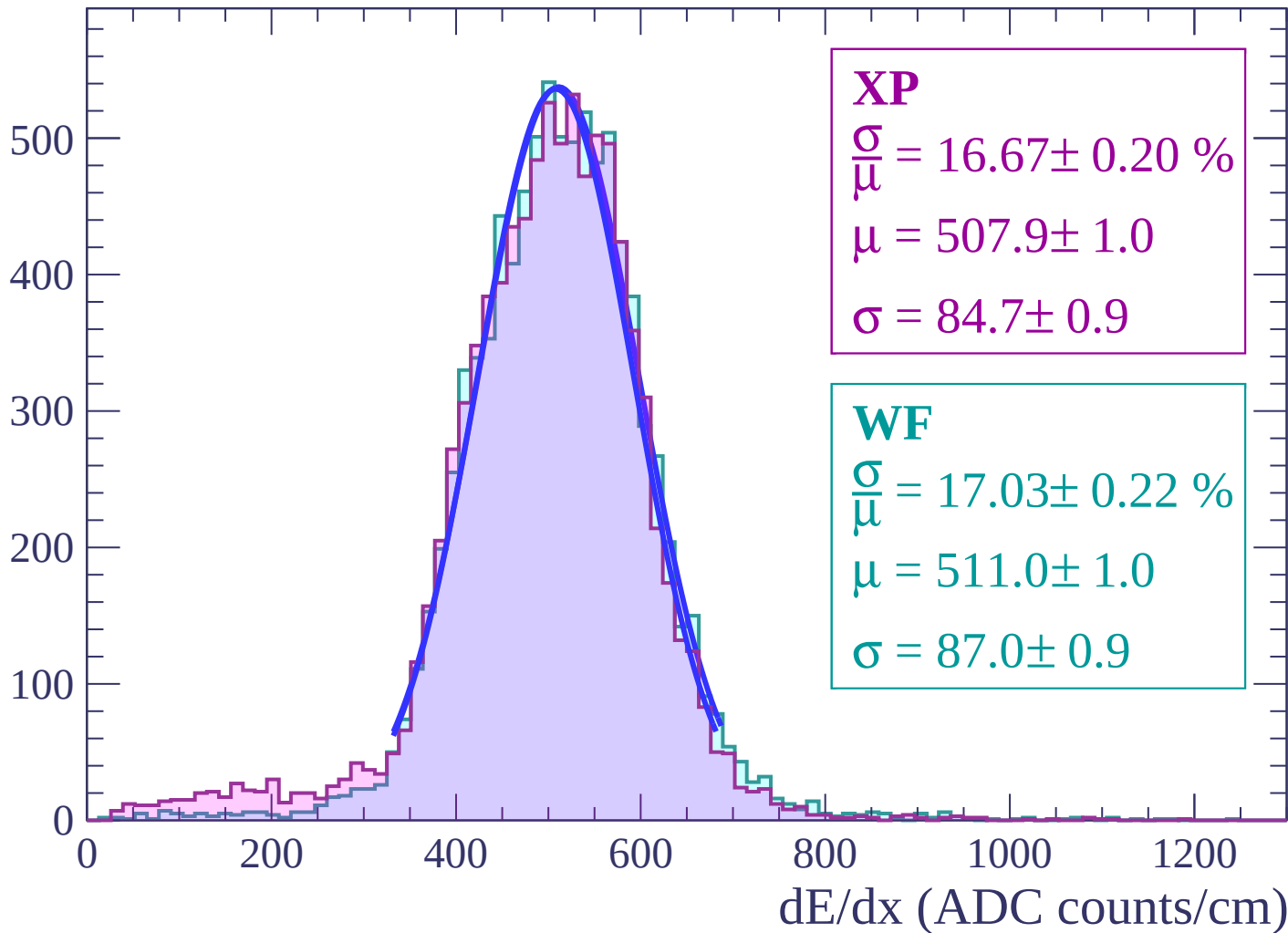
Energy loss | $-38 < \theta < -36$

Count



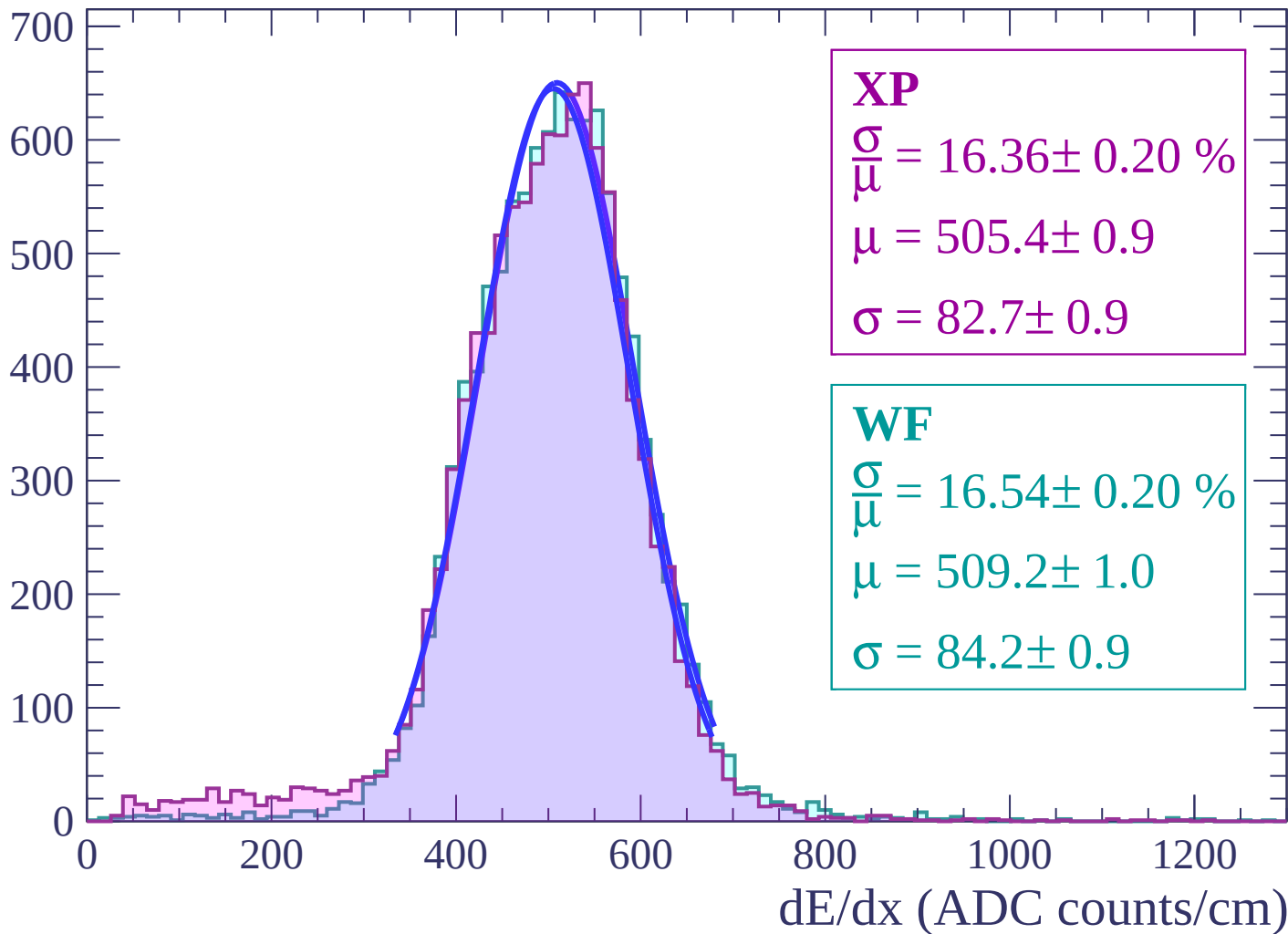
Energy loss | $-36 < \theta < -34$

Count



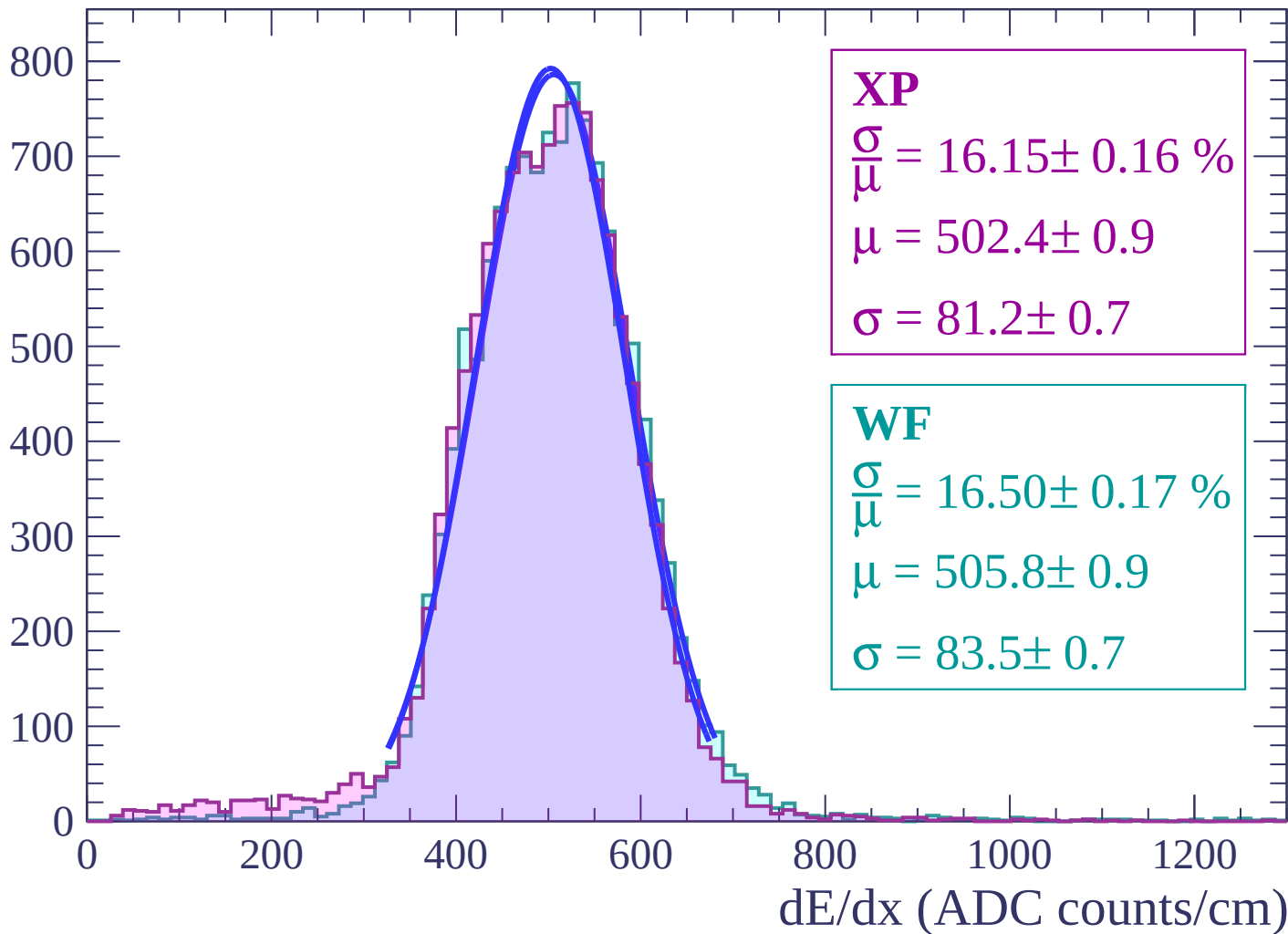
Energy loss | $-34 < \theta < -32$

Count



Energy loss | $-32 < \theta < -30$

Count



Energy loss | $-30 < \theta < -28$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 16.18 \pm 0.16 \%$$

$$\mu = 496.9 \pm 0.8$$

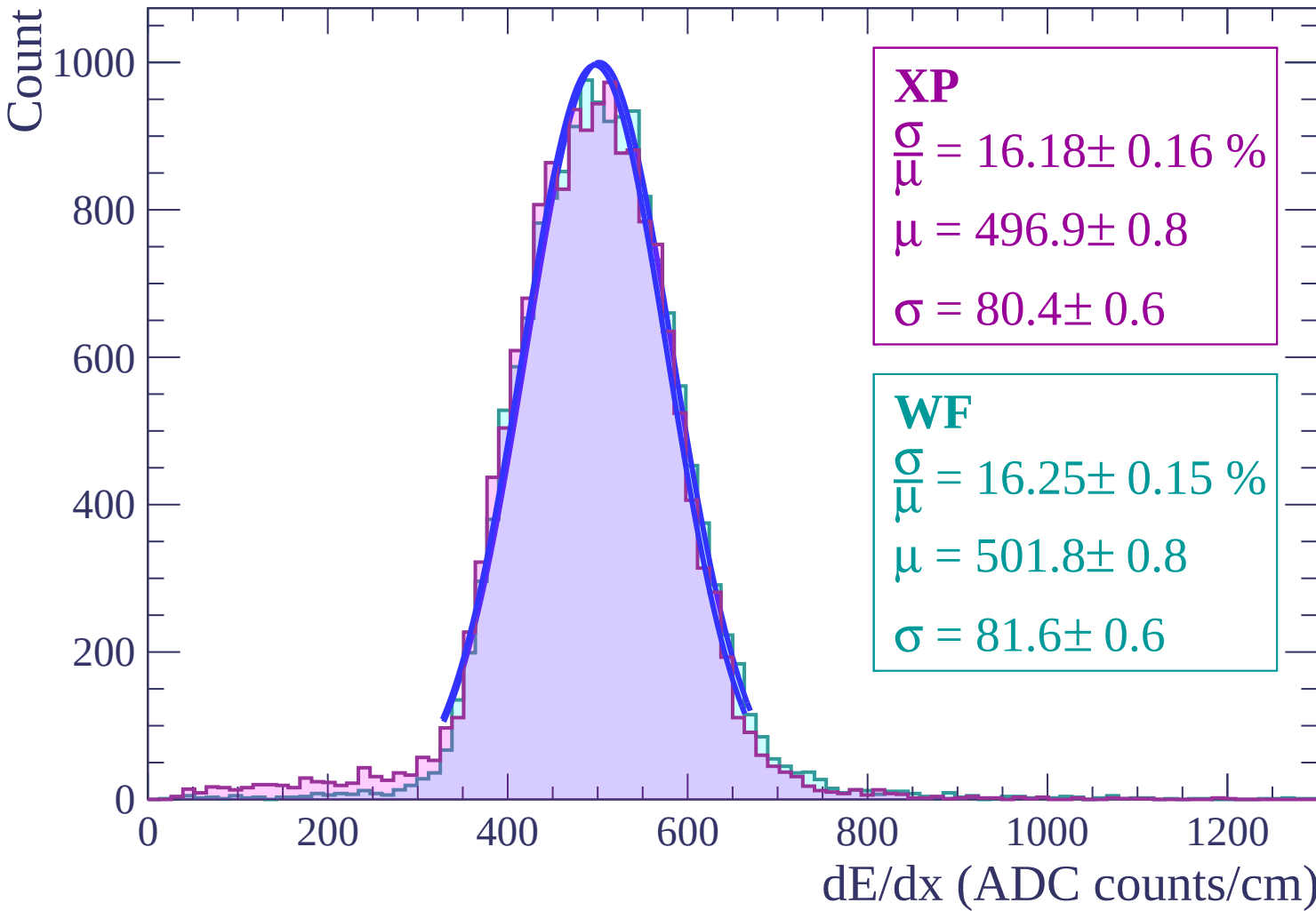
$$\sigma = 80.4 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 16.25 \pm 0.15 \%$$

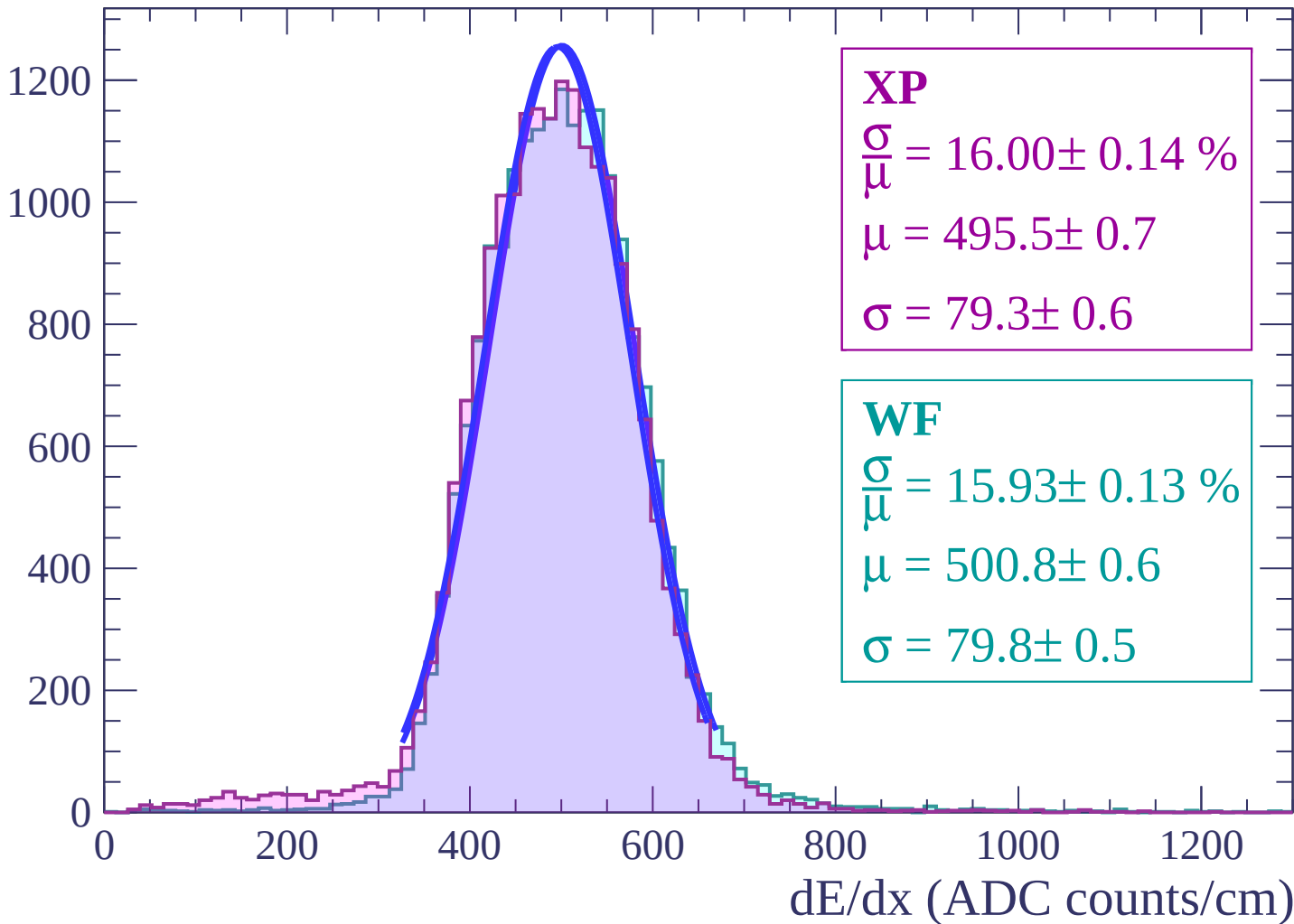
$$\mu = 501.8 \pm 0.8$$

$$\sigma = 81.6 \pm 0.6$$



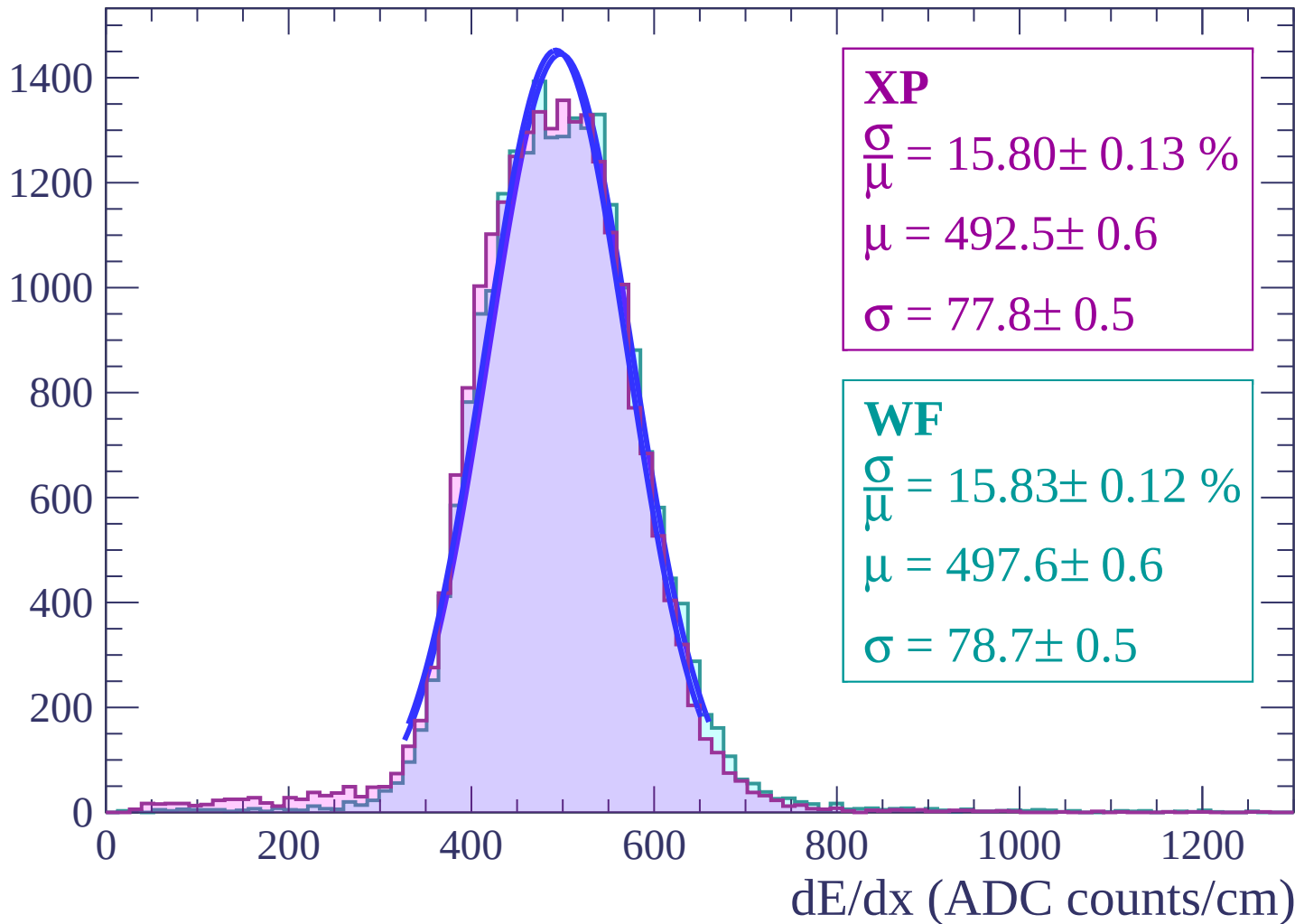
Energy loss | $-28 < \theta < -26$

Count

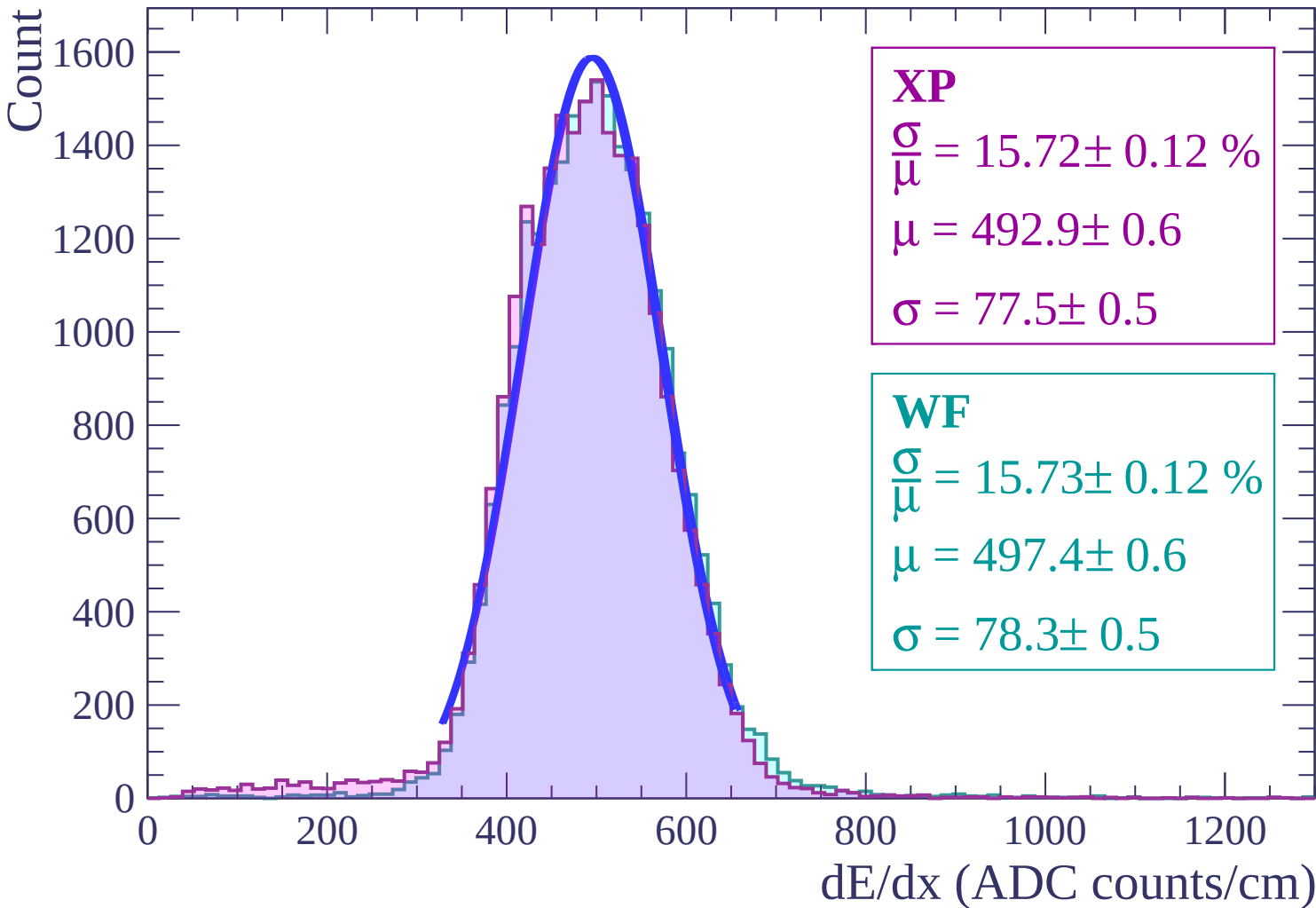


Energy loss | $-26 < \theta < -24$

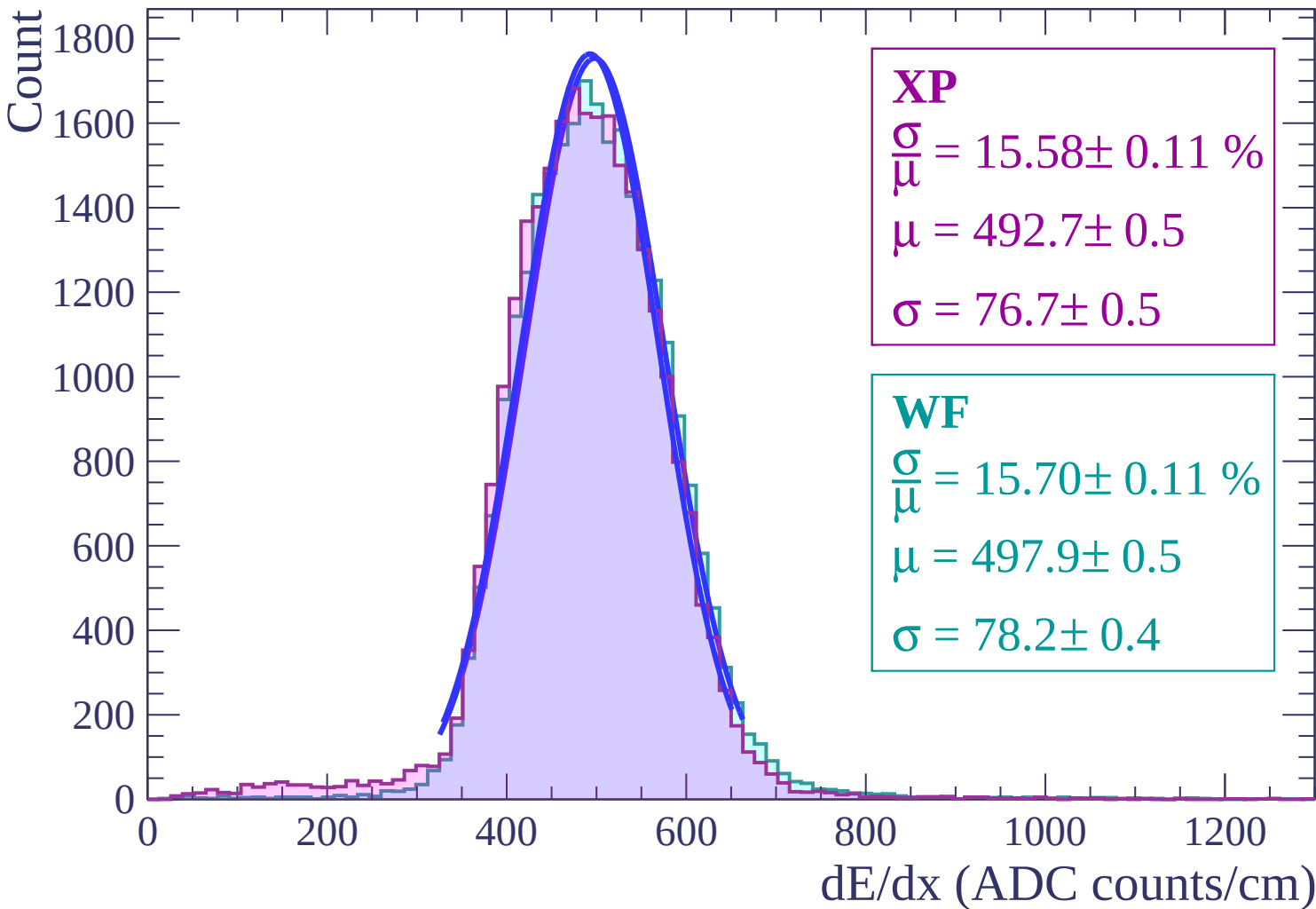
Count



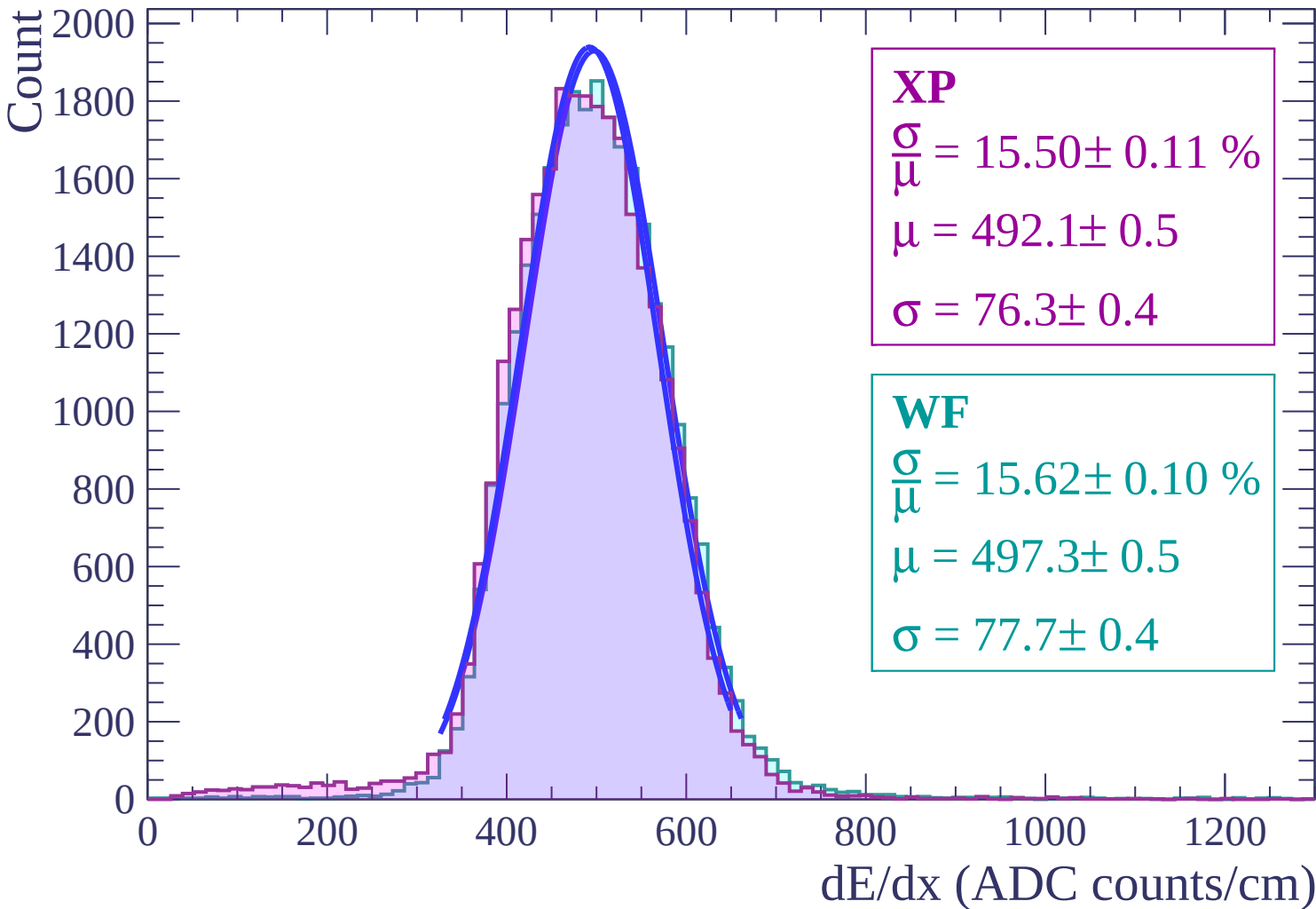
Energy loss | $-24 < \theta < -22$



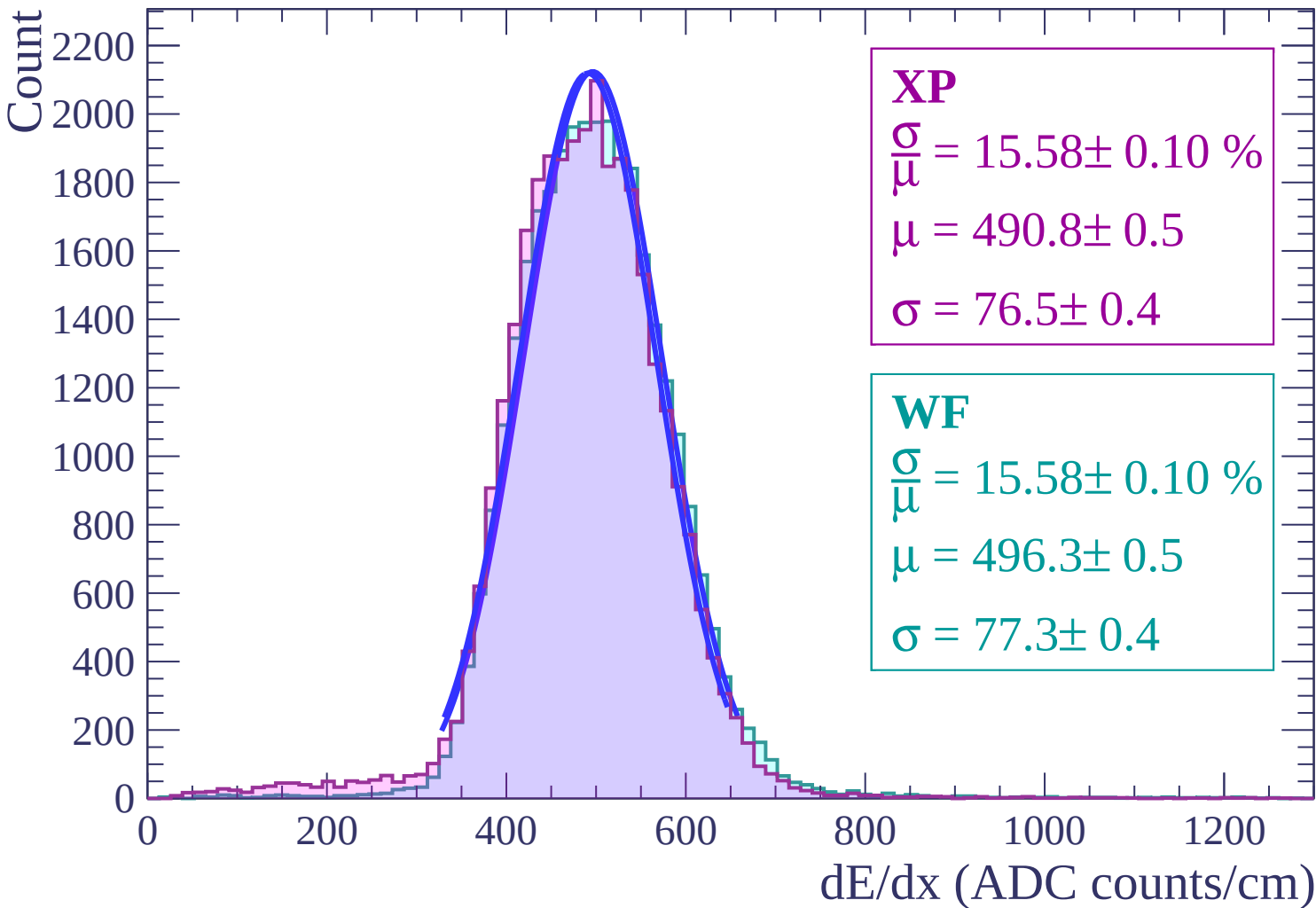
Energy loss | $-22 < \theta < -20$



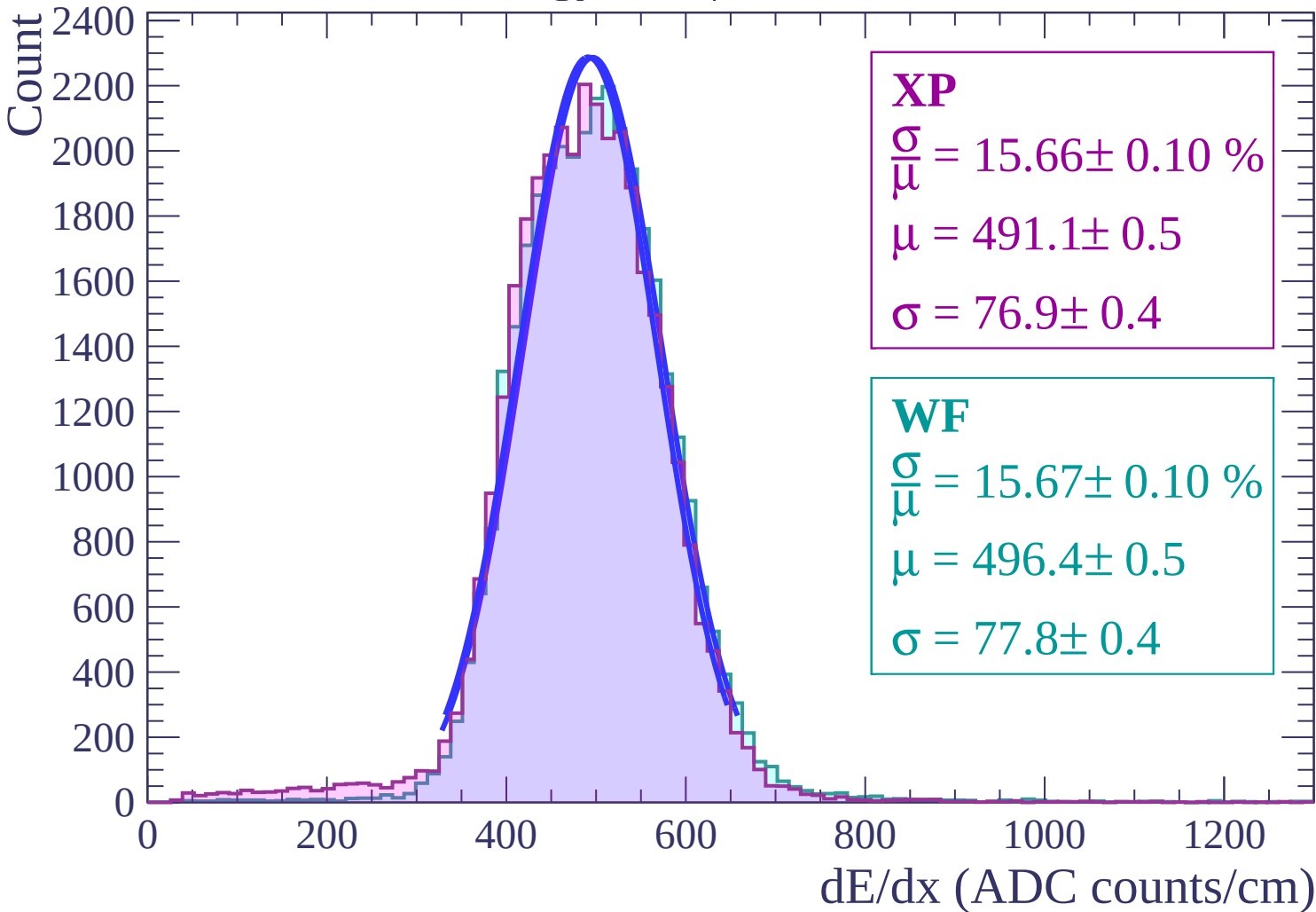
Energy loss | $-20 < \theta < -18$



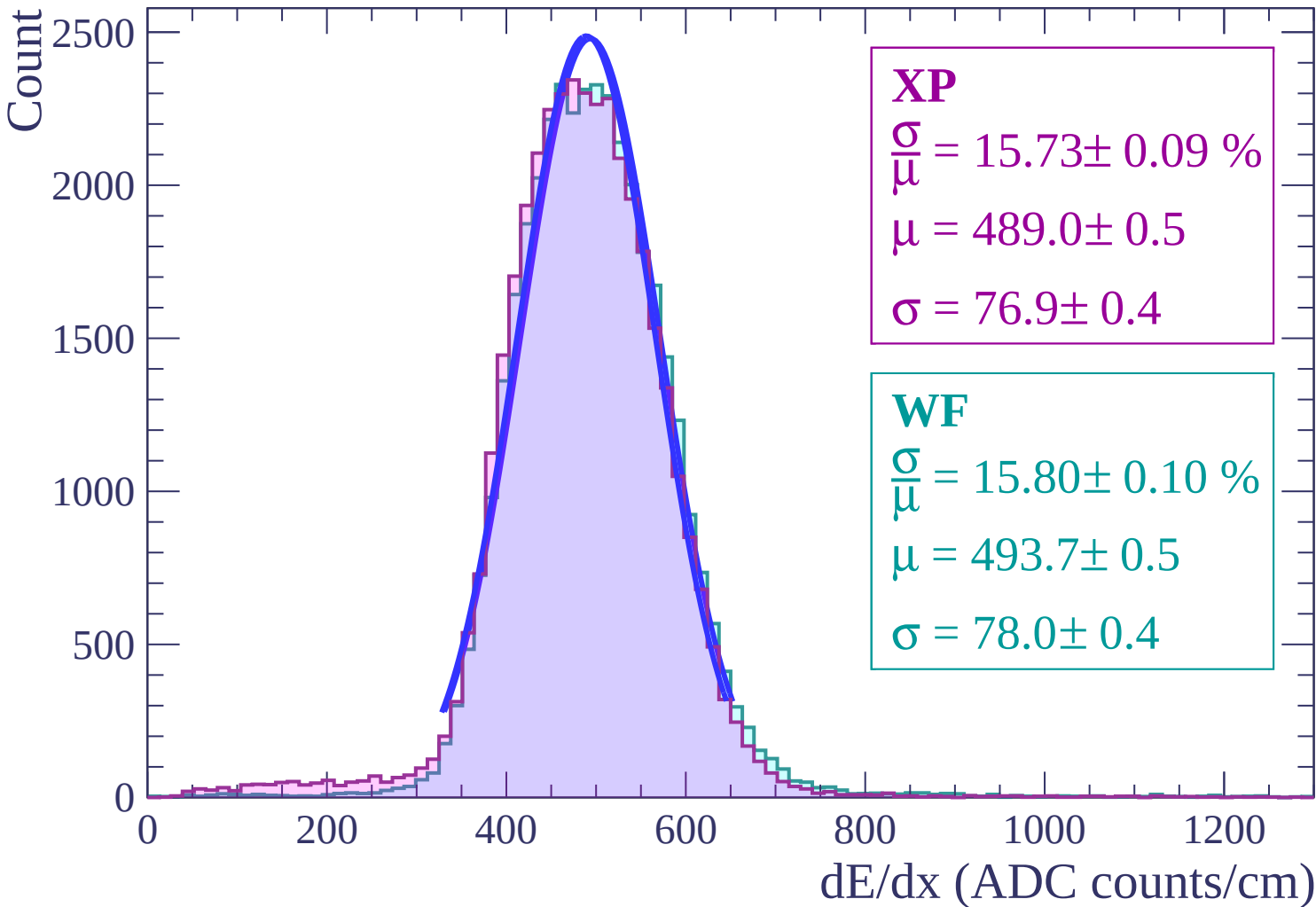
Energy loss | $-18 < \theta < -16$



Energy loss | $-16 < \theta < -14$



Energy loss | $-14 < \theta < -12$



Energy loss | $-12 < \theta < -10$

Count

2500

2000

1500

1000

500

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 15.56 \pm 0.09 \%$$

$$\mu = 488.5 \pm 0.4$$

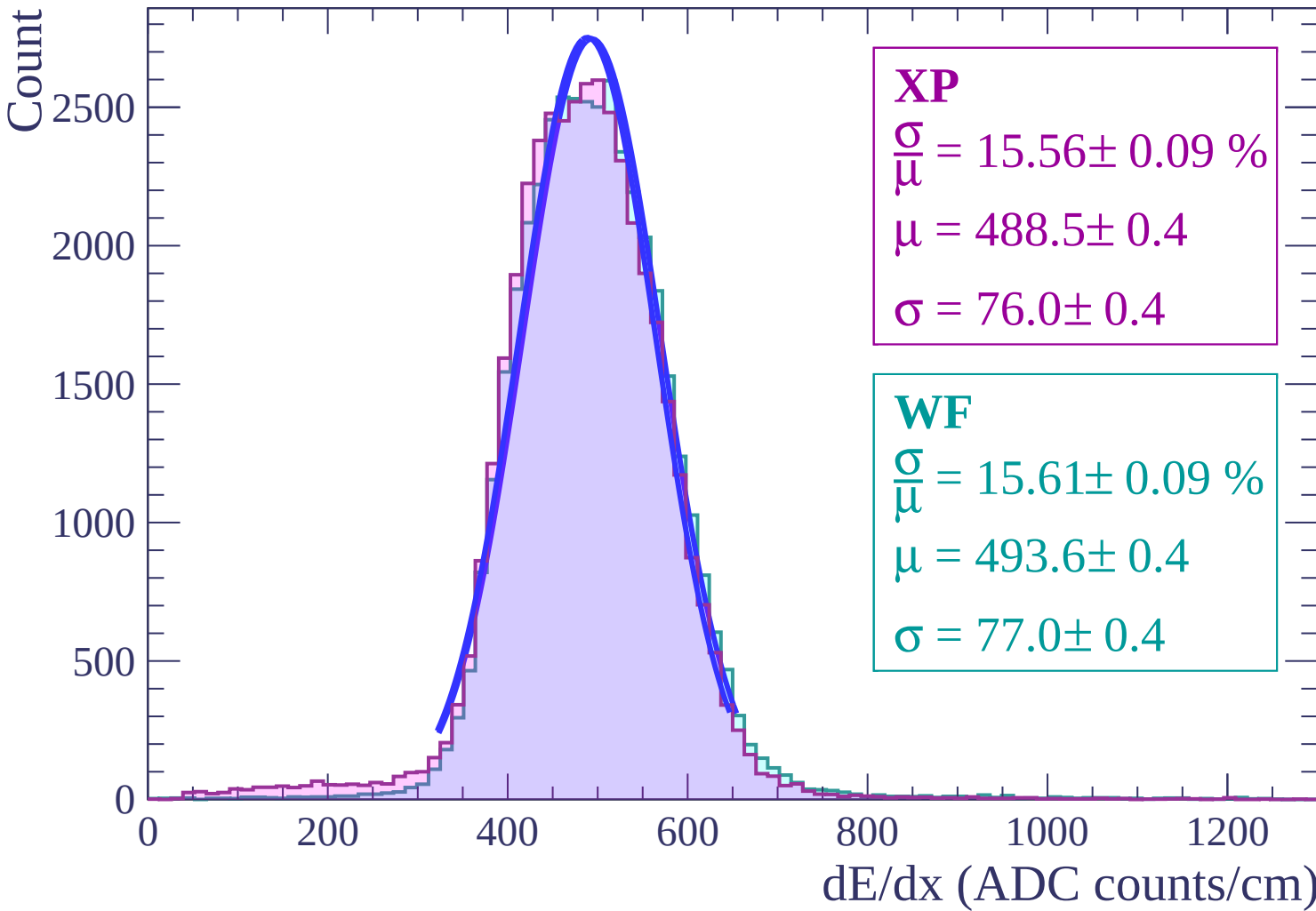
$$\sigma = 76.0 \pm 0.4$$

WF

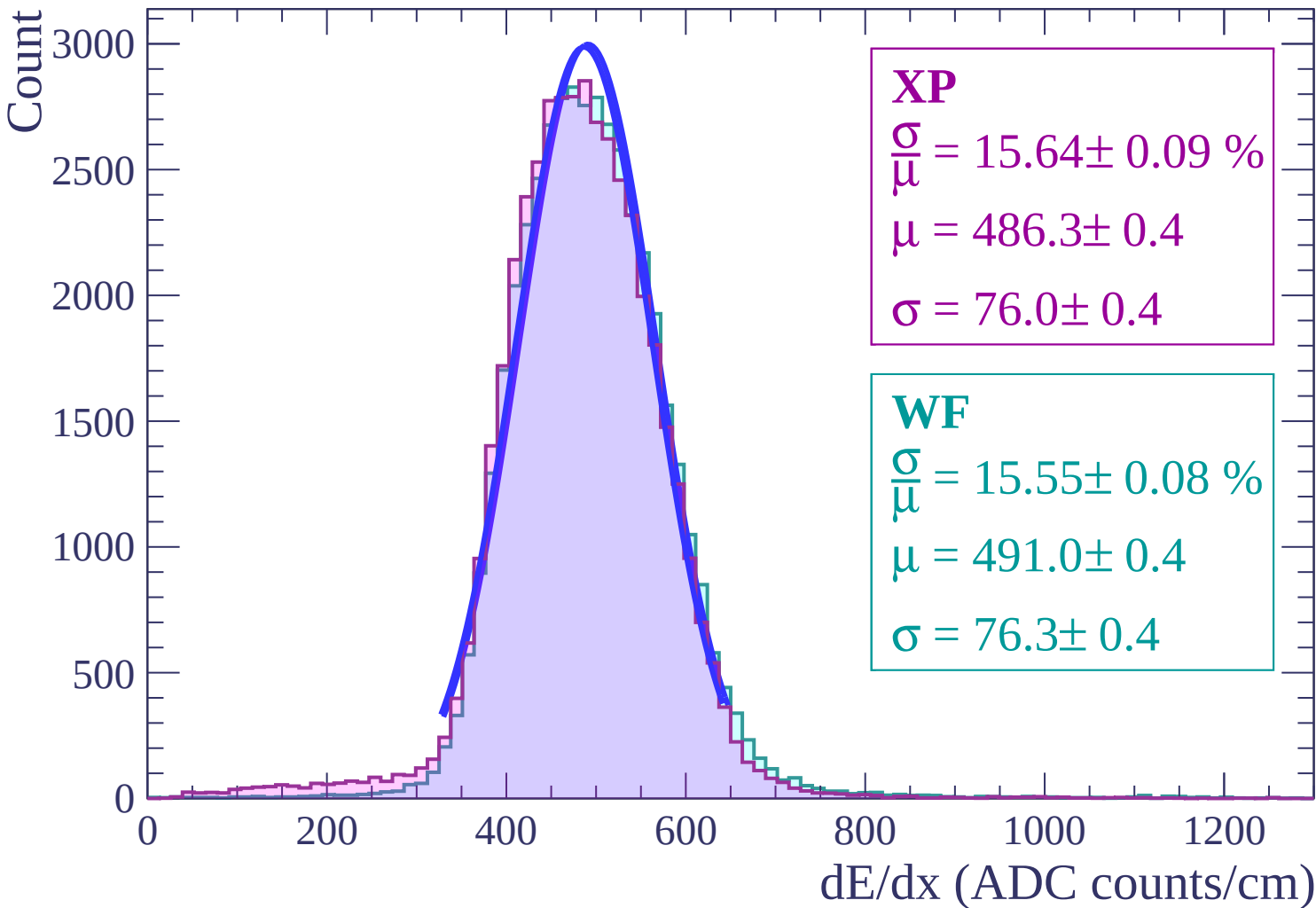
$$\frac{\sigma}{\mu} = 15.61 \pm 0.09 \%$$

$$\mu = 493.6 \pm 0.4$$

$$\sigma = 77.0 \pm 0.4$$



Energy loss | $-10 < \theta < -8$



Energy loss | $-8 < \theta < -6$

Count

3000

2500

2000

1500

1000

500

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 15.73 \pm 0.08 \%$$

$$\mu = 486.5 \pm 0.4$$

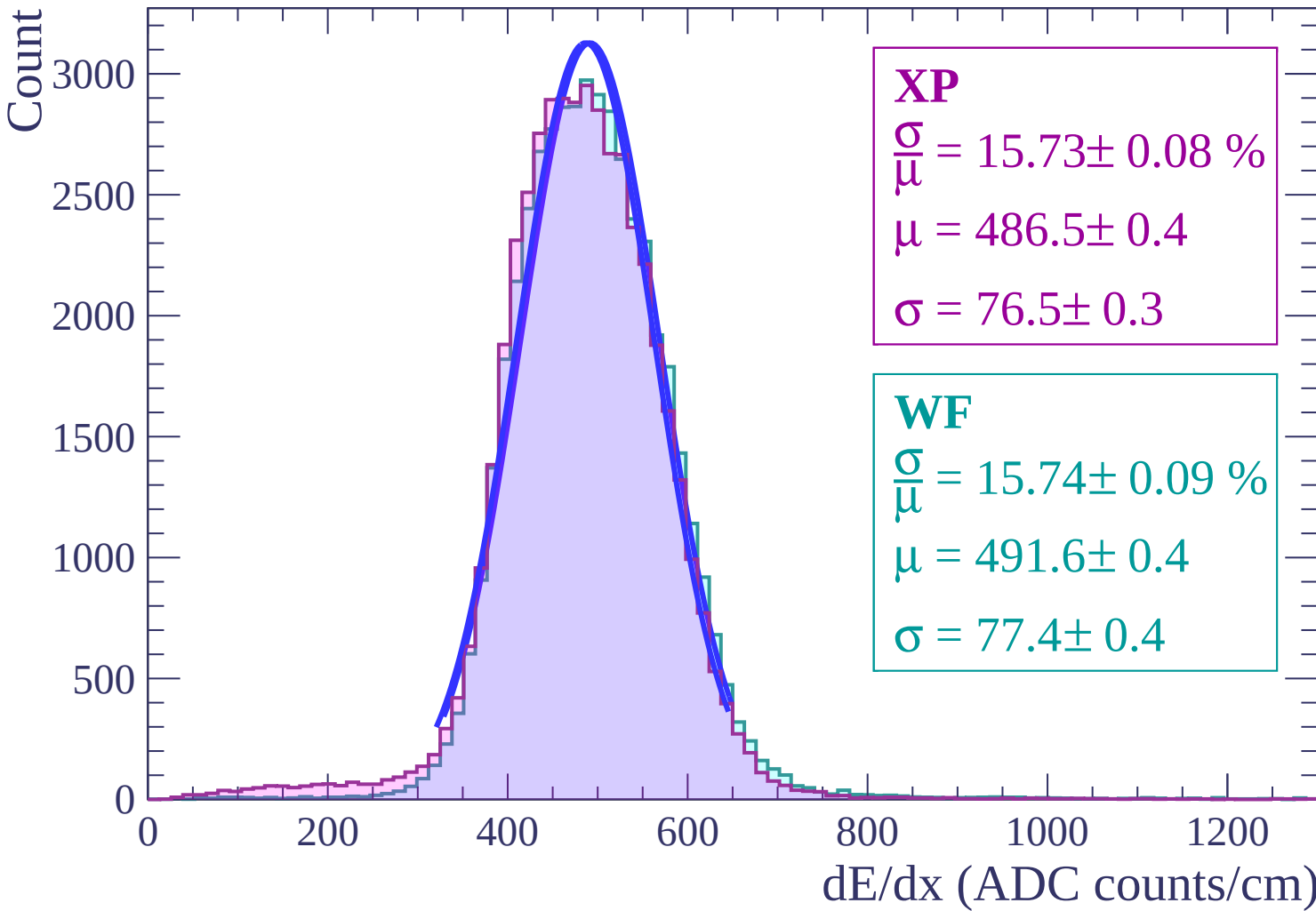
$$\sigma = 76.5 \pm 0.3$$

WF

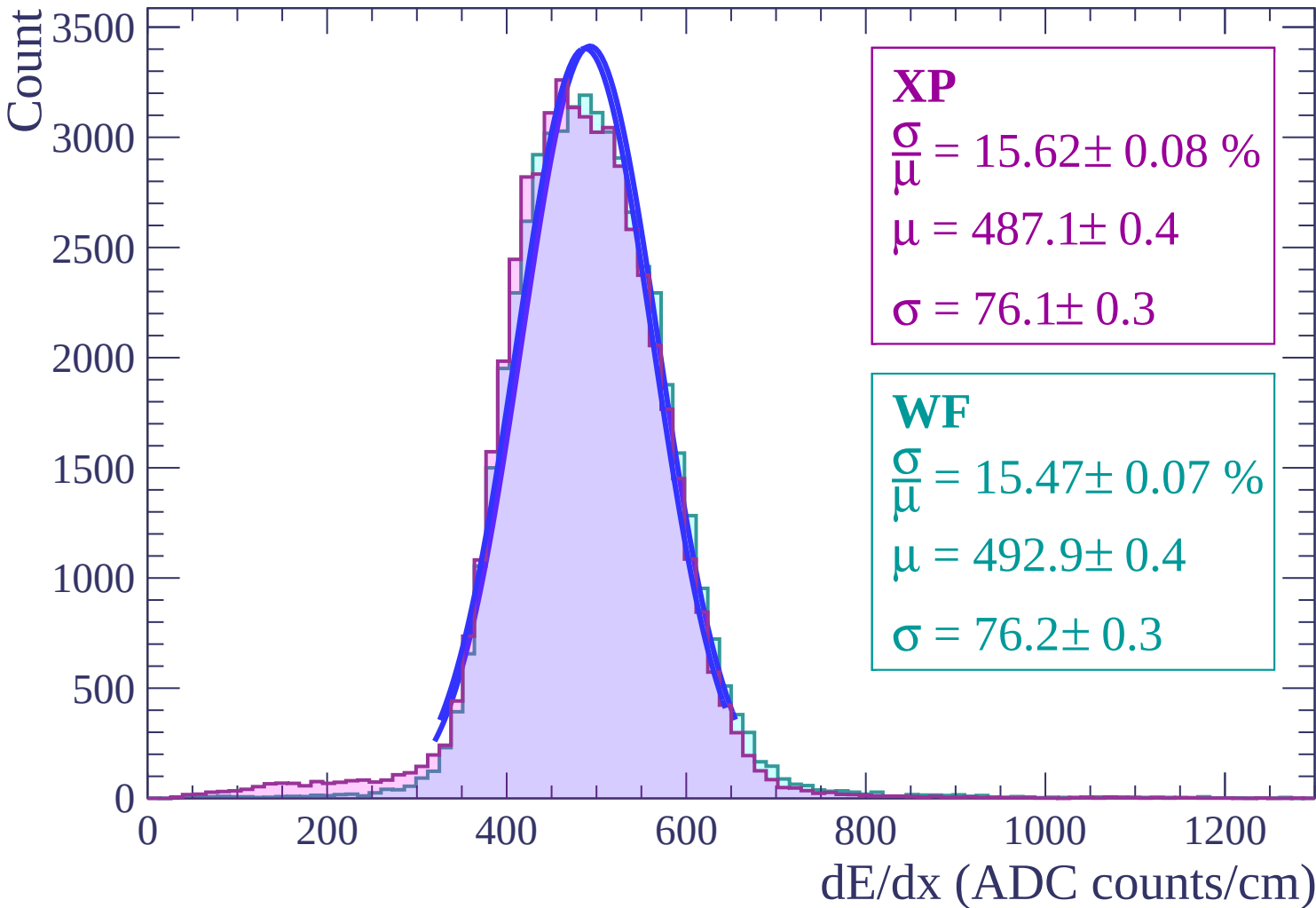
$$\frac{\sigma}{\mu} = 15.74 \pm 0.09 \%$$

$$\mu = 491.6 \pm 0.4$$

$$\sigma = 77.4 \pm 0.4$$

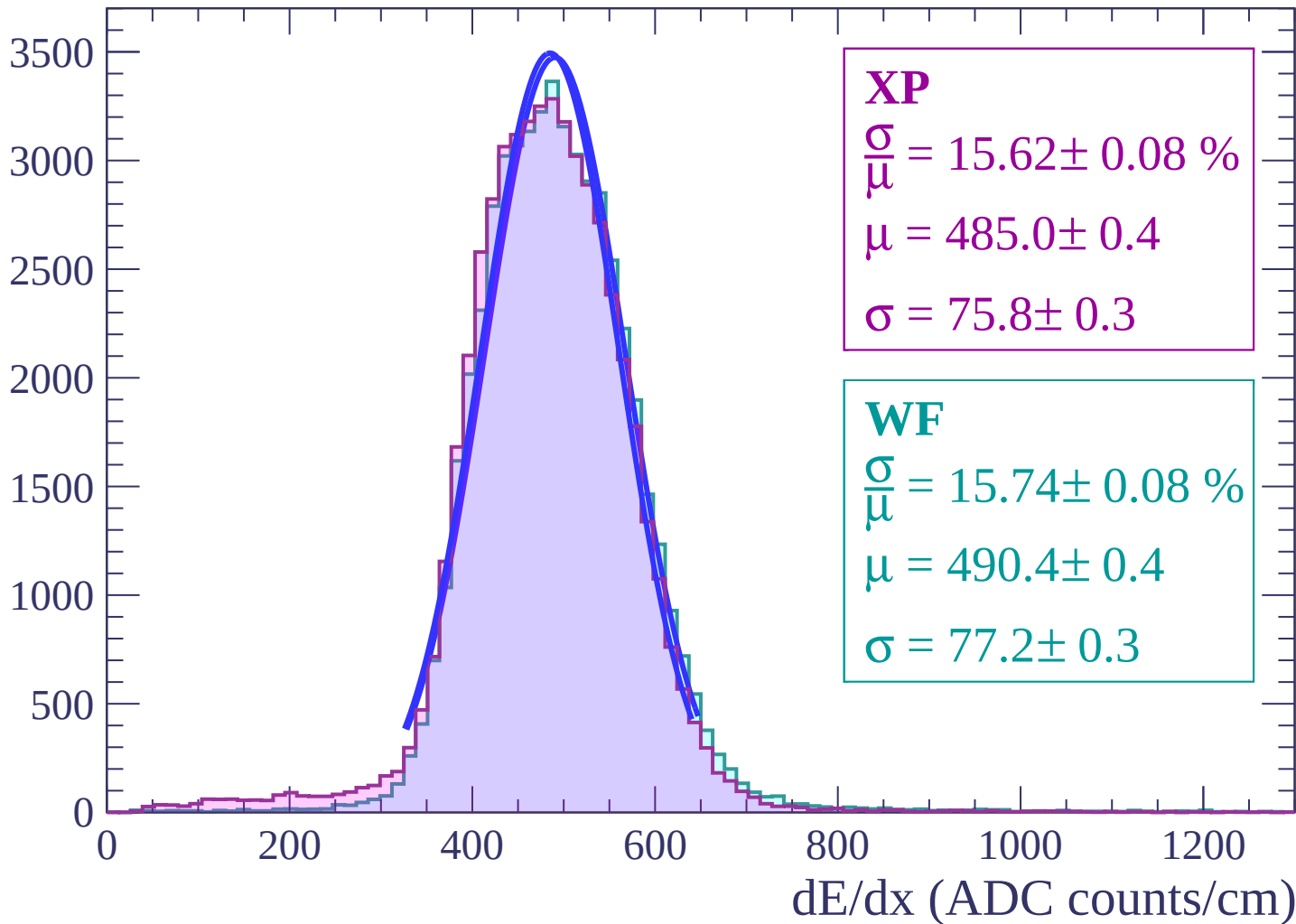


Energy loss | $-6 < \theta < -4$

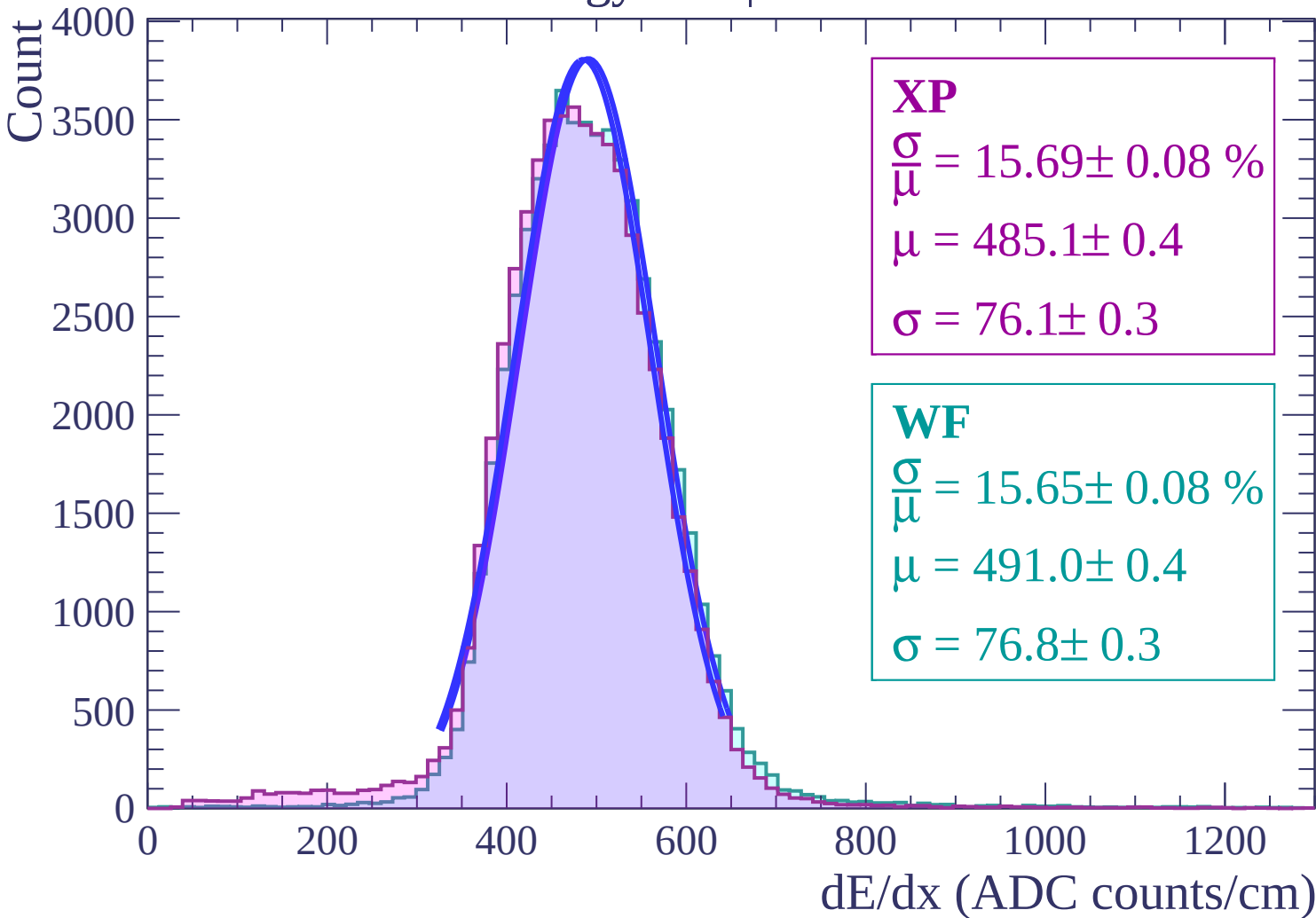


Energy loss | $-4 < \theta < -2$

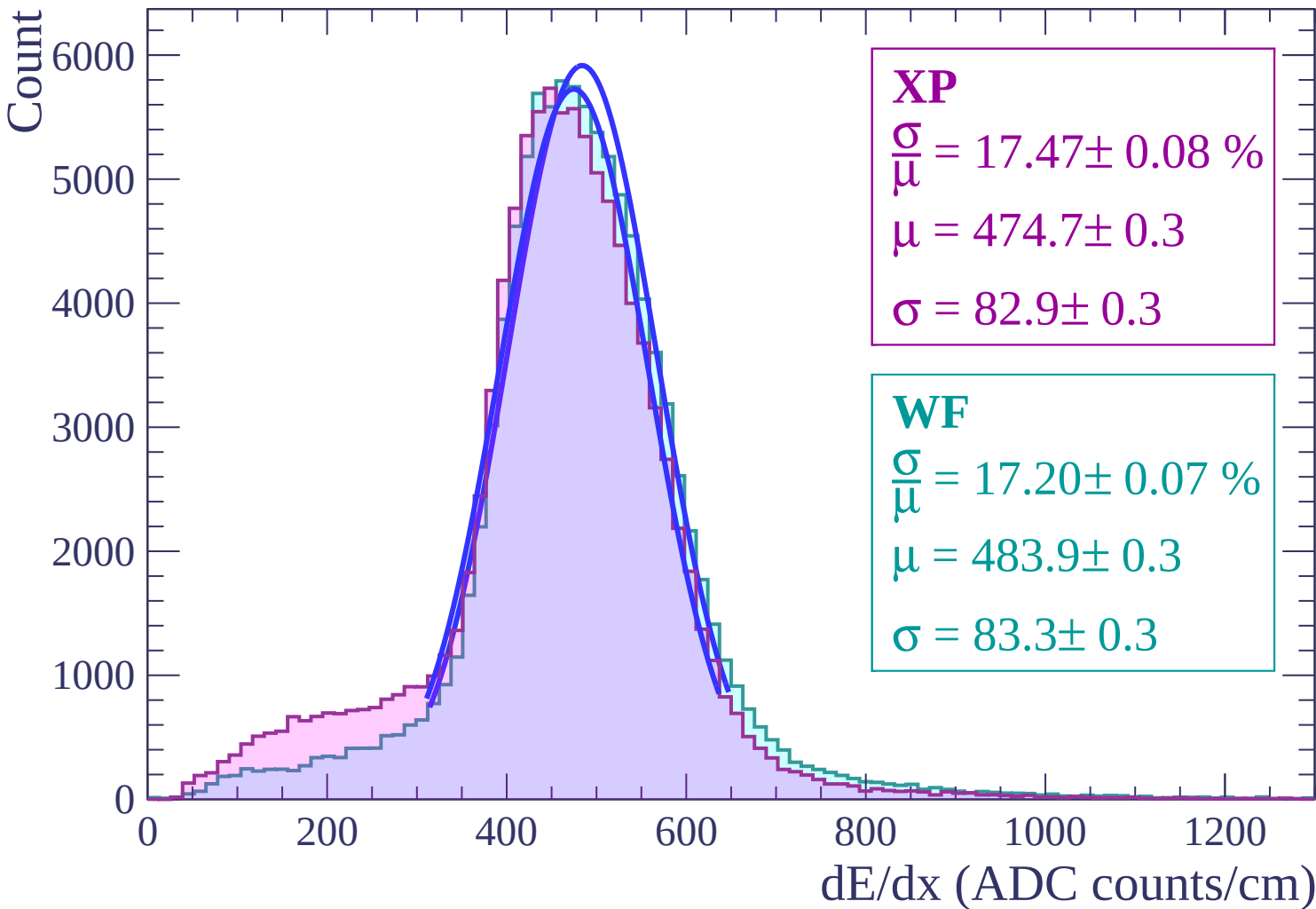
Count



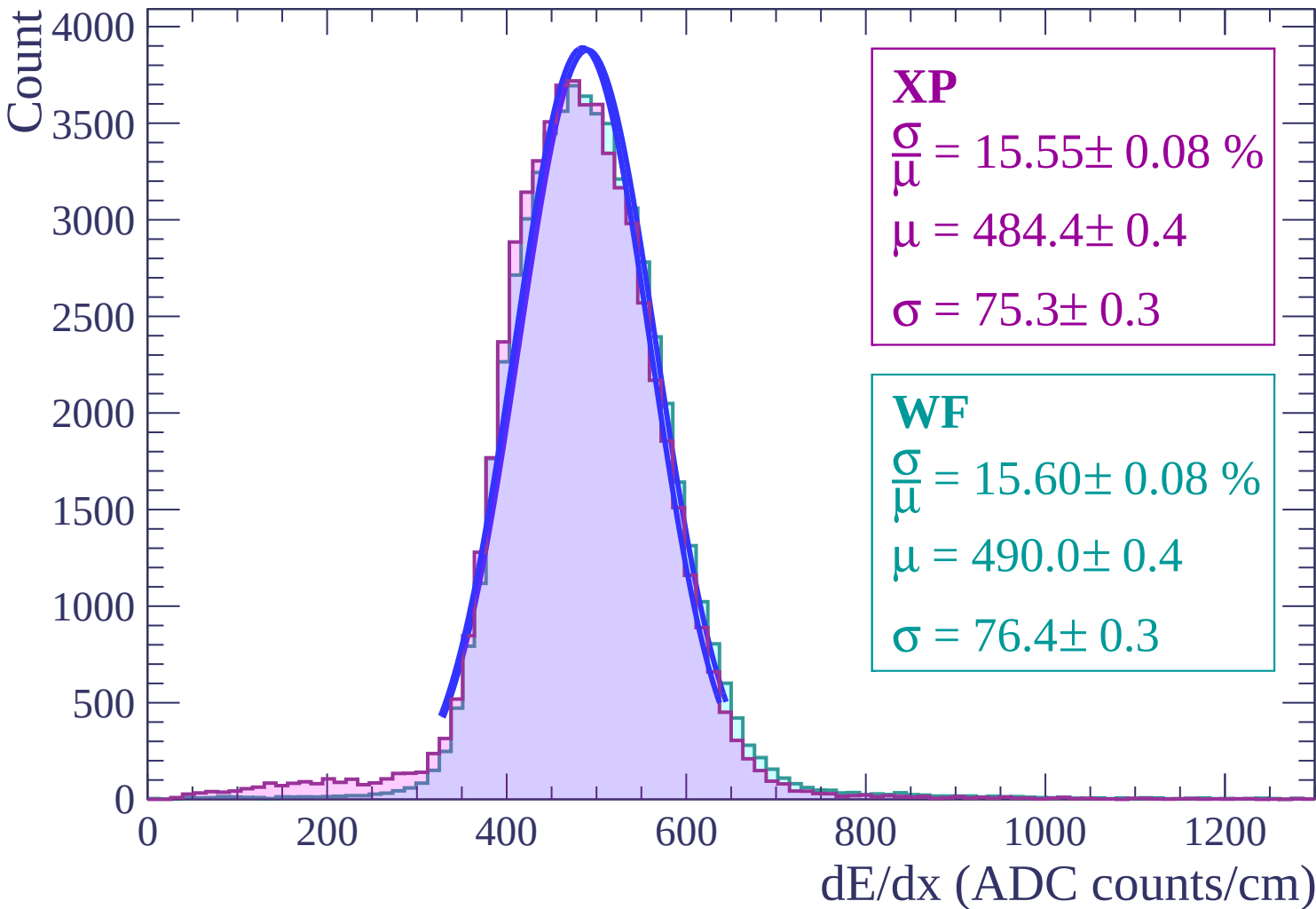
Energy loss | $-2 < \theta < 0$



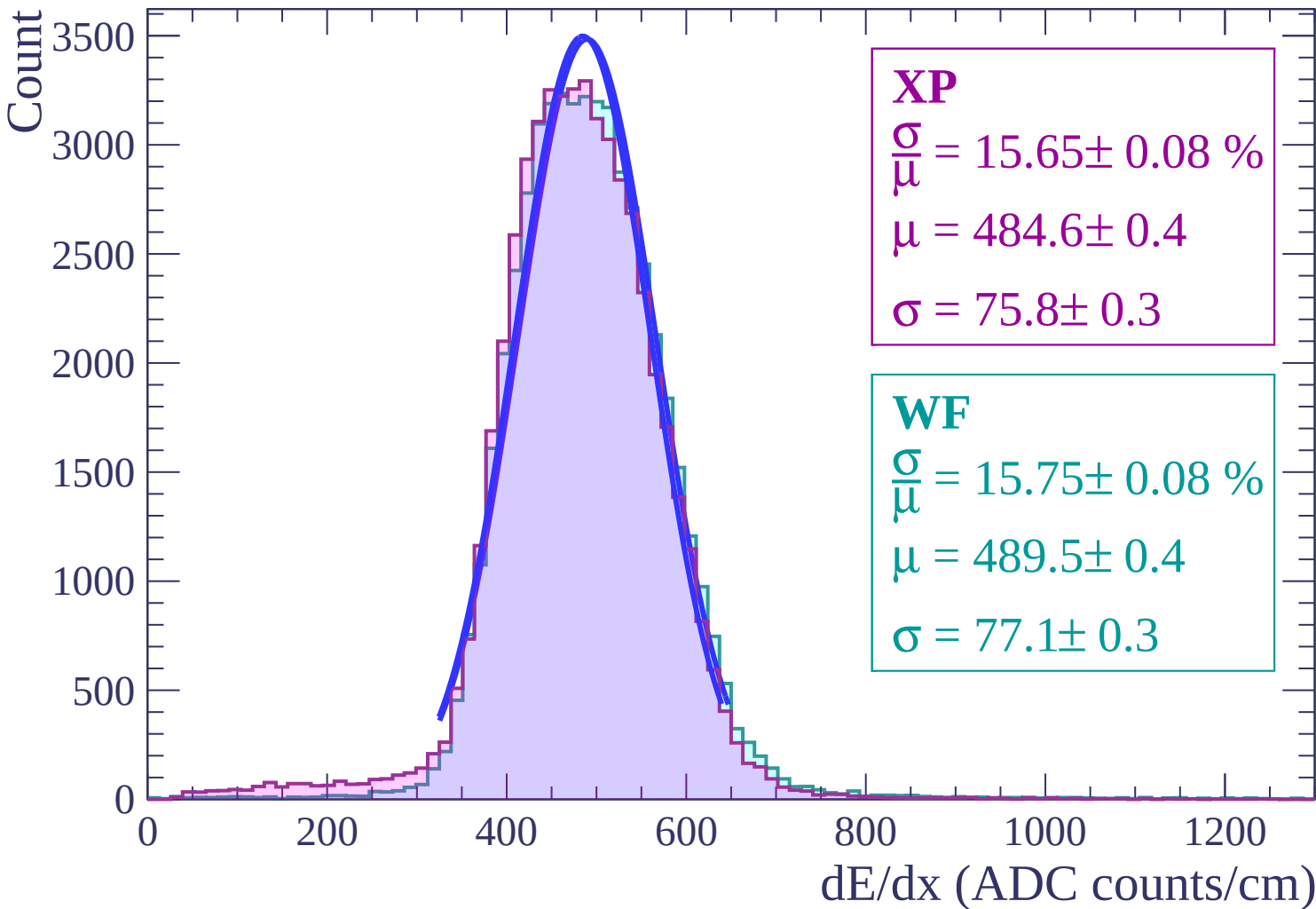
Energy loss | $0 < \theta < 2$



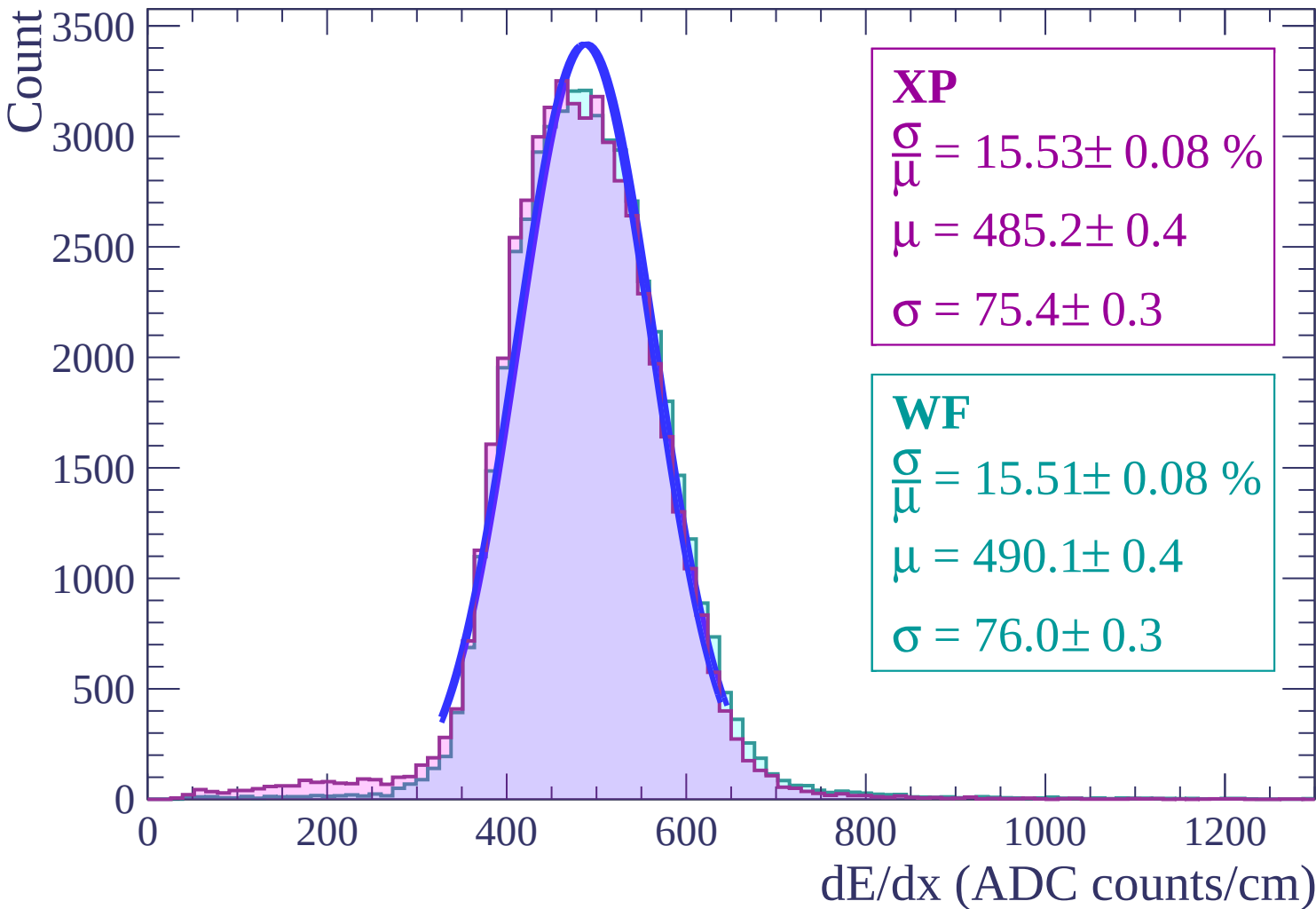
Energy loss | $2 < \theta < 4$



Energy loss | $4 < \theta < 6$

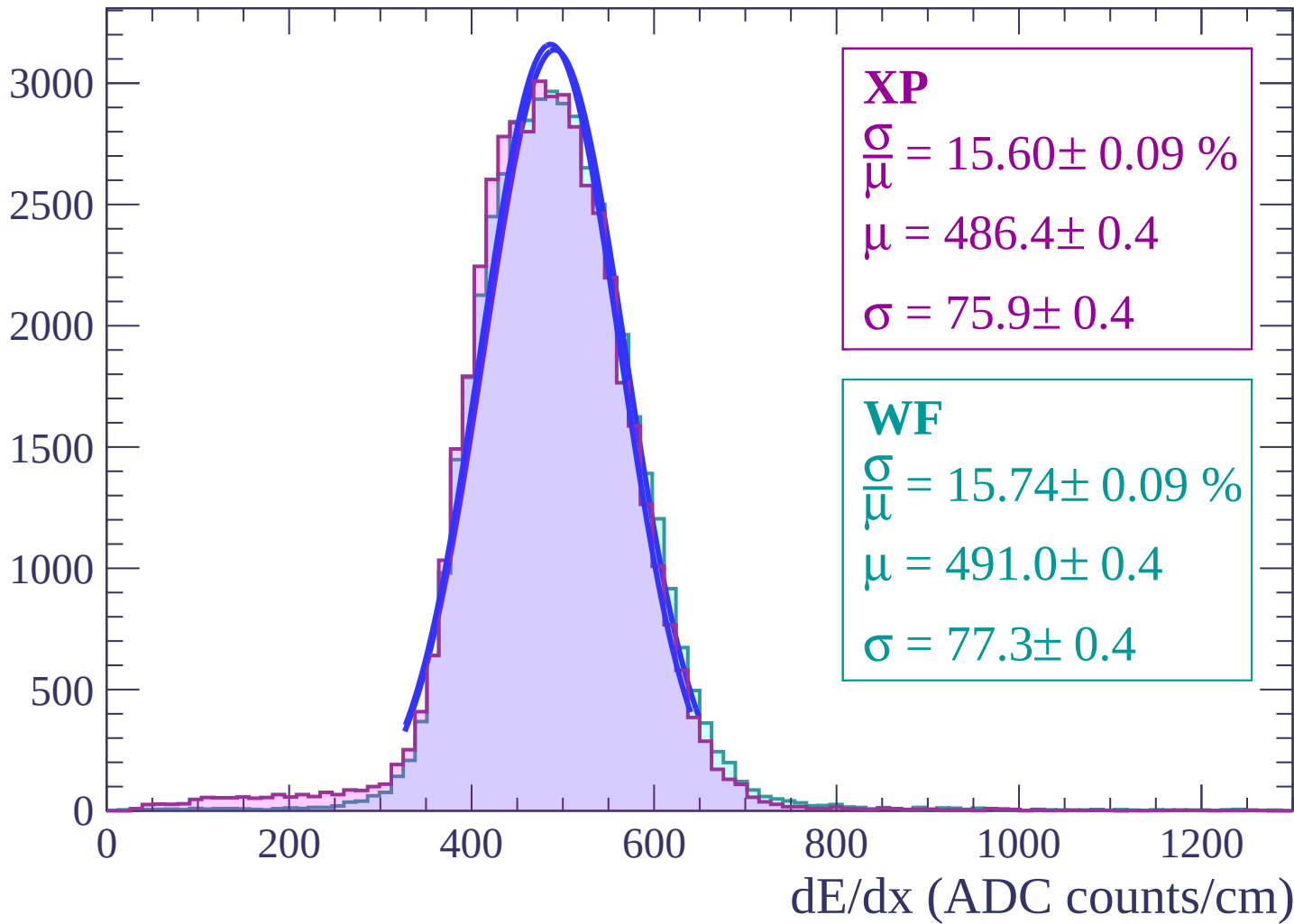


Energy loss | $6 < \theta < 8$

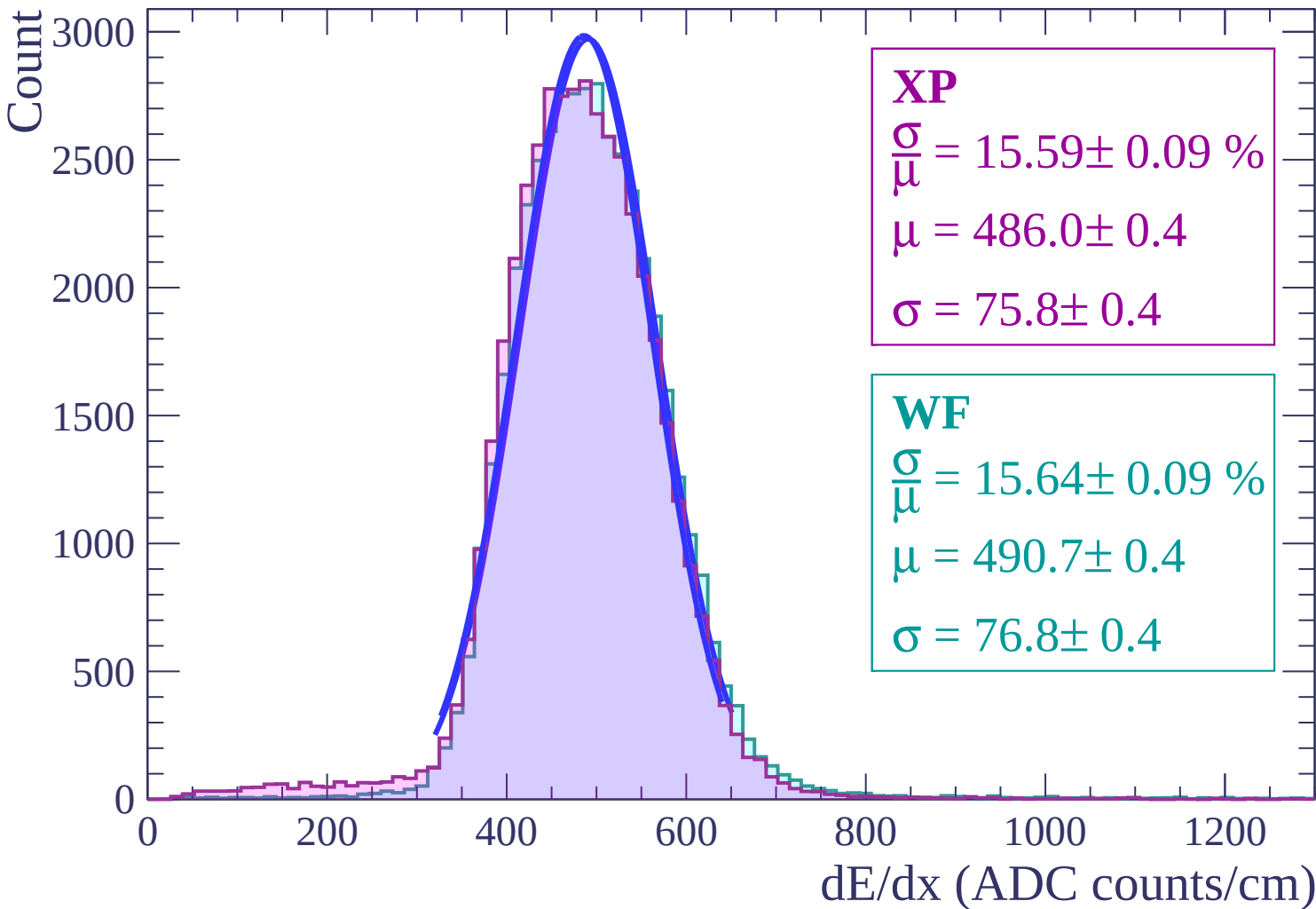


Energy loss | $8 < \theta < 10$

Count



Energy loss | $10 < \theta < 12$



Energy loss | $12 < \theta < 14$

Count

2500

2000

1500

1000

500

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 15.63 \pm 0.09 \%$$

$$\mu = 487.6 \pm 0.4$$

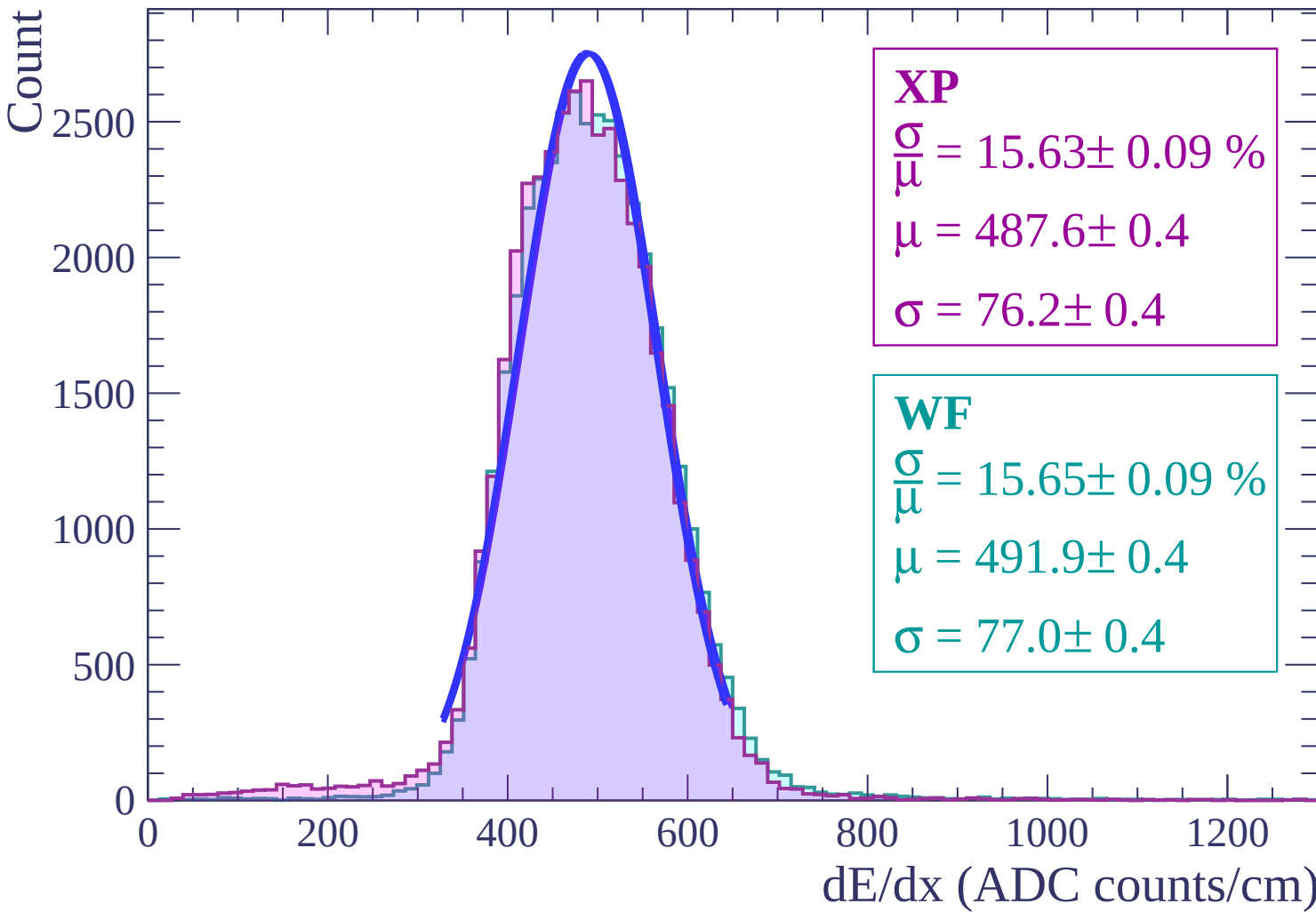
$$\sigma = 76.2 \pm 0.4$$

WF

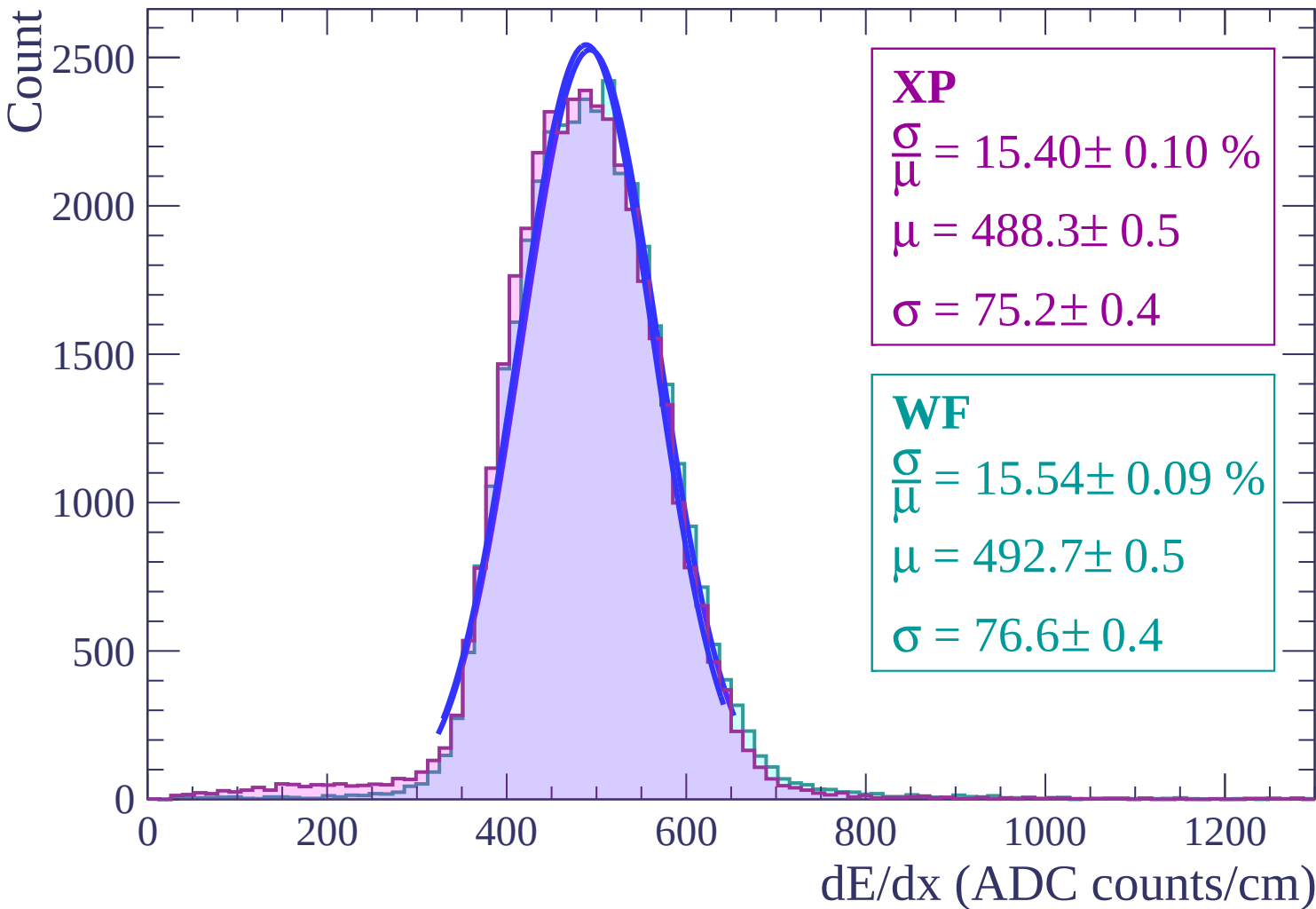
$$\frac{\sigma}{\mu} = 15.65 \pm 0.09 \%$$

$$\mu = 491.9 \pm 0.4$$

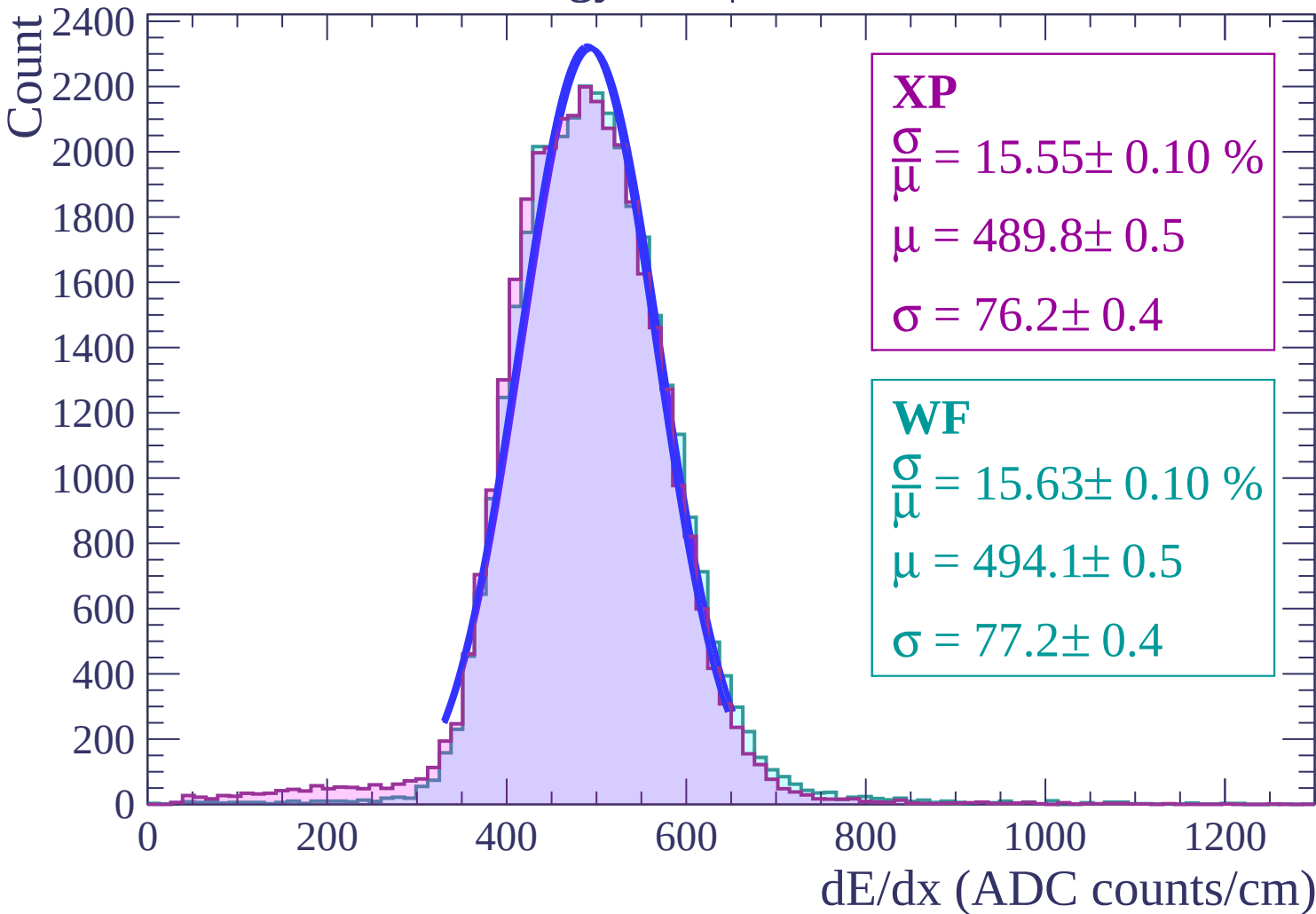
$$\sigma = 77.0 \pm 0.4$$



Energy loss | $14 < \theta < 16$

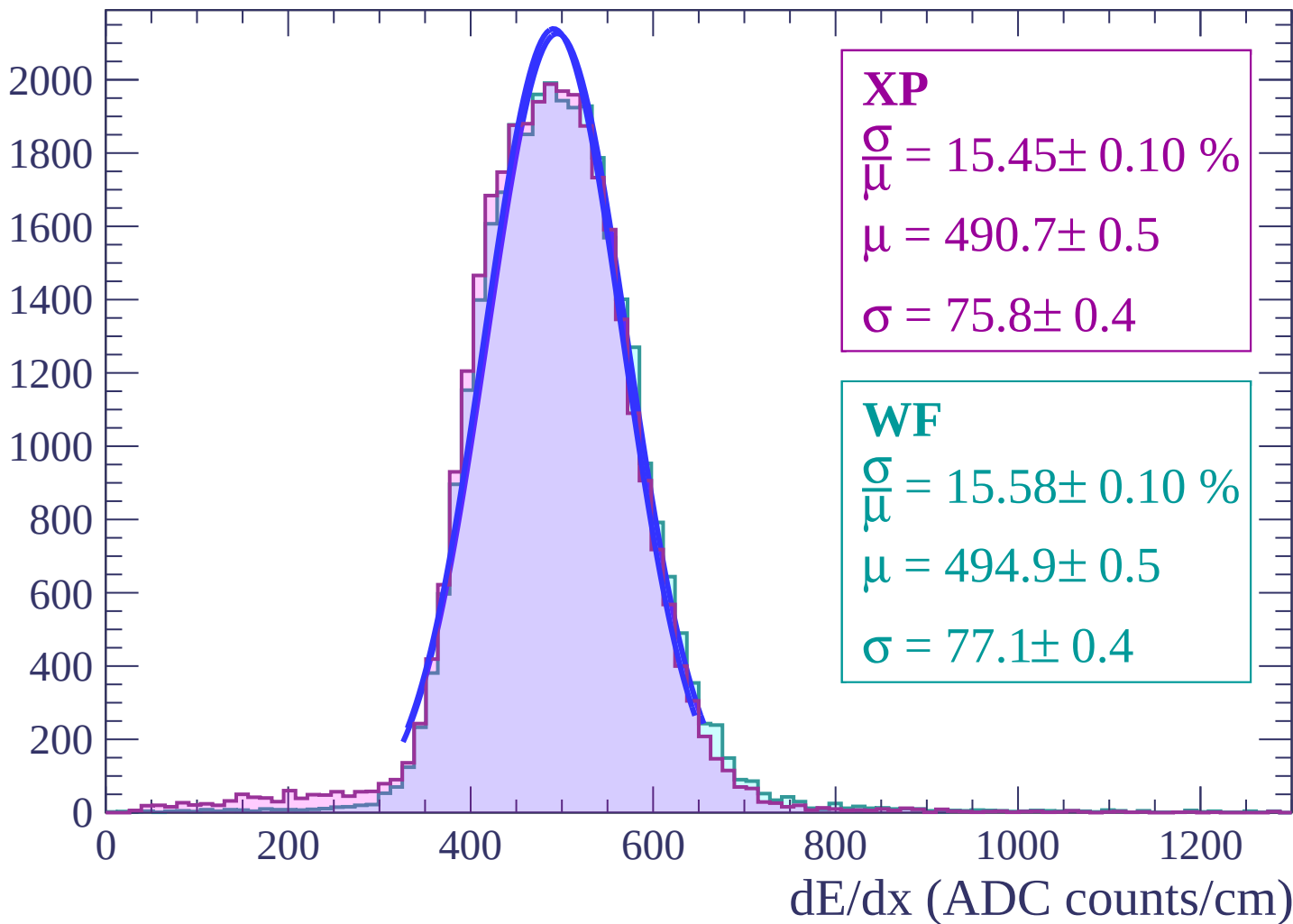


Energy loss | $16 < \theta < 18$

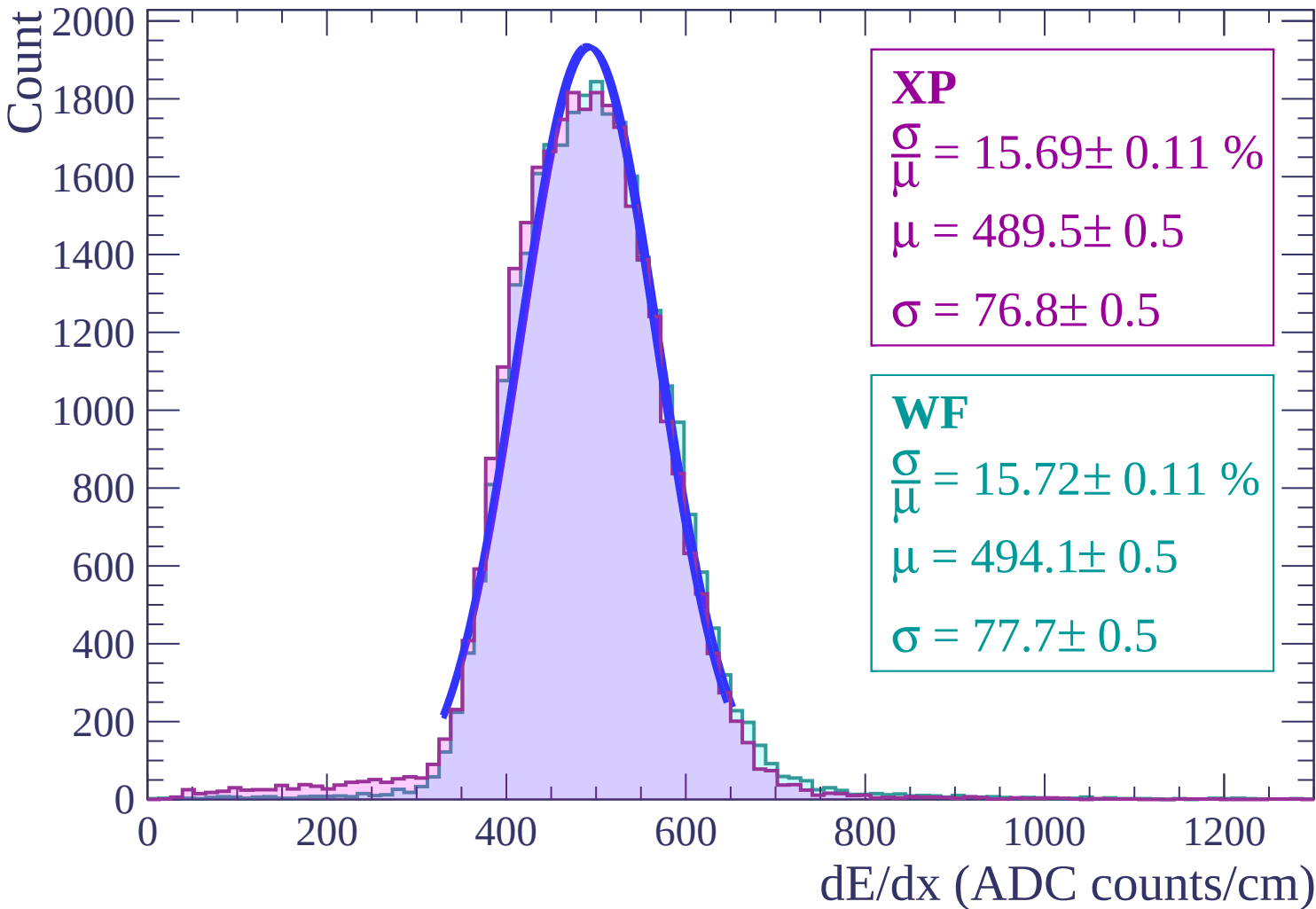


Energy loss | $18 < \theta < 20$

Count



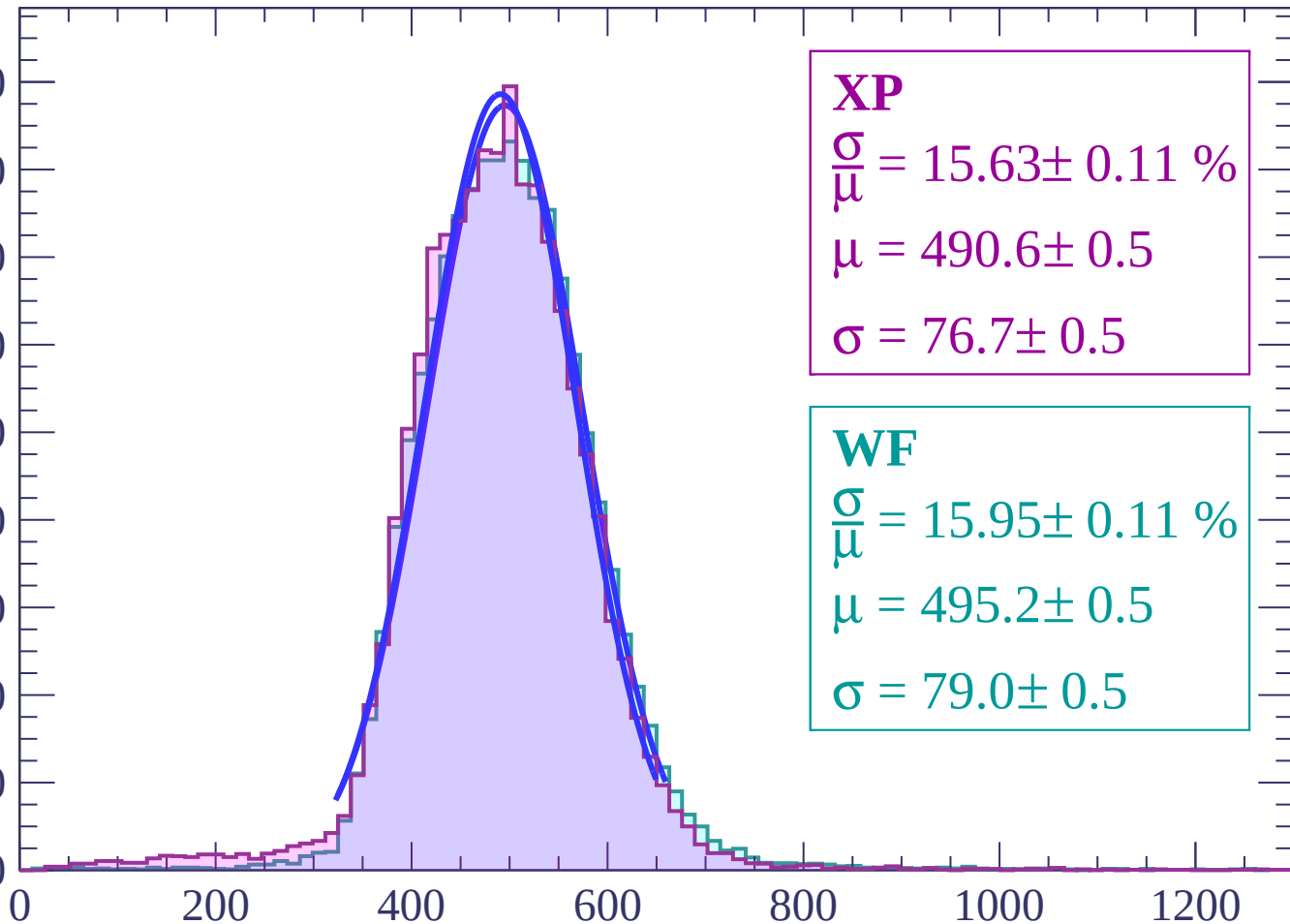
Energy loss | $20 < \theta < 22$



Energy loss | $22 < \theta < 24$

Count

1800
1600
1400
1200
1000
800
600
400
200
0



XP

$$\frac{\sigma}{\mu} = 15.63 \pm 0.11 \%$$

$$\mu = 490.6 \pm 0.5$$

$$\sigma = 76.7 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 15.95 \pm 0.11 \%$$

$$\mu = 495.2 \pm 0.5$$

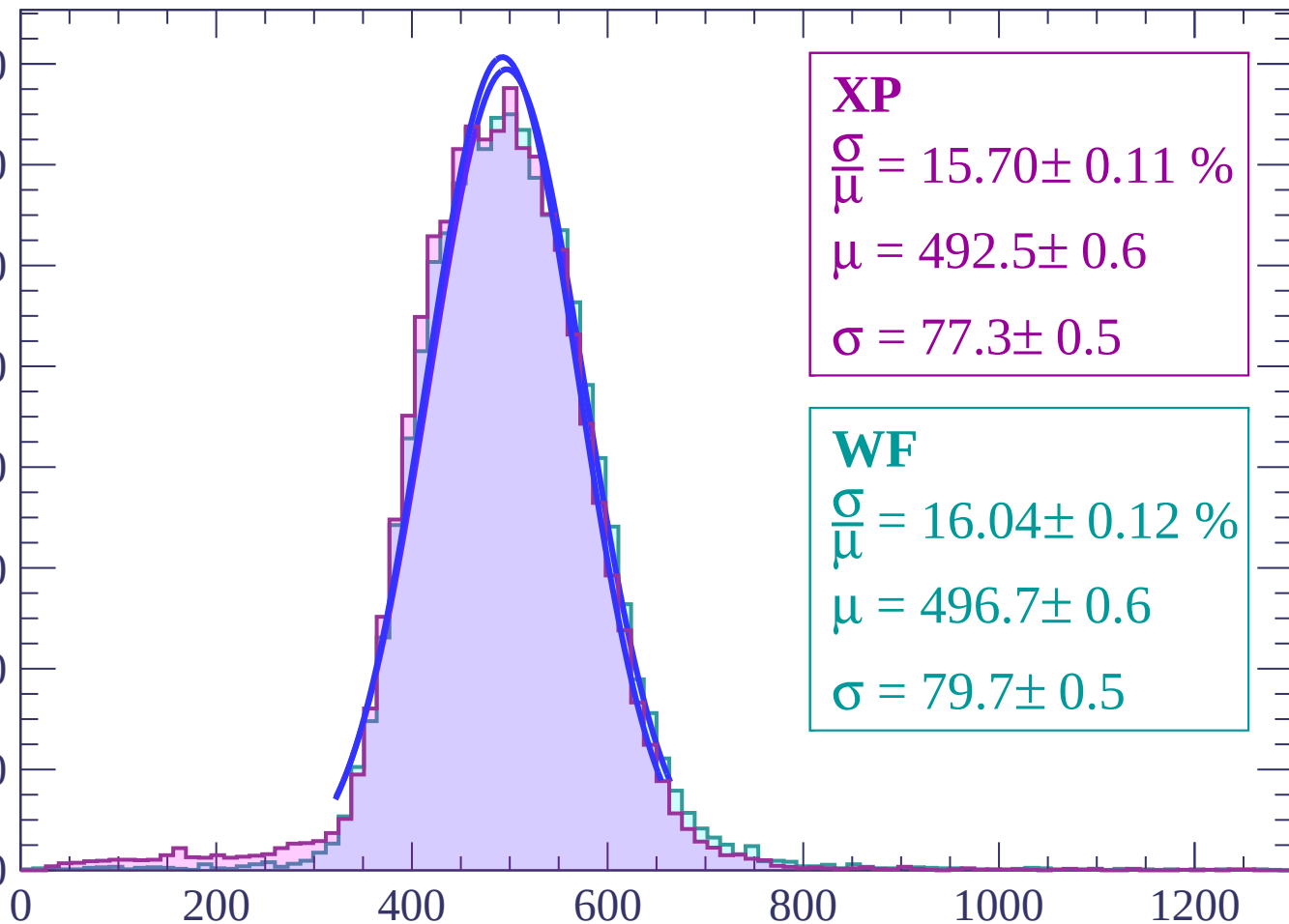
$$\sigma = 79.0 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $24 < \theta < 26$

Count

1600
1400
1200
1000
800
600
400
200
0



XP

$$\frac{\sigma}{\mu} = 15.70 \pm 0.11 \%$$

$$\mu = 492.5 \pm 0.6$$

$$\sigma = 77.3 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 16.04 \pm 0.12 \%$$

$$\mu = 496.7 \pm 0.6$$

$$\sigma = 79.7 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $26 < \theta < 28$

Count

1400

1200

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 15.98 \pm 0.12 \%$$

$$\mu = 493.2 \pm 0.6$$

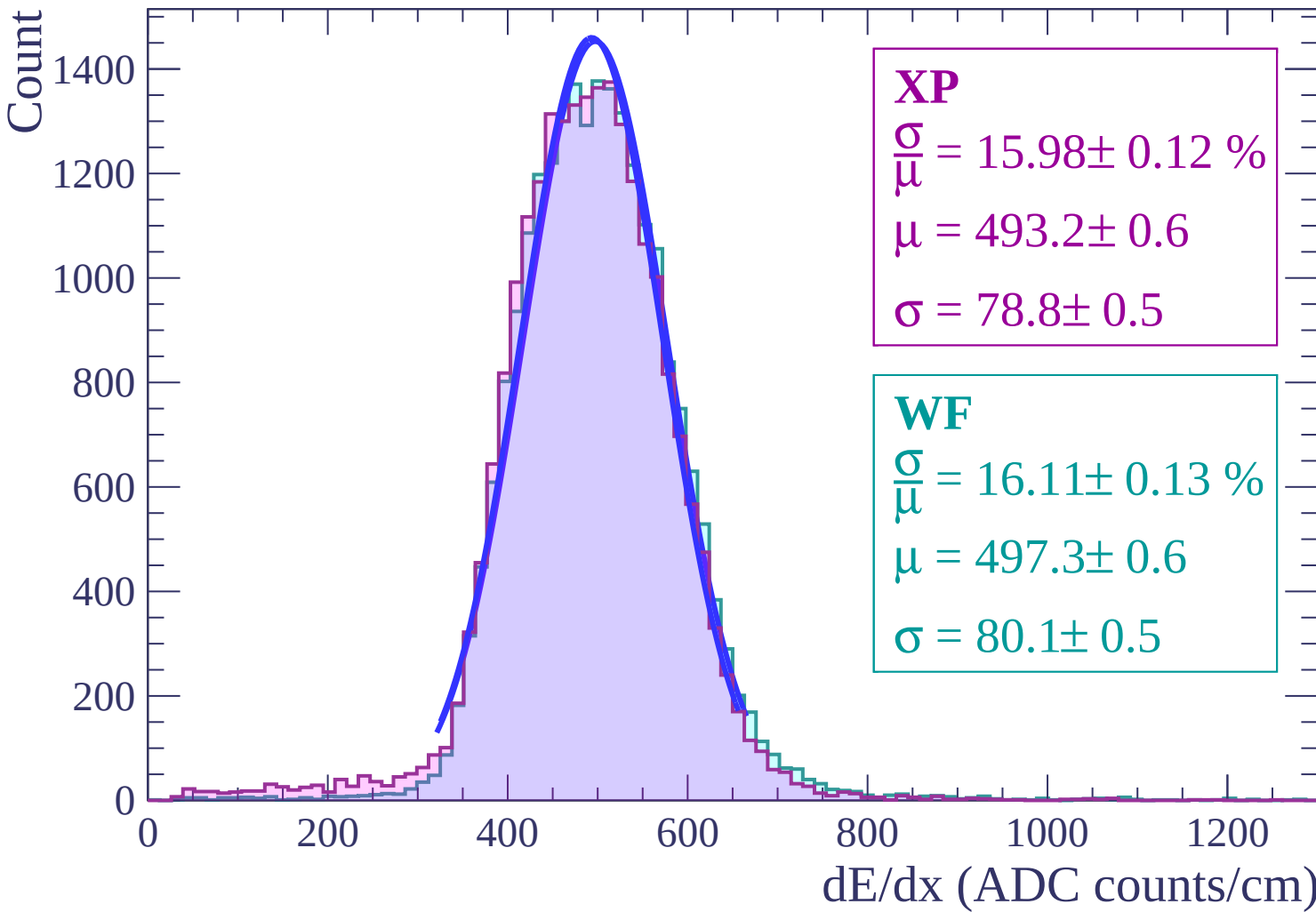
$$\sigma = 78.8 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 16.11 \pm 0.13 \%$$

$$\mu = 497.3 \pm 0.6$$

$$\sigma = 80.1 \pm 0.5$$



Energy loss | $28 < \theta < 30$

Count

1200
1000
800
600
400
200
0

XP

$$\frac{\sigma}{\mu} = 15.95 \pm 0.14 \%$$

$$\mu = 494.1 \pm 0.7$$

$$\sigma = 78.8 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 16.21 \pm 0.14 \%$$

$$\mu = 498.6 \pm 0.7$$

$$\sigma = 80.8 \pm 0.6$$

0

200

400

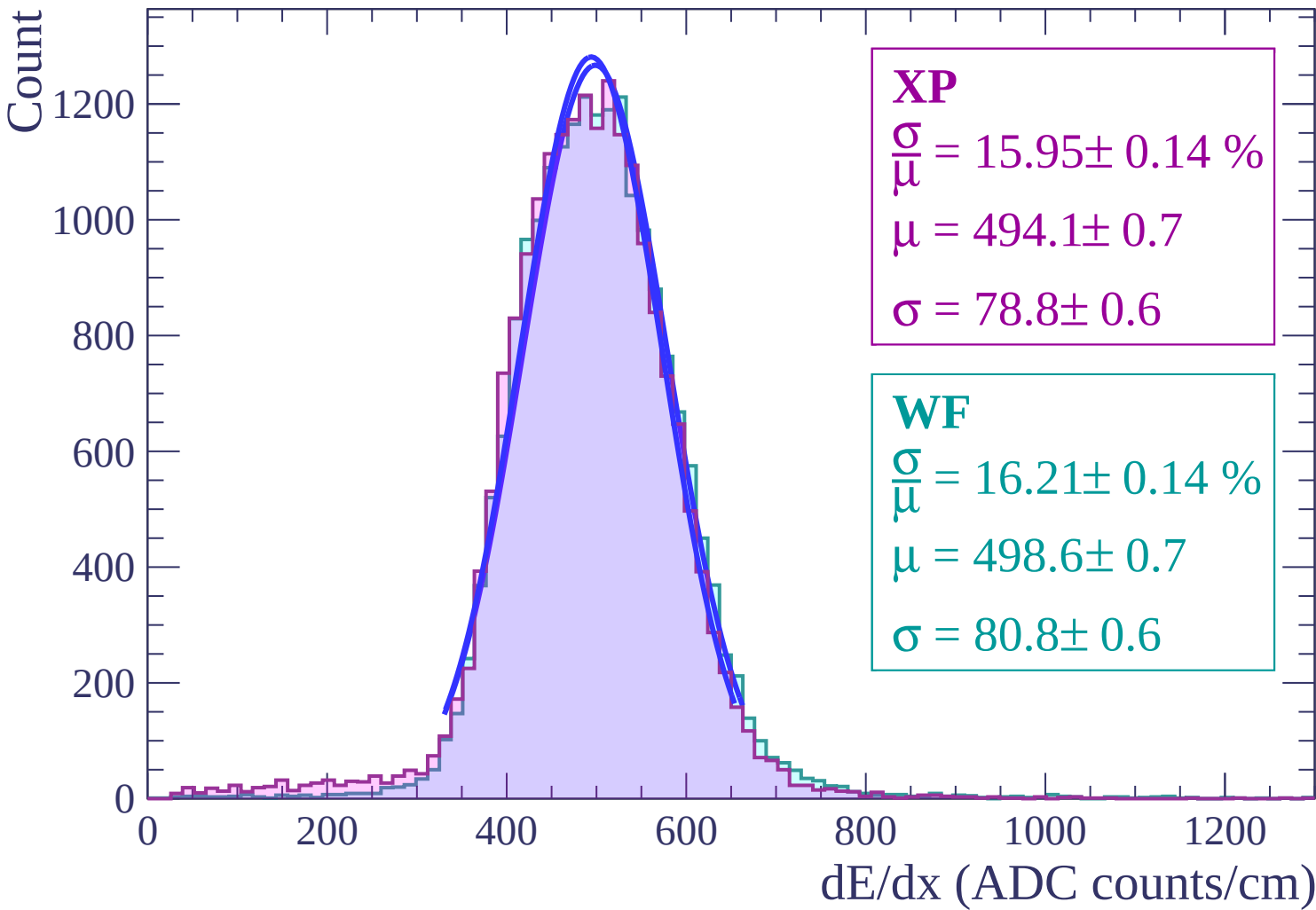
600

800

1000

1200

dE/dx (ADC counts/cm)



Energy loss | $30 < \theta < 32$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 16.29 \pm 0.16 \%$$

$$\mu = 497.9 \pm 0.8$$

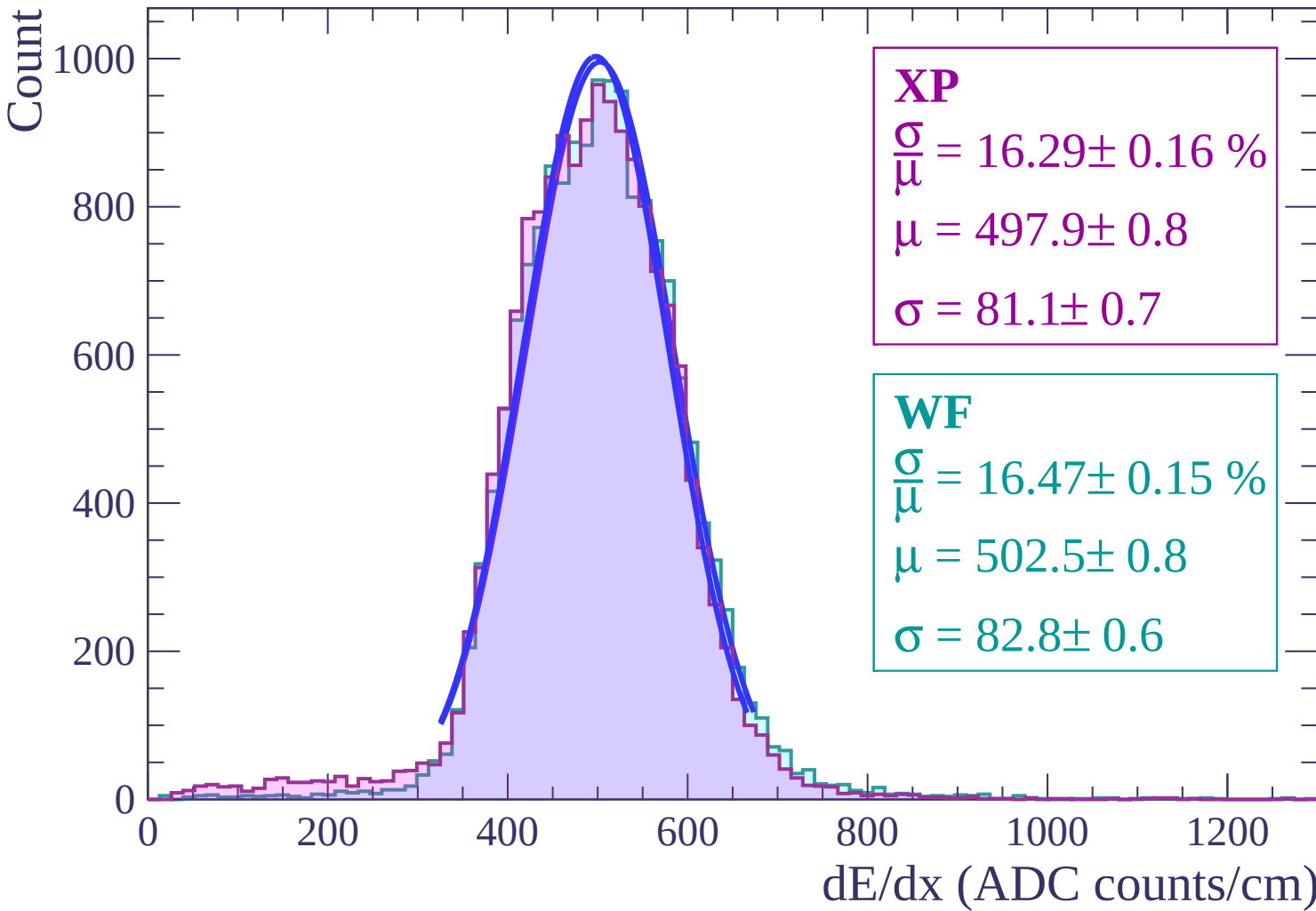
$$\sigma = 81.1 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 16.47 \pm 0.15 \%$$

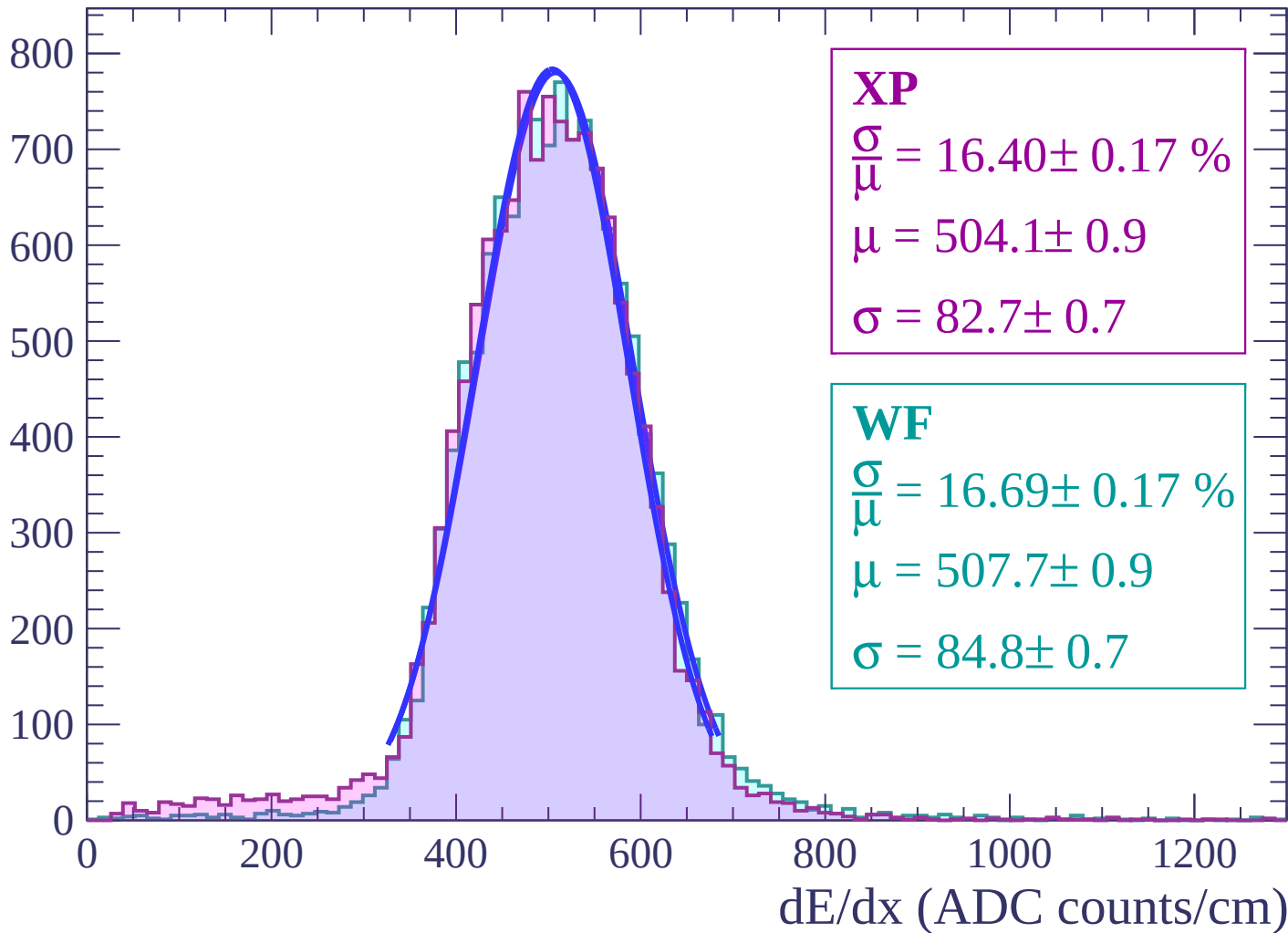
$$\mu = 502.5 \pm 0.8$$

$$\sigma = 82.8 \pm 0.6$$

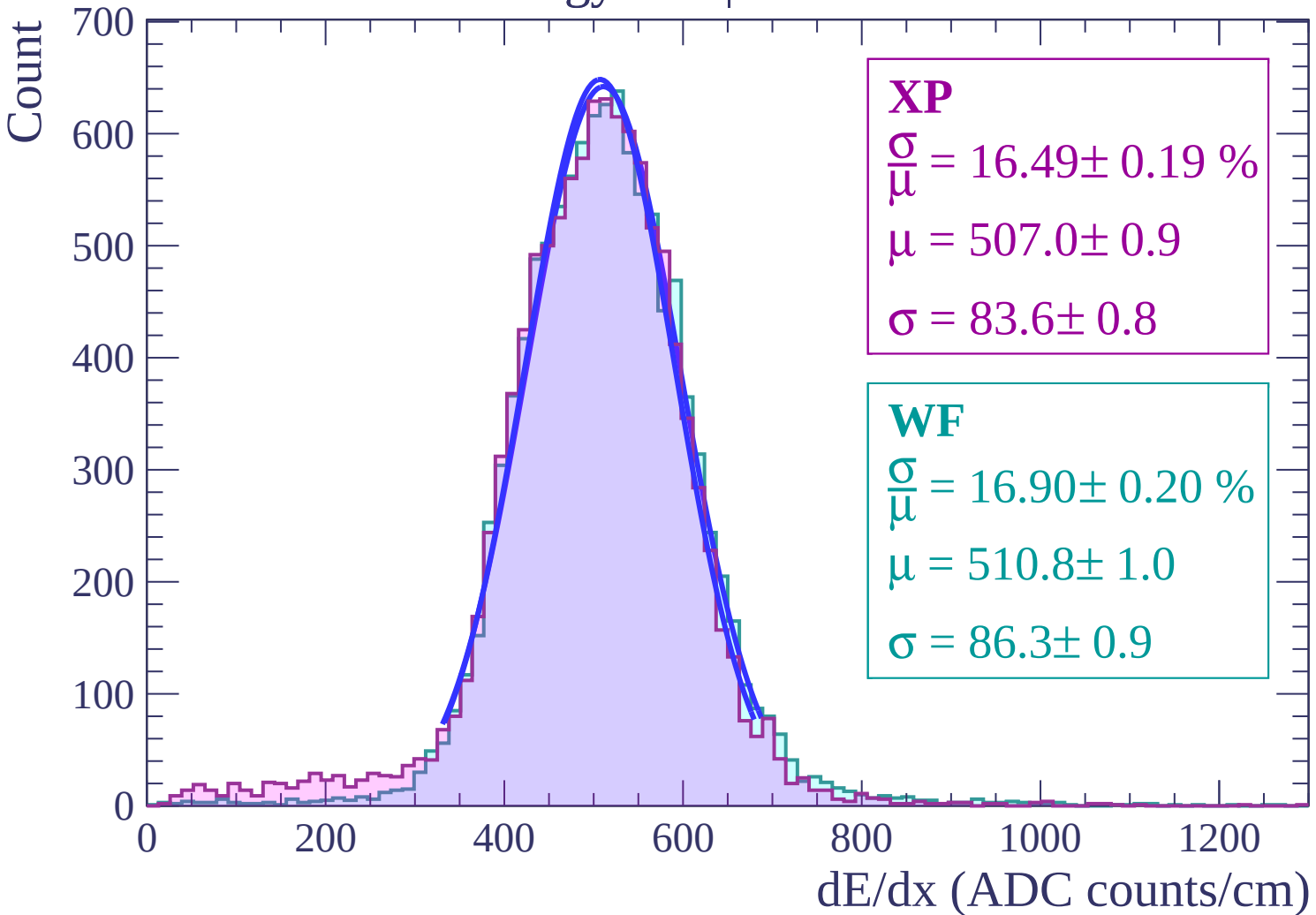


Energy loss | $32 < \theta < 34$

Count

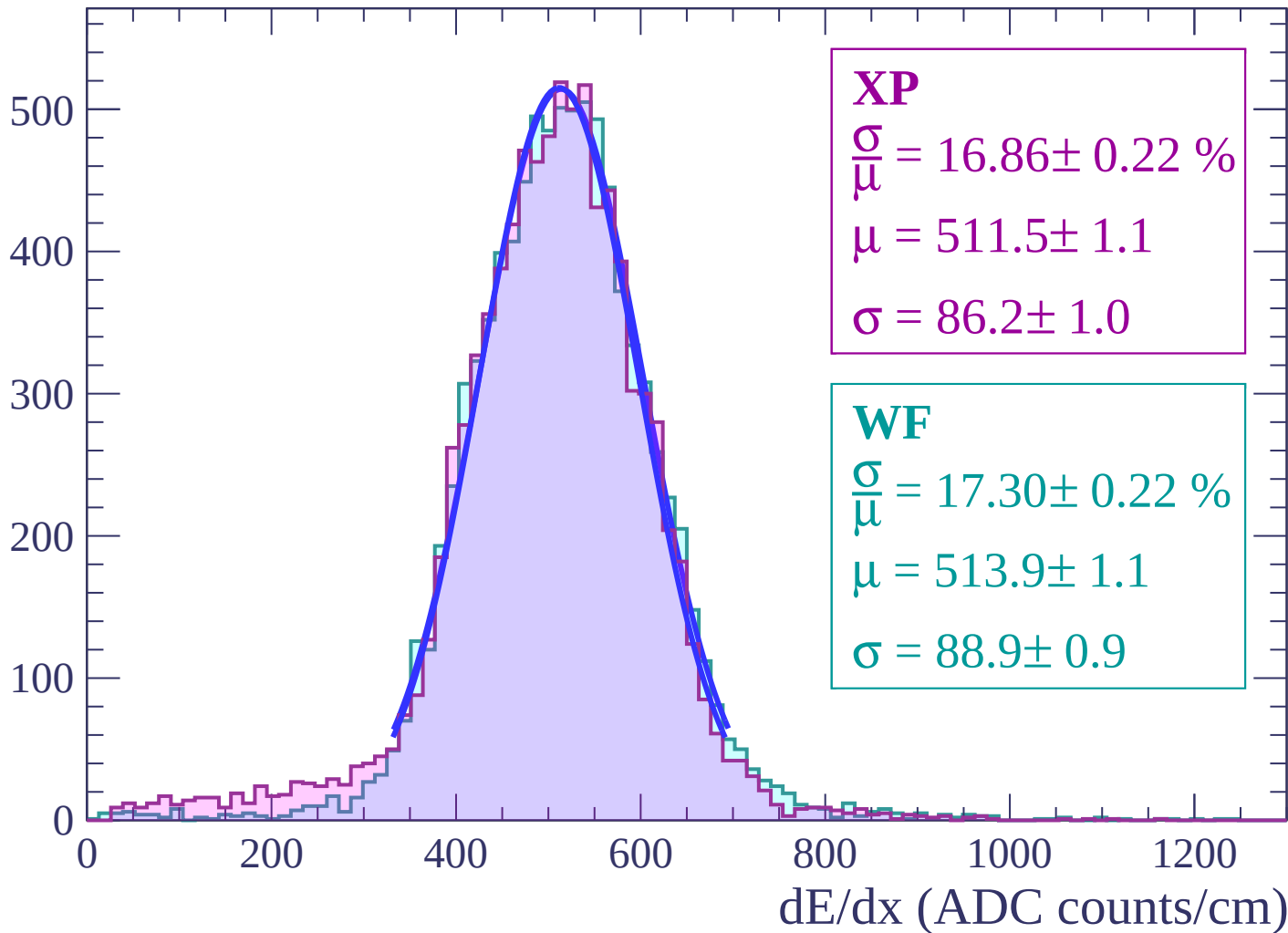


Energy loss | $34 < \theta < 36$



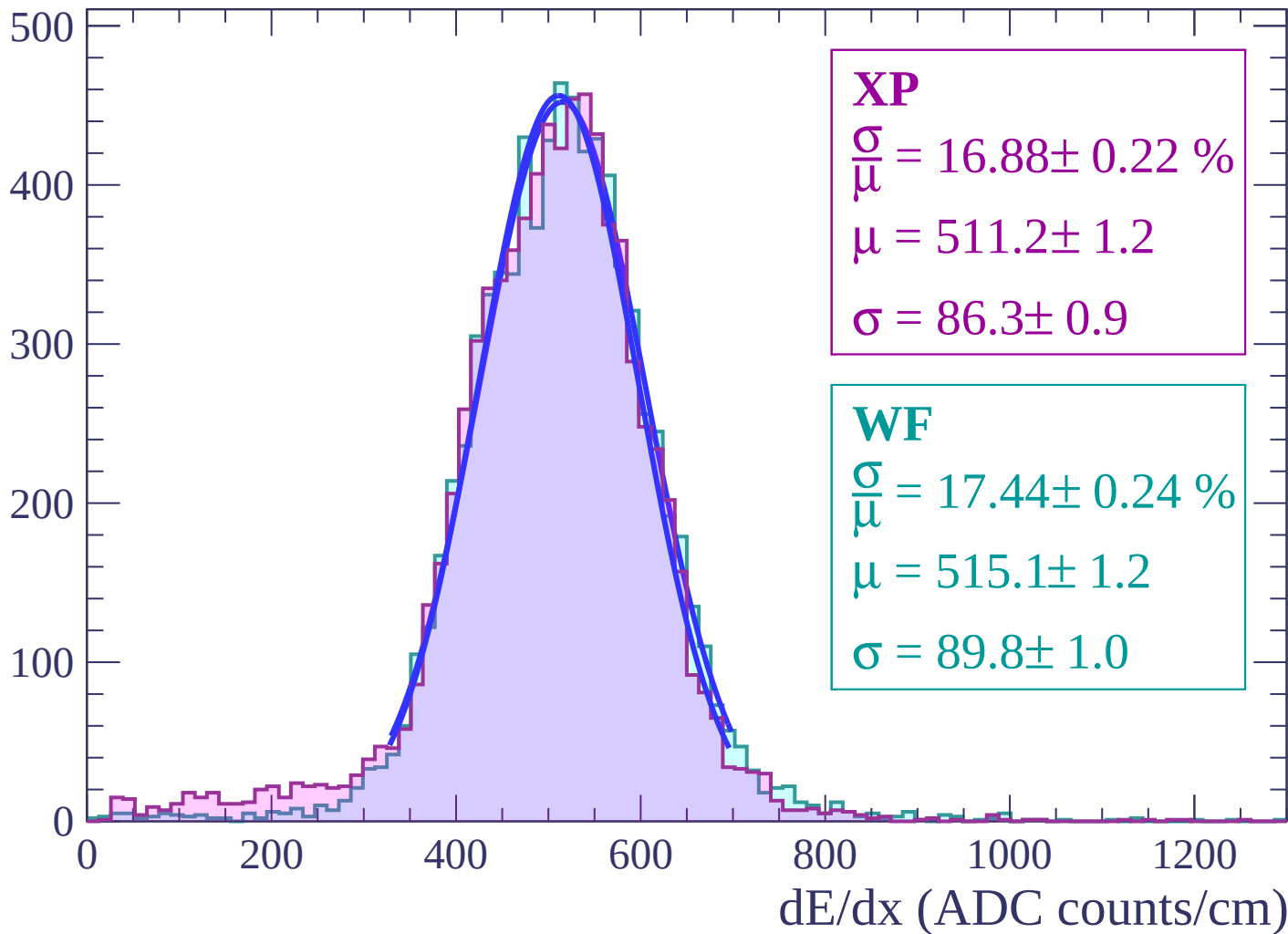
Energy loss | $36 < \theta < 38$

Count



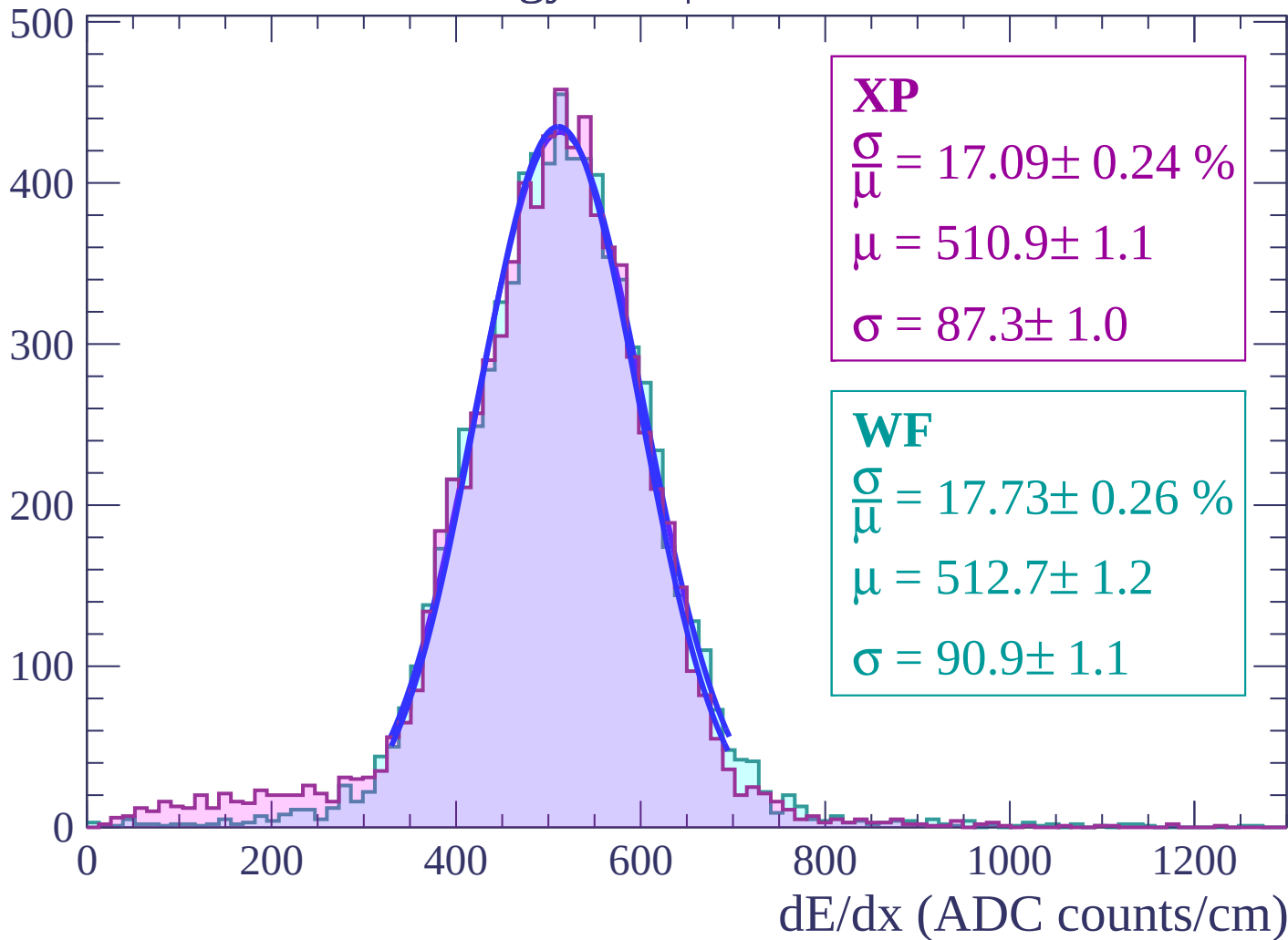
Energy loss | $38 < \theta < 40$

Count



Energy loss | $40 < \theta < 42$

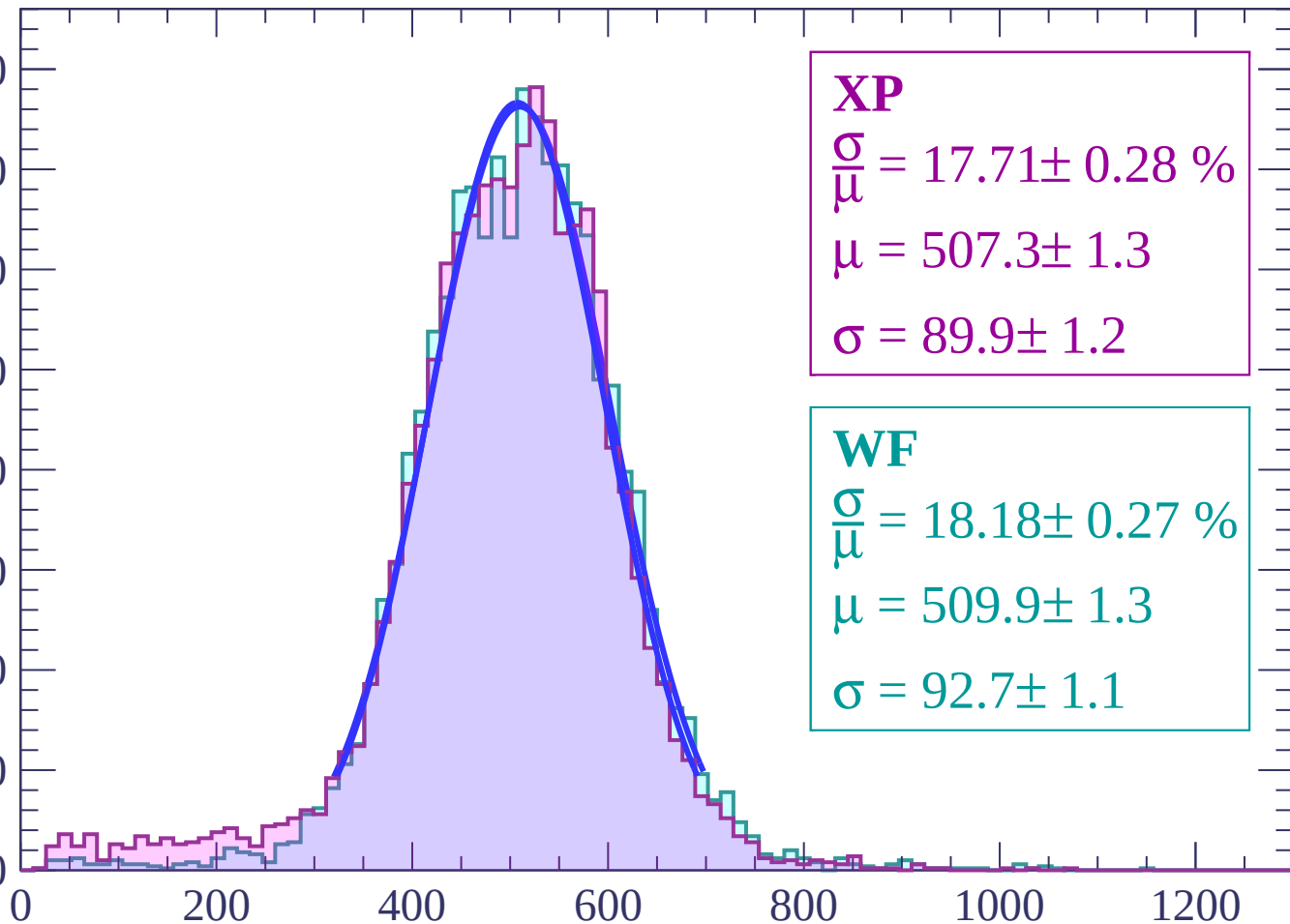
Count



Energy loss | $42 < \theta < 44$

Count

400
350
300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 17.71 \pm 0.28 \%$$

$$\mu = 507.3 \pm 1.3$$

$$\sigma = 89.9 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 18.18 \pm 0.27 \%$$

$$\mu = 509.9 \pm 1.3$$

$$\sigma = 92.7 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | $44 < \theta < 46$

Count

400
350
300
250
200
150
100
50
0

XP

$$\frac{\sigma}{\mu} = 17.58 \pm 0.30 \%$$

$$\mu = 504.9 \pm 1.3$$

$$\sigma = 88.8 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 18.07 \pm 0.30 \%$$

$$\mu = 505.5 \pm 1.3$$

$$\sigma = 91.3 \pm 1.3$$

0

200

400

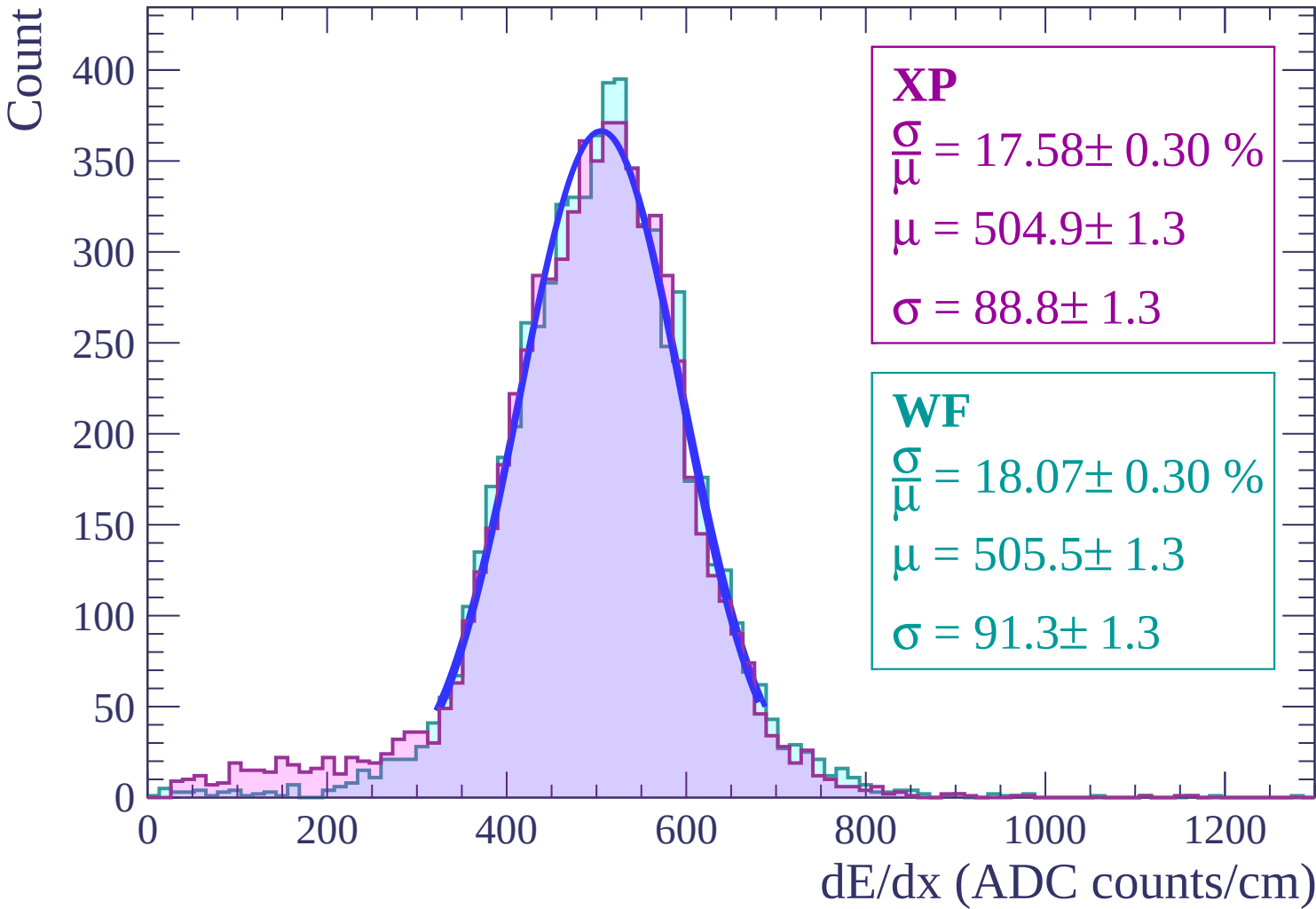
600

800

1000

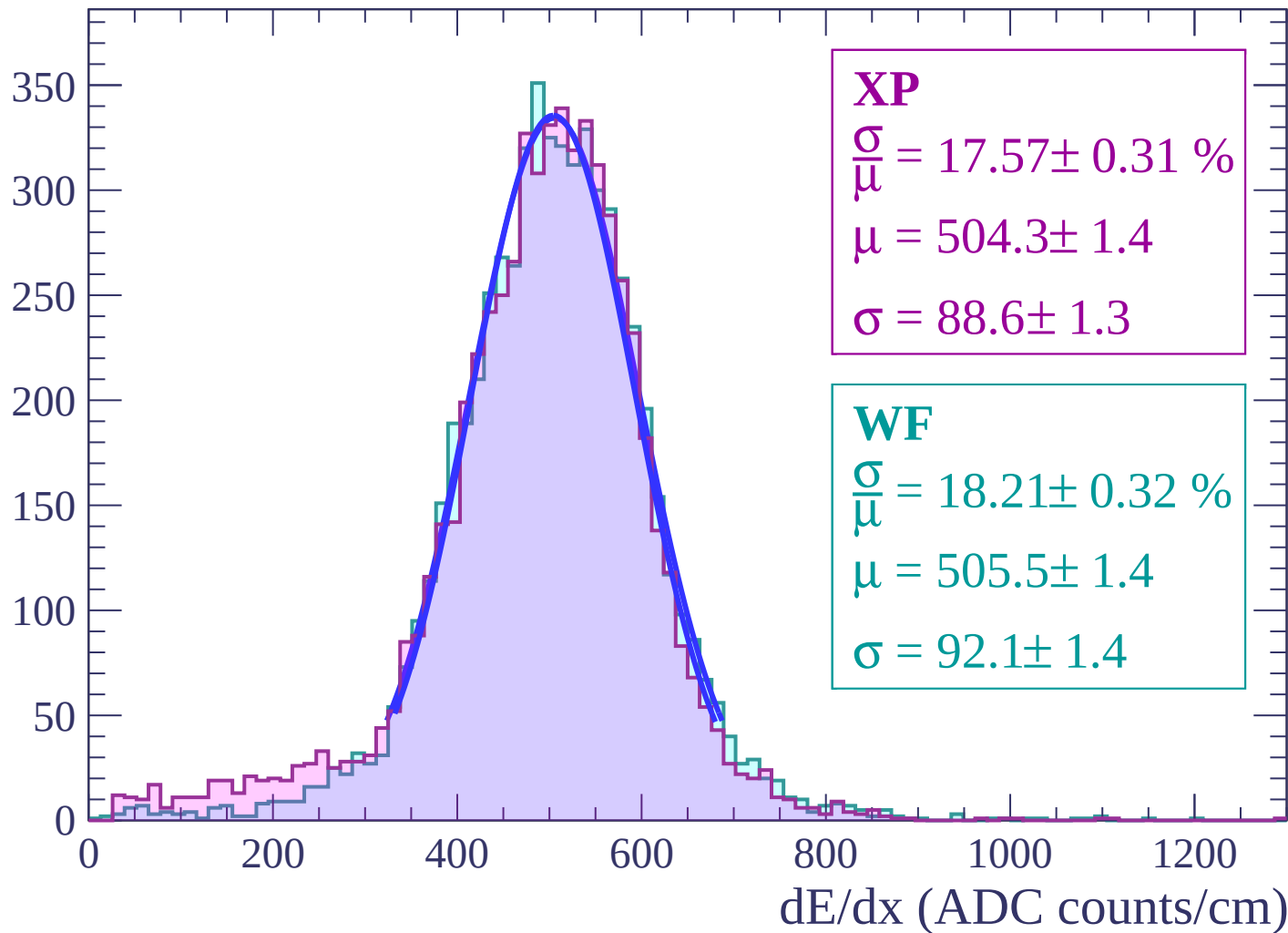
1200

dE/dx (ADC counts/cm)



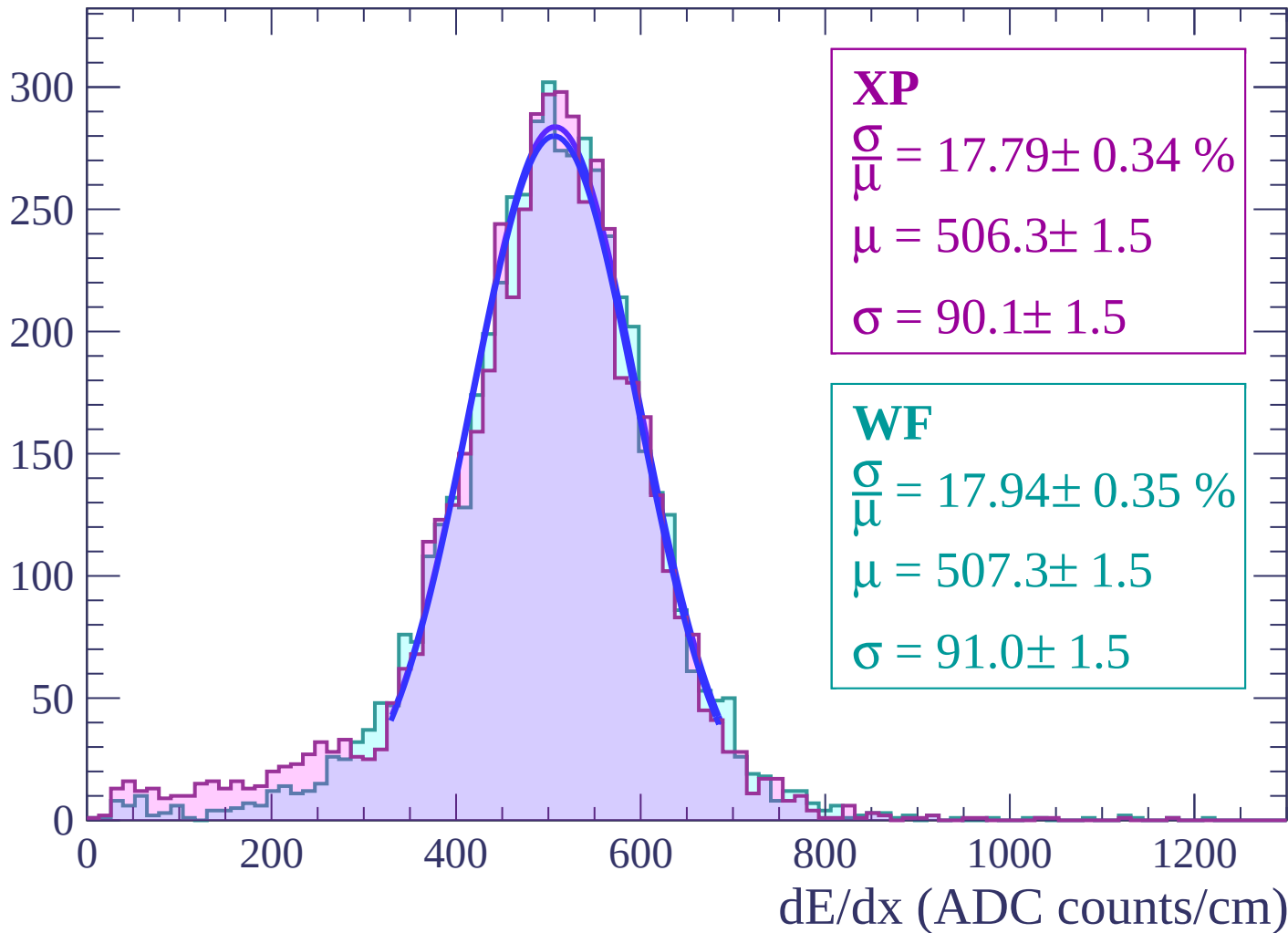
Energy loss | $46 < \theta < 48$

Count



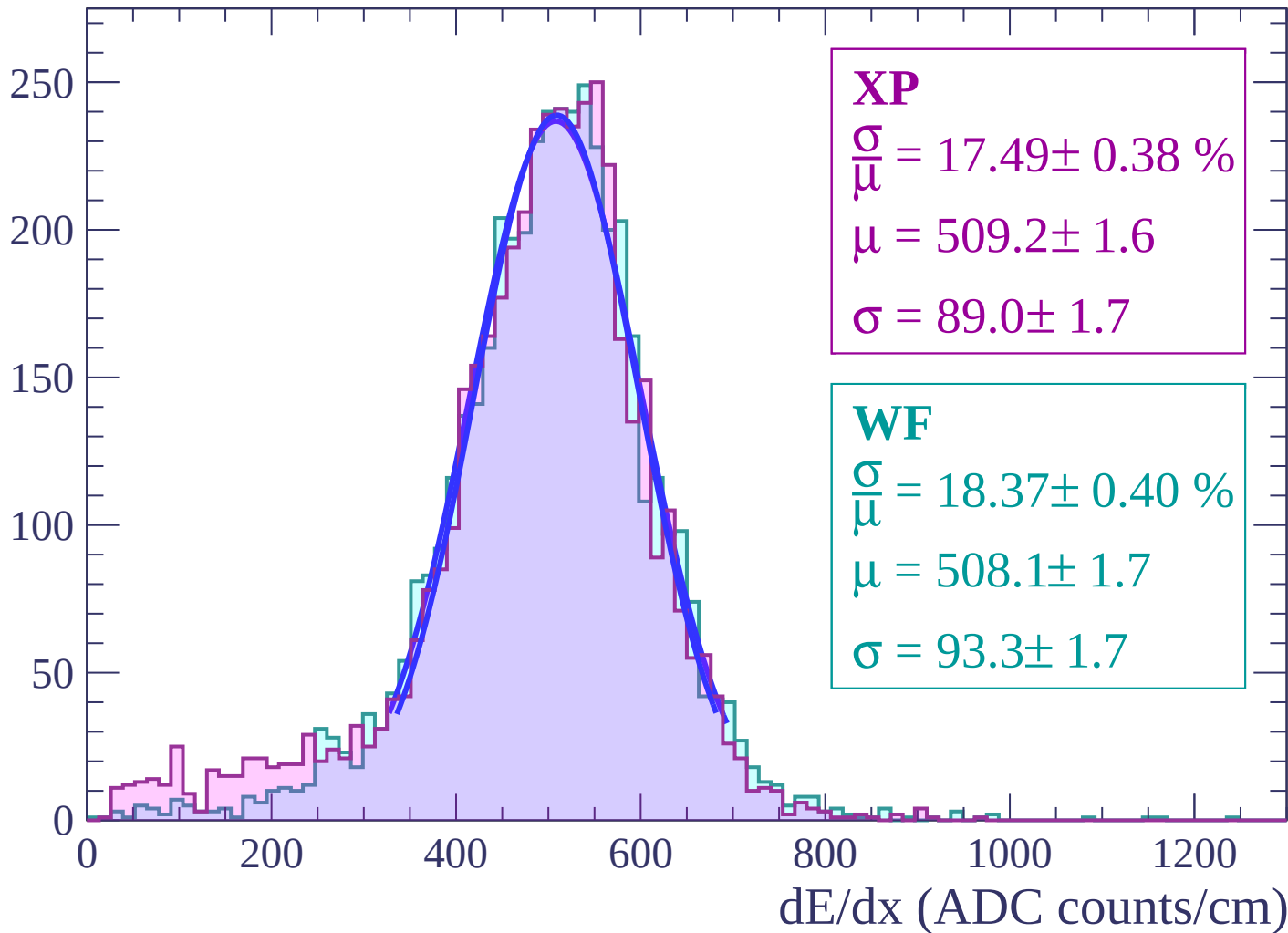
Energy loss | $48 < \theta < 50$

Count



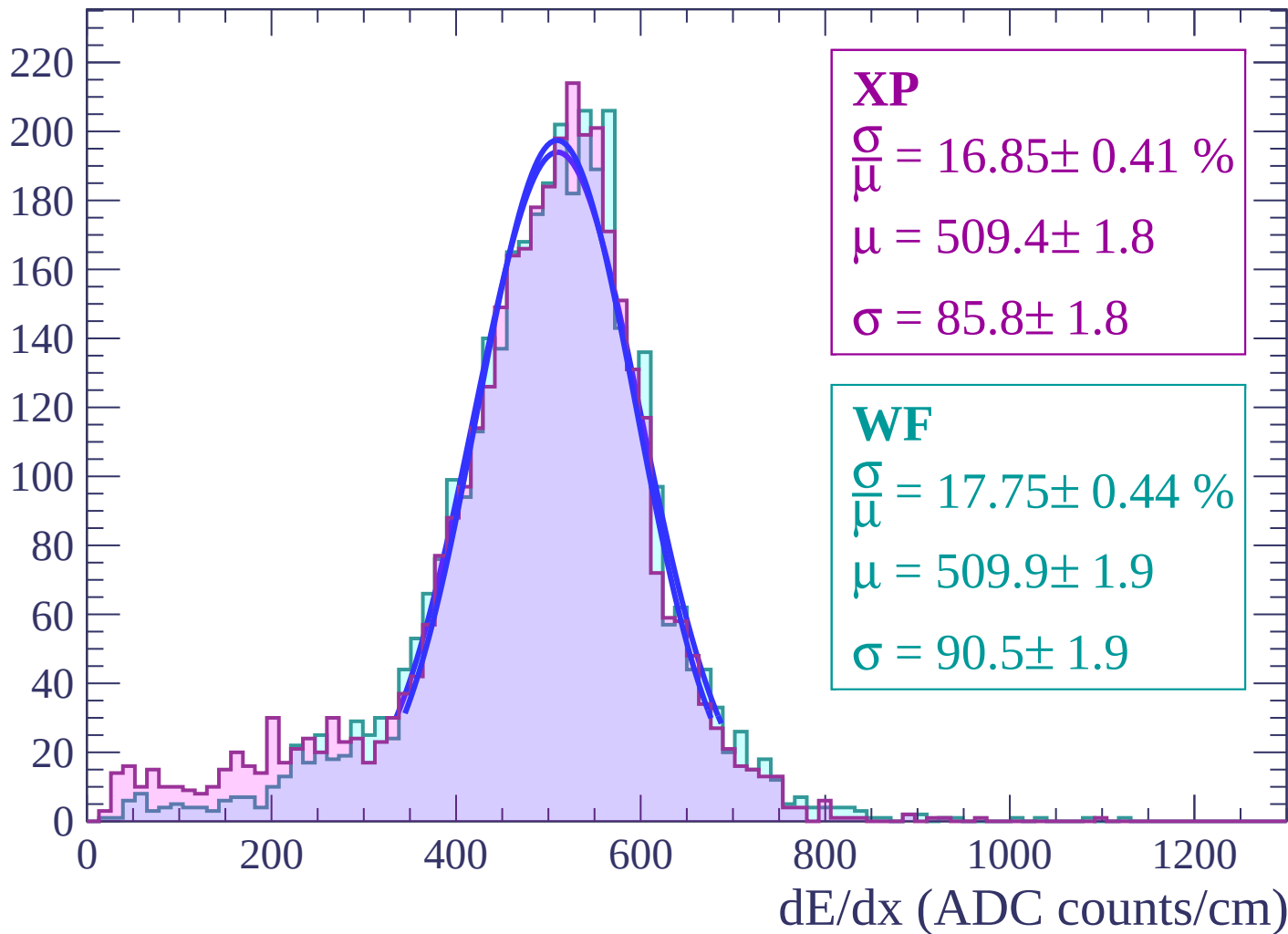
Energy loss | $50 < \theta < 52$

Count



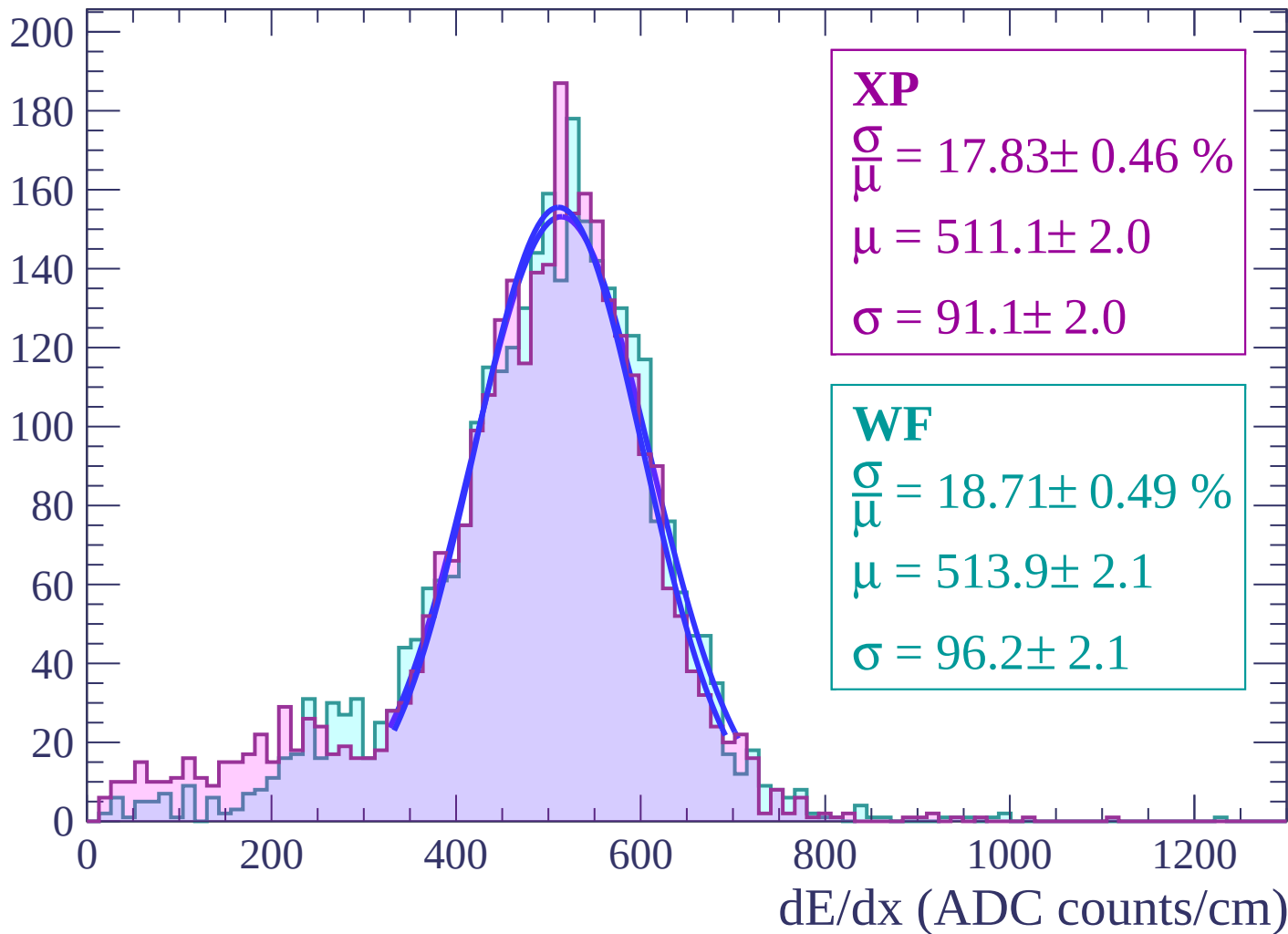
Energy loss | $52 < \theta < 54$

Count



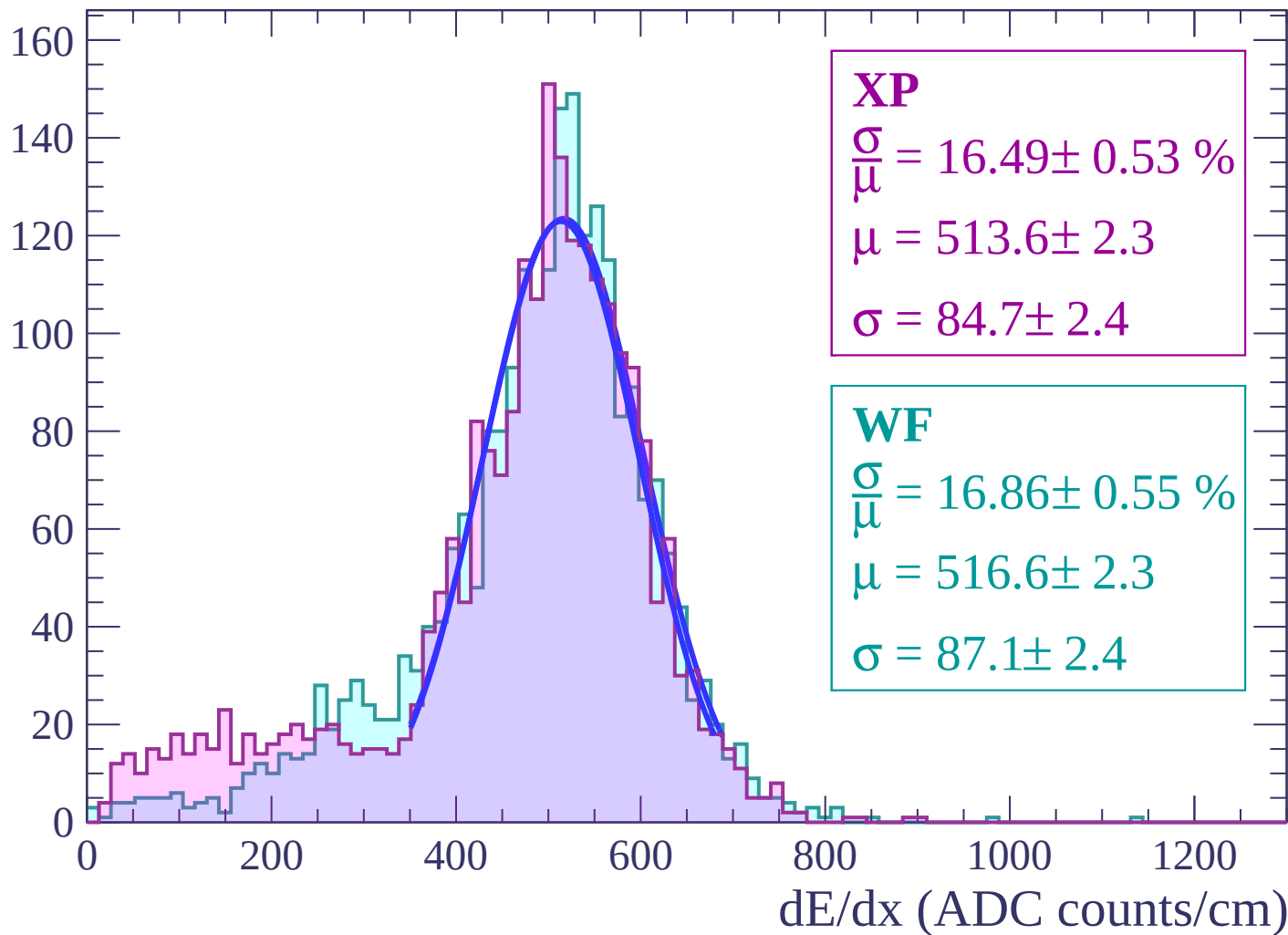
Energy loss | $54 < \theta < 56$

Count



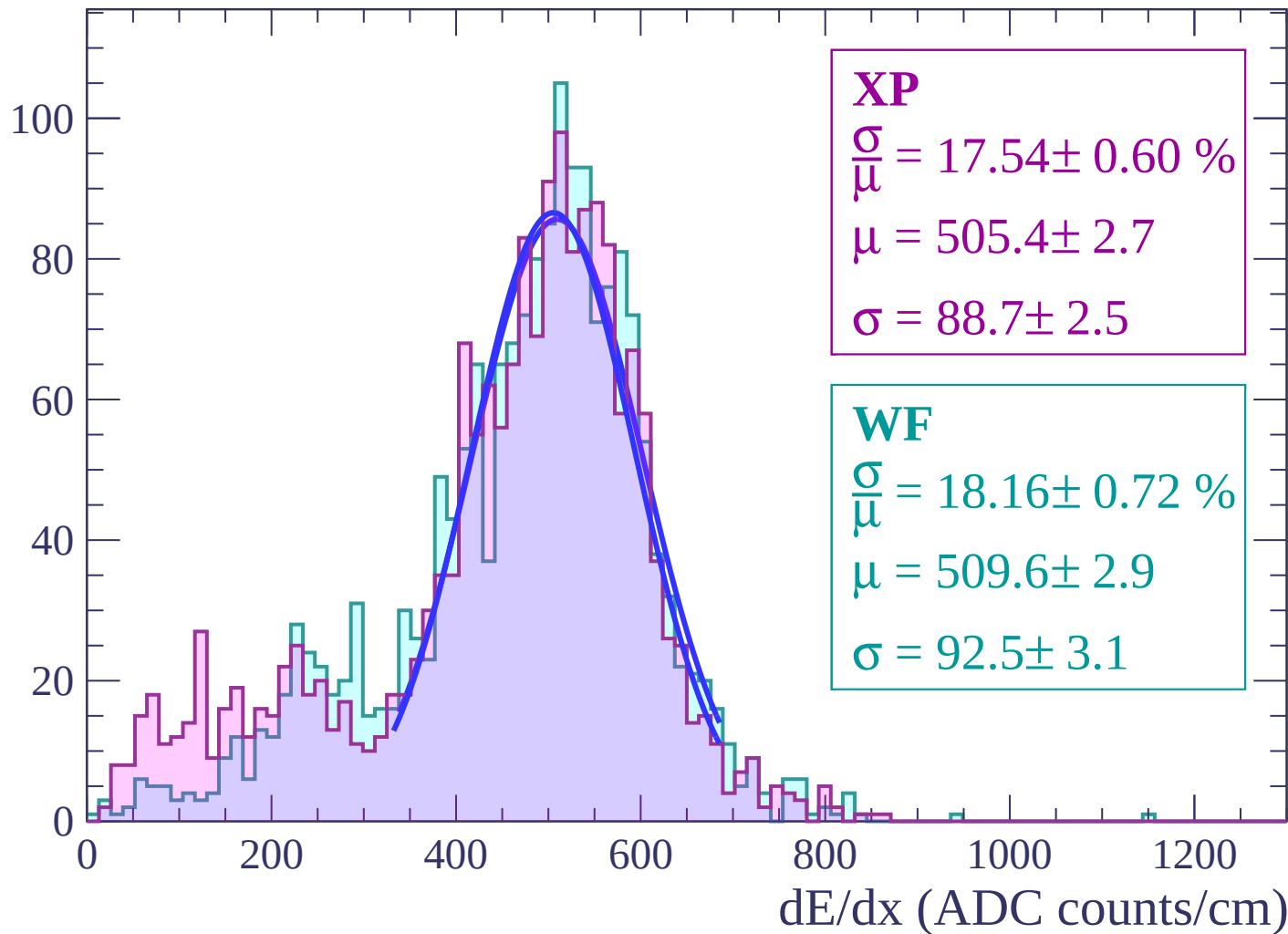
Energy loss | $56 < \theta < 58$

Count



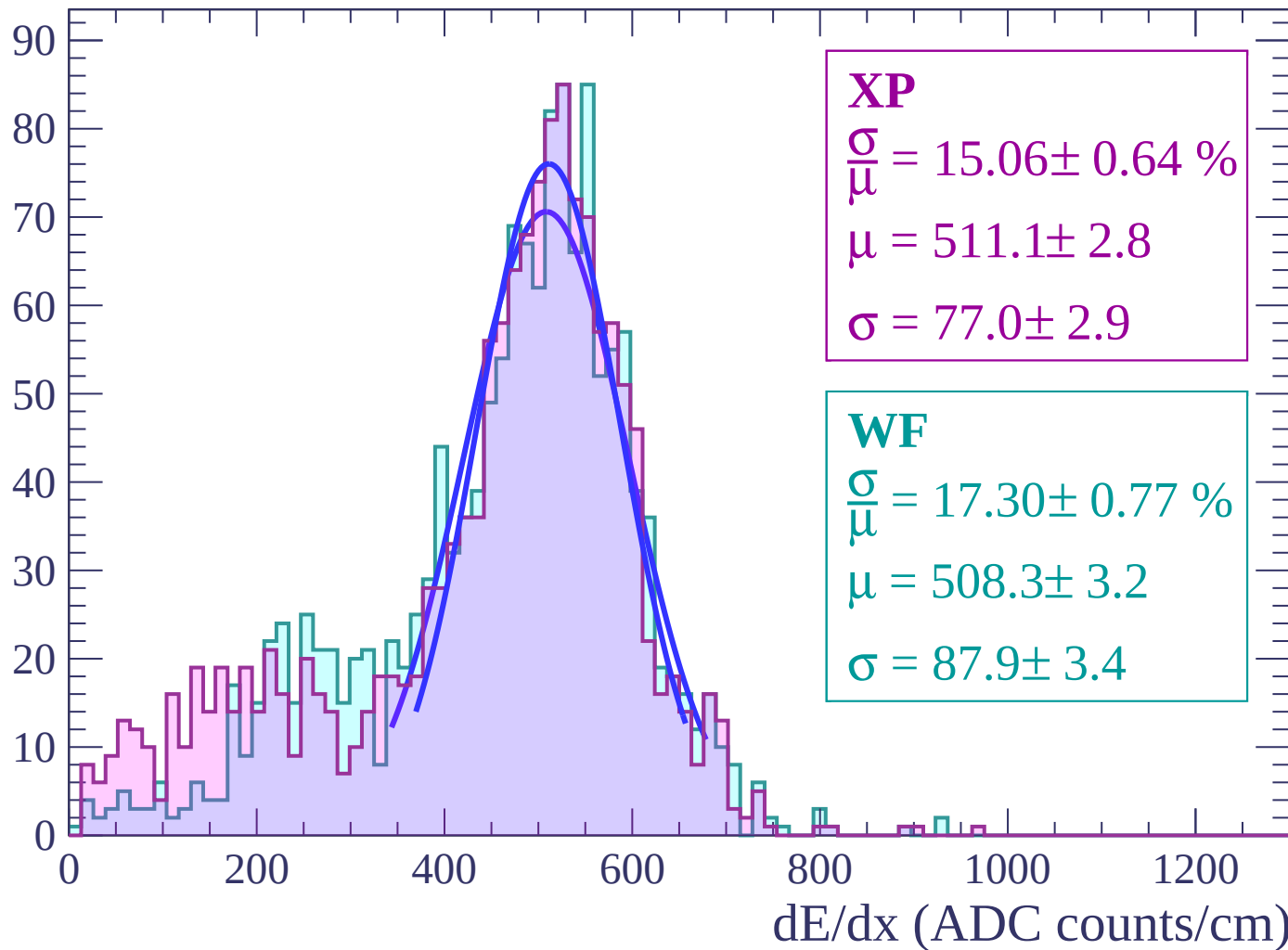
Energy loss | $58 < \theta < 60$

Count



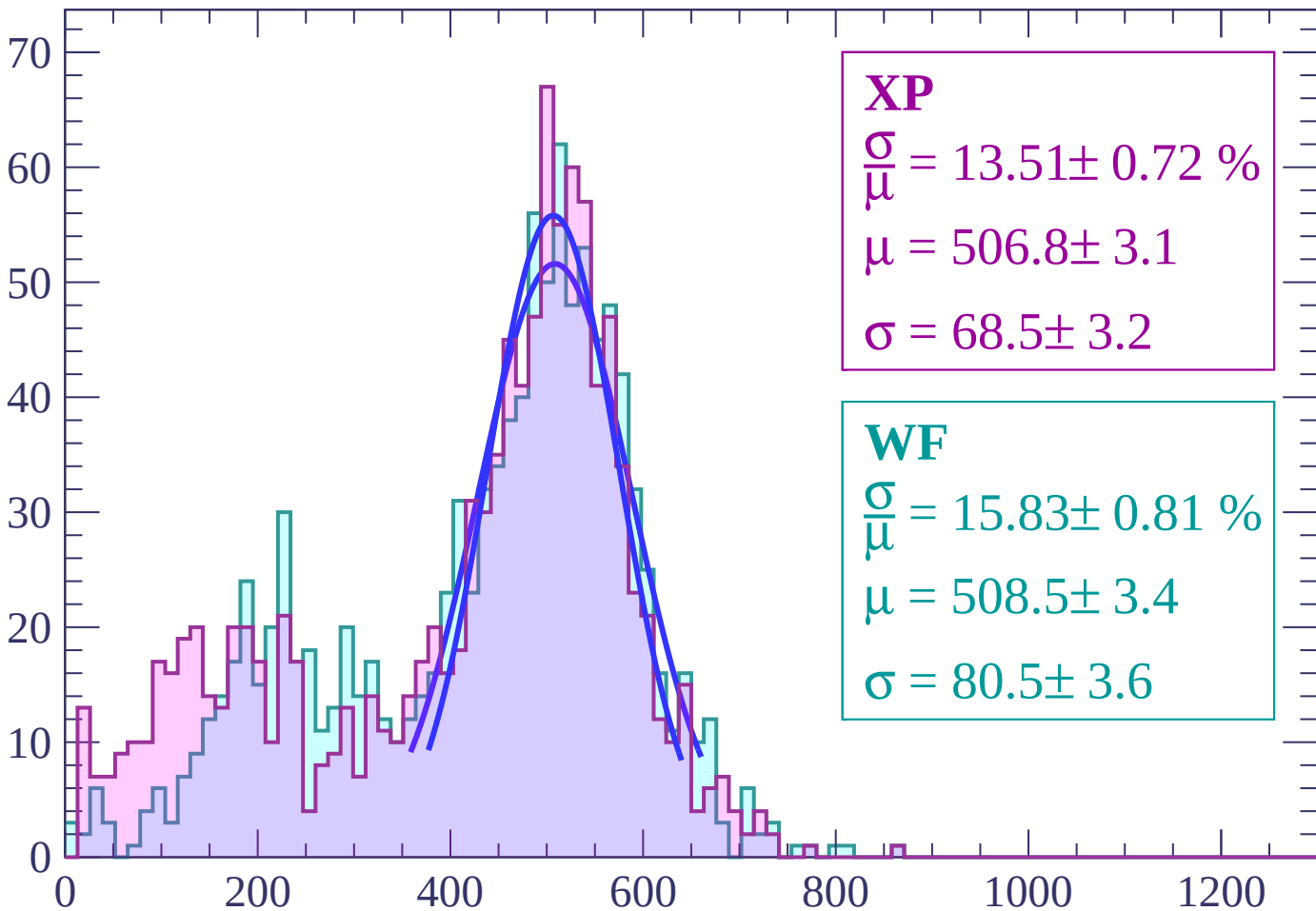
Energy loss | $60 < \theta < 62$

Count



Energy loss | $62 < \theta < 64$

Count



XP

$$\frac{\sigma}{\mu} = 13.51 \pm 0.72 \%$$

$$\mu = 506.8 \pm 3.1$$

$$\sigma = 68.5 \pm 3.2$$

WF

$$\frac{\sigma}{\mu} = 15.83 \pm 0.81 \%$$

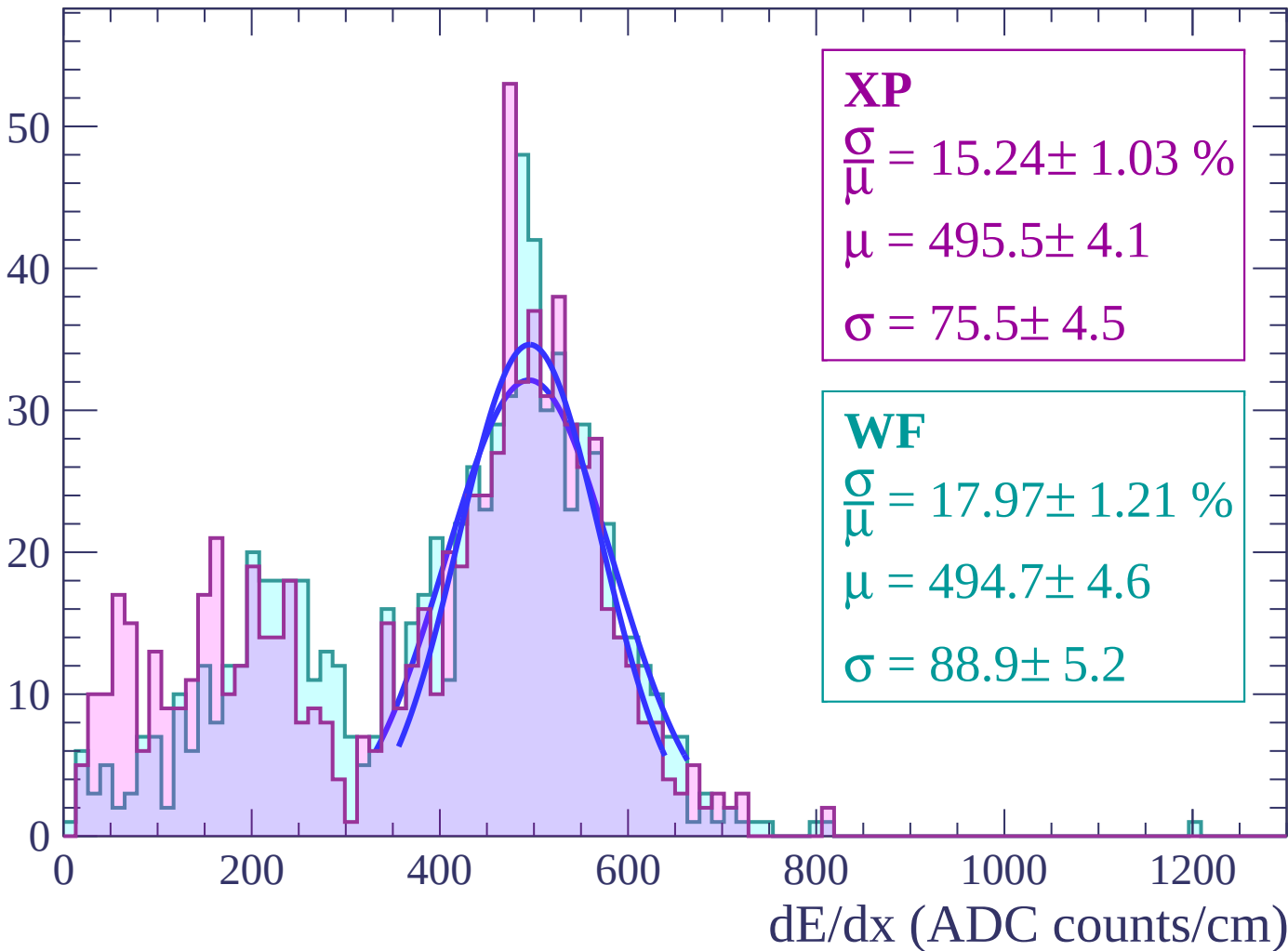
$$\mu = 508.5 \pm 3.4$$

$$\sigma = 80.5 \pm 3.6$$

dE/dx (ADC counts/cm)

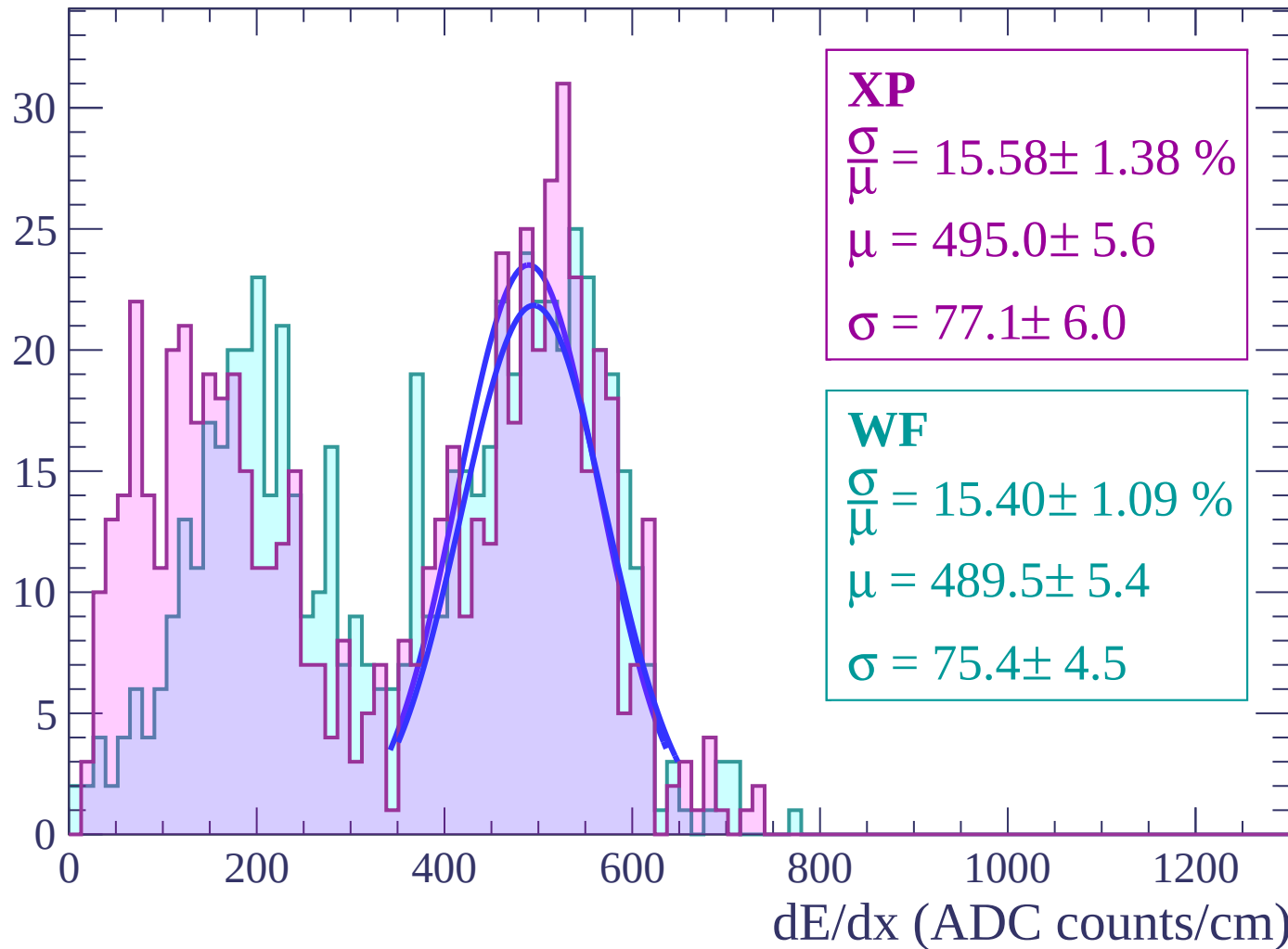
Energy loss | $64 < \theta < 66$

Count



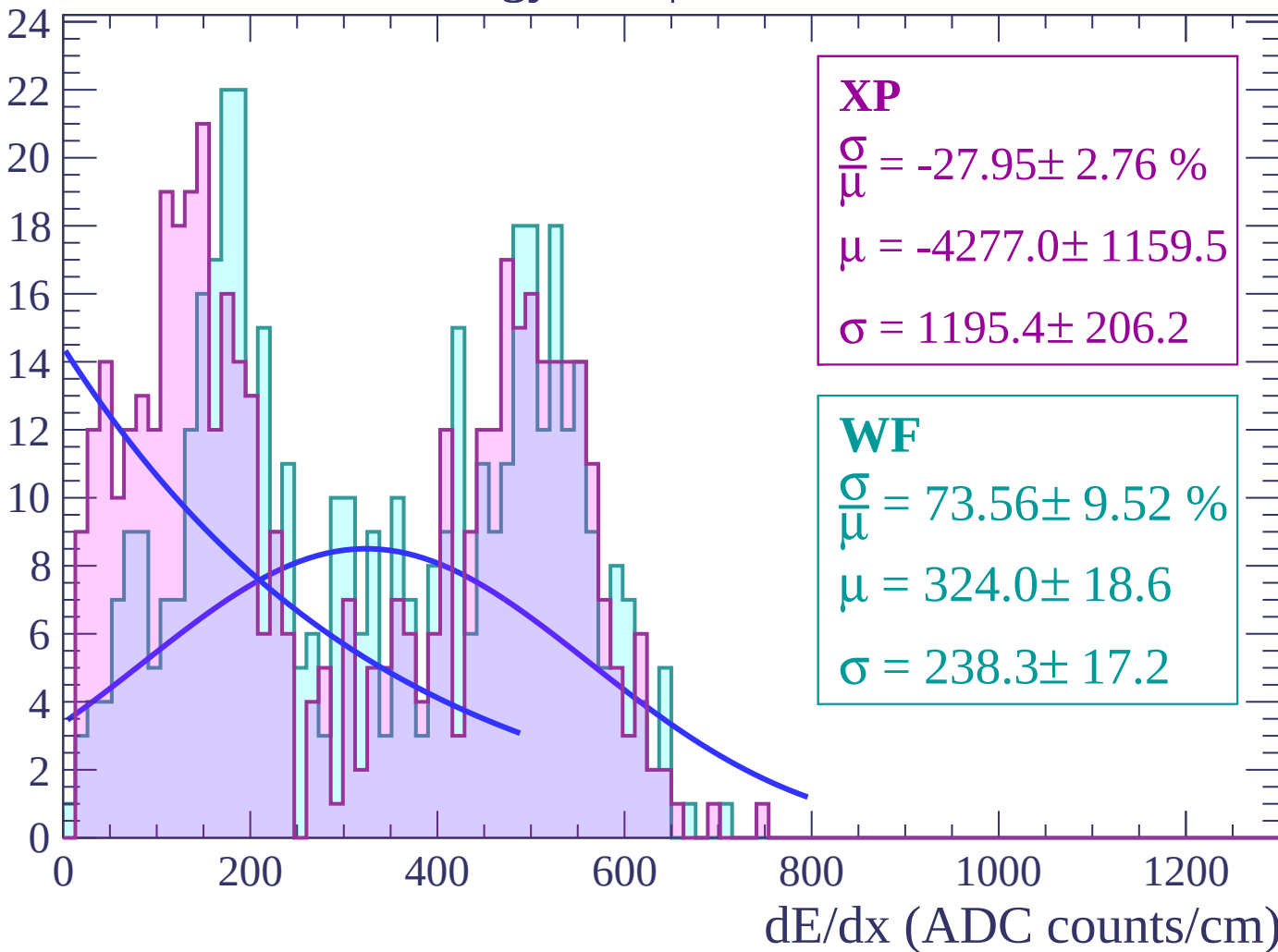
Energy loss | $66 < \theta < 68$

Count



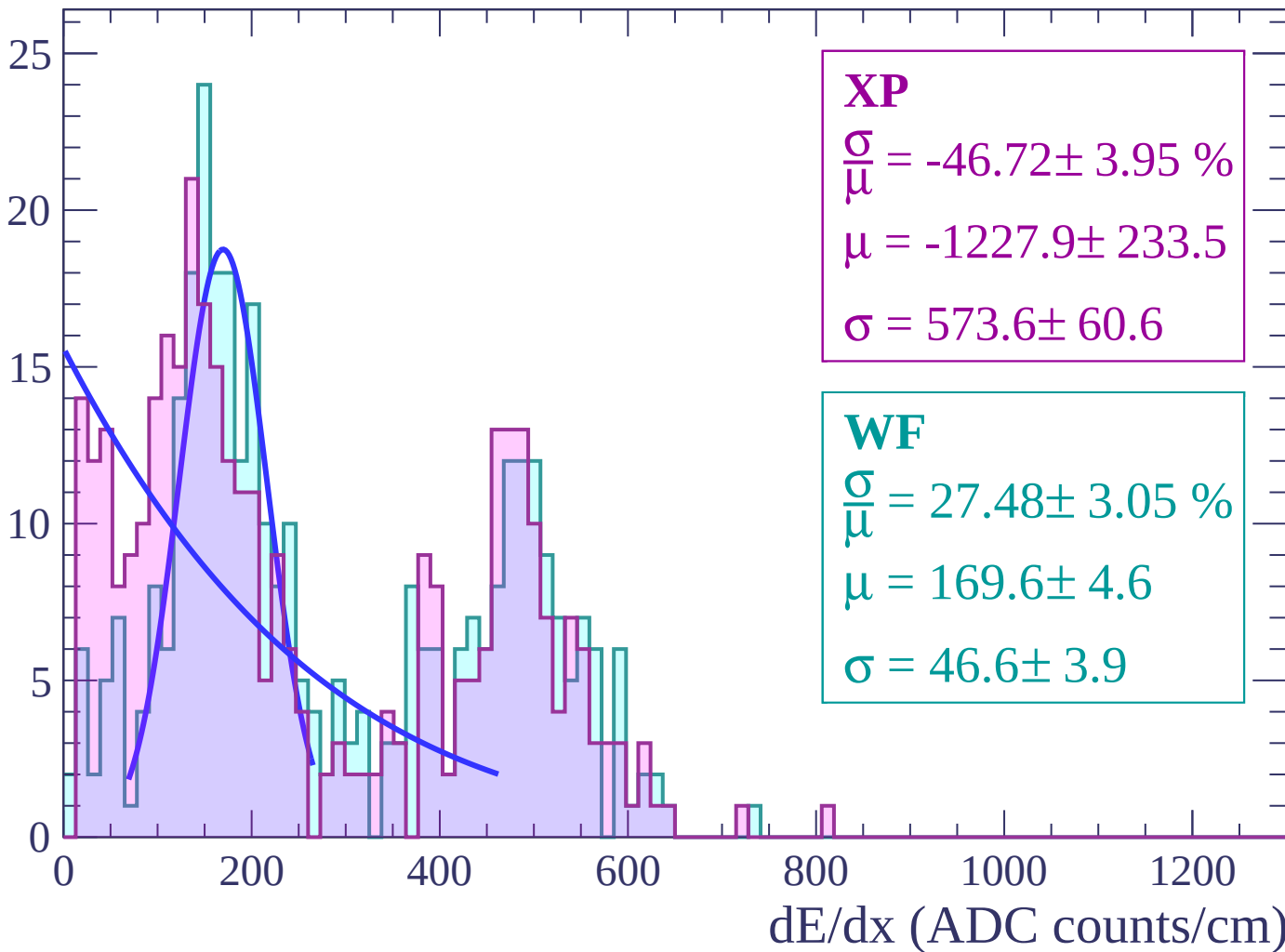
Energy loss | $68 < \theta < 70$

Count



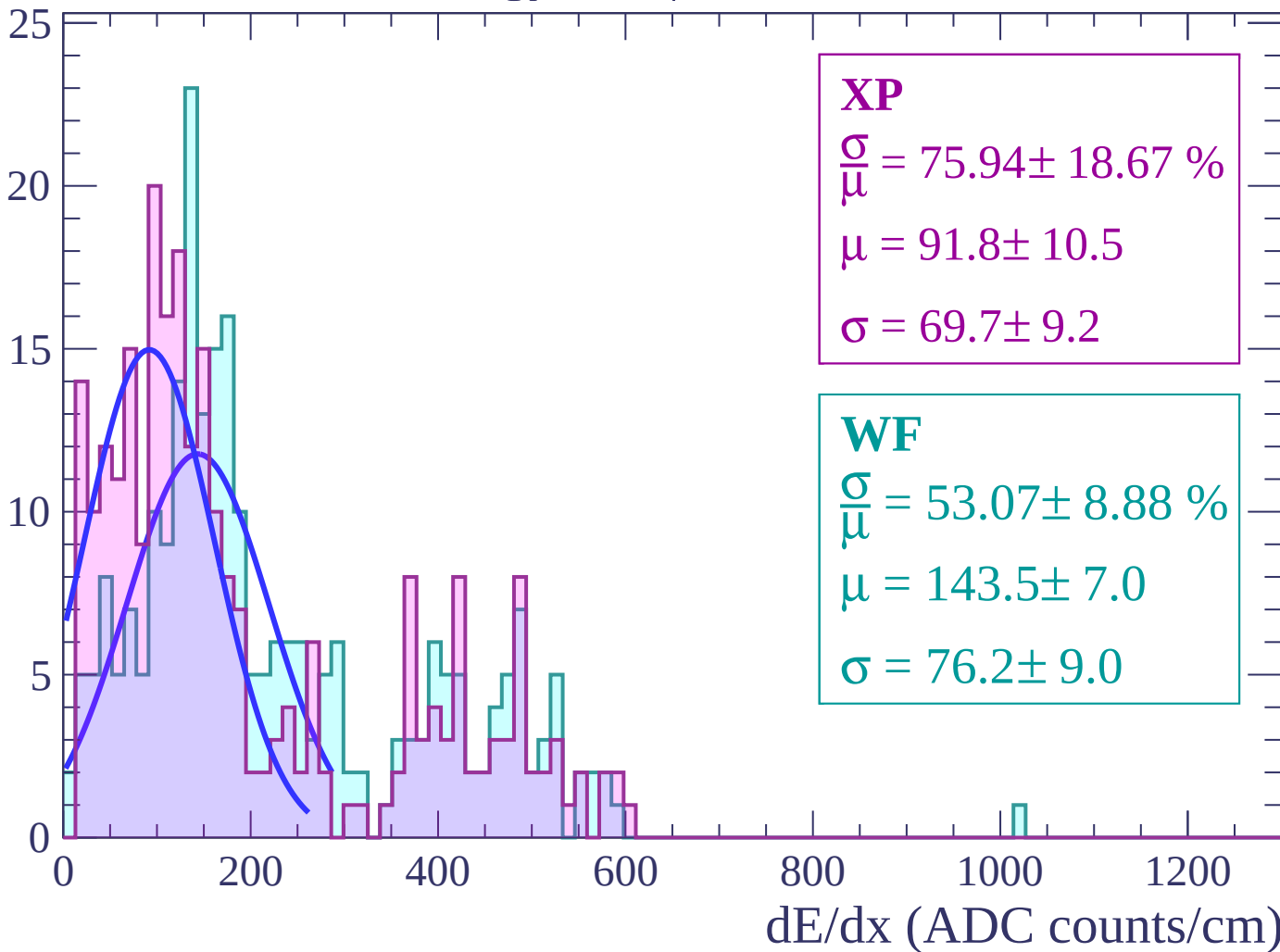
Energy loss | $70 < \theta < 72$

Count



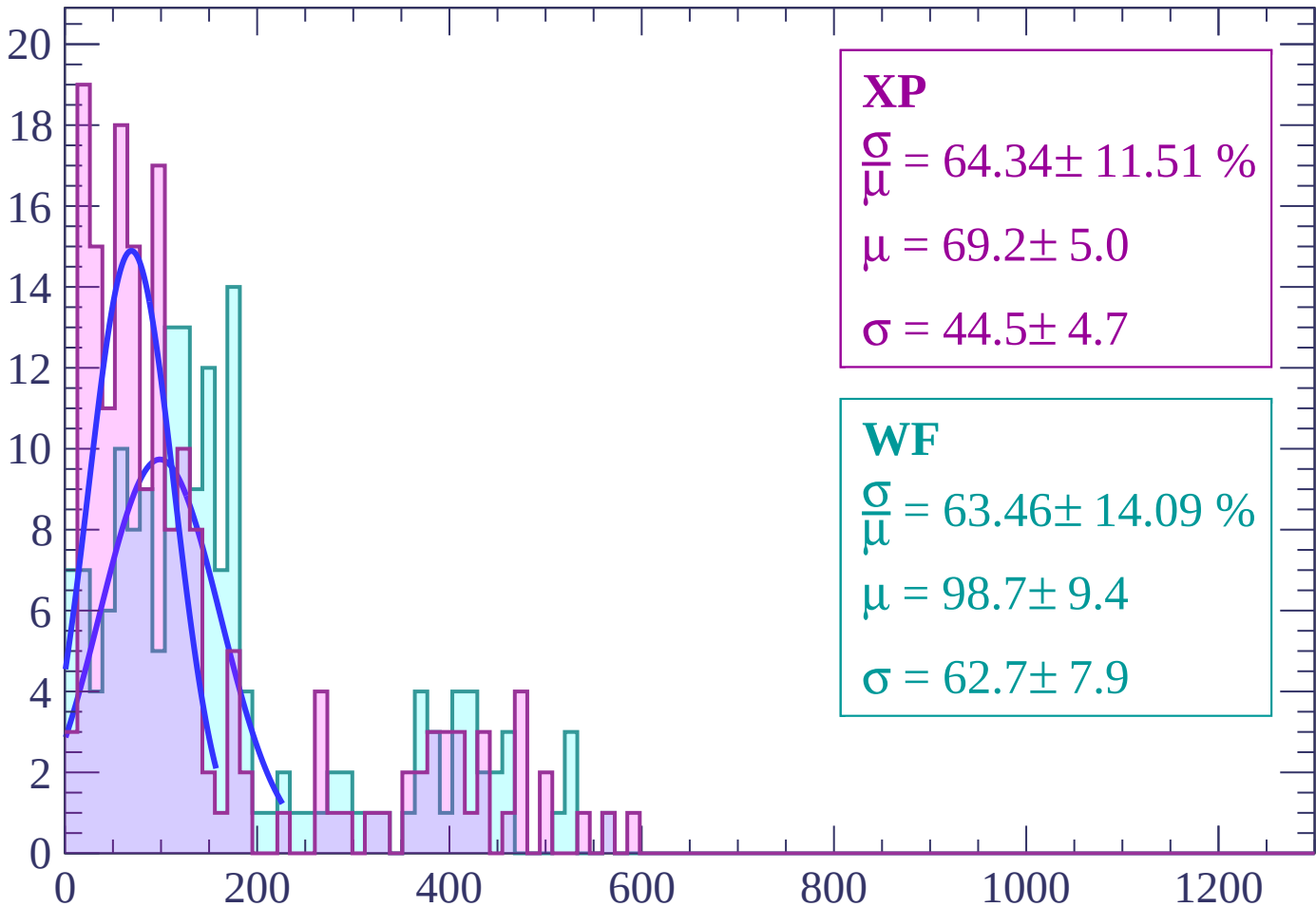
Energy loss | $72 < \theta < 74$

Count



Energy loss | $74 < \theta < 76$

Count



XP

$$\frac{\sigma}{\mu} = 64.34 \pm 11.51 \%$$

$$\mu = 69.2 \pm 5.0$$

$$\sigma = 44.5 \pm 4.7$$

WF

$$\frac{\sigma}{\mu} = 63.46 \pm 14.09 \%$$

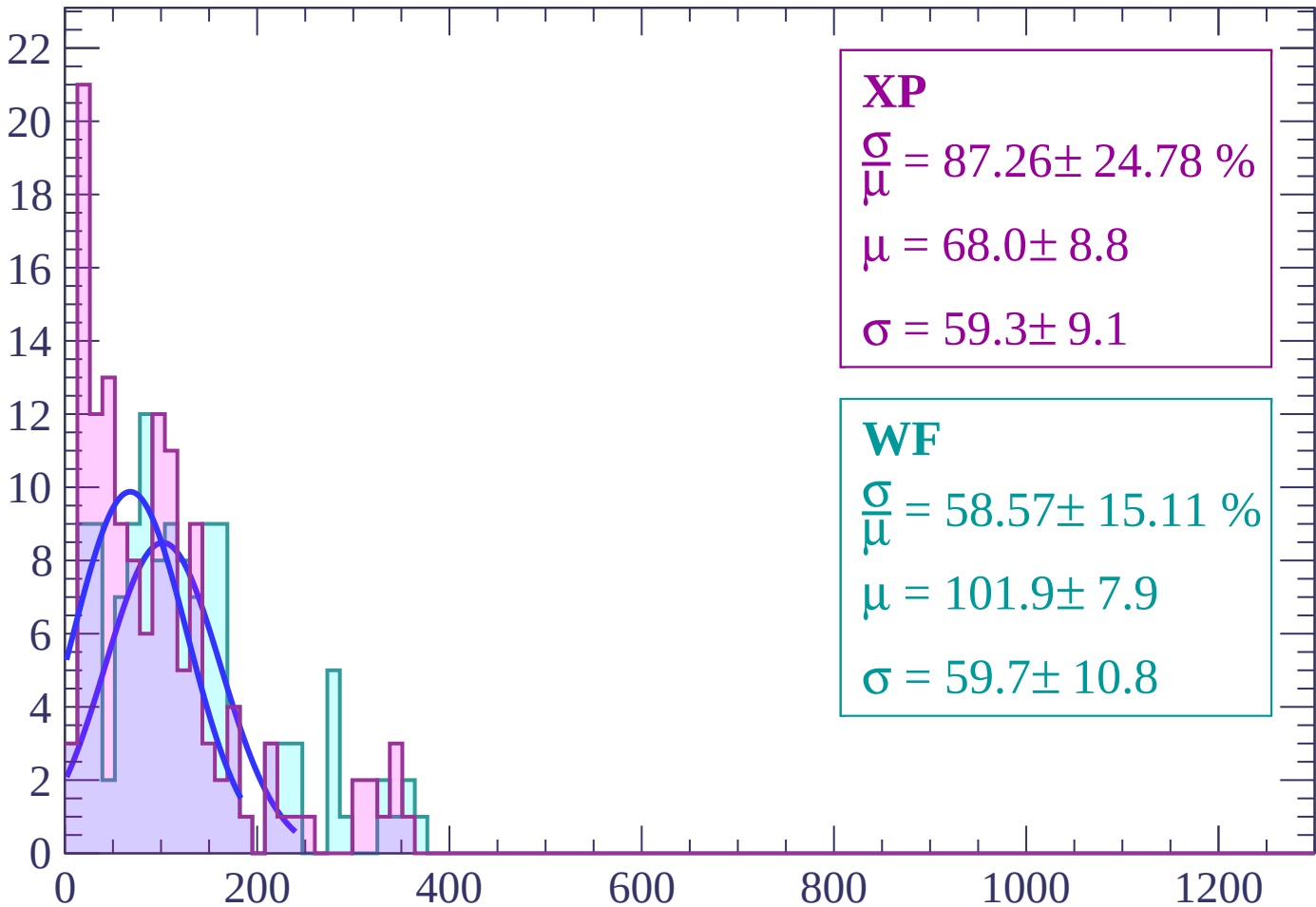
$$\mu = 98.7 \pm 9.4$$

$$\sigma = 62.7 \pm 7.9$$

dE/dx (ADC counts/cm)

Energy loss | $76 < \theta < 78$

Count



XP

$$\frac{\sigma}{\mu} = 87.26 \pm 24.78 \%$$

$$\mu = 68.0 \pm 8.8$$

$$\sigma = 59.3 \pm 9.1$$

WF

$$\frac{\sigma}{\mu} = 58.57 \pm 15.11 \%$$

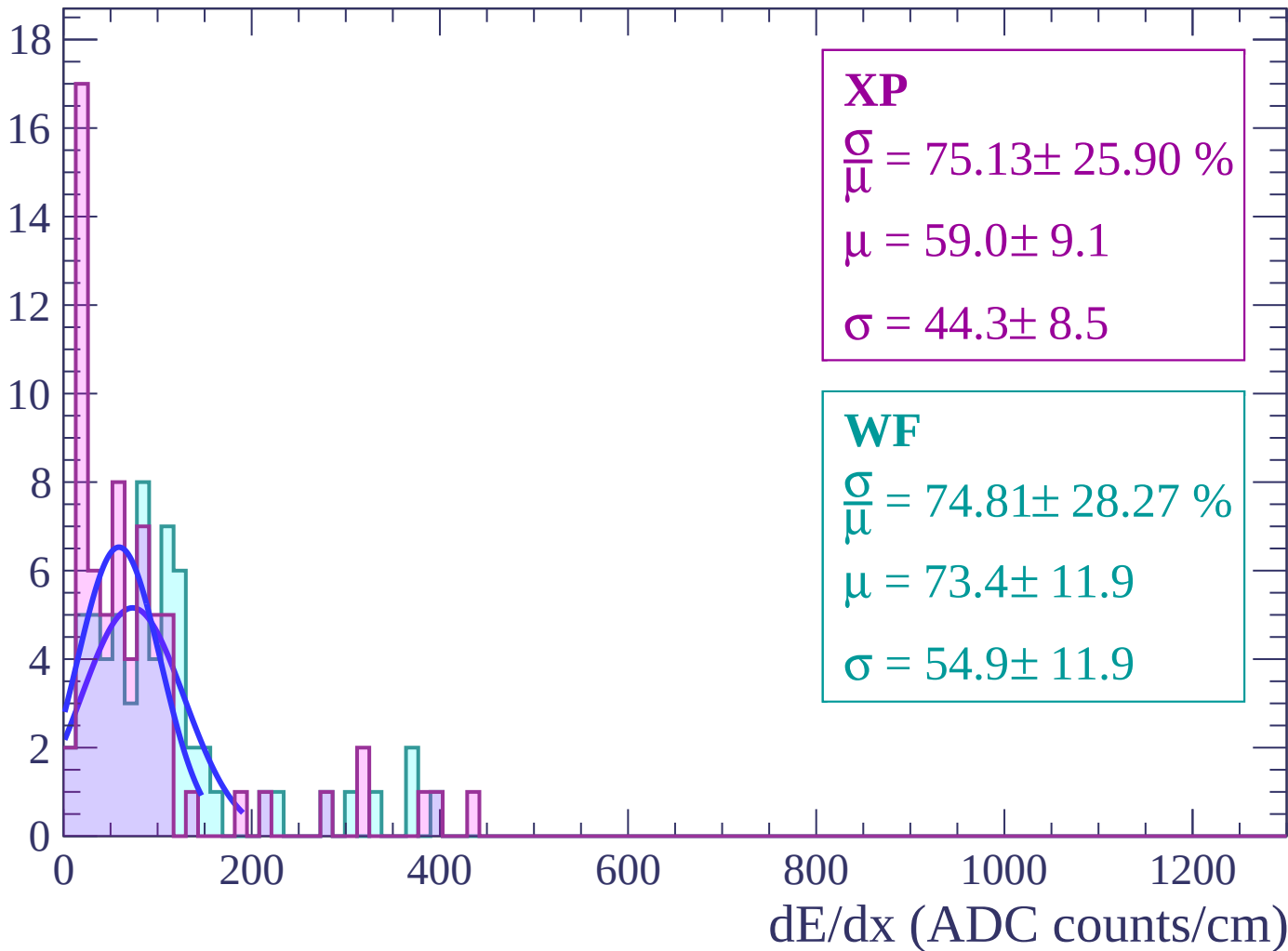
$$\mu = 101.9 \pm 7.9$$

$$\sigma = 59.7 \pm 10.8$$

dE/dx (ADC counts/cm)

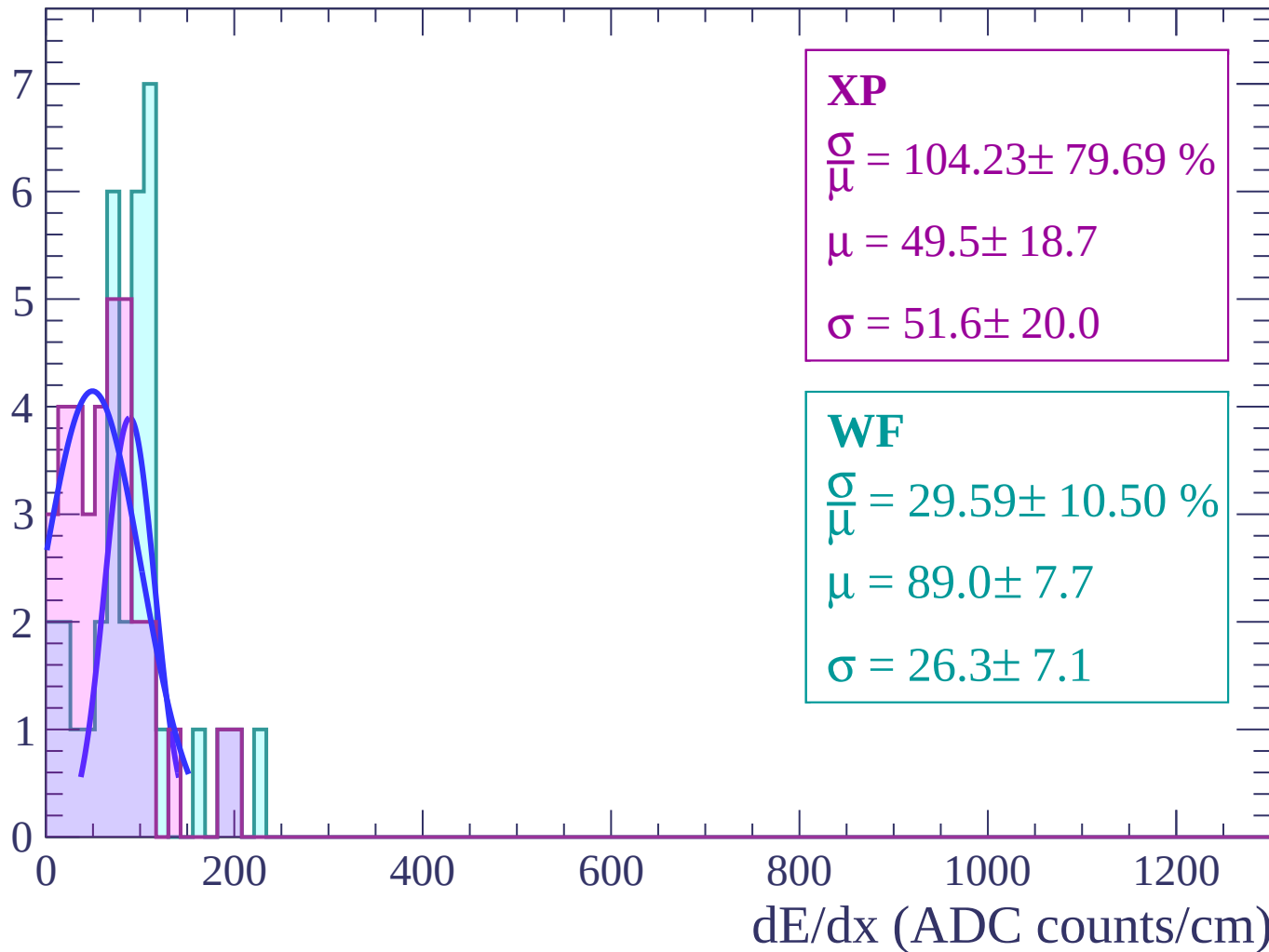
Energy loss | $78 < \theta < 80$

Count



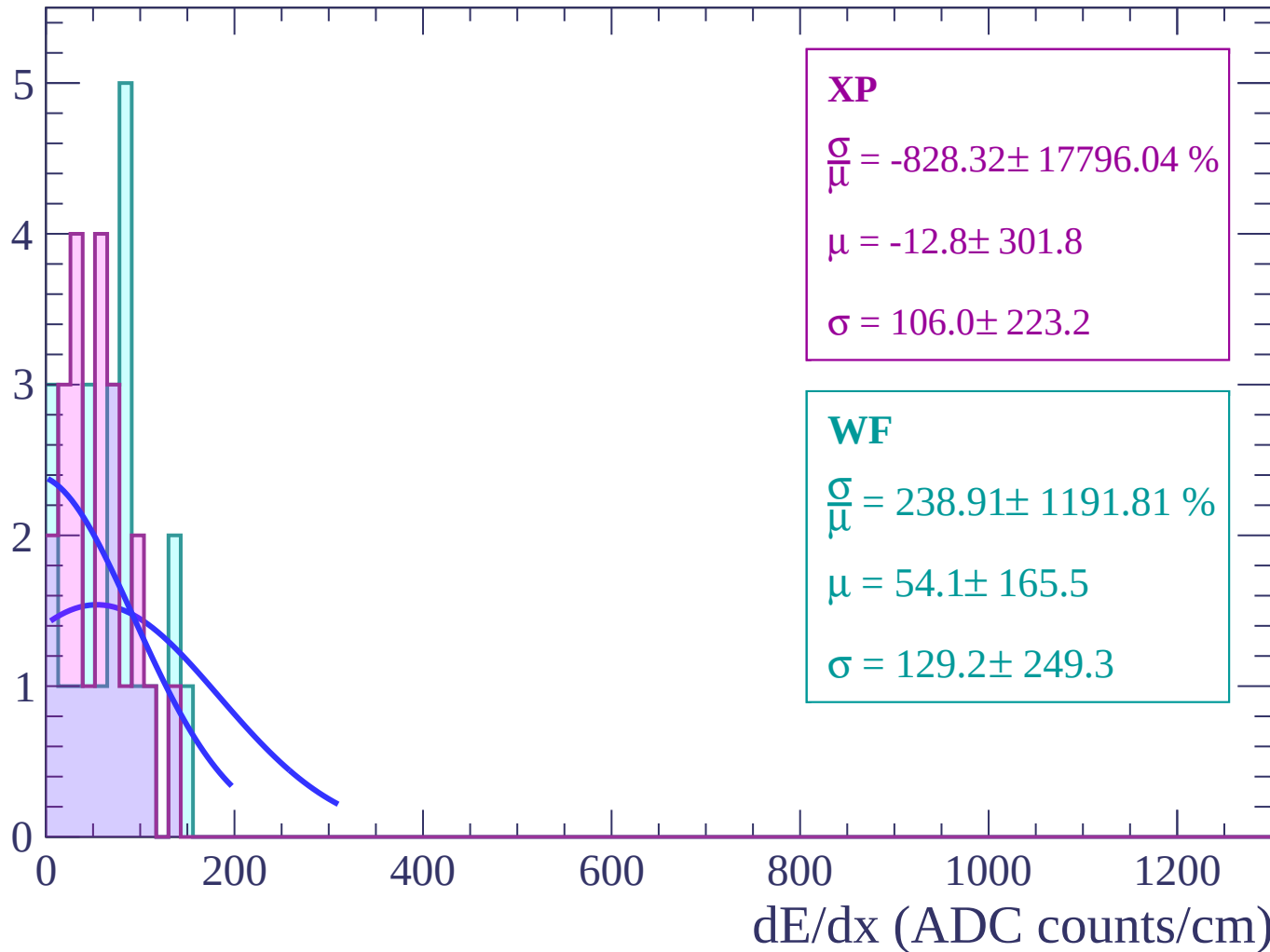
Energy loss | $80 < \theta < 82$

Count



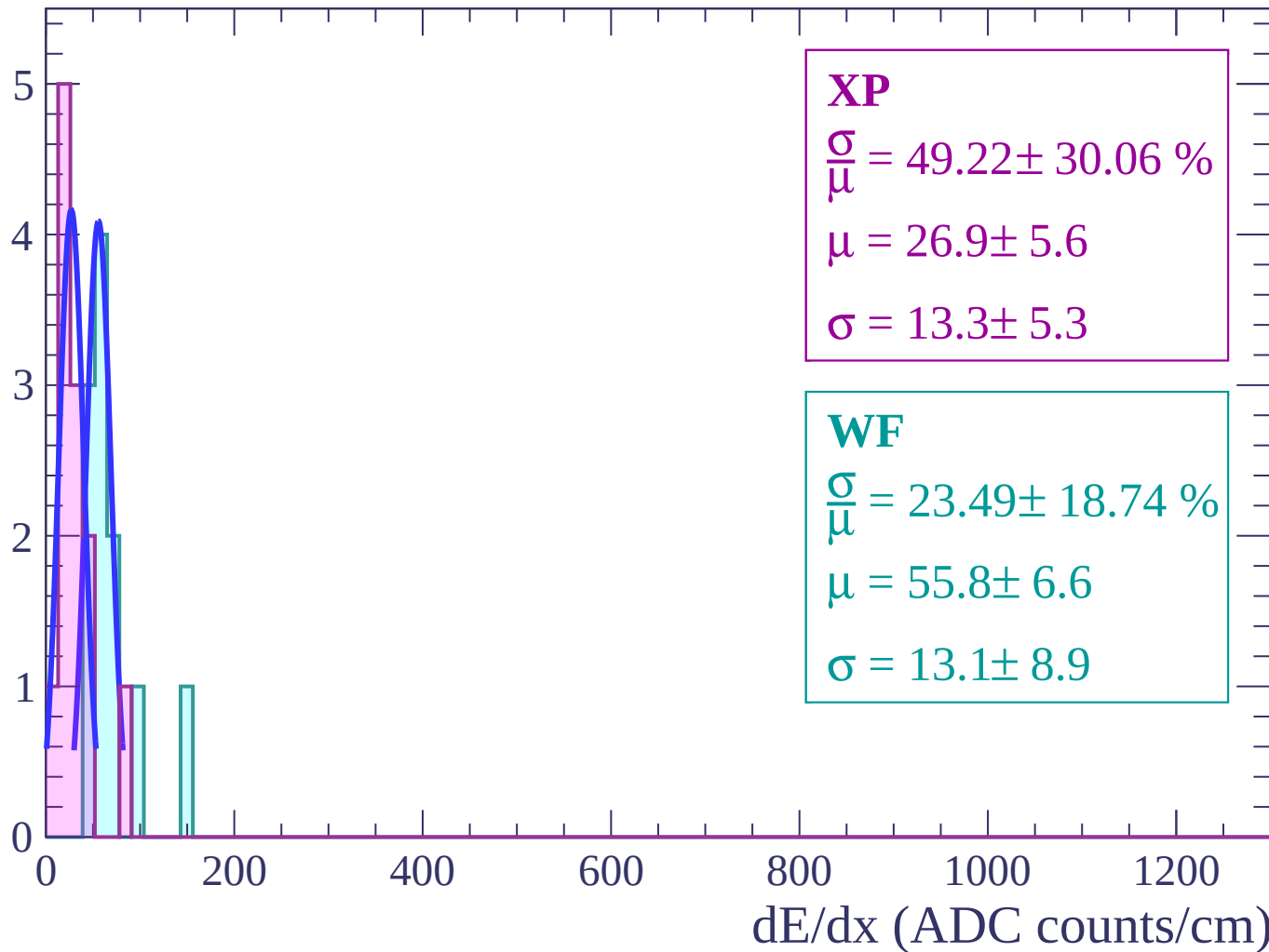
Energy loss | $82 < \theta < 84$

Count



Energy loss | $84 < \theta < 86$

Count



Energy loss | $86 < \theta < 88$

Count

1.0

0.8

0.6

0.4

0.2

0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $88 < \theta < 90$

Count

1.0

0.8

0.6

0.4

0.2

0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 4.16 \pm 21.99 \%$$

$$\mu = 6.2 \pm 2.4$$

$$\sigma = 0.3 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 4.16 \pm 21.99 \%$$

$$\mu = 6.2 \pm 2.4$$

$$\sigma = 0.3 \pm 1.3$$

Energy loss | $90 < \theta < 92$

Count

3.0

2.5

2.0

1.5

1.0

0.5

0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 4.16 \pm 38.30 \%$$

$$\mu = 6.2 \pm 1.5$$

$$\sigma = 0.3 \pm 2.3$$

WF

$$\frac{\sigma}{\mu} = 4.16 \pm 38.30 \%$$

$$\mu = 6.2 \pm 1.5$$

$$\sigma = 0.3 \pm 2.3$$

